Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 — UNIT 1 · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_01 · UNIT 1

1 UE26ME142A · Unit_01: UNIT 1

This unit master organizes source material for Engineering Mechanics 1, Unit_01, officially titled UNIT 1. It includes 16 video transcripts and 16 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26ME142A, Unit_01 (UNIT 1). Bucket folder present: yes. Videos: 16; slide decks: 16.

EV0000001

2 UE26ME142A_Unit 1_Class 1.1_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 1.1_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000002 · EV0000003

Graphs and diagrams



Figure G000006. UE26ME142A Unit 1 Class 1.1 Lecture PPT · source slide page 1.

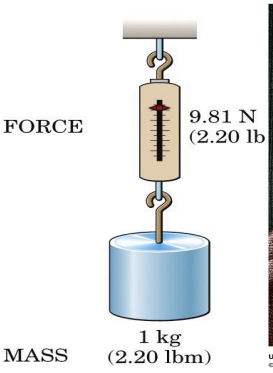
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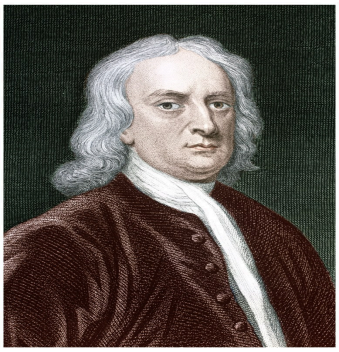
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

Mechanics – Statics and dynamics

Space
Mass
Time
Particle
Rigid body
Elastic body
Plastic body
Force
Types of force
Units of force
Newton's laws
Characteristics of force





Sir Isaac Newton

S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction

Figure G000007. UE26ME142A Unit 1 Class 1.1 Lecture PPT · source slide page 2.

EV0000305

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

Mathematics

Similar triangles
Law of Sines
Law of Cosines
Trigonometry
Integration

Deformation
Displacement
Deflection

Translatory motion
Rotatory Motion

Resultant
Coordinate system

S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction

Figure G000008. UE26ME142A Unit 1 Class 1.1 Lecture PPT · source slide page 3.

EV0000307

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

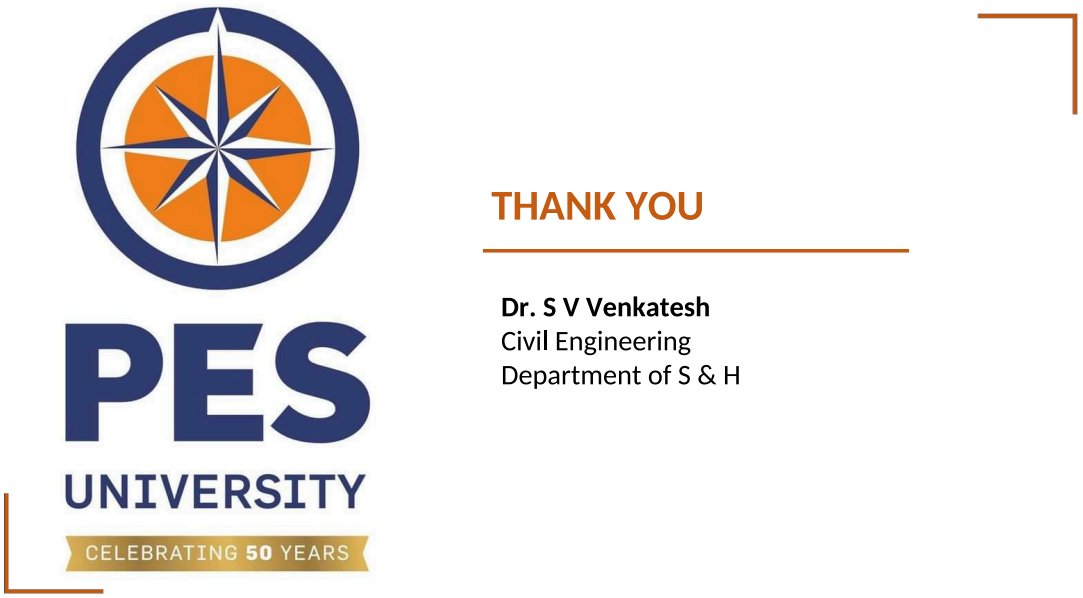


Figure G000009. UE26ME142A Unit 1 Class 1.1 Lecture PPT · source slide page 4.

EV0000308

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000301 · EV0000302 · EV0000303
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Mechanics – Statics and dynamics Space Mass Time Particle Rigid body Elastic body Plastic body Force Types of force Units of force Newton's laws Sir Isaac Newton Characteristics of force S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction EV0000304 · EV0000305 · EV0000306
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS Mathematics Deformation Translatory Displacement motion Similar Deflection Rotatory Motion triangles Law of Sines Resultant Law of Cosines Coordinate Trigonometry system Integration S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction ## Page / slide 4 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000307
4. Source slide page 4: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000308

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.740] Dear students, welcome back to the classes on engineering mechanics statics. This is class number two. I am Dr. Sv. from Civil Engineering under Department of Science and Humanities. [00:00:19.012] under PS University. In the very first class we discussed a few things regarding this subject, that is what is

indicated here in this slide. We came to know what is meant by mechanics, the different between statics and dynamics. [00:00:39.108] Then the definitions, few of the definitions of space, mass, time, particle, rigid body, elastic body, plastic body. Then we define what is a force, what are the types of forces, what are the units of forces in S side. [00:00:57.764] What are Newton's laws? And then we discussed about the characteristics of forces. In between, I also discussed about the prerequisite for understanding these subjects. That is mathematics, then physics, and of course [00:01:18.500] English that was required. Okay. So we'll continue with that in this class. This is class number two. Okay. So let's look at few of the mathematical things that are required for understanding engineering mechanics. [00:01:36.196] First of the foremost thing is similar triangles under geometry. You all know what are similar triangles. So if any two triangles, if the sides of the triangles of these two different triangles, if the sides are proportional to [00:01:54.788] uh each other the three different sides then we call that triangle as a similar triangle or if the three angles are proportional to each other in these two triangles we call them as similar triangles or the combination of this [00:02:10.944] One side and two angles or two sides and one angle, if they are proportional to each other in these two triangles, then we call them as similar triangles. Then let's understand what is law of science. If you take a triangle, any triangle, then [00:02:30.916] If a side is the distance of the side is taken as capital A, the angle opposite to this side in the triangle is taken as small A. Similarly, the other side is taken as capital B, the distance, the angle opposite to this side is taken as [00:02:50.832] Small b, the last side, if the distance is taken as capital C and the angle opposite to this side is taken as small a, then the sides of the triangle. [00:03:03.952] Will be proportional to the sine of the angle opposite to the other two angles or opposite to this side or angle between the other two sides. So that means A by capital A divided by sine of small A, that is the angle made by the [00:03:22.948] Uh other two sides other than capital A is small. So A divided by sine A is equal to B divided by sine B is equal to C divided by sine C. So that's what the law of science talks about. Okay. [00:03:39.664] Next, let us look at what is law of cosine. What is law of cosine? Again, we take a triangle, any general triangle. If the sites are named as the distances of the sites are named as A, B, and C, and the angle opposite to side A is taken as small A. [00:03:59.940] Then we can say that the distance of the side A, capital A, A square is equal to [00:04:09.252] B square plus C square minus 2 B C into cos of angles A. Okay. So that means. [00:04:21.700] When you have a triangle, we can determine the any side of the triangle if we know two sides and one angle which is opposite to the side of the distance for which we have to calculate, then we can do it easily by using law of cosines. Okay. [00:04:37.860] Next comes what is complementary angle. All of you know that complementary angles sum up to 90 degrees. Similarly, what is supplementary angles? Supplementary angle is the sum of two angles will come to 180 degrees. [00:04:56.260] Okay, what are corresponding angles? Corresponding angles will always be equal.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:02.020] In magnitude. Okay. Alternate angles are also will have equal angles between them. The vertically opposite angles also will be equal to each other. So these things have to be known when it comes to [00:05:16.880] geometry and this we will be using exclusively when we are going ahead and learning engineering mechanics statics. And of course tripometry basics of trigonometry is required. Then of course basics of integration is also required to certain extent. [00:05:35.332] Now what are these requirements mathematical requirements is also given in the prescribed test textbook, that is

Engineering Mechanics Statics by Miriam Craig and Bolton, eighth edition, published by [00:05:52.224] while publication in the appendix uh chapter. So we usually ask our students to have a copy of this textbook so that they can refer to this whenever it is required for them to understand internal mechanic statics. Okay. Next comes this [00:06:11.300] Few things that we need to understand. That is first one is deformation. What is deformation? No, when you take a body and there is a force applied onto this body, the body's shape will change. [00:06:29.156] But the position of the body would not change. Such [00:06:35.076] Changes when you apply a force is called as deformation. The position of the body does not change, only the shape of the body changes. The best example is you take a sponge and when you apply force onto this, the shape of the sponge. [00:06:52.592] Will change whereas the position does not change. That's the best example for understanding what is deformation. The next one is displacement. [00:07:01.636] Now, you take a body, and when you apply a force onto body, for example, let me take this pen. Okay. Now this pen, the pen is in a position A, say, and when I apply a force onto this, the pen would move from position A to position B. [00:07:21.092] This change in the position of the object because of the application of the force is called as displacement. The body is getting displaced from one point to another point because of application of the force. Okay. Next. [00:07:38.720] What is defluction? Defluction is the best example for defluction is you can take the electrical wire between two poles. We have two poles and in between we have an electrical wire. Okay. Initially, when [00:07:55.712] The electrical people are tying this, they would have given enough tension to this wire so that it becomes almost horizontal. Over a period of time, because of its own weight, this wire would [00:08:12.260] actually bent or it will take a shape of a [00:08:16.740] uh bow okay when you have a bow inverted or kept it horizontal you can see that it will have a shape uh uh like the uh sector of a s curve okay or it will take the shape of a curve it's a sector of a circle okay [00:08:33.476] Now that is because of the sulfate of the wire itself. So initially it would have been horizontal. No, if you take any point on this wire, the [00:08:47.076] Body has undone the shape change because it was horizontal, now it has changed to a curve one. And again the position of that point would have changed, it would have come down. So there is both deformation and displacement occurring simultaneously in [00:09:04.128] an object, such changes is defined as deflection. Okay next let's define what is resulting. [00:09:14.596] Now if you are if I take an object and if I apply n number of forces on this object, okay, so there could be a [00:09:25.380] There could be a change in the position of the body, or the shape of the body can change, or there could be deflection. Any of this or combination of this can occur. [00:09:36.836] Now, instead of applying n number of forces onto this body, if I apply only one force which can bring in the same change as that of the n number of forces on this body, then that single force [00:09:54.368] is called as a result. Okay. [00:09:58.372] Next, let's look at this coordinate system. In the last class also, I talked about coordinate system, whether it is three-dimensional or two-dimensional. When it comes to three-dimensional coordinate system, we consider all the three axes that is X, Y, and Z.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.656] But I also told you that we use the right hand first three fingers to take or to note the directions of the coordinate system. For example, if I take the thumb as x-axis. [00:10:31.328] Then the forefinger will be y-axis and the middle finger will be the z axis. Now you can observe that all these three axes are going to be mutually perpendicular to each other. The three fingers that I am taking of my right hand is also mutually perpendicular to each other. [00:10:48.112] To each other. Now, if we require all these

three coordinate systems to define the position of a body, then that is called as a three-dimensional force system. [00:11:03.460] Now, if I take an object, say for example I take this pen, okay, this pen has got length and it has got diameter, so it's a three dimensional body. [00:11:12.740] But when I if I apply a force onto this, say I will take this as x-axis and this as y-axis, and I am applying a force onto this pen along the x-axis or only along the y-axis or only in the plane x y. Okay. [00:11:29.520] I'm not going to be applying the force normal to this XY or along the Z axis, then I can represent these forces only in these two planes, that is X and Y. Then [00:11:44.452] The coordinate system that I can consider can be only 2D coordinate system. Now this you can do it only if you have the body in one plane, the forces acting also in the same plane, and the effect of the [00:12:02.404] Forces onto this body will also occur in the same plane. Then we can treat this as a two-dimensional force system. Okay. Next, we will look at what is translatory motion. [00:12:19.140] Now, what is translative motion? I talked about displacement earlier when I told you that if there is a force applied on to a body, if the position of the body changes from one position to another position because of application of the force. [00:12:39.108] that without any change in the the shape of the body, then we call that as translatory motion. [00:12:49.060] The translatory motion can occur along x-axis if I take if I take this, okay. If the body moves only along X, that is translatory along X. If the body moves only along Y, it is Y translatory motion. The body can move along Z also. [00:13:06.628] Forward and backward with respect to the screen, okay. Z axis, that is also translatory motion. Now. [00:13:14.980] There could also be rotatory motion because of the application of the force. What is this rotatory motion? Say, for example, let me take this pen, okay, and I apply a force. Because of which, and I take this as x axis, the thumb is x-axis, the forefinger is the y-axis, and the [00:13:31.344] Middle finger is an axis. If on application of the force, if this pen rotates, rotates about the x-axis, this is x axis, okay, this is x axis, if it rotates about the x axis like this, okay. [00:13:47.908] Then that is translatory motion along X. Y axis, this is the Y axis, if the body rotates along the y-axis, then it is rotation along y. [00:13:59.716] Then you have z axis here, if the body rotates along the z axis, then it is rotation along the z axis. So if I take a point in space, a point can have orc. [00:14:17.156] Any translatory motion of this point because of application of the force can be represented along this three-coordinate system that is translatory motion along X, translatory motion along Y, and translatory motion along Z. [00:14:34.628] Similarly, this point can rotate about x axis, about y-axis, and about z axis. [00:14:43.684] Best thing that we can do is or to understand this is you take a plane. Okay. A plane which is moving. Okay. If it is moving along X direction, then that is a transfer to motion along X. Okay? Then if it rotates like this, this is rotation important.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:03.780] The x-axis, okay. Then if the plane goes up and down, up and down, that is translatory motion along y. If the plane rotates along the horizontal plane, if it rotates like this, if it rotates like this, then this is rotation along y. [00:15:23.140] Then if the plane moves perpendicular to the screen, to and fro, okay, that is transmittery motion along Z, and if it rotates like this, then it is rotation along Z axis. [00:15:41.732] I hope all of you have understood what are the different [00:15:46.372] uh coordinate systems and what is translatory motion and rotatory motion. So if I take any object in space it will have three translatory motion or any translatory motion can be represented along these three axis. [00:16:03.232] Similarly, any rotatory motion of the point in space can be represented along these three axes, that is x, y, and z axis. Okay. Now, if we are going to be considering only two-

dimensional, okay, say for example, I'm going to take a plane which is perpendicular to the screen, okay. [00:16:23.108] Then if I take the x axis that is horizontal axis and the vertical axis as y, so the plane is perpendicular to our screen. Okay. So what are the different uh translatory motion and rotatory motion that we can have? [00:16:39.620] So the body cannot come out of the plane. Okay, it has to be in the plane itself. So it can move along X. Okay? It can move along Y? So it can have [00:16:51.204] It cannot move along the Z because once we say it is moving along Z, then it is it is coming out of the plane. Okay? So it is not possible for a two dimensional. Okay? So it can move along X, it can move along Y, it cannot move along Z. Next. [00:17:07.236] Now, those are the two transverse motions that the body can have. Now coming to rotatory motion. [00:17:14.212] Say I take this uh pen itself, which is in this plane, okay? So can it rotate about x? If it has to rotate about x, then it has to come out of the plane. So that is not possible. Can it rotate about the z axis? [00:17:29.488] No, this is the rotation along about z axis. No, it is not possible, it is coming out of the plane, no. But can it rotate about the z axis? Now if it rotates about the z axis, this is the rotation that it can have. So it is not coming out of the plane, XY plane, so that is possible. [00:17:47.204] So for a two-dimensional object, it can have [00:17:52.100] Two transfer motion, one along X and one along Y, if I am taking XY plane, and it can rotate about the Z axis. It cannot rotate about the X axis, it cannot rotate about the Y axis. Okay. So I hope all of you have understood the [00:18:09.200] whatever I've discussed till now. Of course in addition to this mathematics, all of you should be knowing about addition, subtraction, [00:18:19.588] Division and multiplication. Along with this, you should also know what is dot product when it comes to vectors because all forces are vectors. So we should know what is addition of vector, subtraction of vector, what is dot product of vectors. [00:18:36.848] vector and cross product of vector. Okay. These things have to be known by you before we start to understand engineering mechanics statics. Okay. So with this we will now [00:18:55.172] Close today's discussion. This is class number two on engineering mechanics statics, which is a subject which is common to all students who are entering B Tech program in the first year. I am Dr. SV Venkatesh, a civil engineering. [00:19:14.916] from Department of Science and Humanities under PS University. Hope to see you in the next class. That is class number three. Thank you very much. Good day.

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Full transcript: 2712 normalized words, 77 timestamped segments, 1167.381 seconds duration. Low-confidence segments flagged: 0.

EV0000002 · EV0000003 · EV0000004 · EV0000005

3 UE26ME142A_Unit 1_Class 1_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 1_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000018 · EV0000019

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.516] Dear student, welcome all of you for engineering mechanics classes. I am Dr. Svinkitesh, civil engineering from the Department of Science and Humanities. I welcome all of you for [00:00:19.716] the first year courses under PS University. I will be teaching you the subject engineering mechanics statics. [00:00:34.628] which is a common subject for all students who are entering into B Tech program under PS University. So let's look at this subject in detail. So [00:00:51.972] I will start with the first chapter, which is Introduction to Statics. For this subject, we are using a textbook by the title Engineering Mechanics Statics. And we are using the eighth edition. [00:01:10.212] This addition is in SI version. [00:01:16.548] for this book is JL Meriam and LG Craig. So we will start with this. And you can see here the front [00:01:33.604] covering page or the face page of the book. It comes under two different colours, but the contents are same. The eighth edition is adopted by [00:01:50.884] J and Bolton and that's what you can see here. The author's name is LJ Craig. Sorry, J L Miriam, L G Craig, J and Bolton, and it's the same thing here also. [00:02:06.084] But only the difference between these two is the colour. One is in red colour, the other one is in green colour. You can buy whichever is available in the market. Okay? Both of them, the content inside is same, so you can take up whichever book that is available in the market. [00:02:25.764] Okay. So in the first introduction chapter, we will be dealing with this. First thing is what is mechanics? Mechanics is that branch of engineering which deals with the study of bodies which are under rest. [00:02:46.148] Which are in motion. There are two aspects for this. The first one is the bodies which are at rest, which is called as statics. The bodies which are in motion. [00:02:59.172] comes under dynamics. So mechanics is divided into two parts. One is the steady of bodies which are at rest which is called statics. The other one is the bodies which are in motion which is comes under dynamics. Okay. [00:03:15.332] Few of the basic definition that we need to know is listed here. Of course, all this you would have gone through in your earlier classes, which is before joining your MTEC, you would have gone through this. [00:03:34.532] Now what is the basic subject or the prerequisite that is required to understand this subject is of course one is English and uh we know that the medium of uh communication here is English itself. [00:03:51.716] And all of you have studied in English medium, or you would have changed from some other medium to English medium. Okay? So you should know English in total. Then of course physics. [00:04:06.628] This subject is actually an extension of whatever you have learned in your physics. There, you would have learned the basics. Here you are going to learn the application of the basics with respect to engineering. [00:04:23.396] Then you should know mathematics. In mathematics, all the basics that you have learned is what is required. For example, arithmetic, which is addition, subtraction, multiplication, and division must ensure. Then algebra, linear algebra. [00:04:42.276] Then trigonometry, then integration, which is calculus. And of course very importantly geometry. The geometry plays a very important role when you have to solve some problems. As we go on, you will come to know what are the things that we should know.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:01.604] Okay. So with this background, we will start. Now we will look at what is the definitions for space. All of you would have known what is space. How do you define space? You should know what is space. Space is the object. [00:05:20.228] which occupies the point which can be measured with respect to the three coordinate systems. We have three coordinate systems for defining any point on this earth or in

surrounding [00:05:39.652] We need to, we or we ref we refer to three coordinate systems that is x-axis, y axis, and the z axis. To know this, we use the right hand first three fingers, that is the thumb, the four finger, and the middle finger, and this uh [00:05:58.916] are mutually perpendicular to each other. If you take the thumb, the thumb if you take it as the x-axis, the forefinger represents the y axis and the middle finger represents the z axis. [00:06:15.780] Now if I want to define a point, okay, I will use these three coordinate systems to define this point. [00:06:24.036] So, space is the region occupied by an object which can be measured with respect to the three-coordinate system by the distance or by angular measurements. So we can define a point in space. [00:06:41.220] By the three coordinate systems, the distance as X, Y, and Z, which talks about what is the X distance of that point with from the origin measure along the x-axis. Then y distance is what is the distance measured from the origin. [00:07:01.124] Along the y axis, then Z distance is the distance measured from the origin along the z axis. This is one way of identifying a point in space with respect to the three different axes. We can also locate [00:07:19.172] The point by using the angular measurements. That is what is the angle made by this point in space with respect to the x axis that is theta x, then with respect to y axis that is theta y, then with respect to the z axis that is theta z. [00:07:39.172] Either we can have this three distance measurement or three angular measurement or the combination of this distance measurement or the angular measurement. Okay. So difference [00:07:59.108] Between this is 3D. 3D is when we use all the three coordinate systems, we go for what is called as three-dimensional objects. Now, if we can represent the given object and the forces acting on them along any one plane. [00:08:19.044] Then we can define that or we can represent it as a two-dimensional object. Okay. So that's the difference between 3D and 2D. 3D is when we have to define. [00:08:30.564] The object by all the three coordinate systems, then we go for 3D, or it is called as three dimensional objects. If we can represent the object and the forces acting on them along any one plane using only two coordinate systems, for example, only X and Y. Okay, then we call that as [00:08:50.404] 2D. Okay. The next in the definition is mass. What is mass? It is the amount of substance that is present in a body is called as mass. Okay. Then how do we define time? [00:09:09.156] Time is defined as the measure of succession of events. For example, if we have to measure the or we have to know the time elapsed between [00:09:27.396] The instant that we leave our house and the instant that we reach the college, that measure between the instant that we leave our house and the instant that we reach the college, that measure of succession of events is called as time. [00:09:46.436] Now time comes into picture only when we are discussing the objects which are in motion. [00:09:58.340] We don't consider time when we are talking about objects which are at rest, that is, when we are dealing with statics. So we don't talk about time in statics, but time is a very good measure when it comes to dynamics. Okay. Next, how do we define a particle?

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:18.212] A particle is defined as an object which occupies a single point in space which has mass, which has negligible dimension. [00:10:34.692] With respect to naked eyes. Of course, when you go to nano level, you can certainly measure the dimensions of a particle. Now, what do we call or what do we say that it is the uh [00:10:50.372] Point which is occupied by the object in space, one single point. That means we can measure it with respect to the coordinate system X, Y, and Z. Then when we say it has mass, the particle will settle down. [00:11:10.372] So that is why it has got mass. Okay. Then it is very tiny and we cannot measure it with the naked eyes. Such objects are defined as particles. Next, the

three different types of bodies. Okay, one is the rigid body. [00:11:30.564] What is a rigid body? A rigid body is one in which when we apply force onto this, there is no change in shape and size of the body. Such bodies are defined as rigid body. [00:11:48.580] Next, elastic body. What is an elastic body? When a force is applied on an elastic body, it changes its shape and size. [00:11:59.076] But on removal of the force causing this change, the body will regain its original shape and size. Such bodies are called as elastic bodies. Next is a plastic body. What is a plastic body? A plastic body is one. [00:12:17.028] Which will change its shape and size on application of a force, but it will not regain its original shape and size. Such bodies are called as plastic bodies. Okay. Next, how do we define a force? [00:12:35.140] A force is something which changes the state of rest of uniform motion of an object. So, when we apply a force, a body which is at rest will start to move. If [00:12:54.084] the object or the body is already in motion, it can stop that motion or it can increase the speed of the body or it can reduce the speed of the body. So such [00:13:10.372] For such application or such thing is called as a force. Now, next comes what are the different types of forces. Forces can be broadly divided into two types. One is [00:13:27.268] Forces which are having forces which are in one plane, coplanar forces. The other one is non-coplanar forces. So coplanar forces are two dimensional. [00:13:46.660] Which can be represented only along two dimensions. Non-co-planar forces are forces which can be represented along all the three coordinate systems that is required. Now coming under this coplanar and non-coplanar, you can have again two categories. One is [00:14:06.084] Concurrent forces and non-concurrent forces. Similarly for non-copy also concurrent forces and non-concurrent forces. What are concurrent forces? Concurrent forces are forces which will have the same point of application. Non-concurrent forces which will have different points of application. [00:14:25.284] Under non-concurrent force system, we can have parallel force system and non-parallel force system. Under parallel force system, all the forces are going to be parallel to each other. Non-parallel force system, they need not have to be parallel. [00:14:42.724] Under parallel force system, we can have two categories. One is like parallel force system and unlike parallel force system. What is like parallel force system? Like parallel force system, where the sense of the force are same for all. Unlike parallel force system, the sense of the forces are different.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:00.868] So this is what is different types of forces that we have. Next coming to what is the you what are the units of forces. Unit of forces in SI, that is system of interaction. SI [00:15:16.708] is Newtons. And you can see the picture of Sir Isaac Newton. Okay, in this. Okay. So one Newton is nothing but one kg of mass [00:15:35.492] Which is pulled by the gravitational force. Okay. So it is 1 kg meter per second square. Okay. So that's the unit of a force. So you can see here force. [00:15:52.164] 9.81 newtons means it is one kg of force which is pulled by a gravitational force that is 9.81. So it comes up to 9.81 newtons. 1 into 9.81 meters square meter per second square. Okay. Next. [00:16:11.748] What are Newton's laws? There are three Newton laws. All of you have studied this. Newton's law one, two, and three. What is Newton's law one? Newton Law 1 states that [00:16:27.748] Any body would continue to be in its state of rest or of uniform motion unless and until an external force is applied onto it. Is what you have studied. But here, instead of saying external force, we say that unless and until it is [00:16:48.356] It is unbalanced force that is acting. So the force can be internal or external. For example, a person who is standing, he can walk. Where is that force coming? It is because the internal that is developed by the person. So we call it as unbalanced force. So the

definition is [00:17:07.844] A body will continue to be in its state of rest or of uniform motion unless and until there is an unbalanced force acting on it. The second law states that the force is proportional to the mass of the body. [00:17:27.268] So force is equal to the mass of the body and acceleration. Acceleration is the proportionality constant. F is equal to M. A. Of course, the Newton's third law, all of you know every action has an equal and opposite reaction. So these are the three laws of laws defined by Newton. [00:17:47.588] Which you already know or studied. Okay. Now comes very important aspect. That is what are the characteristics of forces. [00:17:55.780] So according to the book, he says three characteristics, but I would like to have it as four characteristics. Let's look at this. What is the characteristics of force? Force one is the magnitude. Magnitude is defined as the amount of force that is acting. [00:18:15.620] Okay, is called as magnitude. The amount of quantity of the force is called as magnitude. Next is the direction. The direction is the angle made by the force with respect to a reference axis. The reference axis can be anything. The third one is the [00:18:34.148] Point of application. Where are we supposed to apply this? Force. That is the point of application. The fourth one is the sense of the force. Whether it is acting away from the point or towards the point. [00:18:50.404] In the book, it clubs the direction and the sense together. Garney says that line of action of the force. Now if you take all the forces to be moving away from the point. [00:19:07.652] Then your direction if it is theta and if it is moving towards the angle will become 180 plus theta. Okay. So that is the difference that you can think. So I would refer to it as 4 with the sense and in the book it drops the direction and the sense together. [00:19:27.012] Okay, so this is what is the characteristics of force. So this much we can understand in the first class. Thank you very much. We will continue this in the next class. Okay, thank you very much. And I am Dr. Svang Tesh, civil engineering. [00:19:46.884] from the Department of Science and Humanities, PS University. Thank you. See you in my next class where we will be discussing further into the introduction of statics. Thank you.

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Full transcript: 2499 normalized words, 69 timestamped segments, 1206.101 seconds duration. Low-confidence segments flagged: 0.

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4 UE26ME142A_Unit 1_Class 2.0_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 2.0_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000034 · EV0000035

Graphs and diagrams



Figure G000010. UE26ME142A Unit 1 Class 2.0 Lecture PPT · source slide page 1.

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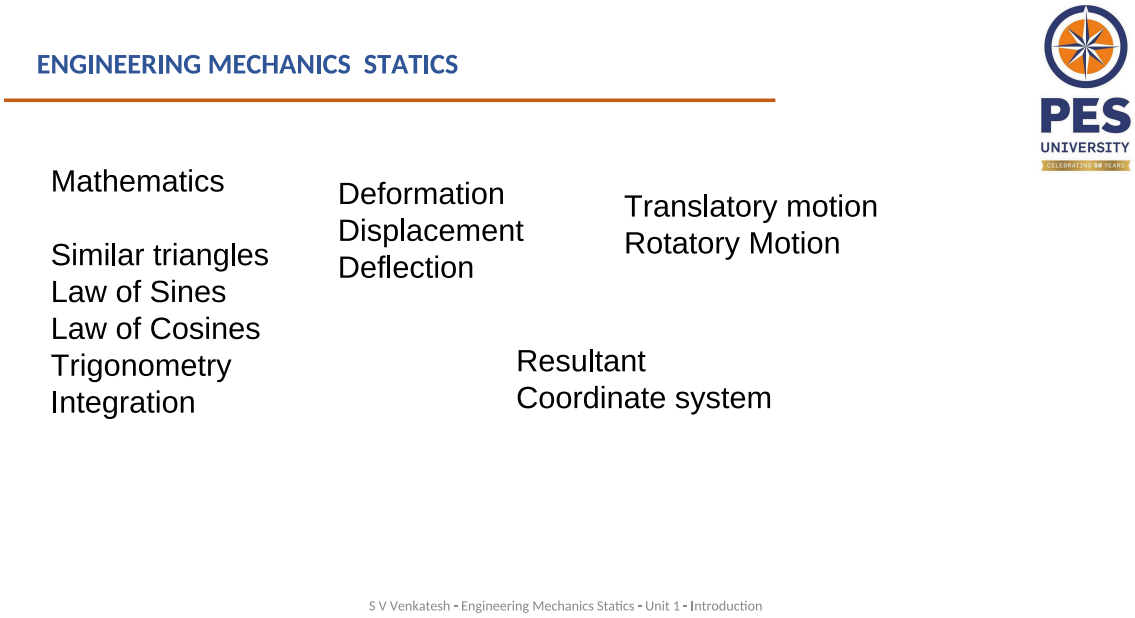


Figure G000011. UE26ME142A Unit 1 Class 2.0 Lecture PPT · source slide page 2.

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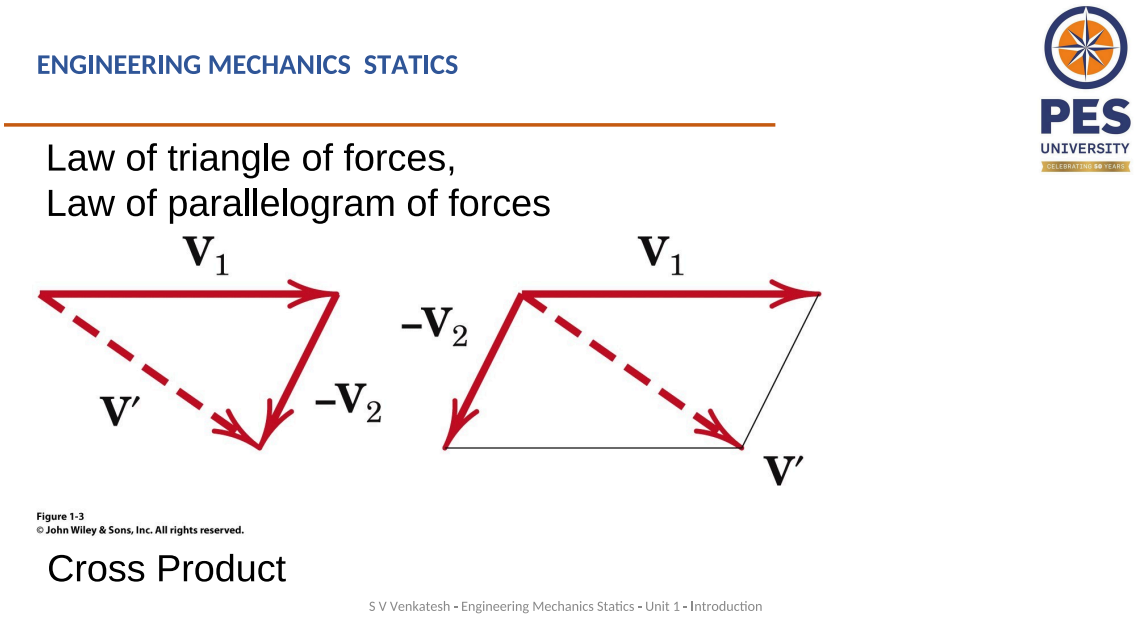


Figure G000012. UE26ME142A Unit 1 Class 2.0 Lecture PPT · source slide page 3.

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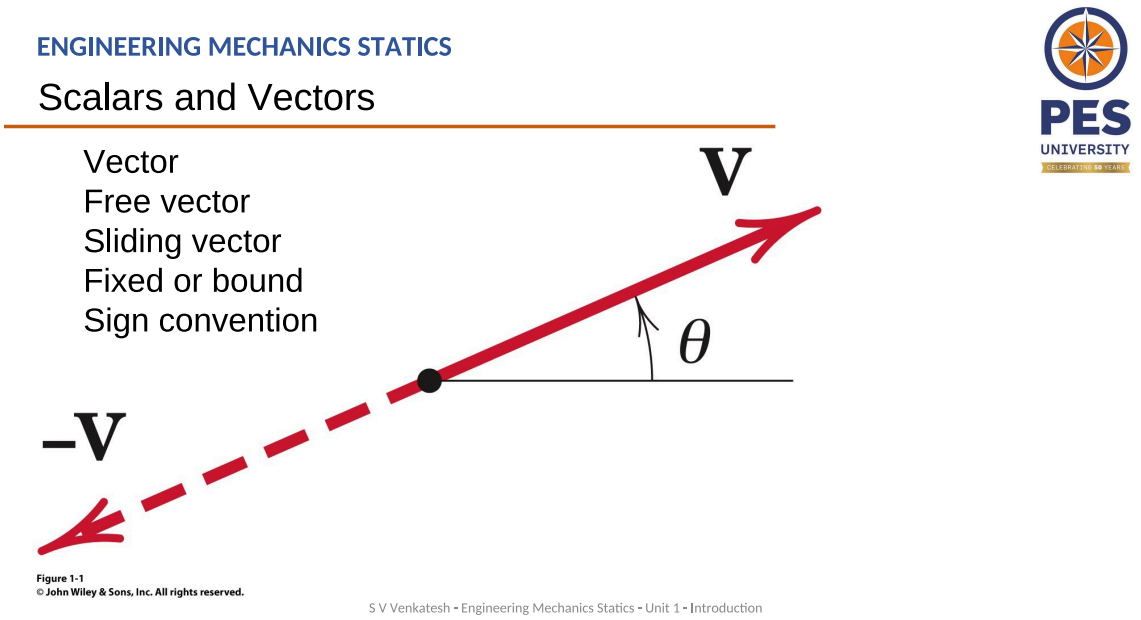


Figure G000013. UE26ME142A Unit 1 Class 2.0 Lecture PPT · source slide page 4.

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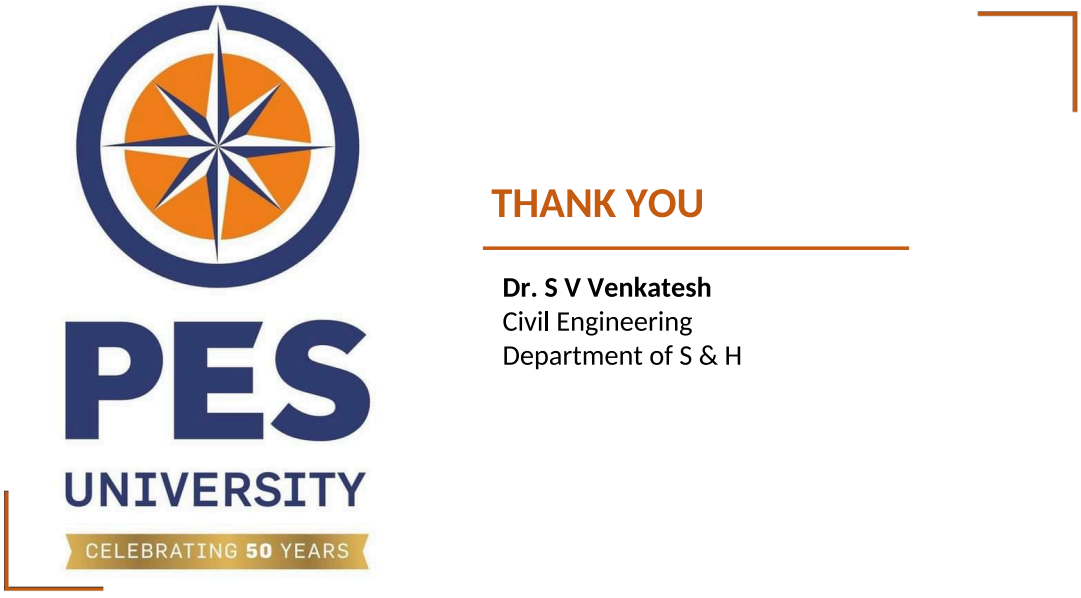


Figure G000014. UE26ME142A Unit 1 Class 2.0 Lecture PPT · source slide page 5.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000309 · EV0000310 · EV0000311
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Mathematics Deformation Translatory Displacement motion Similar Deflection Rotatory Motion triangles Law of Sines Resultant Law of Cosines Coordinate Trigonometry system Integration S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction EV0000312 · EV0000313 · EV0000314
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS Law of triangle of forces, Law of parallelogram of forces Cross Product S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction EV0000315 · EV0000316 · EV0000317
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS Scalars and Vectors Vector Free vector Sliding vector Fixed or bound Sign convention S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction ## Page / slide 5 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000318
5. Source slide page 5: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000319

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.324] Dear student, welcome to Engineering Mechanics Static classes. My name is Dr. Swivin Kitish from Civil Engineering under Department of Science and Humanities, PS University. This is class number five. [00:00:15.812] In the last class we discussed about the dot product, we also looked about how to use the law of parallel of forces and to get the resultant [00:00:28.644] As well, we looked at how do we resolve the given force along XY direction, determine their components, and then how do we use this component to determine the result. Now let's go about and understand few more things. [00:00:45.764] Now let's look about what do we mean by resolution of a force. Now in the last class we I talked about component of the force, that is once a force is given one force, how do we resolve that along x direction and y direction? [00:01:01.828] Now that is also a resolution, but in general, resolution of a force means splitting up of a force into N number of forces such that that N number of forces [00:01:16.336] would produce the same effect as that of a single force. That means you have one force, it will have some effect on the body. It can be translatory motion or it can be rotatory motion or it can be both translatory as well as rotatory. [00:01:32.324] Now I will split this force into n number of forces. Now if I apply this n number of forces on a body, then the effect of this n number of forces will be the same as that of that single force which was acting earlier. [00:01:51.204] So that is called as resolution of forces. Next look at moment. What is moving a force? Earlier I had discussed about cross product. Okay. Cross product is the position vector cross the force vector. [00:02:10.660] It is not a force vector cross the position vector. That means the distributory property is not valid in case of finding out the cross product. It should always be the position vector cross the force vector. [00:02:28.068] The magnitude, whatever we get, is nothing but moment. Now there are two ways of getting this moment, or first I will define what is the definition of moment. The rotating effect produced by a force on a body is called as moment. So moment is equal to [00:02:46.084] Force into perpendicular distance. When I say perpendicular

distance, the distance between the point about which we want the movement and the point about which the force is acting such that the force is acting at 90 degrees between the point where the movement is required. [00:03:03.696] On the point where the force is acting, or we find out what is the perpendicular distance between these two. So that is called as a moment. So cross product will give us the moment of a force about a point. [00:03:19.556] Okay. Next look let us look at what do we mean by tension. So here you can see we have uh [00:03:28.452] A cable that is given here, which is attached to this. The cable, okay, when we apply external forces such that it is moving away from each other, the body would try to elongate. [00:03:41.828] Now this elongation that happens because of the external forces that is applied is called as tension. Now when we cut the cable and look at what is the force that is acting, the internal force, here the external force will be acting away, the internal forces will be acting. [00:03:59.216] towards each other. Towards each other. The internal forces when it acting towards each other, the body won't get elongated. So such forces are called as [00:04:08.740] Tensile force. Now we can see here we have represented this. When we look at from this end, how do we look at this is point A, where the cable is fixed to this clamp. So that is point A. The cable is indicated by an arrow here. [00:04:25.792] The force, that force is moving away from point A and on the other side, if it is point B, that would also be moving away. So you can see that the two forces will be acting towards each other. That is the internal force. Okay now. [00:04:42.020] How where do we get this tension? Now you can see here the simple example is given here. We have this cable. [00:04:49.124] Which is used to carry some load here. This load is put put up on a platform. There is some load, and this this entire load with the platform is lifted by the crane. So when we try to lift this, this tension here, it is going to be pulled.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:08.900] Upwards, whereas here, because of this weight, this will be pulled downwards. So internal forces would be acting. [00:05:16.804] towards each other so the the cable will experience tensile force whereas if you see the external upwards it is being pulled upwards because of the weight it is not being pulled downwards so external forces is acting away from each other like this [00:05:32.304] Okay, so that is called as tensile in K. But the vice versa if you see compression. [00:05:40.036] When you apply forces which is moving, the external forces which is moving towards each other, the body would try to get compressed. So when the body tries to get compressed, the force experienced by the body is called as compressive force. [00:05:54.512] Okay, the internal forces when you see the internal forces will be moving away from each other, the body would try to get compressed, so the body would be under compressive force. Right. Next, we will define what is called as principle of transmissibility of forces. [00:06:13.028] Now to explain this, I have taken a simple body which is supported at O and C. O is an hint support. We will come back to this type of support later when we take up an equilibrium chapter. And at C we have a roller support. This body is [00:06:32.548] Force P at B point. Now you can see that there are this is the line of action of the force and there's a point here here also and there also the force is apply. Now what does this principle of transmission P of forces states? It states that [00:06:50.660] The point of application of a force can be along the line of action, anywhere along the line of action of the force. [00:07:03.172] Such that it produces the same effect on the body, provided the point is in contact with the body. Now you need to understand the difference between a free vector or rather a sliding vector and the principle of transportative forces. [00:07:21.220] Sliding vector states that the point of application of the force can be anywhere along

the line of action of the force, provided it does not change the effect of the force. [00:07:34.212] Okay, so the line of action of the force, the point of application can be along this anywhere, minus infinity to plus infinity. [00:07:42.916] But when it comes to principle of transmitter of forces, if we are talking with respect to a body, when we apply a force on the body, if the force is in contact with the body, then only there will be some effect. Otherwise, there won't be any effect. So principle of transmitter of forces states that the point of application of a force is [00:08:02.916] Can be along the line of action of the force, the body would still experience the same effect, provided the point of application of the force is in contact with the body. This is the difference between a [00:08:19.620] Sliding vector and principle of transmitter of forces. Right. Next, this is a very important slide. [00:08:28.516] There are several problems that we will be solving with respect to what you mean by projection and what you mean by components. I have explained both of this earlier also, but not as projection. [00:08:43.792] Okay, of course, we have looked at components of a force. Okay. Now let's look at what is the difference between a projection and a component. Projection is the cosine component of the force on any axis or any plane. Okay. For example, if we take this. [00:09:03.524] Okay, say I'll take this force R, there are two axis, one is this axis, small A, axis is representative as small a, and there's an axis B that is small b. [00:09:16.420] Okay. Now the projection of this force on B, if you want to find out, it will be equal to R, the force, into the cause of the angle made by the force R and the axis B, that is this angle. [00:09:34.596] Okay, so that is what is called as a projection. Now how do we get that? We will draw a line from the extreme end of the force. [00:09:46.500] to the axis B such that this line that is R from the end to this B axis, whatever line that is drawn, that line will make it 90 degrees with the B axis.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.060] So, whatever we get from this A till this point, this is called as a projection. Similarly, if I want the projection on A axis, then we will draw a line from the extreme end of the force. [00:10:18.020] Till this axis A, such that this line, this line is at 90 degrees with respect to the axis A. So from A till this point is called as a projection. The projection is represented as FA. [00:10:35.008] And F B. It is just the cost component of the force. [00:10:43.780] Axis whatever we are considering. Okay. So that is what is called as projection. When it comes to components, components are splitting up of the force along the axis such that [00:11:00.964] This would this component when it is acted on a body would produce the same effect as that of the single force. [00:11:12.164] Now projection as such does not have any role to play with respect to the effect of this projection on the body. It just [00:11:22.084] Projection. We are projecting the force on some axis. That's all. Just like how we watch a movie in a theater. It is projection, but in a movie there will be only one light source. [00:11:33.284] Which will fall on the film and an image is formed on the screen which will be larger than the film itself. So as we move the screen, the image will increase in size or decrease in size. Whereas in case of projection, [00:11:48.420] There will be n number of light sources, but these light sources will always be at 90 degrees to the screen. Always it will act at 90 degrees to the screen, and we will take the light source which will pass through the edges of the force and whatever the image that is formed. [00:12:07.620] Will be either equal to the length of the force or to be lesser than the length of the force, irrespective of the position of the screen or the axis on which we are considering. I hope it is very clear what is the difference between. [00:12:24.336] projection and component. Now how do we find out the component of force? Very simple, use either law of triangle of forces or law of parallelogram of forces. Now here in the first diagram they have given you how to get the

[00:12:41.776] Component of the force R on A axis and B axis by considering it as a parallelogram. So F1 and F2 are the components of the force along A and B. [00:12:55.268] Similarly, this also if they are not acting at the same point, of course you can bring back them and make it to be acting at one point, point of application, and then considering it or drawing a parallel and then finding out F1 and F2. [00:13:11.888] Now here in this one you can see that you have R the force and you can use this if there are two axes A and B draw a line parallel and take the forces [00:13:25.248] So use law of triangular forces and you can get the components of the force F1 and F2. Similarly, in this case also you can use the law of triangular forces. Now this diagram indicates both projection and component. [00:13:41.412] FA and F B are the projection, whereas F1 and F2 are the components. Now you can see that this forms a parallel. Whatever. [00:13:56.644] The sides that we are going to get or the components. When you use law parallel of forces or of triangle of forces, you get the component of the force. But when you simply draw a line which is perpendicular to the axis, you get the [00:14:14.308] Projection of the force onto that axis. I hope this is very clear. We'll be solving a lot of problems on this, so you need to understand this very carefully. Okay. Next [00:14:26.820] We had already discussed about resultant, this slide indicates how do we get the resultant of two forces which are parallel to each other. [00:14:36.324] When we have a force F1 and F2, we want to find out the resultant of these two forces. So in such case, what we do is we apply a force F [00:14:47.908] In the positive direction and in the negative direction also. Consider F1 and F, find out the resultant of these two as R1. [00:14:58.756] Similarly, consider minus F and F2. Find out the result end of these two. When we clog these two, this F1 and minus F1 will get cancel. So we will have R2 and R1. If I find out the result end of this, I would get the result end as R

EV0000042 · EV0000043 · EV0000044 · EV0000045

Transcript block 4 (source interval 000900-001199 seconds): [00:15:15.568] This is one way of solving. Of course, you can go ahead and use vector addition as F1 vector plus F2 vector, the dot product, okay, you will get the resultant of this two forces. Okay. Next. Let us look at this rectangular compound. [00:15:34.532] I had discussed about components where there was axis A and access B. I had also discussed this earlier when I talked about resultant how to determine the resultant or use the dot products. How do we use the dot product to get the resultant? When I explained that I have already considered [00:15:51.424] The x axis and y axis I have discussed about ij what are these unit vectors along x and y. How do we get the components f x and f y? Okay. Now the components and the projection when the two axes [00:16:05.856] on which we have to find out the component and the projection are mutually perpendicular to each other or mutually perpendicular to each other, there is no difference between projection and component because [00:16:19.748] When I try to draw a parallel, this forms a rectangle. A rectangle is also a type of parallelogram. Okay. Rectangle is a parallel, but parallel is not a rectangle. Please remember that. So [00:16:36.228] If I use the trigonometry here, I will get Fx, or if I use parallel law, I will also get Fx. So when the two axes are mutually perpendicular to each other, I can use [00:16:50.148] Either projection or law of parallelogram of forces or law of triangle of forces to get the components. The magnitude, whatever we get, would be the same. Okay. However, to get the sense, we have the force. [00:17:06.852] Uh F1 acting like this. Okay. So rotate it and merge it with x axis. Whatever sense that we get, we I rotate this and merge it here. I get the sense to be this. That would be the sense of your Fx. Then I will have this force, I rotate this and merge it with this, I get the sense of FY. [00:17:25.732] Now, I can also rotate this and get it here, right? Now one thing that you have

to remember is when you are rotating to determine the sense of the components, you can rotate it not more than 90 degrees. You can rotate it by between 0 and [00:17:43.940] 90 degrees, not greater than 90 degrees. That you have to remember. So if I rotate this and merge it here, the angle would be more than 90 degrees, so that is not permit. So I can rotate it and merge it like this only. Similarly, I can rotate this and merge it like this. So I get the sense, I get the magnitude, the direction is along X, the direction is along Y. [00:18:03.876] The point of application will be the same as that of the force F itself. Okay? Next, these are several. [00:18:11.940] Uh indication of the diagram which indicates X and Y taken other than the horizontal and the vertical axis. So you can easily get the component of the force along this axis X and Y by considering this angle beta. [00:18:29.504] Everywhere they have given beta. So by doing this, you can get the components of the force along X and Y. Now be careful in this case your Fx and FY depends upon your alpha as well as beta. [00:18:43.268] The angle made by the force with the x axis, when I draw an x axis here, this total angle will be beta. So the angle made by the force with the x axis will be beta minus alpha, that's why you're getting here as beta minus alpha. And here also it is uh if I draw a line x-axis here. [00:19:03.460] And extend that this angle made by the force with x axis will be total angle is one spite that is 180 minus beta I get this angle. So f [00:19:15.204] Cause of pi minus beta, I'll get y, and it is it is negative f y. And f x, if I rotate this and merge it here, I get x to be this. So it is f sine pi minus beta. The other things you can do it, I think. Okay. Now here. [00:19:34.980] If there are four two forces that is acting, how do we get the resultant is what is indicated. So you can get the components of each force separately. F1 as F1x. [00:19:47.396] And F2 as F to X, add these two kit to get the component of the resultant or the force R. [00:19:55.844] Along yes. Similarly, I will find out what is F1Y and F2Y using this and getting the dot product of this, I can easily find out what is R.

EV0000046 · EV0000047 · EV0000048 · EV0000049

Transcript block 5 (source interval 001200-001499 seconds): [00:20:09.316] Now, when we are solving problems, what needs to be done? These are the several steps that you have to follow simply. State the given data, state the desired result that you want, state any assumption and approximation that you are going to make while solving the problem. [00:20:26.212] Okay, the solution to get the solution, draw any diagram you need to understand the relationship. You can draw n number of diagrams till you understand. Then state the governing principle. What are the principles that you are going to use? Make the calculations, whatever calculation that you're doing. [00:20:43.556] Ensure that your calculations are consistent with the accuracy justified by the data. If the data is given as two decimal points, three decimal points, you have to also get the result in two decimal points or three decimal points. [00:21:00.260] Be sure that you have used consistent units. If the units are given in kilonewtons for one force and newtons in the other force, make sure that you will convert the kilonewt into Newton or Newton into KN and use appropriately. [00:21:14.308] Ensure that your answers are reasonable in terms of magnitude, direction, common sense, etcetera Use that to get and then draw the conclusions. [00:21:23.940] Okay. Then how do we formulate is what is given. This is what we follow when we have to solve problems. So we will end today's class here. We'll continue in the next class. Thank you very much. My name is Dr. Sventesh from PS University. Thank you very much. See you in the next class.

EV0000050 · EV0000051 · EV0000052 · EV0000053

Full transcript: 3322 normalized words, 87 timestamped segments, 1303.403 seconds duration. Low-confidence segments flagged: 0.

EV0000034 · EV0000035 · EV0000036 · EV0000037

5 UE26ME142A_Unit 1_Class 2.1_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 2.1_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000054 · EV0000055

Graphs and diagrams



Figure G000015. UE26ME142A Unit 1 Class 2.1 Lecture PPT · source slide page 1. EV0000321

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

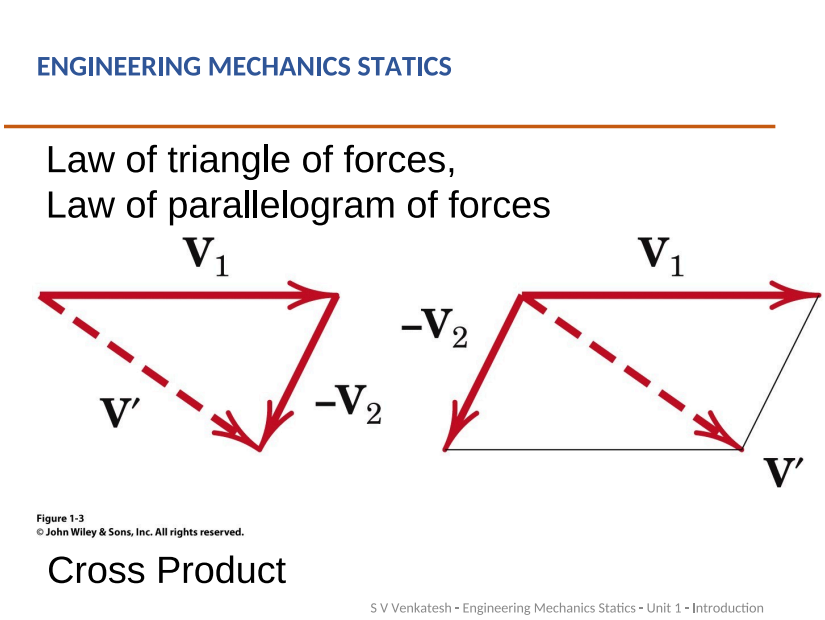


Figure G000016. UE26ME142A Unit 1 Class 2.1 Lecture PPT · source slide page 2.

EV0000324

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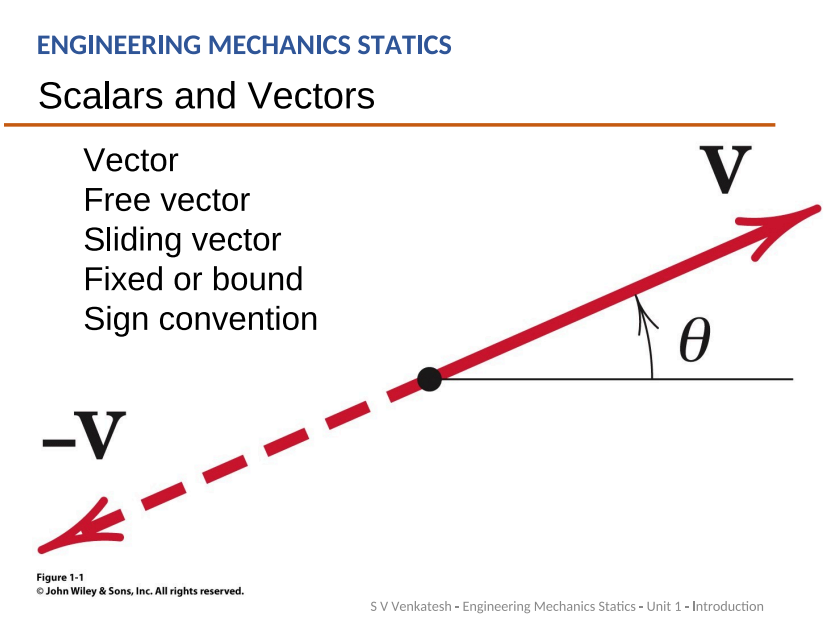


Figure G000017. UE26ME142A Unit 1 Class 2.1 Lecture PPT · source slide page 3.

EV0000327

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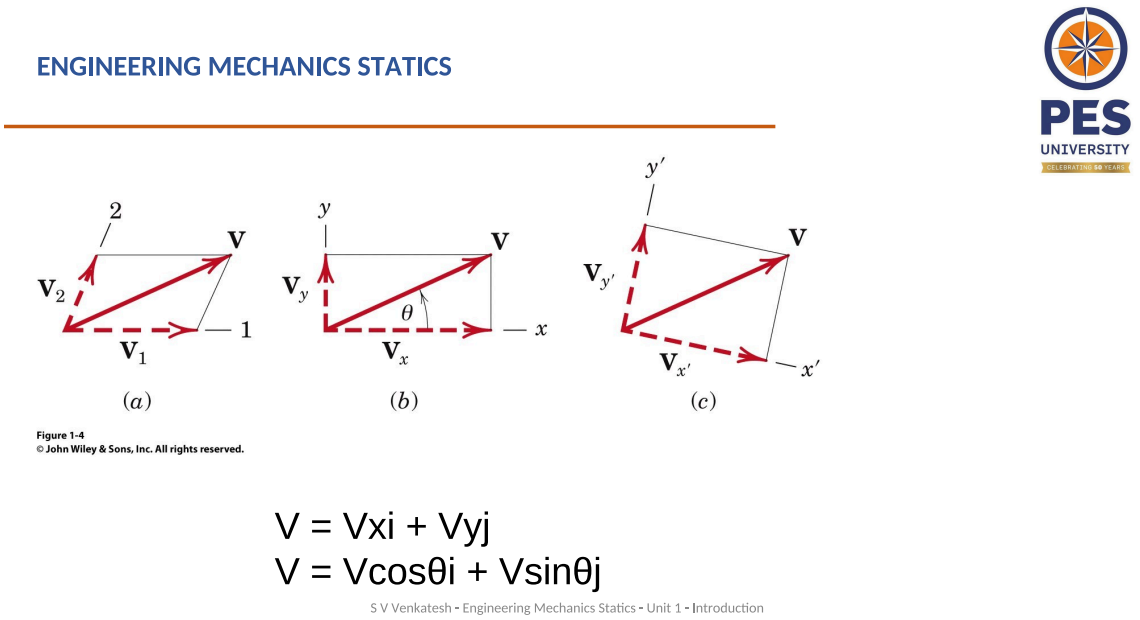


Figure G000018. UE26ME142A Unit 1 Class 2.1 Lecture PPT · source slide page 4.

EV0000330

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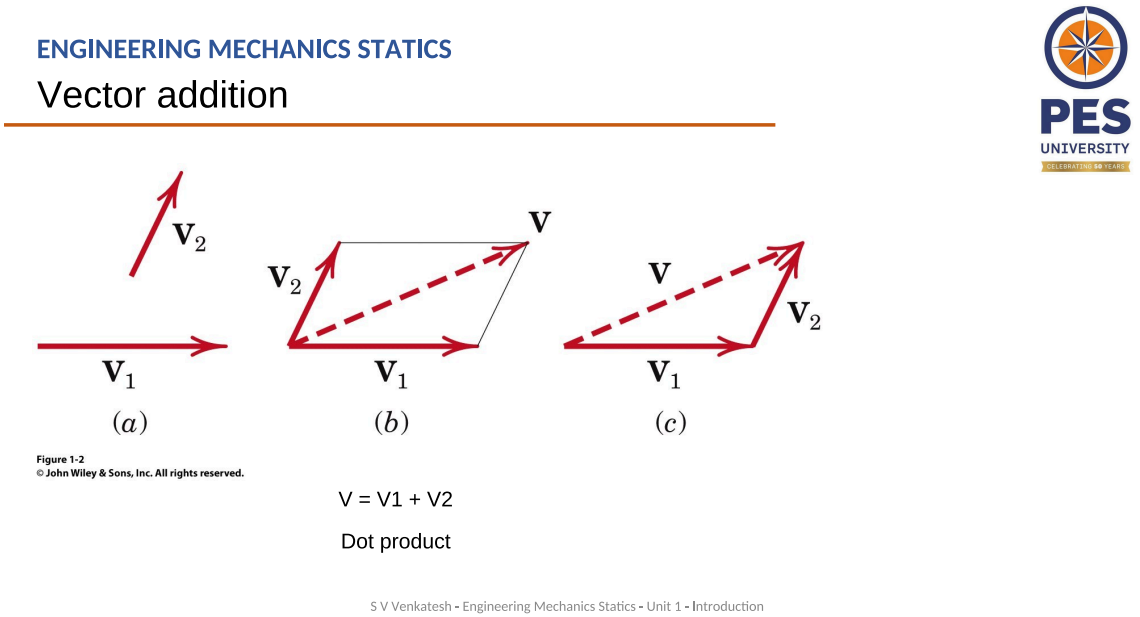


Figure G000019. UE26ME142A Unit 1 Class 2.1 Lecture PPT · source slide page 5.

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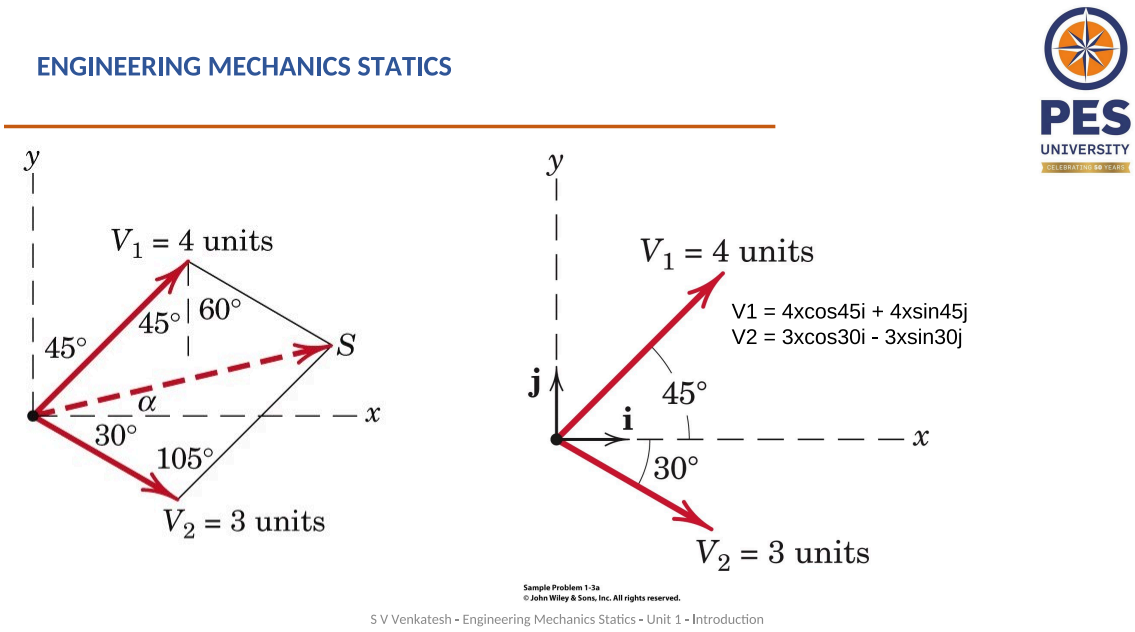


Figure G000020. UE26ME142A Unit 1 Class 2.1 Lecture PPT · source slide page 6.

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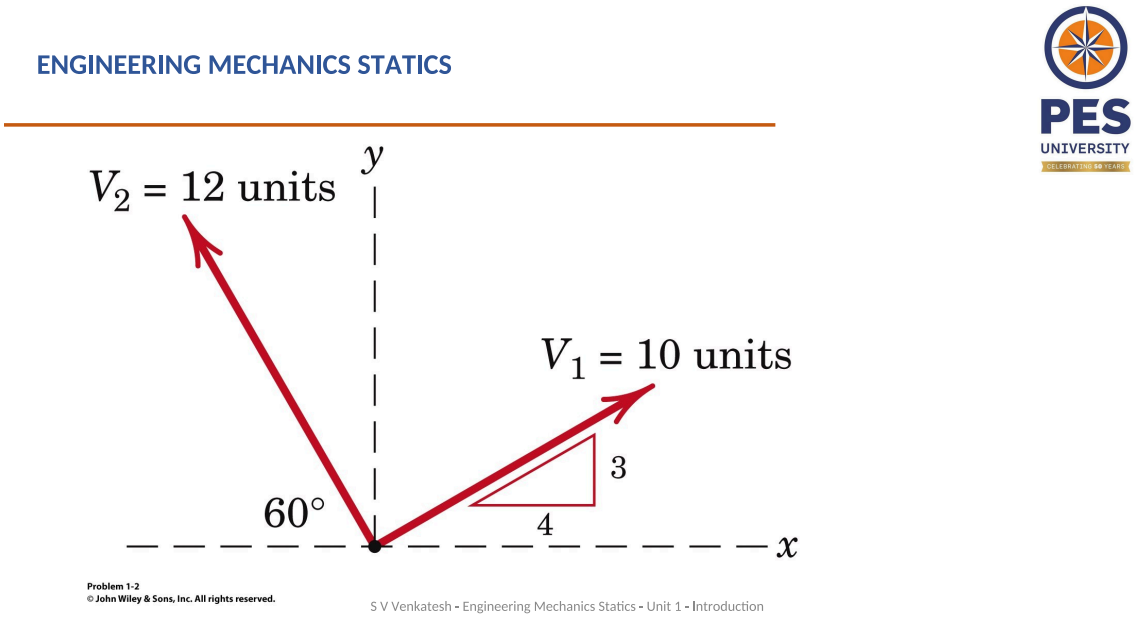


Figure G000021. UE26ME142A Unit 1 Class 2.1 Lecture PPT · source slide page 7.

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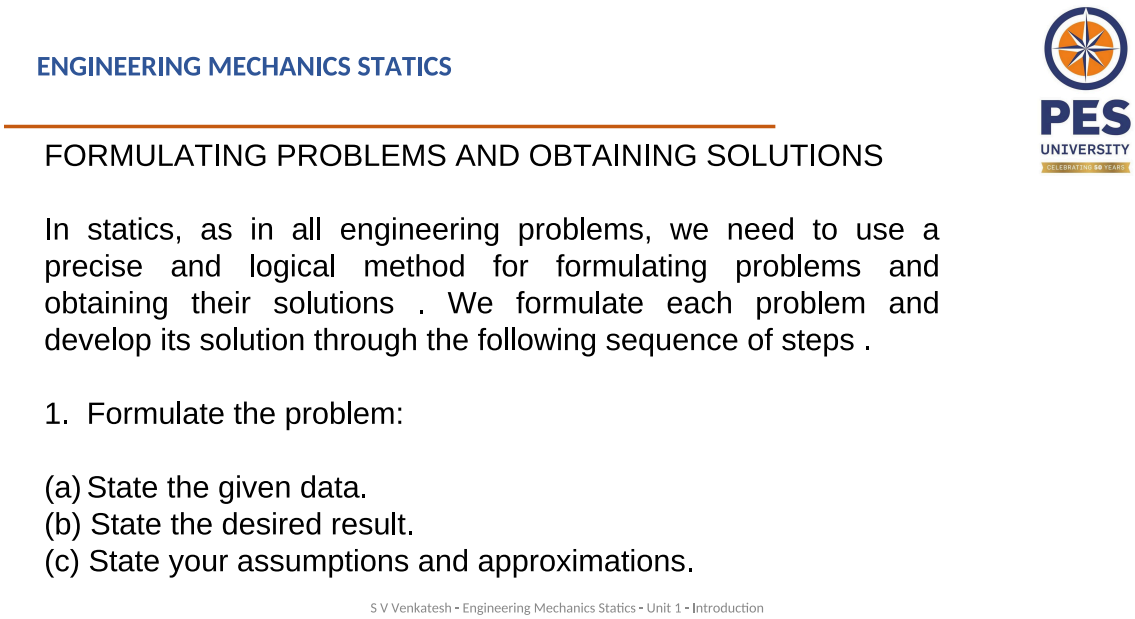


Figure G000022. UE26ME142A Unit 1 Class 2.1 Lecture PPT · source slide page 8.

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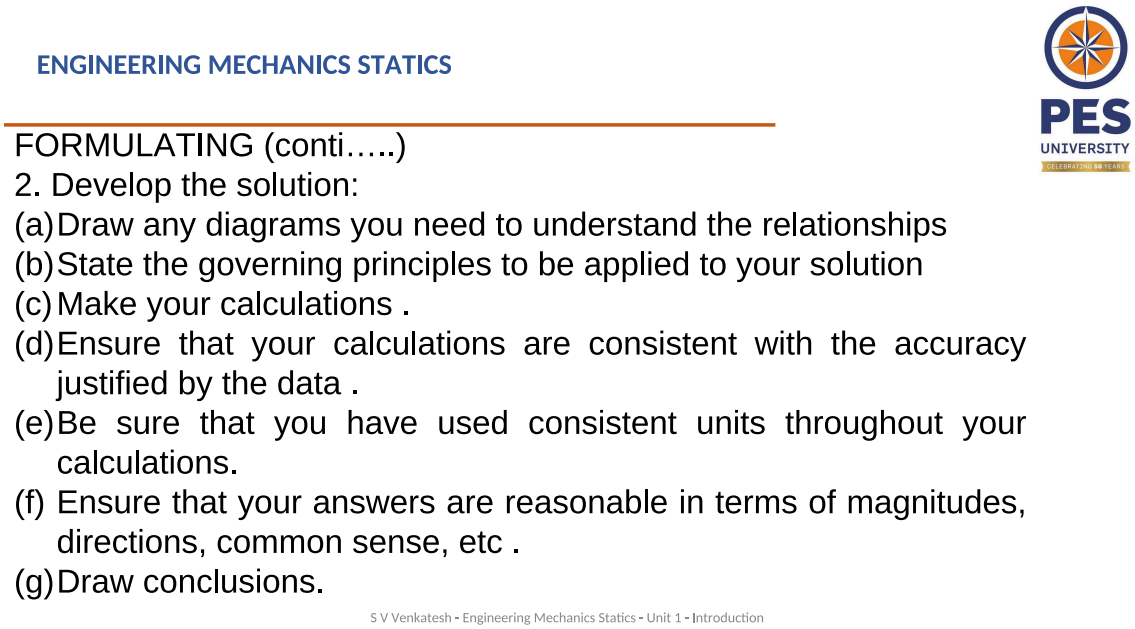


Figure G000023. UE26ME142A Unit 1 Class 2.1 Lecture PPT · source slide page 9.

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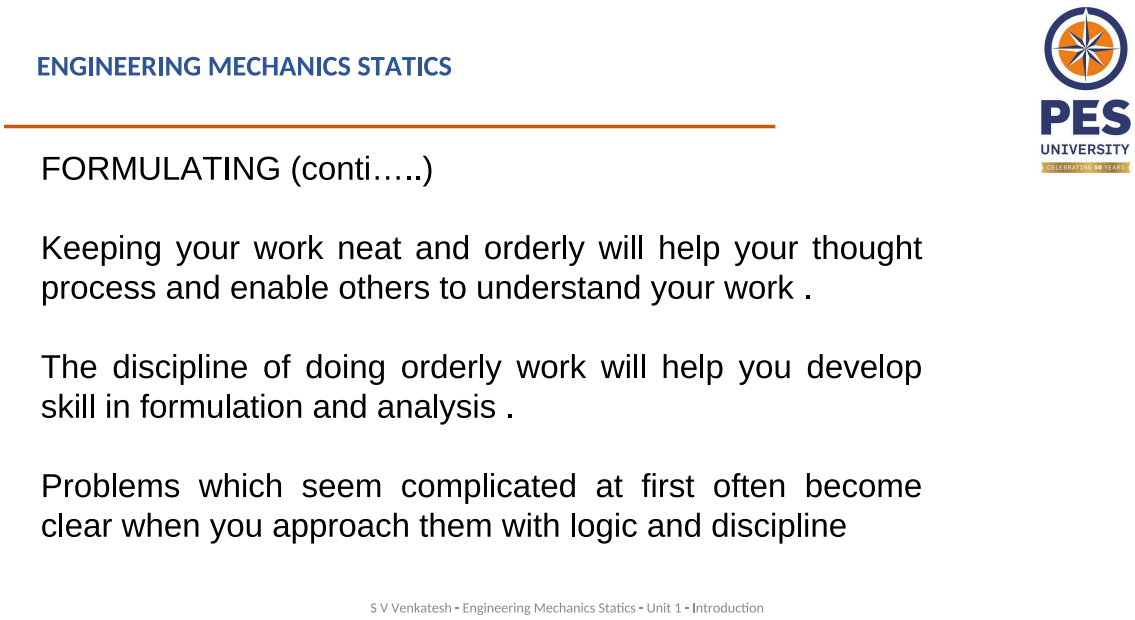


Figure G000024. UE26ME142A Unit 1 Class 2.1 Lecture PPT · source slide page 10.

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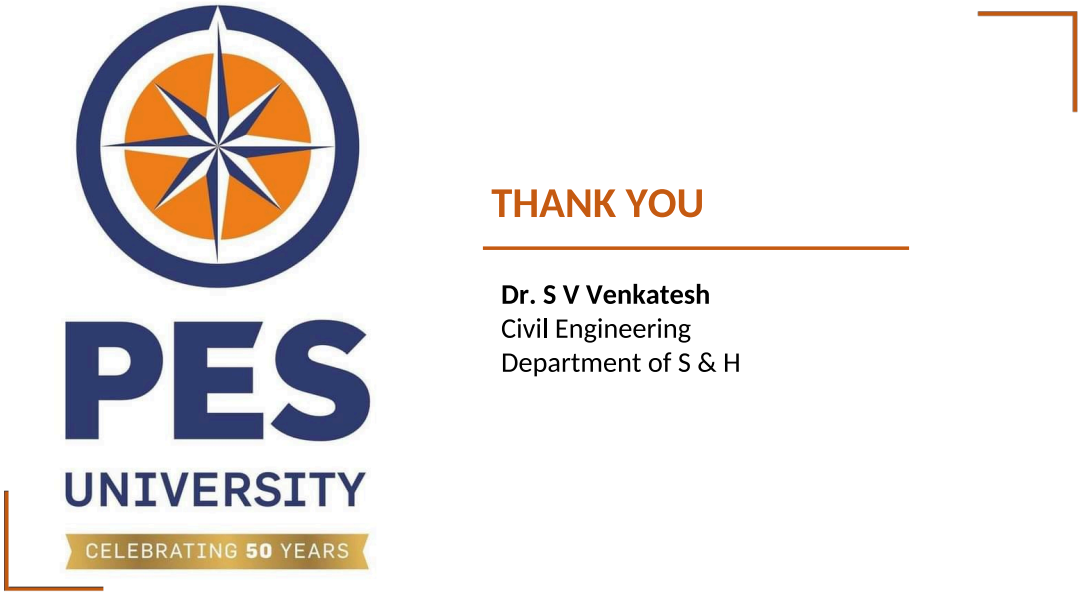


Figure G000025. UE26ME142A Unit 1 Class 2.1 Lecture PPT · source slide page 11.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000320 · EV0000321 · EV0000322
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Law of triangle of forces, Law of parallelogram of forces Cross Product S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction EV0000323 · EV0000324 · EV0000325
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS Scalars and Vectors Vector Free vector Sliding vector Fixed or bound Sign convention S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction EV0000326 · EV0000327 · EV0000328
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS V = Vxi + Vyj V = Vcosϑi + Vsinϑj S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction EV0000329 · EV0000330 · EV0000331
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS Vector addition V = V1 + V2 Dot product S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction EV0000332 · EV0000333 · EV0000334
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS V1 = 4xcos45i + 4xsin45j V2 = 3xcos30i - 3xsin30j S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction EV0000335 · EV0000336 · EV0000337
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction EV0000338 · EV0000339 · EV0000340
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS STATICS FORMULATING PROBLEMS AND OBTAINING SOLUTIONS In statics, as in all engineering problems, we need to use a precise and logical method for formulating problems and obtaining their solutions . We formulate each problem and develop its solution through the following sequence of steps . 1.Formulate the problem: (a)State the given data. (b) State the desired result. (c) State your assumptionsS V Venkatesh - Engineeringand Mechanicsapproximations.Statics - Unit 1 - Introduction EV0000341 · EV0000342 · EV0000343
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS STATICS FORMULATING (conti.....) 2. Develop the solution: (a)Draw any diagrams you need to understand the relationships (b)State the governing principles to be applied to your solution (c)Make your calculations . (d)Ensure that your calculations are consistent with the accuracy justified by the data . (e)Be sure that you have used consistent units throughout your calculations. (f)Ensure that your answers are reasonable in terms of S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introductionmagnitudes, directions, common sense, etc . EV0000344 · EV0000345 · EV0000346
10. Source slide page 10 (extracted text): ENGINEERING MECHANICS STATICS FORMULATING (conti.....) Keeping your work neat and orderly will help your thought process and enable others to understand your work . The discipline of doing orderly work will help you develop skill in formulation and analysis . Problems which seem complicated at first often become clear when you approach them with logic S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introductionand discipline ## Page / slide 11 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000347

11. Source slide page 11: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000348

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.484] Dear student, welcome back to the classes on engineering mechanics statics. My name is Dr. Sv. Venkadesh from Civil Engineering in the Department of Science and Humanities under PS University. [00:00:17.988] This is class number three. In the last class, we had discussed few of the basics that is required with respect to mathematics for understanding the engine mechanic statics. [00:00:32.964] Which are listed here. Again, the same slide. Then we also discussed about what is the difference between deformation, displacement, and deflection. [00:00:42.884] Then we defined what is meant by resultant and what are the different coordinate systems three-dimensional, three D and two-dimensional two D. Then we also discussed about what is meant by translatory motion and what is meant by rotatory motion. [00:01:00.132] So we will continue from here, we will look further what is required to understand the subjective material mechanics statics. Okay. Next, we should understand what is called as law of triangle of forces. Now what is the statement of law of triangle of forces? [00:01:18.628] So it states that if any two forces are acting on a particle, and these two forces can be represented on the adjacent sides of a triangle taken in order, then the closing side of the triangle. [00:01:36.676] Taken in the opposite order will give the result of these two forces. What does this mean? Now let's take a force V1 and then V2. These are the two forces which are acting on a particle. [00:01:54.724] And these two forces are represented on the sides of a triangle taken in order. What does that mean? So I've represented this V1 force along this side, and V2 force along this side. So you can see that. [00:02:10.244] V1 force sense is moving in this direction and it is V2 force is coming like this. So if you take the sense of V1 and V2, both of them are moving in the clockwise direction. So those two are the sides of a triangle. [00:02:26.928] On the adjacent sides of the triangle, which are represented, the forces have been represented. Now the closing side of the triangle is this side. [00:02:34.884] Which is being represented as V dash, which indicates the resultant of the two forces that is V1 and V2. But if you observe this sense of this V dash is in the anticlockwise direction. Now V1 is moving. [00:02:52.932] In the clockwise direction, V2 is also moving in the clockwise direction, whereas V dash, which is a resultant of V1 and V2, is in the opposite sense. So only if you consider the closing side of the triangle in the opposite sense of the other two. [00:03:11.716] sites or the forces, then that gives you the resultant of the two forces V1 and V2 with respect to the magnitude of the resultant and the direction or the sense of the resultant. [00:03:27.492] So that is what is called as law of triangle of forces. Next we need to understand what is meant by law of parallelogram of forces. [00:03:40.164] Now what is law of parallelogram of forces states? Law of parallel of forces states that if two forces are acting on a particle, and if these two forces can be represented on the adjacent sides of a parallelogram. [00:03:59.428] Then the diagonal of the paragram passing through the intersection of the two forces gives you the magnitude and the direction of the resultant. So what does that mean? [00:04:16.484] Say V1 is one force and V2 is another force, and if we can represent these two forces on the sides of a parallelogram, so this is one side of the parallelogram, that is V1. [00:04:30.308] Whereas this is another side of the paragraph representing V2, okay, these two are the sides of the paragraph, then the diagonal passing through the intersection of the two forces. What is the intersection of these two forces? This point. [00:04:46.480] Which is a diagonal, this is the diagonal passing through the

intersection of these two forces, passing through the intersection of these two forces would give you the resultant of the two forces. Okay, so you can see that this forms a parallel. Okay.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.212] And this is the diagonal of the parallelogram. But there are two diagonals. This also would be the diagonal. But this diagonal does not give you the resultant. Only the diagonal which passes through the intersection of the two forces would give you the [00:05:26.436] Resultant of the two forces. No. Here you can see that the two forces are taken such that both of them are moving away from the point. [00:05:37.700] Now you can take it that way, or both of them should be moving towards the point. You cannot take as I discussed in the law of triangle of forces, both of them are having or will rotate in the same direction. No, you cannot do it in case of law of parallel forces. If you want to. [00:05:54.944] To it or find out the resultant using law of parallel of forces, then you should take both the forces either moving away from the point or toward the point, both of them. Then only it is possible for you to determine the resultant of two forces. If you take this force to be acting here. [00:06:14.276] Then the sense of the force should be towards that. If the sense is towards this point itself, then this would be the diagonal which represents the result. Kindly understand this carefully. [00:06:25.668] The the this problem will not come in law of triangular forces, but in case of law of triangular forces, the two forces that you take should have the same sense. It should have or it should rotate the in the same direction. Then only you will get the [00:06:42.992] Resultant as the closing side of the triangle in the opposite sense. Okay. So [00:06:49.828] In case of law of parallel ram forces, the two forces should either move from each other or should move towards each other. Then you can take the resultant as the diagonal passing through the intersection of these two forces. Okay, both in magnitude and induction. So we get the magnitude of the four of the resultant. [00:07:09.156] By taking this as the force. Now the sense of the resultant can be taken as if these two are the forces that is acting. So if it is acting like this, and this is acting like this, this is acting like this, then these two, when you merge it together, you will get the sense of the resultant. [00:07:29.252] Okay. So I hope you have understood the difference between the law of triangle of forces and law of parallelogram of forces. Okay? Now if you take this law of triangle of forces, I can also take it as this. [00:07:43.364] And this, then what happens? Now these two will not be in the same sense, but in the opposite sense. So what I should do I if I take this as the sense of one uh side of the four of the triangle, then this has to be taken at [00:07:59.460] Here in the downward direction here, so that both of them will have the same sense, then the closing side will give you the resultant. If it is a law of triangle of forces, if it is long parallel forces, you have to consider it like this. I hope you have understood this. Now. [00:08:16.996] Next is in fact I told you that forces are vectors. And I also told you if you remember in the very first class, you should know what is the difference between a dot product and a cross product. Now what is a cross product? A cross product is nothing but [00:08:34.148] The multiplication of the position vector of the force into the force represented in terms of vectors. So when I say in terms of vectors, what does it mean? It means that [00:08:49.316] It is the component of the force along the x axis, say as f x and i, where i is the unit vector of the force along x direction, then f y [00:09:03.844] J where f y is the component of the force along y direction and j is the unit vector of the unit vector along y direction. [00:09:13.604] So that is 2D. If it is three dimensional, you will get all the three axis, so you get it as FXI, and if I assume that all the three will be positive, then it will be FYJ plus F Z K. [00:09:29.124] Where Fx, FY, and FZ are the component of the force along X, Y and Z

direction. Whereas Ij K are the unit vectors along the axis X, Y, and Z. [00:09:45.316] Then what is position vector? Position vector is the component of the point from the starting of the force to the end of the force along x, y, and z, which is represented as x i plus yj plus k.

EV0000058 · EV0000059 · EV0000060 · EV0000061

Transcript block 3 (source interval 000600-000899 seconds): [00:10:03.460] uh z where or z k where x y and z are the component of the position of the vector along x y and z axis and your i j k or unit vectors along [00:10:21.092] X, Y, and Z axis. Okay now, when it comes to cross product, what is it that we have to remember? We should remember this. I cross I is equal to 0. That means you cannot cross multiply the [00:10:38.532] Position and the magnitude of the component which are in the same axis that is x axis. If it is Xi, cross FX I, then you get it as I cross I, I cross I is 0. Okay, whereas similarly [00:10:57.988] K cross k is also zero, j cross j is also zero when it comes to cross product. Whereas I cross j will give you k which is positive. [00:11:08.164] J cross K will give you positive I and K cross I will give you positive J. Okay. Whereas if it is I cross K you get as negative J, if it is K cross J, you get us negative I. [00:11:23.504] If it is J cross i you can as negative k. You have to remember this. Okay? So that is what is cross product. Okay. [00:11:32.036] Alright, next let's look at what are the different types of vectors. Now before going to know about different types of vectors, let's look at what are scalas and vectors. What is scalars? Scalas are such quantities. [00:11:47.140] Which have only magnitude, but they don't have direction. For example, distance, speed, etc. Next, vectors. What are vectors? Vectors are physical quantities which will have both magnitude and direction. For example, velocity. [00:12:06.308] Okay? And acceleration. Right? No. I had also talked about when I talked about characteristics of the forces in the very first class, I told you according to the textbook, [00:12:17.872] It defines the characteristics of a force as three characteristics, whereas I am going to be considering it as four characteristics. What are these characteristics? I'll repeat it. One is the magnitude of the force, the direction of the force. [00:12:32.708] The point of application of the force and the sense of the force. Now the magnitude is given by V here. [00:12:39.620] The direction is given by the angle theta with respect to a reference axis. The point of application is here where the force is acting. Now whether the force is moving away from the point or towards the point will give you the sense. In the textbook he combines this with [00:12:55.908] The sense and the direction as one, so he divides it into three parts or three characteristics, whereas I consider as four characteristics. Based on the point of application of the vector, we can divide the vector into three categories, namely free vector. [00:13:15.364] Sliding vector, fixed or bound vector. What is this? What is a free vector? A free vector is a vector in which the point of application of the vector can be anywhere in the space. [00:13:33.412] But it will not have any difference on the effect of this vector on a body. Okay? If the point of application of the vector gets changed to any point in space, the effect will not change. [00:13:51.332] Such vectors are called as free vector. For example, a couple. I will talk about it when I take up the uh vectors. When I take up the couple. [00:14:06.084] Okay, I will talk about this in detail. Okay, as of now, you understand that example for a free vector is a couple. [00:14:14.212] Okay, what is sliding vector? A sliding vector is a vector in which the point of application, that is this point of application, can be anywhere along the line of action of the force. Only along this I can change the point of application. Either I can take it as this [00:14:31.504] Or this, or this, or this, or this, anywhere I want. But I cannot change it anywhere I want. For example, I cannot shift it here, or here, or here. Or here. Only along this line I can shift. [00:14:46.692] Then the effect of this vector on the body will not change. But if I change it

anywhere else, the effect is going to change. Such vectors are called as sliding vector. Next, coming to fixed or bound vector, fixed or bound vectors are vectors in which there is only one vector.

EV0000062 · EV0000063 · EV0000064 · EV0000065

Transcript block 4 (source interval 000900-001199 seconds): [00:15:05.924] point of application for a given vector. For example, if this is the point of application of the vector, if I change this point anywhere [00:15:15.524] Along the line or anywhere I want, the effect is going to change. Okay. So only this is the point of application of that vector. So it is fixed, you cannot change it. Okay. So the difference between the three is free vector, the point of application can be anywhere, the effect does not change. [00:15:32.228] Sliding vector, the point of application can be chang only along the line of action of the force, then the effect is going to be the same. [00:15:39.780] Whereas fixed vector the point of application is only one, only then the uh effect is going to be the same. If I change the point of application anywhere, the effect is going to change. Hope this is clear. Next let's look at what is a sign convention. [00:15:55.652] Now, the sign convention for vector or forces is usually similar to how we plot a graph. Okay? So how do we plot a graph? We take x-axis horizontal and y-axis vertical. The intersection of these two is taken as origin. [00:16:15.076] The point which we have to plot along X, if it is positive, we take it from the origin to the right. [00:16:22.820] That is positive. How do we plot the graph? Similarly, positive y means we will take it above the origin. Negative y, below the origin. Negative x, left of origin. Similarly, if a vector is moving to the right. [00:16:42.436] Okay, if it is moving to the right, for example, if I take this and the component of this is going to be along this, moving from left to right, we take it as positive, whereas if it is moving from right to left, right to left, we take it as negative. That is along the x-axis. [00:16:59.012] Similarly along y-axis, if the force is moving upwards we take it as positive, if it is moving downwards we take it as negative. This is the usual sign convention for forces. [00:17:08.644] But why vice versa is also true. If I take the component of the force along X as positive, moving from right to left as positive, then moving from left to right will be negative. Similarly, if I take downward forces as positive or [00:17:24.304] or downward forces as positive, upward forces will be negative. So you can change the sign convention in whichever way that you want, but usually the sign convention that is followed as [00:17:35.364] If the force is moving from left to right along X, it is positive. From right to left, it is negative. If it is moving upwards, it is positive. If it is moving downwards, it is taken as negative. I hope this is clear to everybody. [00:17:52.580] So, in this in today's class, I have defined what is law of triangle of forces, what is law of parallel of forces, what do you mean by a cross product. [00:18:08.004] What is the difference between a scalar and vector? The definition along with example. Then we have defined three different types of vectors, that is free vector, sliding vector, and fixed or bound vector. [00:18:24.036] Then we have also looked at what is the sign convention for forces. How do we consider positive sign convention for force and negative sign convention for force? [00:18:36.100] What is the usual sign convention that is used for vectors or forces, but you can also change it vice versa is also true. [00:18:45.892] Okay, so with this, we have come to the end of today's class. This is class number three. Okay, so far we have defined several important aspects for understanding engineering mechanics, which you have learned probably. [00:19:02.560] before or in your POC up to plus two you have learned these things and this forms a basics for understanding engineering mechanics okay so we will stop at this point [00:19:17.536] Uh I thank everybody. My name is Dr. Sviving Tesh. I'm from the civil engineering under Department of Science and Humanities under

PS University. Thank you very much. See you till the next class. The next class is class number four. Thank you. Good day.

EV0000066 · EV0000067 · EV0000068 · EV0000069

Full transcript: 2919 normalized words, 77 timestamped segments, 1176.832 seconds duration. Low-confidence segments flagged: 0.

EV0000054 · EV0000055 · EV0000056 · EV0000057

6 UE26ME142A_Unit 1_Class 2.2_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 2.2_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000070 · EV0000071

Graphs and diagrams



Figure G000026. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 1.

EV0000350

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

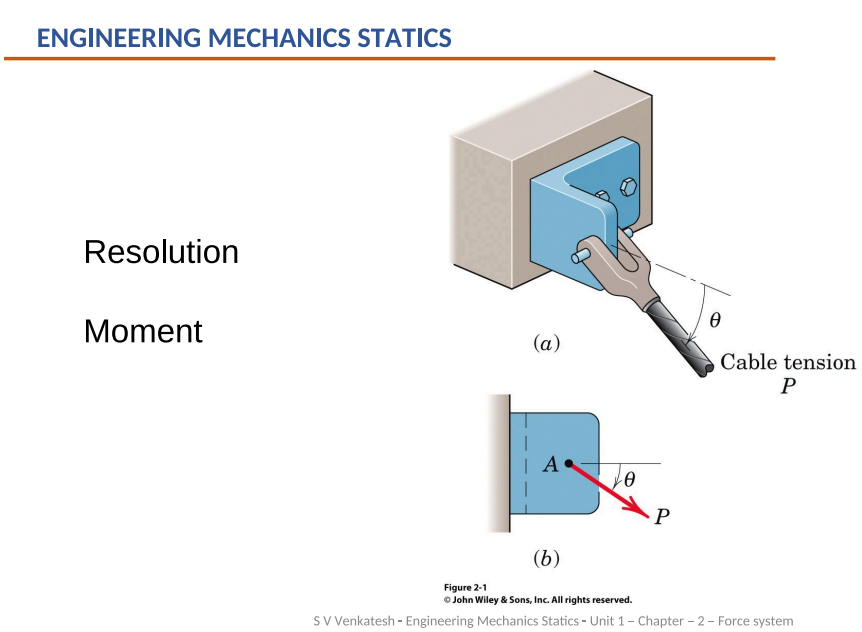


Figure G000027. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 2.

EV0000353

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

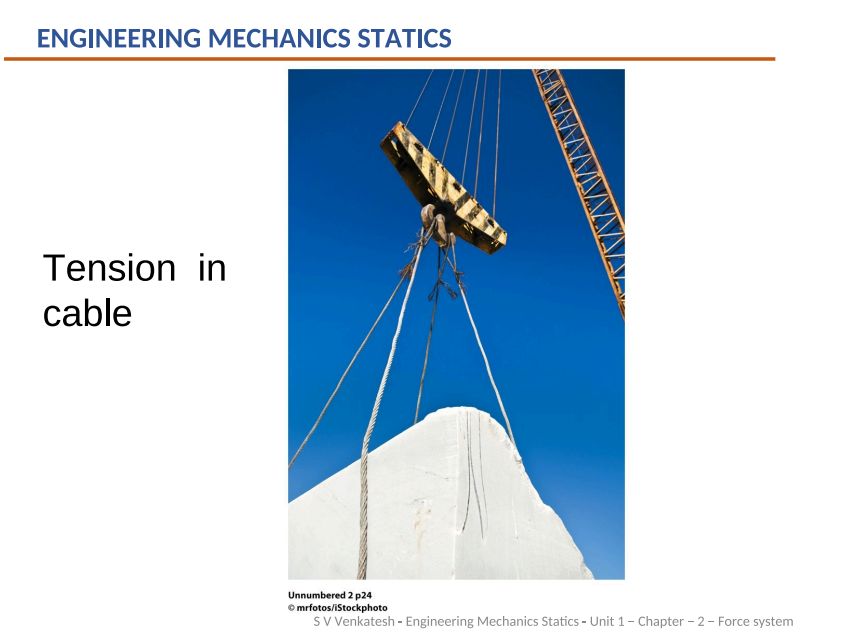


Figure G000028. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 3.

EV0000356

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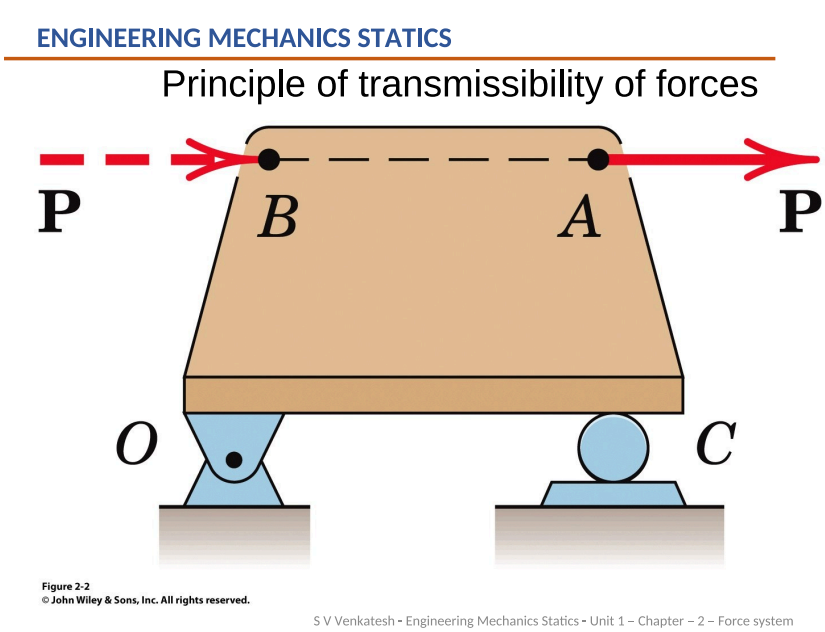


Figure G000029. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 4.

EV0000359

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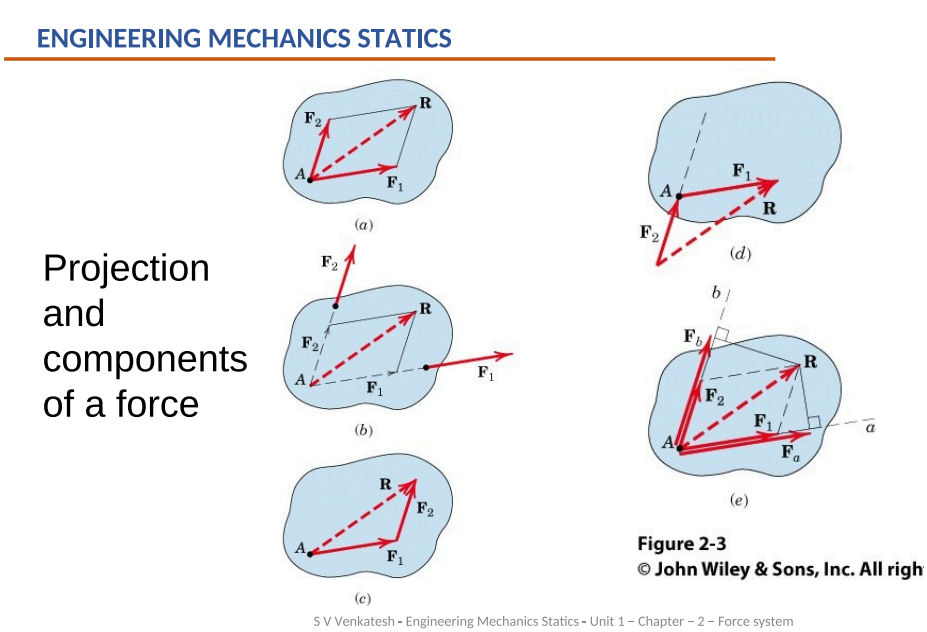


Figure G000030. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 5.

EV0000362

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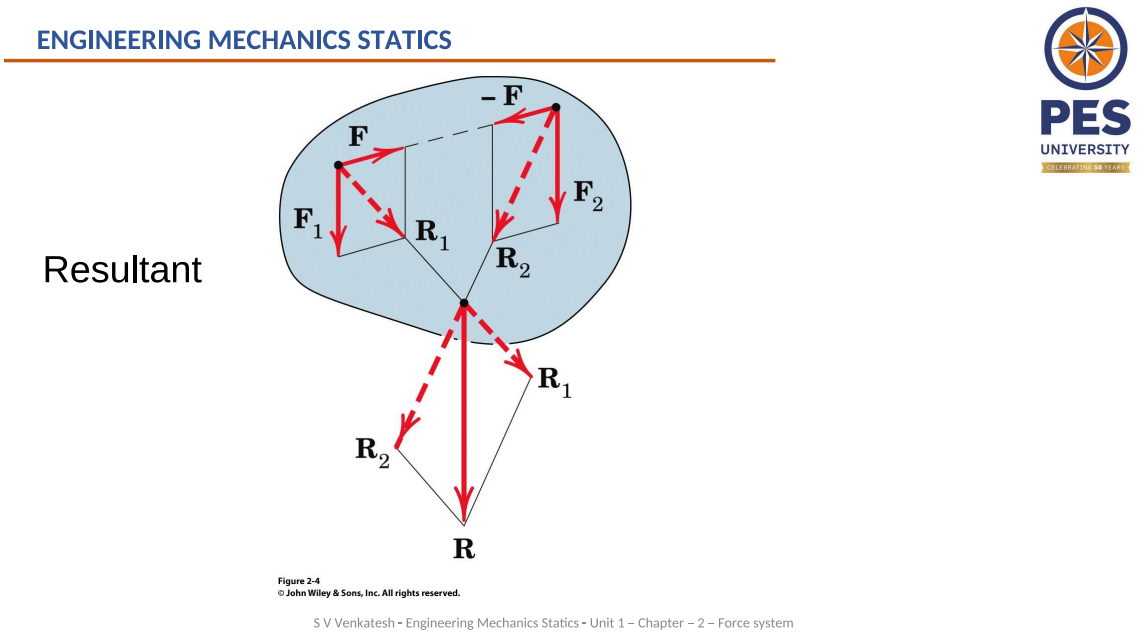


Figure G000031. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 6.

EV0000365

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

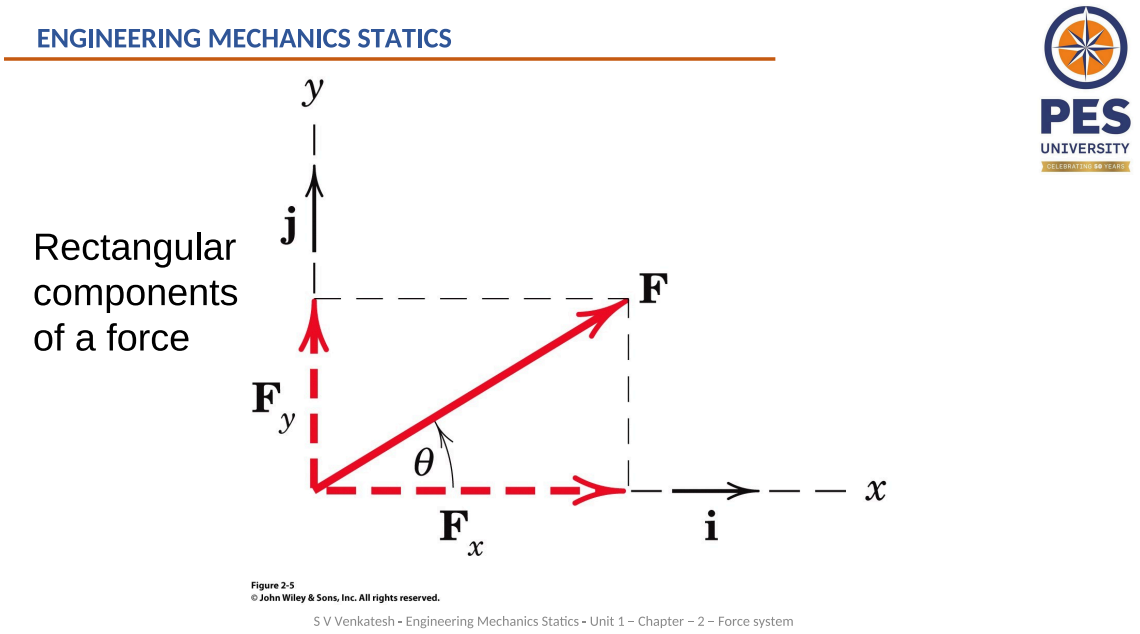


Figure G000032. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 7.

EV0000368

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

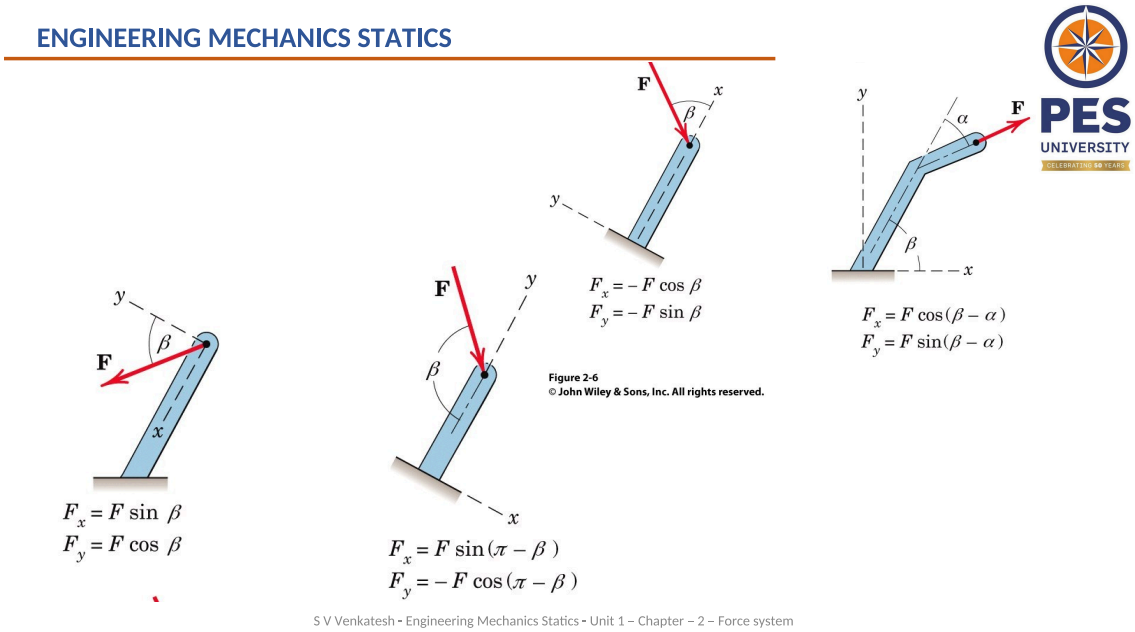


Figure G000033. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 8.

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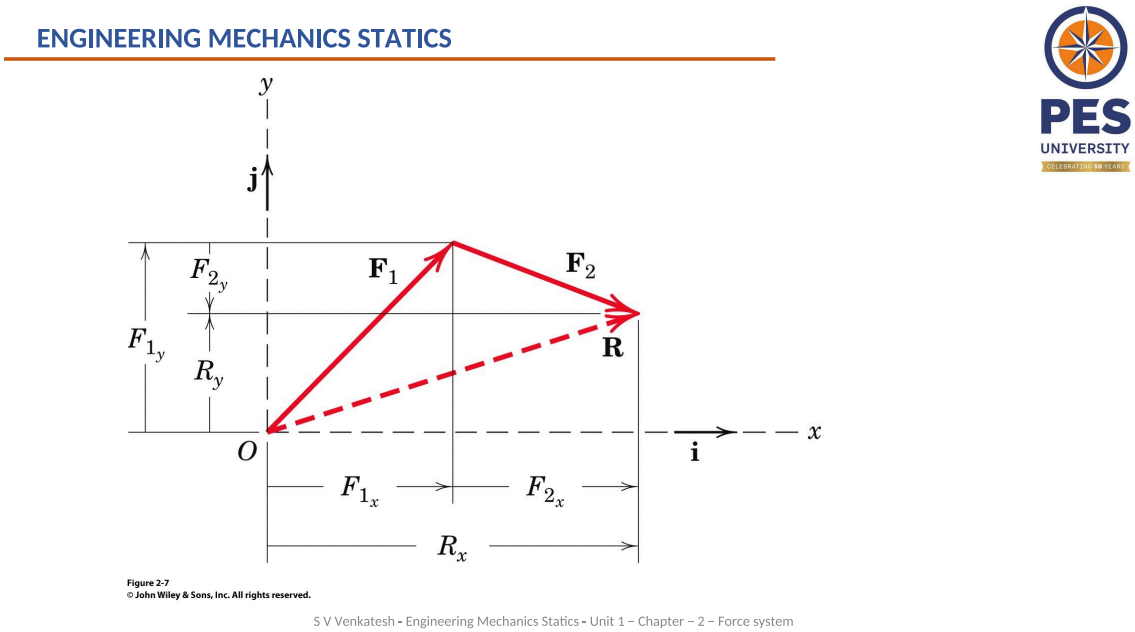


Figure G000034. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 9.

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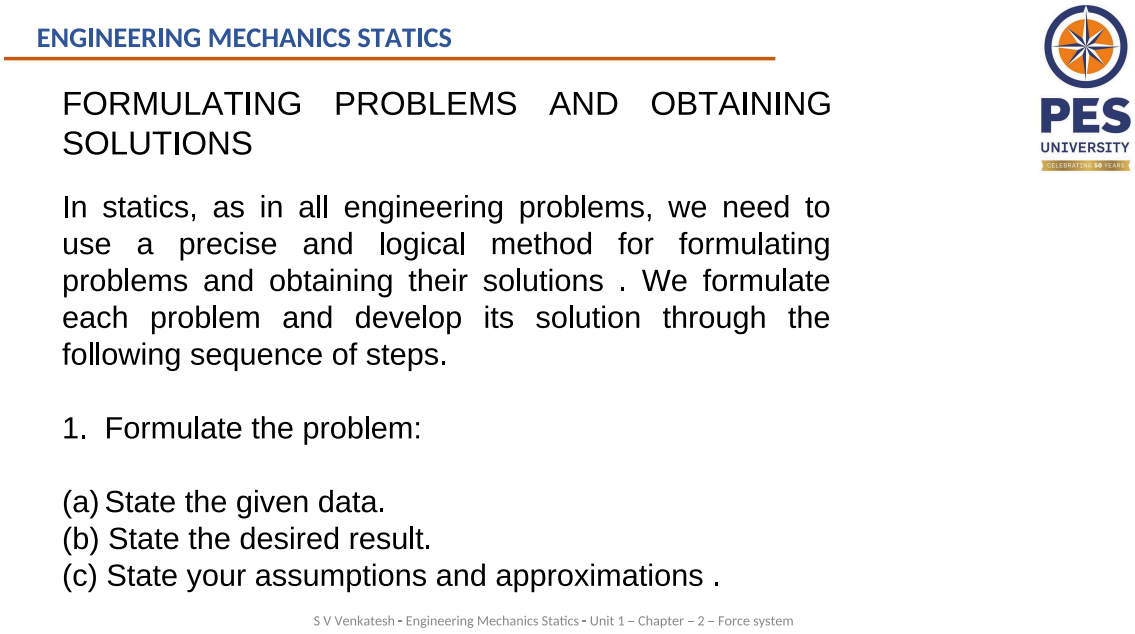


Figure G000035. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 10.

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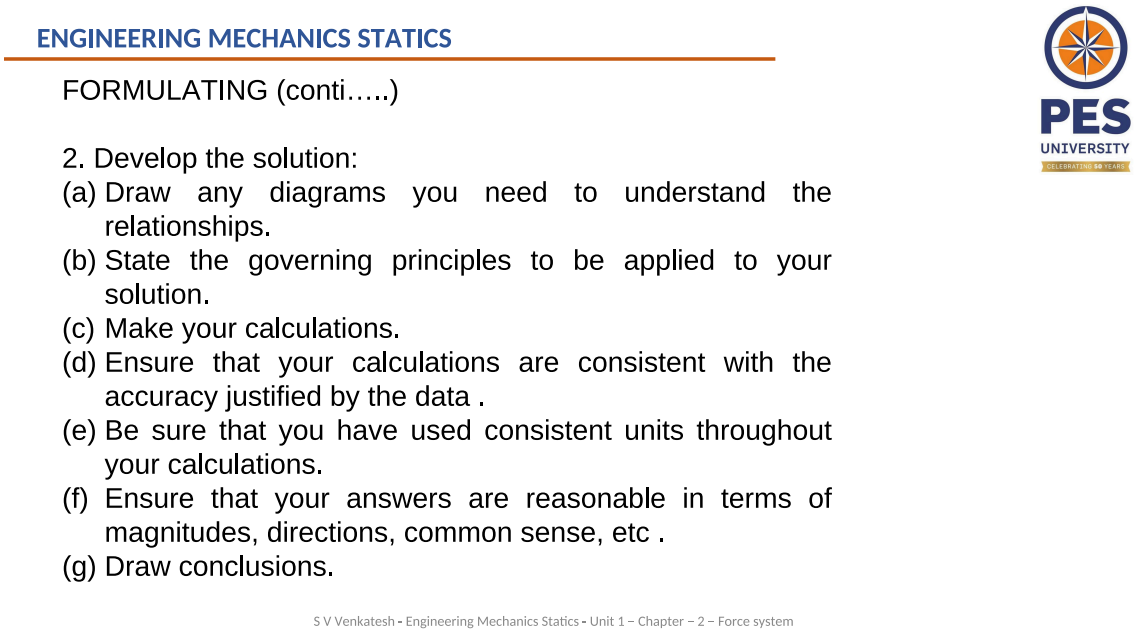


Figure G000036. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 11.

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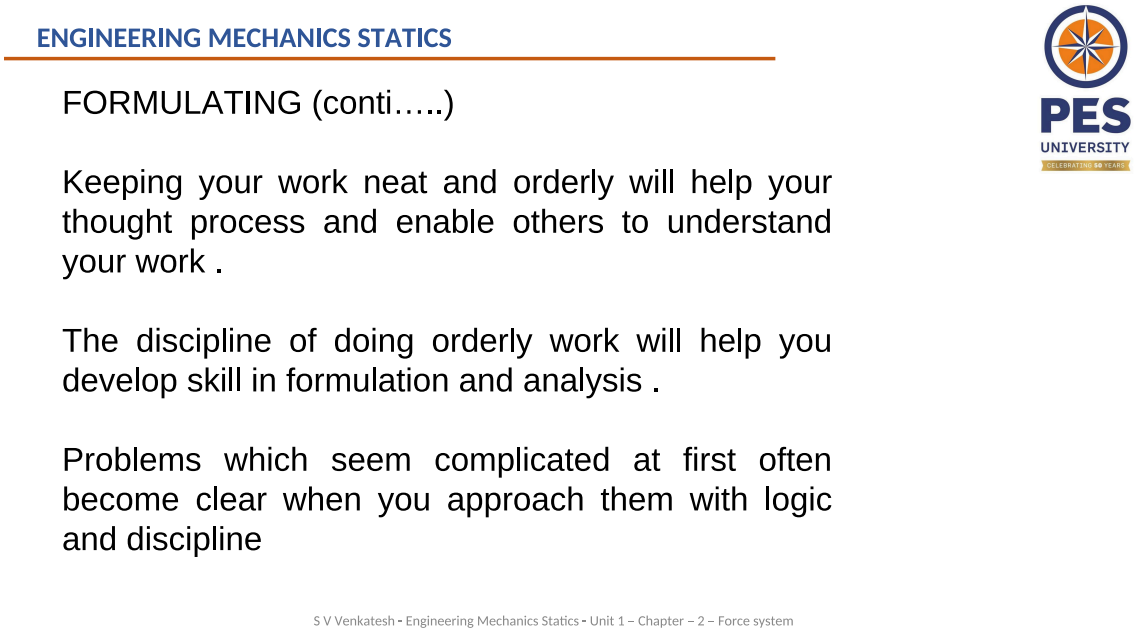


Figure G000037. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 12.

EV0000382

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Figure G000038. UE26ME142A Unit 1 Class 2.2 Lecture PPT · source slide page 13.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000349 · EV0000350 · EV0000351
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Resolution Moment S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000352 · EV0000353 · EV0000354
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS Tension in cable S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000355 · EV0000356 · EV0000357
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS Principle of transmissibility of forces S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000358 · EV0000359 · EV0000360
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS Projection and componen ts of a force S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000361 · EV0000362 · EV0000363
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS Resultant S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000364 · EV0000365 · EV0000366
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS Rectangula r component s of a force S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000367 · EV0000368 · EV0000369
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000370 · EV0000371 · EV0000372
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000373 · EV0000374 · EV0000375
10. Source slide page 10 (extracted text): ENGINEERING MECHANICS STATICS FORMULATING PROBLEMS AND OBTAINING SOLUTIONS In statics, as in all engineering problems, we need to use a precise and logical method for formulating problems and obtaining their solutions . We formulate each problem and develop its solution through the following sequence of steps. 1.Formulate the problem: (a)State the given data. (b) State the desired result. S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force (c) State your assumptions and approximations . system EV0000376 · EV0000377 · EV0000378
11. Source slide page 11 (extracted text): ENGINEERING MECHANICS STATICS FORMULATING (conti.....) 2. Develop the solution: (a)Draw any diagrams you need to understand the relationships. (b)State the governing principles to be applied to your solution. (c)Make your calculations. (d)Ensure that your calculations are consistent with the accuracy justified by the data . (e)Be sure that you have used consistent units throughout your calculations. (f)Ensure that your answers are reasonable in terms of magnitudes, directions, common sense, etc . (g)Draw conclusions. S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000379 · EV0000380 · EV0000381

12. Source slide page 12 (extracted text): ENGINEERING MECHANICS STATICS FORMULATING (conti.....) Keeping your work neat and orderly will help your thought process and enable others to understand your work . The discipline of doing orderly work will help you develop skill in formulation and analysis . Problems which seem complicated at first often become clear when you approach them with logic and discipline S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ## Page / slide 13 (OCR) THANK YOU Dr. S v PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000382
13. Source slide page 13: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000383

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.452] Dear student, I welcome all of you for engineer mechanics static classes. My name is Dr. Sv Venkadesh from Civil Engineering under Department of Science and Humanities, PS University. This is class number four. [00:00:19.876] In the last class, we had discussed uh law of triangle of forces, law of paragraph of forces, cross product, then scalars and vectors, then what are the different [00:00:36.164] Types of vectors based on the point of application of the vector. And then what is a sign convention that we use to represent the vectors? This was discussed in the last class, that is class number three. Now we will continue in this class number four from here. [00:00:55.780] Now here you can see there are two vectors, vector v1 and v2, and these two are being represented on the adjacent sides of a parallelogram. [00:01:09.764] So, the diagonal of the paragram passing through the intersection of these two vectors would give us the resultant of these two vectors v1 and v2 as v, both in magnitude as and in direction. [00:01:28.388] Now, how do we get this vector? So the formula that is used to get the magnitude of the vector is V1 plus V1 square plus V2 square plus 2 times V1 and V2 into V [00:01:46.532] Cause of the angle made by the two vectors that is V1 and V2, that is this angle. Say theta. So I repeat, the formula to get the magnitude of the resultant V is given by V1 square plus V2 square. [00:02:02.192] plus 2 V1 into V2 into cos of the angle between the two vectors V1 and V2 which I am considering this angle as theta. Okay. So that's the magnitude of the resultant force V. Now to get the [00:02:18.628] Direction of the resultant that is the angle made by the resultant with respect to one of the vectors, say V1, if this is the angle that I am considering, say alpha. [00:02:32.868] Then tan alpha will be equal to we will to get that alpha we will make a construction here. So I am going to drop a perpendicular from this end perpendicular to V1. So we will get this. So then what happens? This becomes a rectangle triangle, whereas this side [00:02:52.900] will be equal to uh V1 sorry, it is going to be equal to V two sine of angle theta that is this angle which is same as the angle between V one and V two. [00:03:06.788] So it is V two sine theta which will be this vertical, whereas this horizontal will be equal to V one plus this distance, which is going to be equal to V two. [00:03:18.372] cos of uh theta okay. So this angle alpha tan of this angle alpha that is tan alpha will be equal to this side which is V2 sine theta divided by V1 plus V2 cos theta. [00:03:35.556] So we get the direction of the resultant with respect to V1. Now to get the sense we have V1 horizontal, V2 slightly inclined. So both of them are moving away from the point. So to get the [00:03:52.240] uh sense of the resultant we'll mark these two and whatever is going to be the sense of that would be the sense of even so when I merge this and march this this would be the sense of the resultant. So we got the magnitude of the force [00:04:07.684] The direction of the resultant, the sense of the resultant, and of course the point of

the resultant will be the same as the point of application of these two forces, that is this point. So that's how we get the magnitude, direction, sense and point of application of the resultant. [00:04:26.404] Now, without using the law of parallelogram of forces, can we get the resultant of the two forces? Or we will go for what is called as rectangular components of a force. So to get the rectangular components of the force, [00:04:43.152] You this diagram you can see. What does this diagram indicate? We have a force V which is making an angle of theta with respect to the x axis. Now our aim is to determine what are the components of this force V along X and along Y.

EV0000070 · EV0000071 · EV0000072 · EV0000073

Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.108] No, to get that we will draw a line parallel to Vx, which is [00:05:08.868] Along x axis passing through the edge of V, that is from here, so an horizontal line. Similarly, parallel to y, we will draw a line from this edge that will be perpendicular to the x. So this forms a [00:05:23.808] Rectangle, and that's why it is called as a rectangular component. Now a rectangle is also a parallelogram. [00:05:31.876] So we can take this as a parallel also. However, we will get the component of the force V that is Vx and Vy, which is along the x-axis and along y-axis as V [00:05:48.996] Cos of this angle theta, we get Vx. Similarly, V sine of this angle theta we get as Vy. [00:05:58.468] So whether we have the two axis along horizontal or vertical or about any other which is indicated in this diagram, that is diagram number C. [00:06:09.956] Now we can see that there is an axis x dash and there's an axis y dash. These two are mutually perpendicular to each other, and if we have a force V, we can still get the component V x dash and Vy dash as the cause of this angle. [00:06:28.388] And V sign of this angle as V y 1. Okay. Now the other way of getting whether [00:06:37.668] We have Vx and Vy and then if we want to get V which is nothing but the result n we go for what is called as a dot product which I will be explaining it later. So V can be represented as VXI plus Vyj. [00:06:54.160] This is the vector representation of a force. What is Vx? It's the component of the force along x axis. Vy is the component of the force along y axis, whereas I and J are the unit vectors along. [00:07:13.604] X axis and along Y axis. This is for a force which is two dimensional. If we have a three dimensional one, then we will also get plus [00:07:26.916] V ZK. Okay. Now what is V x? Vx as I have already informed you that Vx is nothing but V cos theta, that is the component of V along X, and Vy is the component of V along y axis, that is V sine theta. [00:07:46.980] Now in this case it will become V equals VX dash I dash plus V Y dash [00:07:59.236] j dash where i dash and j dash will be the unit vector along x dash and y dash says and v cos theta if I take this as theta that will be equal to v x dash whereas v sine theta will be equal to v y dash [00:08:16.228] Okay, so I hope you have got what is the difference between the forces if they are not mutually perpendicular, then we have to go for parallaxis theorem only. But if the two axes are mutually perpendicular to each other, we can get the result and by simply doing this one. [00:08:34.532] Okay, whether it is along x y axis or whether it is along x dash and y dash axis, we can get the resultant or the component of the forces along this axis. Right. Next let us see what this vector addition is. Now we have [00:08:54.148] Taken a force V1 and F force V2. Now simply adding these two vectors, additional adding adding, that is vector addition, we will get the resultant of these two forces V1 and V2. [00:09:10.884] Now, this diagram indicates law of parallelogram of forces, whereas this diagram indicates law of triangle of forces. Whatever that you are going to use, simply V will be equal to V1 plus V2. It is a dot product of [00:09:28.128] The two vectors will give us the magnitude of the resultant of these two forces. Okay? Right. Now there is a simple example that I have taken here. We have a force V1 of [00:09:47.076] 4 units that is the magnitude of the force which makes an

angle of 45 degrees with the y-axis. The sense of V1 is moving away from the origin, the origin is this, the intersection of the x-axis and the y axis is the origin, the force is moving away from that.

EV0000074 · EV0000075 · EV0000076 · EV0000077

Transcript block 3 (source interval 000600-000899 seconds): [00:10:05.956] We also have one more force which is also moving away from the origin, which is V2. The magnitude of V2 is three units, but it makes an angle of thirty degrees with the x-axis below the x-axis. [00:10:21.636] Now, our intention is to determine the resultant of these two vectors, V1 and V2, for which you can see there is a construction that is done. The construction is we have constructed a parallel where there is a line. [00:10:41.732] Which is drawn parallel to V1 passing through the edge of V2. Similarly, there is a line which is drawn parallel to V2 passing through the edge of V1. So this forms a parallelogram. [00:10:56.132] And according to the law of parallelogram of forces, the diagonal passing through the intersection of the two forces would give us the magnitude both in direction. [00:11:06.756] And in magnitude. So S indicates the resultant of these two forces, and alpha is the angle made by this with respect to the [00:11:18.500] X axis. Okay. Now, how do we determine the resultant of this? Okay. So one way is we can use the law of parallel grammar of forces. So yes will be equal to V1 square plus V2 square plus 2 times [00:11:38.180] V1 into V2 into cos of the angle between the two forces that is this, which is simply equal to 45 degrees, that is this one, 45 plus 30 degrees, which would be 75. That gives us the magnitude of the resultant, yes. [00:11:57.604] So to get the direction of yes with respect to V2, we will simply draw a line from here, which will be perpendicular to this. So probably if I extend this and draw a line, it will be somewhere here. So and if I take the alpha angle as this angle total. [00:12:17.584] Or if I take only this one, then this total angle will be equal to 30 plus alpha, 10 of 30 plus alpha will be equal to uh V1, that is this V1, [00:12:32.484] Sine of this angle, which is the total angle, that is 45 plus 35, which is 75, divided by uh V2, which is 3, plus this component, which is V1. [00:12:45.380] Uh cos of 75. So I'll get angle alpha plus thirty from that or tan of alpha plus thirty from that I can find out what is alpha. That is one way of solving or getting the [00:12:57.668] Magnitude, direction, the sense of course will be when you merge these two, you will get the sense. Okay, and the point of application will be same as that. That is one way of determining the resultant of these two forces. The other way of determining the resultant of these two forces is [00:13:17.316] Simply determine the component of V1 along X and Y and determine the component of V2 along X and Y. The dot product of these two would give us the magnitude of the resultant. So expressing [00:13:36.612] V1 in terms of the unit vector i and j will be v1 is equal to 4. This angle is going to be 45 because they have given this angle as 45. So x and y are always at 90 degrees. So 90 minus 45 will be 45, so that is why this is 45. [00:13:53.924] Directly this angle is thirty, so we can see that this is thirty. So V1 cause of [00:14:02.532] 45 will give us the component of this force V1 along X, and that's what is indicated here. This is 4 into cos 45 i is the 4 into cos 45 is the component of V1 along X. I is the unit vector along X. [00:14:19.364] Now that would be positive because the sense of this when I rotate it and merge it along x axis, the sense is going to be along X, so that is positive. Similarly, when I rotate this force V1, this force V1 to Y axis. [00:14:34.340] Okay, I will get the sense to be acting along y itself, so that is positive. So it is going to be 4 sine of this angle 45. So that is 4 into sin 45, which will be a component of V1 along y, and it's positive, and J is the unit vector along y axis. [00:14:54.244] So similarly to find out the component of V2, V2 into cos of thirty degrees will give me the component of V2 along X. When I rotate V2 and merge it with the Y the X axis.

EV0000078 · EV0000079 · EV0000080 · EV0000081

Transcript block 4 (source interval 000900-001199 seconds): [00:15:09.412] Then I'll get the sense to be same as that of x, so it is positive, so it is 3 into cos thirty degrees, my component of v2 along x, whereas i is the unit vector of along x axis. [00:15:23.812] Next to find out along y, when I rotate this and merge it along y, I get it to be acting downwards, so that's why it is negative. Next to determine the component of V2 along y, it is V2 into sine of 30 degrees, so I get it to be acting downwards. [00:15:41.412] This would be the component of V2 along y and j is the unit vector along the y axis. Now to get the magnitude [00:15:54.596] Of the result, and we simply add V1 and V2. The dot product of V1 and V2, we will get the magnitude yes. So S will be equal to 3 cos 45 plus 3 cos 30 degrees. I [00:16:10.372] Plus four sine forty five minus three sine thirty degrees J would be the magnitude of the resultant of these two forces. I hope you can [00:16:24.196] Find out the magnitude by law of parallel of forces as I have explained, and by resolving the two vectors along their components X and Y, and then adding them or finding out the dot product of these two, and then getting the [00:16:41.392] magnitude of the resultant of these two forces. Okay. Now we will this is a other problem that I have given here for you to work out. However I will be explaining how to get the [00:16:57.700] Resultant of these two forces. Now what is given here is we have a force B1 of 10 units. The point of application of the force is the origin, that is the intersection of X and Y. The force is moving away from the origin. [00:17:15.524] The inclination of V one is given as three. [00:17:21.156] By 4, this triangle indicates the slope of this vector. We can easily determine what would be this angle, or we know that this is a right angle triangle. So if this is 3 and 4, this [00:17:37.984] The hypotenuse is gonna be five, three, four, five method. Okay. [00:17:42.980] You can use that or you can simply determine this angle by taking this right angle triangle, you can find out this angle theta 1 if I take this angle as theta 1. That would be equal to tan inverse of 3 by 4. You can find out the magnitude of that. Next we have uh V2 force. [00:18:00.128] Of 12 units, the origin is still the intersection of X and Y, and again the force is moving away from the origin. The inclination of this V2 with the negative x axis is given us sixty degrees. So we need to find out the resultant, the magnitude of the resultant of these two forces. [00:18:18.820] So one is simply find out what is the component of this force along X and Y. So you express V1 as equal to V1 XI plus V1 Y J. So that would be equal to 10. [00:18:38.436] Cos theta 1, then this is theta 1, I plus 10 sin theta 1 j. So that is V1. Next coming to V2, the component of V2 along X would be in the negative X direction. [00:18:53.648] So it will be negative, so it will be minus 12 cos 60 degrees I. Whereas for the y, if I rotate this and merge it, I get the sense of the component of this force and the positive y direction, so it will be plus, so that will be plus 12 cos 60 degrees J. [00:19:13.392] So we have expressed both V one and V two in terms of the components of uh V one and V two in terms of I and J. So simply [00:19:23.204] Take the dot product of these two, so you will get the magnitude of the result end. So I request you to kindly go through and solve this problem and find out the result end. Okay. So in the today's class, we have discussed [00:19:39.140] uh three problems or two problems in two different ways. One is by using law paragraph of forces, the other one by resolving the given forces along their components and then getting the results. So kindly solve this two. We'll stop at this point and we will continue in the next class. [00:19:56.196] And this is the fourth class. I'll come back to this problem formulation and obtaining solution in my next class. However, explain how to get the this. So we will stop here and we'll continue in the next class. Thank you very much. And my name is Dr. Sv Vengtesh.

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Full transcript: 2947 normalized words, 77 timestamped segments, 1214.379833 seconds duration.
Low-confidence segments flagged: 0.
EV0000070 · EV0000071 · EV0000072 · EV0000073

7 UE26ME142A_Unit 1_Class 3_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 3_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000086 · EV0000087

Graphs and diagrams



Figure G000039. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 1.

EV0000385

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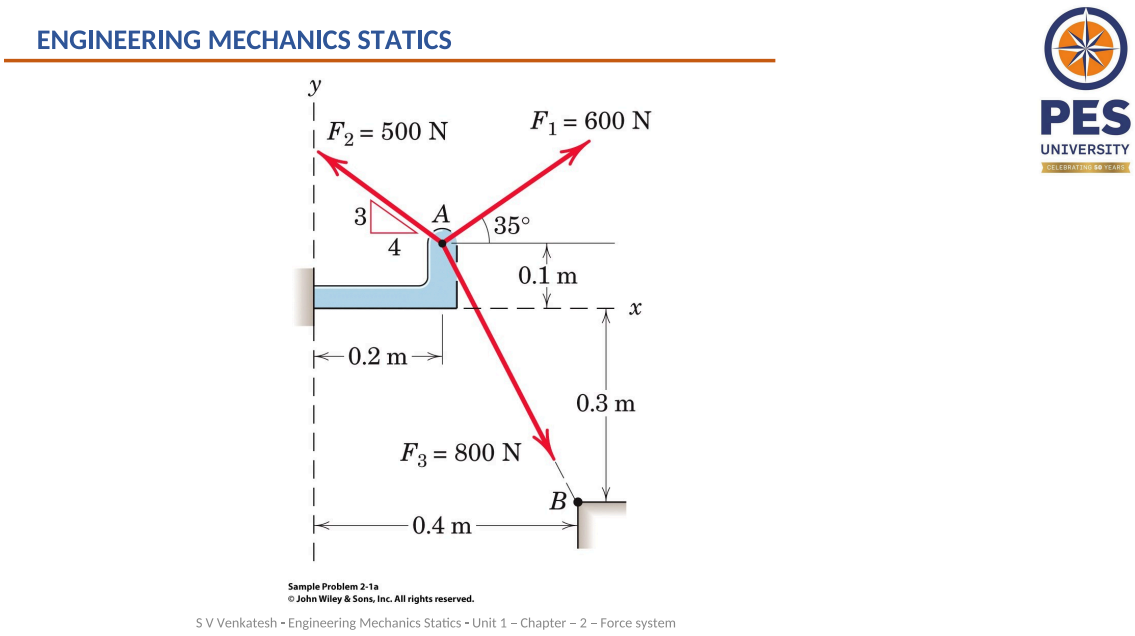


Figure G000040. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 2.

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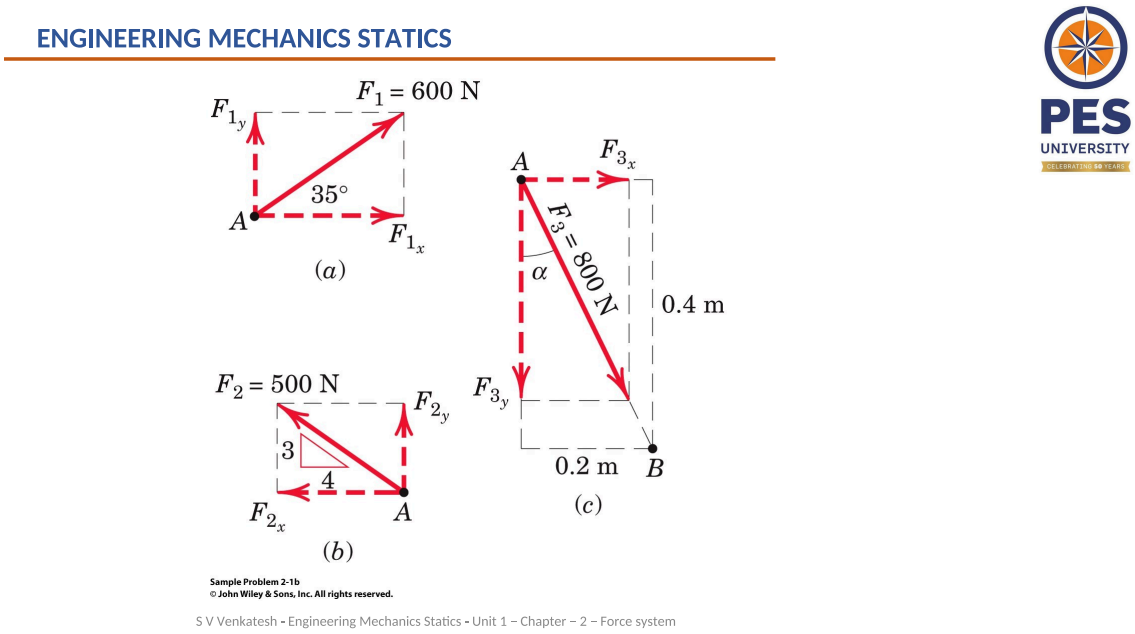


Figure G000041. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 3.

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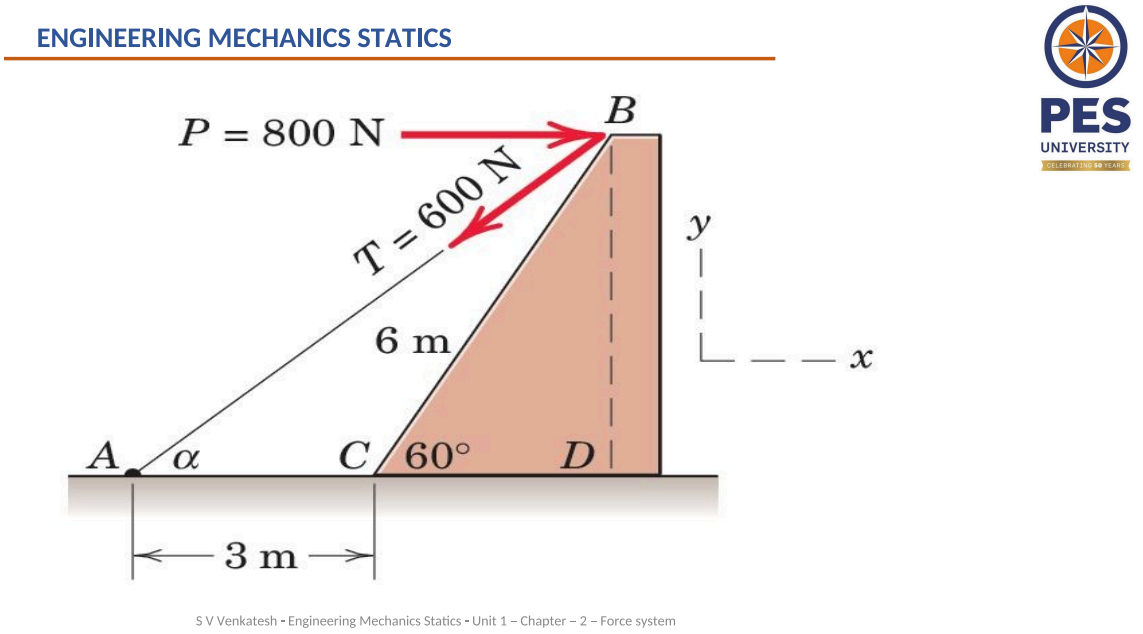


Figure G000042. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 4.

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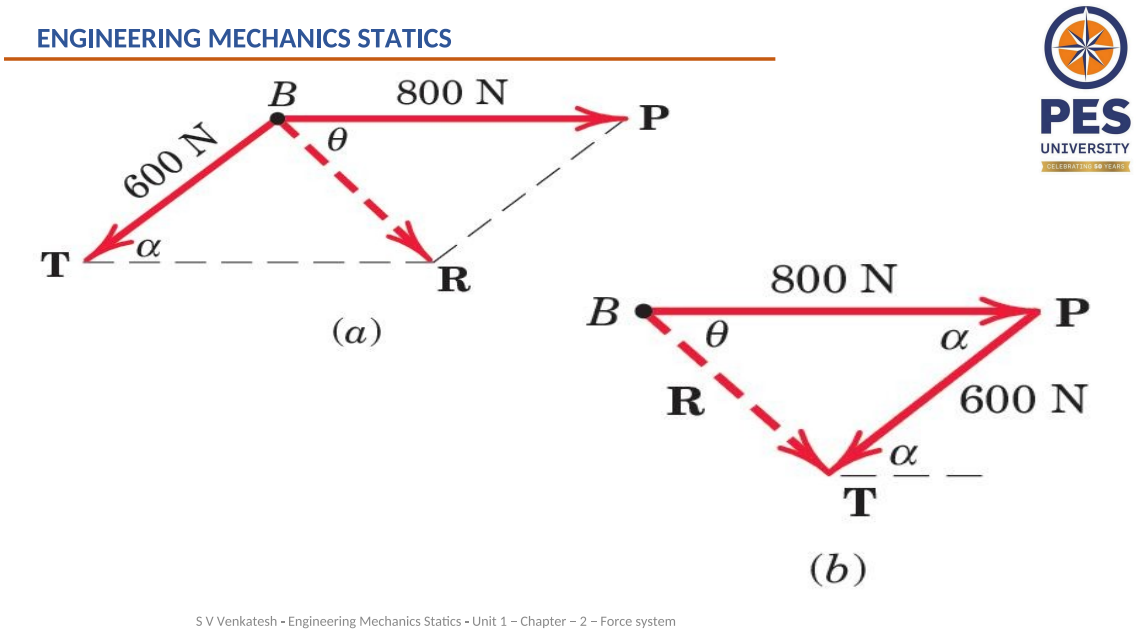


Figure G000043. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 5.

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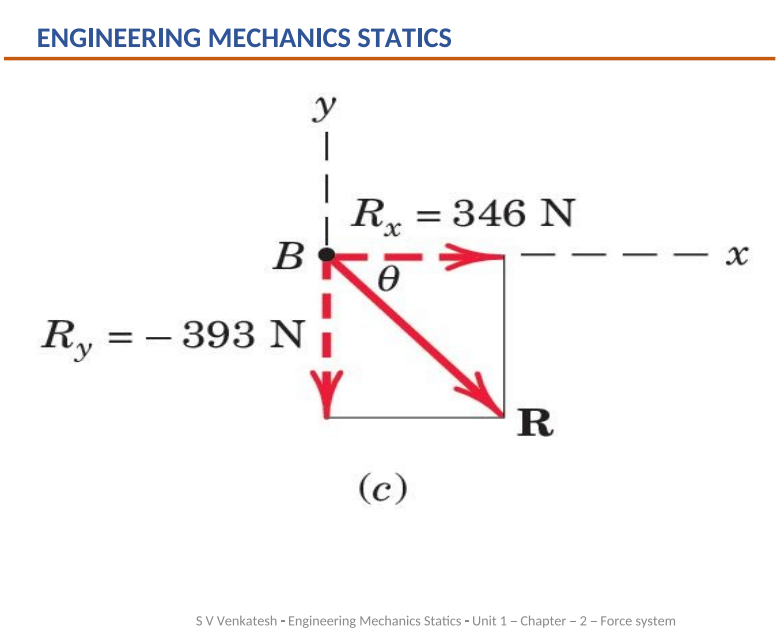


Figure G000044. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 6.

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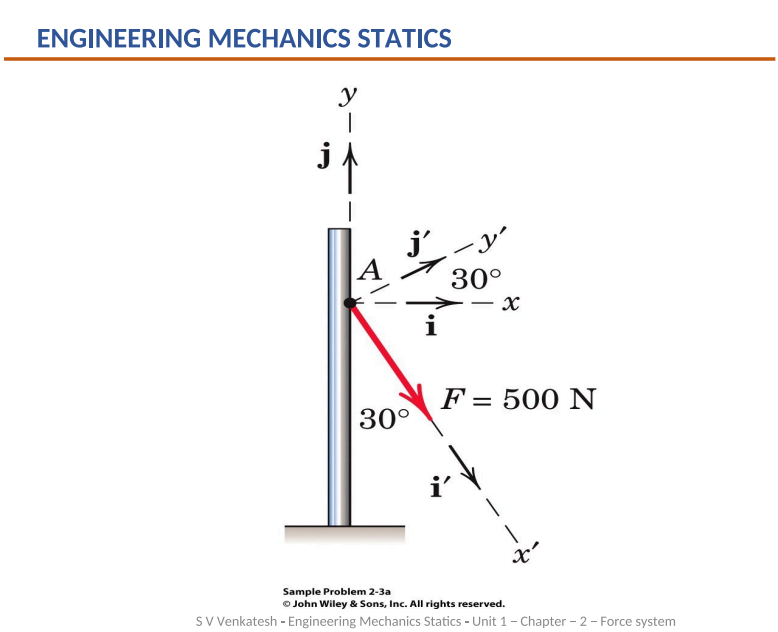


Figure G000045. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 7.

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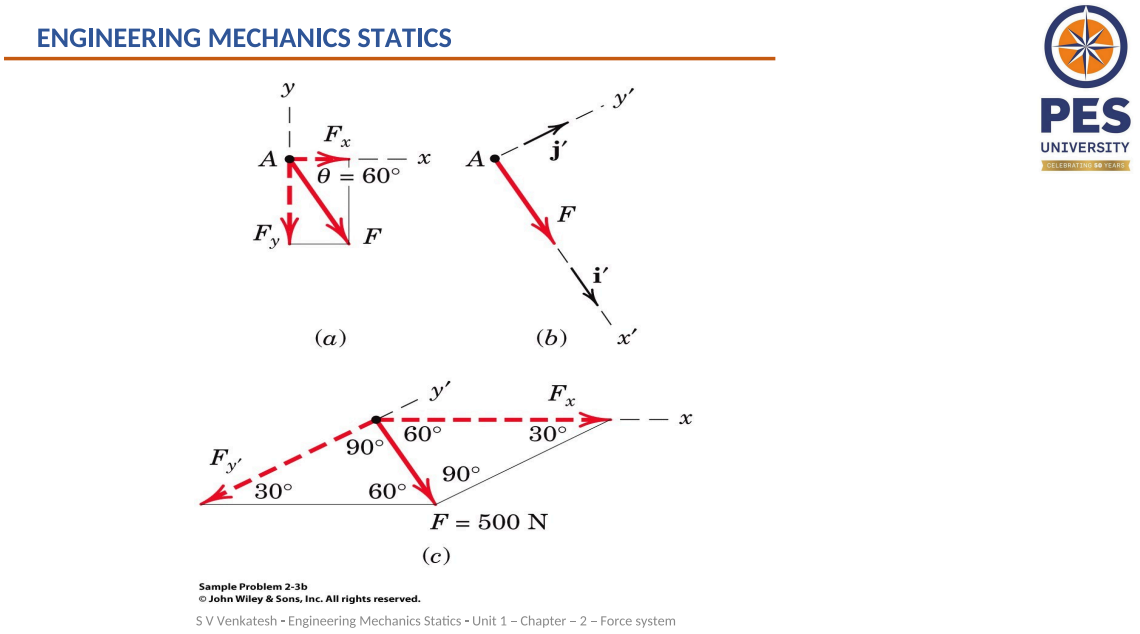


Figure G000046. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 8.

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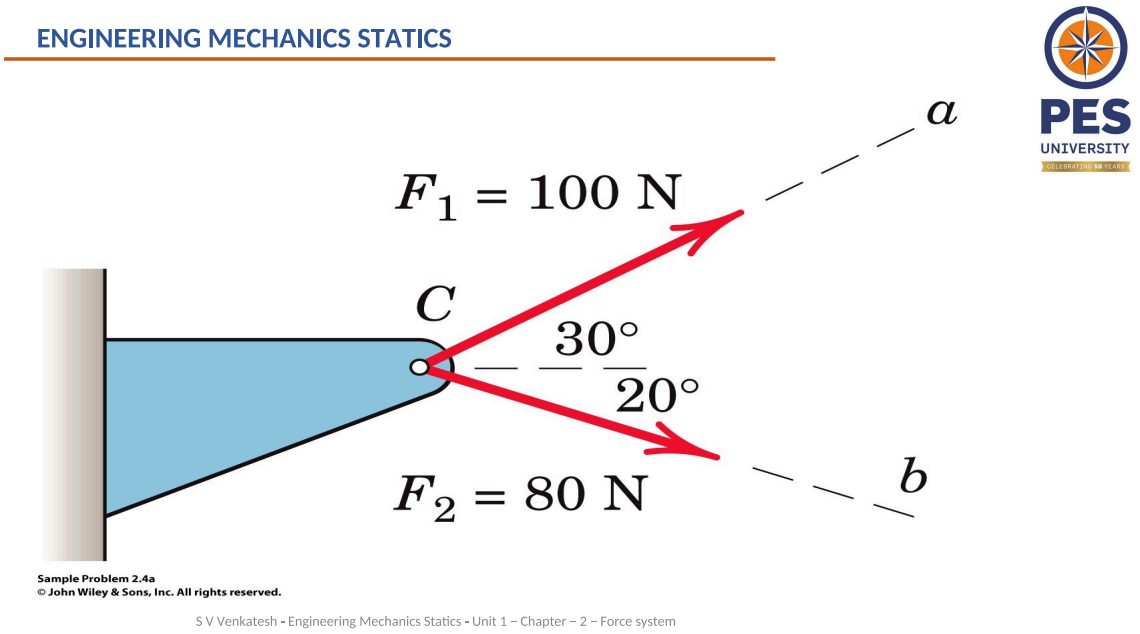


Figure G000047. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 9.

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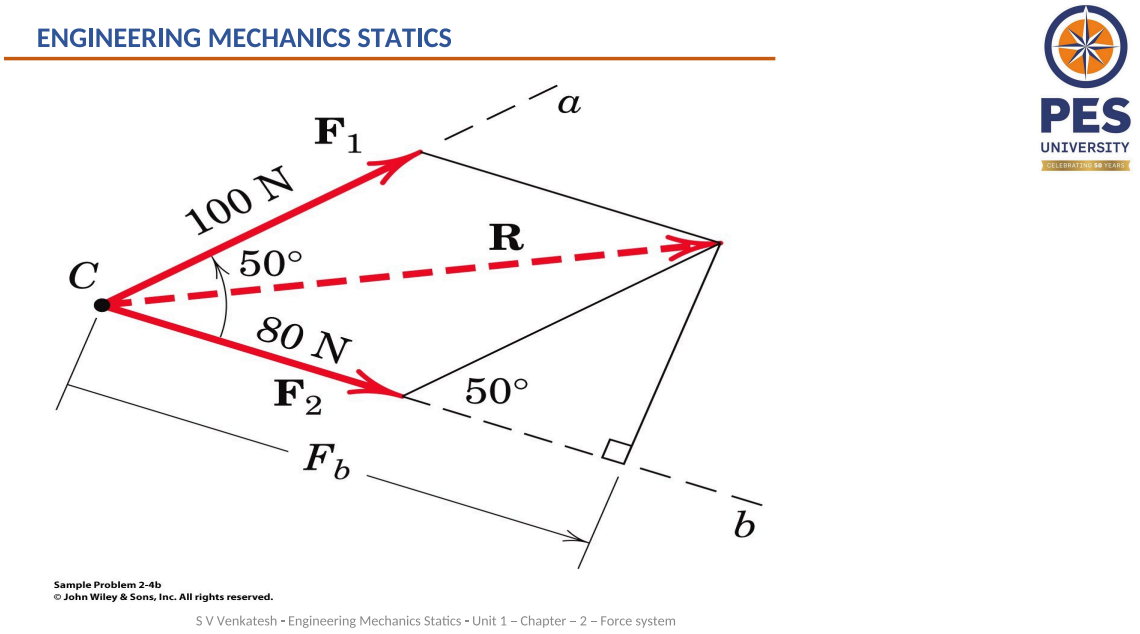


Figure G000048. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 10.

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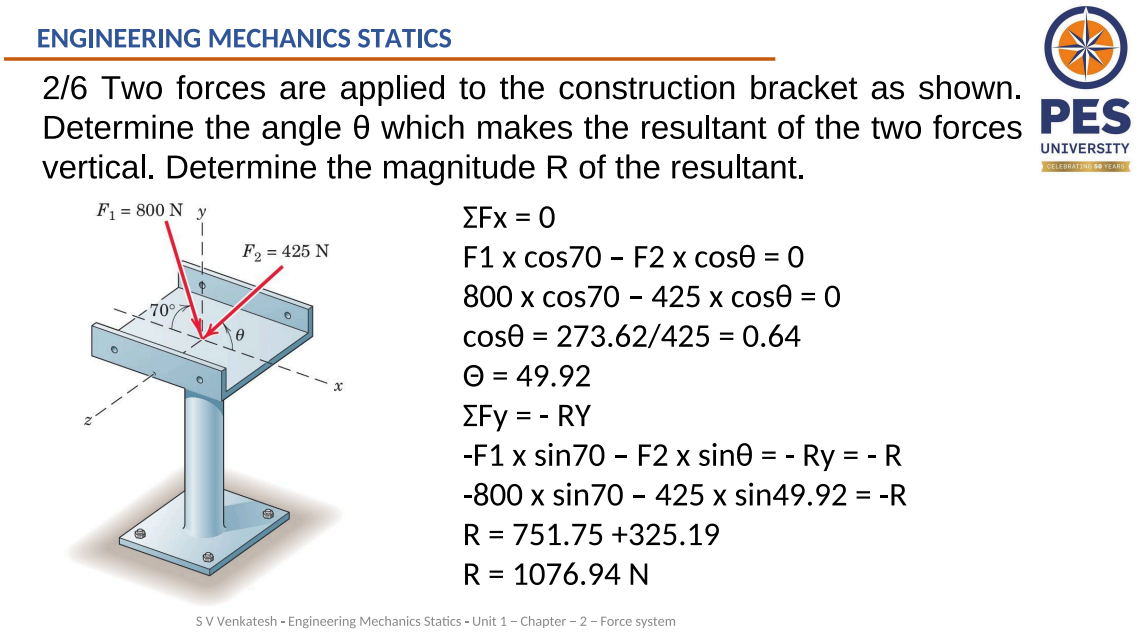


Figure G000049. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 11.

EV0000412

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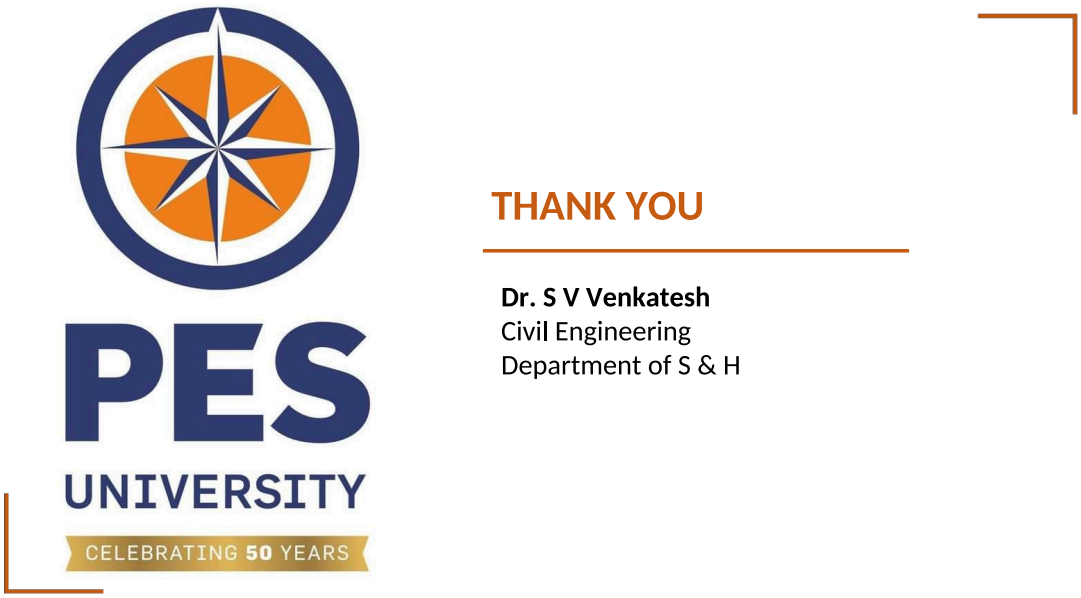


Figure G000050. UE26ME142A Unit 1 Class 3 Lecture PPT · source slide page 12.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000384 · EV0000385 · EV0000386
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000387 · EV0000388 · EV0000389
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000390 · EV0000391 · EV0000392
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000393 · EV0000394 · EV0000395
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000396 · EV0000397 · EV0000398
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000399 · EV0000400 · EV0000401
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000402 · EV0000403 · EV0000404

8. Source slide page 8 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ## Page / slide 9 (OCR) ENGINEERING MECHANICS STATICS PES a UNIVERSITY CELEBRATING50YEARS F1 = 100 N C 30° 20° b F2 = 80 N SampleProblem2.4a JohnWiley&Sons,Inc.Allrightsreserved. S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Chapter - 2 - Force system EV0000405 · EV0000406 · EV0000407
9. Source slide page 9: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000408 · EV0000409 · EV0000410
10. Source slide page 10 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000411
11. Source slide page 11 (extracted text): ENGINEERING MECHANICS STATICS 2/6 Two forces are applied to the construction bracket as shown. Determine the angle ϑ which makes the resultant of the two forces vertical. Determine the magnitude R of the resultant. ΣFx = 0 F1 x cos70 – F2 x cosϑ = 0 800 x cos70 – 425 x cosϑ = 0 cosϑ = 273.62/425 = 0.64 Θ = 49.92 ΣFy = - RY -F1 x sin70 – F2 x sinϑ = - Ry = - R -800 x sin70 – 425 x S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force sin49.92 = -R system 32 9 ## Page / slide 12 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000412
12. Source slide page 12: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000413

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.452] Dear students, welcome to the classes on engineering mechanics statics. My name is Dr. Svink Tish, Civil Engineering, Department of Science and Humanities, PS University. This is class number six. In the last class, we had discussed about the [00:00:20.516] a difference between components and projection. How do we determine the components? How do we determine the projection? What is rectangular? And several definitions that we were supposed to know to solve problems of engineer mechanics. [00:00:37.828] In this class we will look at what are the problems that we can solve. I've taken a few example problems to understand how do we go about solving these problems and let's see depending upon the time [00:00:52.656] We will try to solve one unsolved problem also. Okay. So we will take up this problem. The statement of the problem is the forces F1, F2, and F3, all of which act on point A of the bracket. [00:01:10.084] Are specified in three different ways. Determine the X and Y scalar components of each of these three forces. So what is given here is F1, the magnitude of the force is given. Now you can see that we have to read the statement of the problem. [00:01:28.996] We should read the diagram that is given. From the statement of the problem, what is given is there are three forces which is acting at point A, that we can make out from the diagram. [00:01:39.716] And what is asked is to determine the X and Y scalar components, only the components of the forces along X and Y. Now from the diagram, we can make out what is the magnitude of the forces, all the three forces, what is the sense of all the three forces, the heroid. [00:01:59.492] The point of application is given as point A. This is point A. Now the direction of the forces. He says three different ways. The direction of the forces is specified in three different ways. What are these three different ways? One is for [00:02:18.052] Force F1. They

have mentioned the angle made by the force with respect to the x axis. X axis is horizontal, y axis is vertical. This we have to make out from the diagram. Okay. Now you can see that there is a line here. [00:02:34.404] Or this says a line which is parallel to X axis. So the angle made by the force F one is thirty five degrees. [00:02:41.380] Next, for the force F2, they have not given the angle made by the force with respect to X or Y. Instead, they have given you what is the slope made by this force F2 by specifying what is the vertical and horizontal distance. [00:02:57.824] The vertical distance is three, the horizontal distance is four. So this forms a right angle triangle. So from that I can make out that this the hypotenuse distance is five. [00:03:08.388] Or we can also determine what is the angle made by F2 with respect to x axis or y axis. Say with respect to x axis, it will be equal to tan inverse of 3 by 4. Next. [00:03:21.924] The direction of the force F3 is not mentioned. Instead, they have mentioned the starting point of F3, the ending point of F3 as B, A and B, starting point is A, ending point is B. They have also given you the distances of point A. [00:03:40.676] from the origin as point two along x axis. Similarly from origin the point P as is at a distance of 0.4 meters along Y axis [00:03:52.772] They have also given you what is the distance of point A along y axis from the origin, which is point 1, and the distance of the point B along y ax from the origin as point 3, that is below. [00:04:07.172] So, using this distances, we can make out what is the vertical distance between A and B, that is the X distance as 0.4 minus 0.2, which is 0.0. Similarly, what is the y distance between A and B which will be equal to [00:04:22.576] 1 plus 0.3 point one plus point three, which is 0.4. So these each forces we will now draw a separate diagram to indicate the [00:04:35.844] magnitude of the force, the direction of the force, then the components of these forces along X and Y. So first force F1 is indicated here [00:04:45.956] This is F1 which makes an angle of thirty five degrees with the x axis. So the component of this force along X is indicated as F1x and the component along Y is indicated. [00:04:58.276] F1 y. Now how do we find out F1x? It should be equal to F cos 35 degrees straight away and F1y will be equal to F1 sine 35 degrees.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:12.484] Okay, so they have asked you to the scalar components. So F1x is equal to 600 cost 35, F1Y will be equal to 600 sine 35. [00:05:23.172] Now coming to the second force F2, which is 500 newtons. Now they have given you 3 and 4, and I told you the tripod is going to be 5. So F2X will be like this, F2Y will be like this. To determine F2x, if I take this. [00:05:41.892] It is gonna be equal to F two cause of this angle, cause of this angle is [00:05:49.284] 4 divided by 5 into 500 will give me F2x. Similarly, F2y will be equal to 500 into sine of this angle, which is 3 by 5. So I get F2y. Now coming to the [00:06:06.308] Force F3. F3 is acting here. Now if I take that angle made by the force F3 with respect to y axis as alpha, I can easily calculate what is the angle alpha as tan inverse of 0.2 by [00:06:23.620] uh point four because this forms a right angle triangle. So in this triangle if I take tan alpha, tan alpha will be opposite pi [00:06:33.120] Adjustant opposite is 0.2, adjustant is 0.4, so I can get alpha. So once I get alpha, now I can resolve this force F3 along X as F3 sin alpha, which is 800 sine alpha, then F3Y will be [00:06:50.096] eight hundred cause alpha. These are the [00:06:53.636] scalar components of the forces F1, F2, and F3. I hope this is clear. Now let us move on and take one more example. This example uh the statement of the problem states that combine the two forces P and T. [00:07:10.948] which act on the fixed structure at B into a single equivalent force R. So in this problem they have asked you to determine the [00:07:21.476] Resultant of the two forces that is force P and T, what is the resultant of these two forces? We can solve this in two different

approaches, or whichever approach you want, you can take several approaches. So the two different approaches that I am talking is one is using the law of [00:07:40.132] Parallel of forces, the other one is using the law of triangle of forces, and one more is by resolving P under T [00:07:49.760] Along X and Y, and then combining out finding out the dot product of this, you will get the result. Let's look at all these three different ways of solving the problem, one at each time. So let's take up this diagram. Now [00:08:03.396] You can see that P is represented here, which is 800 newtons. The T is given here in the problem, which you can see, where that is being taken and applied at this point from the origin B. So this. [00:08:22.884] forms or this two forces are represented now on the adjacent sides of a parallelogram. Okay? So 800 newtons and 600 newtons. So P is this and D is this. Both of them are moving away from point B. So the [00:08:38.160] Resultant is given by the diagonal of the parallelogram passing to the intersection of these two forces. So if I draw a diagram here, if I draw the diagonal here, that gives the resultant of these two forces. And how do I find this magnitude of the resultant? That would be equal to P square. [00:08:56.932] Plus T square plus 2 times P into T into cos of the angle made by these two. How do I find the angle? So that is given in the problem as theta, and this is given as alpha. So I can find out what is this. [00:09:14.160] Total angle. So if this is alpha, this would also be alpha. Okay. So this total angle, if this is alpha, this total will be one eighty plus. [00:09:27.620] Uh or rather 180 minus alpha. So this total angle will be 180 minus alpha. I can find out what is the result and r from that. Now similarly I can use the law of triangle of forces where P is represented like this, T is represented like this. [00:09:43.588] The two forces are taken on the adjacent sides of a triangle, taken in order, you can see that this is in order. The closing side of the triangle taken in the opposite order gives us the resultant. So you can find out what is the resultant force by using law of triangle of forces as well.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.884] The other way of solving this is by considering or finding out the component of the two forces along X and Y, and then finding out the resultant R. Now here Rx value we have obtained by taking R1x plus. [00:10:19.536] R2 X. Similarly, R y is taken as R1 Y plus R2Y. So the values are given here. So you get X and Y, Rx and R y. So now I can complete this rectangle and [00:10:34.980] So I get the result at R from here by taking this. How do I get Rx square plus Ry square? I get the magnitude of R. So both of them you have you have. [00:10:48.516] R X coming this, R Y coming this. So these are the two forces. When I merge these two, I get the resultant force acting like this. This is how we can determine the resultant of the of these uh two forces. So I hope you have understood. Let's go at [00:11:05.312] and try to solve another problem. Now this is a this is one unique problem a very uh important one when we are going to be applying or considering [00:11:16.388] Vector approach to solve. In the two previous problems that I have explained, what we have looked at is how do we solve it by going ahead and solving it in scalar approach. Now we will look at how do we solve a problem using vectors. Okay. So the problem statement states that [00:11:33.636] The 500 newton force F is applied to the vertical pole as shown. This is a vertical pole. A 500 newton force that is this one is applied to this vertical pole as shown. This is acting at point A, which makes an angle of 30 degrees with the [00:11:50.704] Y-axis or with the pole, the forces are acting at an angle of thirty degrees. The magnitude of the force is 500 newtons. Now what has to be determined is one. [00:12:01.988] Write F in terms of the unit vectors I and J and identify both its vector and the scalar components means what? We have to find out the component of the force f along x and

along y. [00:12:21.668] How do I make out that? It is i and j. So we have to get with respect to i and j, i and j are the unit vectors along x and y. So that's why we have to determine the component of this force along x and y. [00:12:35.684] Then we have to determine both scalar and vector approach. So vector approach is we have to find out what is FXI plus FYJ. Scalar means only Fx and Fy, we have to determine that. [00:12:49.092] The second part of the problem is determine the scalar component of the force vector along the x-dash and the y-dash axis. Which is the x-dash axis? [00:13:01.348] The axis along the force itself is x dash and perpendicular to that is y dash. You have to find out the component of the force along this x dash and y dash. That is second one. The third one is determine the scalar components of force. [00:13:19.492] Yes, along the x and the x-dash axis, that means what? I have to find out the scalar component of this force along x and along y-dash axis, along this axis. That is what we have to find out. So how do we go about solving this? [00:13:36.196] So let's look at that. So first thing we have taken the first case where we have to determine the [00:13:43.492] A component of the force along X and Y. So the force is represented here, X and Y axis also are represented. The component of the force is also represented along X and Y. So the angle is given as 60 degrees. So here of cost 60 degrees I get fx. [00:13:59.492] Whereas F sine sixty degrees with a negative sign, I'll get it as F y. Why it is negative? Because the sense of the force F y is acting downwards, so I get it as negative. [00:14:10.116] Now to express this as a vector, it is nothing but fx i minus f y j because f y is acting downwards, so it is negative. The scalar component is just fx which is equal to f cos 60 degrees. The scalar component of y that is f y is equal to f cos 60 degrees, that's it. [00:14:30.052] Okay. The next one is to determine the component of the force along x dash and y dash axis. Now x dash axis is acting along f itself and y dash axis is perpendicular to f. So the component of the force along x dash is f itself. [00:14:50.212] The component of the force along Y will be F, cos ninety degrees, and we know that cos ninety degrees is

EV0000094 · EV0000095 · EV0000096 · EV0000097

Transcript block 4 (source interval 000900-001199 seconds): [00:15:01.252] uh 0. So your component of the f along y is 0, only the component along x dash you will get, whereas the component along y dash is 0. Next coming to determination of the scalar component of the force f along x and y dash axis. [00:15:20.612] So you can see F is acting here, X is this axis, Y dash is this axis, so Y dash is perpendicular to F, so that is the component you will get it here. [00:15:34.244] Where you rotate it and merge it up to ninety less we can rotate it. So you get F Y dash in the negative sense, whereas F x you get as positive. So complete the [00:15:45.476] Parallel of forces. They have asked you to determine the components. Component is when we have to determine by using law of triangle of forces. Component will produce the same effect as that of the force. So it should follow law of parallel of forces. So you can complete the parallelog here and [00:16:05.508] We can determine the component of F along F x and FY. For that, we have to determine the angles. So what is given is this is 60 degrees, and we know that F y is perpendicular to F, so this would be 90 degrees. So [00:16:23.044] This is parallel to this, so this forms the same angle, 60 degrees and 60 degrees. [00:16:32.132] Ah whereas these two will be the same, so they are alternate angles, so 90 and 90. So in this triangle, if you take the sum of the angles will be 180, so 90 plus 60, and this angle plus this angle should be equal to 180. So I get this as 30. Similarly, for this triangle also. So now [00:16:51.748] We know that this forms a triangle. So go for sign rule to determine the components. So I can write Fx. [00:17:00.964] by sine 90 is equal to f pi sine 30. So using that I will get f x. Similarly if I take this triangle, F y by sine 30 will be equal to f pi sine sorry f y dash

[00:17:17.184] By sine 60 will be equal to f by sine 30, so I can get f y dash. So I hope this is clear too. Now let us look at the another problem. The statement of the problem is forces F1 and F2 act on the bracket as shown. [00:17:36.196] Determine the projection of FP of their resultant R onto the B axis. Clearly understand the problem. What is asked is there are two. [00:17:46.788] Forces F1 of magnitude 100, F2 of magnitude 80 newtons, the sense of the two is given, the direction of the forces are also given as 30 degrees with respect to X, 20 degrees also with respect to X. This is axis A, this is axis B. [00:18:03.812] They have asked you to determine the projection of these the resultant of these two forces. First we have to find out what is the resultant of these two, then projection of the resultant onto the B axis. Okay, that's what we have to find out. [00:18:20.452] Now, how do we find out the projection? Let's look at this. So to find out the projection of the two, I'll represent these two forces F1 and F2 on the sides of a parallelog. So I'll construct a parallel by taking these two. [00:18:34.468] How do I construct it? We have this force and this force. We know this angles. The total angle is uh 50 degrees. That is 20 plus 30 is 50. So we get this as 50. So this force F1. [00:18:52.804] Is along A and this force F2 is along B. Now I will draw a line parallel to B passing through this edge. This is parallel to this. Similarly, I will draw a line parallel to A passing through. [00:19:07.780] Uh this edge, so these two lines when I draw, it was going to cross each other at this point. So joining C to this point would give me the resultant R. [00:19:19.332] Now what are we supposed to find out? The projection of R onto the B axis. So what I will do from this edge, I'm going to draw a line which [00:19:32.068] Touches B axis such that this angle is at 90 degrees. So this entire line from C to this point would be the projection of the result end on the B axis. Now how do I get that? [00:19:47.908] That would be equal to F P plus this value. This value will be equal to F1, which is same as this. [00:19:56.580] Cause of this will be this value. So your projection of the resultant R onto the B axis will be equal to F2, which is 80, plus the component of this 4 that is F1, on this, which is 100 cos 50 degrees.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:14.000] Which is equal to 144.3 newtons. That would be the projection of the resultant on the B axis. Now the magnitude of the resultant, how do we find it? I have explained it here. So by using law parallel gram of forces, so R squared is equal to. [00:20:33.828] F2 square plus F1 square plus 2 times F1 F2, cause of the angle between the two. That is uh [00:20:42.980] This angle which is 50 degrees, I'm sorry I have made a mistake here. I have taken it as 130, it should be 50 degrees. So if you take it as 50 degrees, you will get it as 163.4 newtons. I hope this is very clear. Okay. [00:21:00.260] Right. So may uh the correction is this is this angle has to be fifty degrees, it is not one thirty degrees. Okay so one more example I've taken. We will come back to this because I'm shooting uh the time. We will look at this problem in the next class. Thank you very much. [00:21:17.924] Uh my name is Doctor Svang Tesh. We'll meet in the next class, class number seven. Thank you.

EV0000102 · EV0000103 · EV0000104 · EV0000105

Full transcript: 3356 normalized words, 87 timestamped segments, 1284.032 seconds duration. Low-confidence segments flagged: 0.

EV0000086 · EV0000087 · EV0000088 · EV0000089

8 UE26ME142A_Unit 1_Class 4_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 4_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000106 · EV0000107

Graphs and diagrams



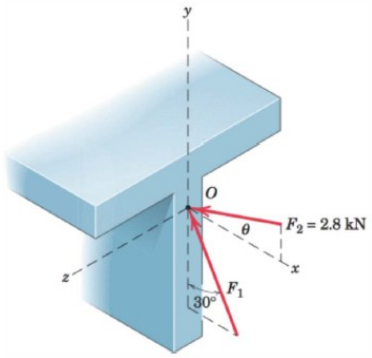
Figure G000051. UE26ME142A Unit 1 Class 4 Lecture PPT · source slide page 1.

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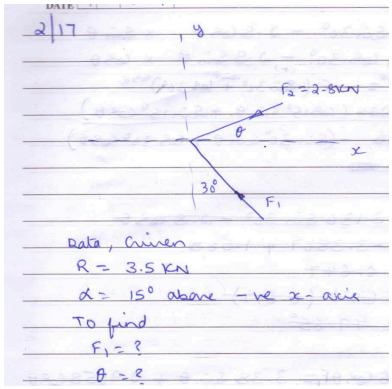
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ENGINEERING MECHANICS STATICS

2/17 The two forces shown act in the X-Y plane of the T-beam cross section. If it is known that the resultant R of the two forces has a magnitude of 3.5 kN and a line of action that lies 15° above the negative x-axis, determine the magnitude of F1 and the inclination θ of F2.



Problem 2/17



S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

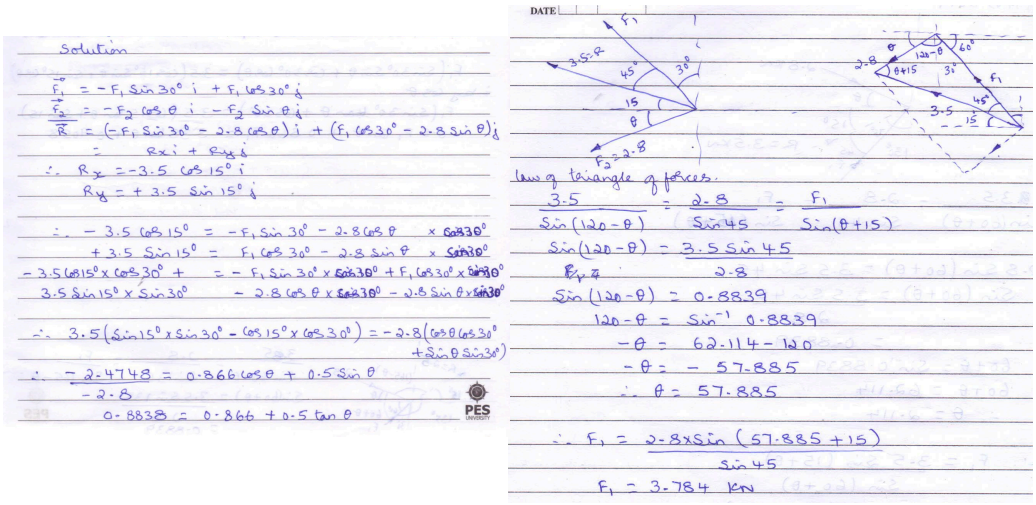
Figure G000052. UE26ME142A Unit 1 Class 4 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS STATICS

2/17 cont..



S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

Figure G000053. UE26ME142A Unit 1 Class 4 Lecture PPT · source slide page 3.

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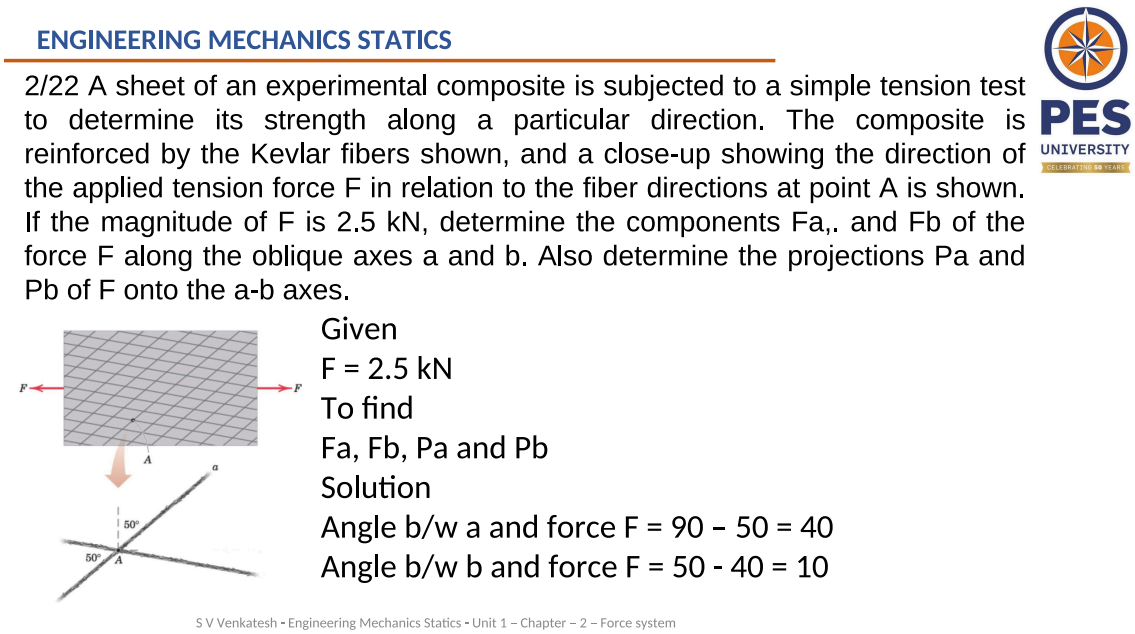


Figure G000054. UE26ME142A Unit 1 Class 4 Lecture PPT · source slide page 4.

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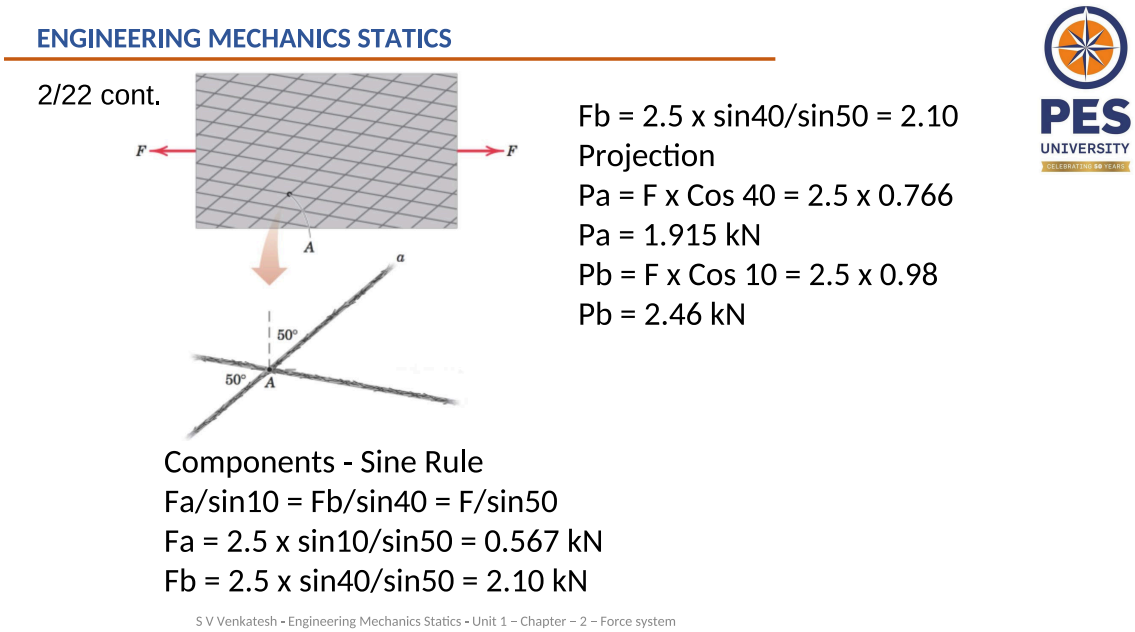


Figure G000055. UE26ME142A Unit 1 Class 4 Lecture PPT · source slide page 5.

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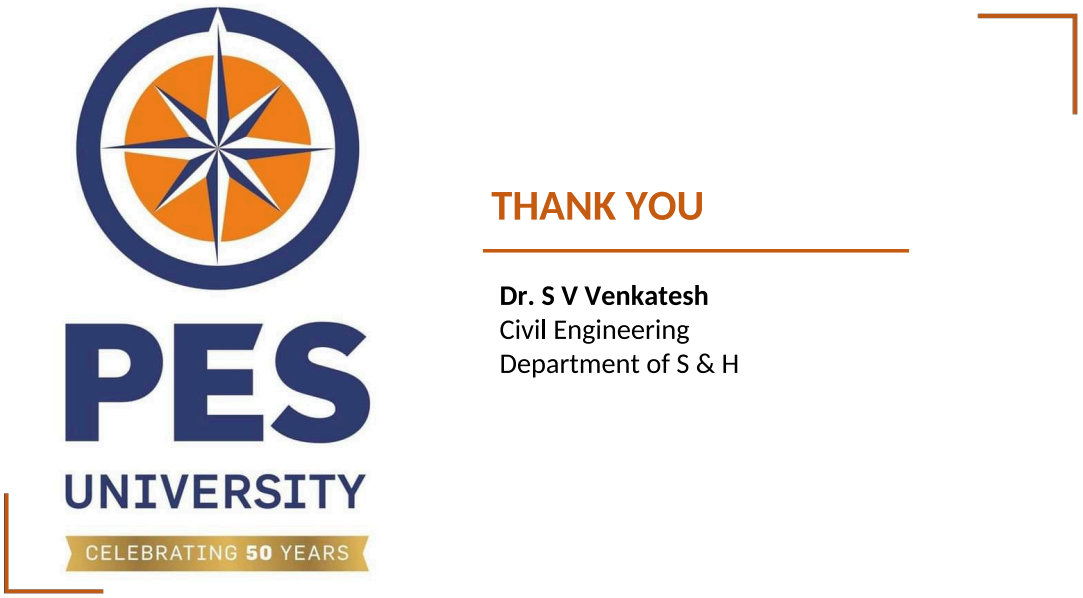


Figure G000056. UE26ME142A Unit 1 Class 4 Lecture PPT · source slide page 6.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000414 · EV0000415 · EV0000416
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 2/17 The two forces shown act in the X-Y plane of the T-beam cross section. If it is known that the resultant R of the two forces has a magnitude of 3.5 kN and a line of action that lies 15° above the negative x-axis, determine the magnitude of F1 and the inclination ϑ of F2. S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ## Page / slide 3 (OCR) ENGINEERINGMECHANICSSTATICS 2/17 cont.. DATE PES soutm 2 UNIVERSITY 0+15 30 CELEBRATING50YEARS S3°+F0° 15 Cϑic-F2Sϑ (F, (030- 2.8Sϑ) S3°-2.8Cϑ)+ Rx+Ry =+3.5S15° 3.5 E S (10- $45 S(ϑ+15) 3.5C15 FS30°-2.8Coϑ xS0830 +3.5S15 FC3-8S×S S<(2D-ϑ) 3.5SM45 -3.5615×C0S30°+ -FS30°xS3ϑ+F,C30°x3ϑ 28 3.5S15xS30° 2.8Coϑ×S3ϑ-.8Sϑx3ϑ S(-) 0-8839 120-9= S 0·8839 3.5 (S15xS30°-Cos15x(S30)=-28Cϑ(30° A 62-114-100 +3 57.885 2. 4748=0.866oϑ+0.5S˜ϑ 2.8 ϑ- 57.885 8838=0-86b+0-5tanϑ PES 2- 8xS.(57.885 +15 S45 E-3.784CN S VVenkatesh-Engineering Mechanics Statics- Unit 1 - Chapter - 2 - Force system EV0000417 · EV0000418 · EV0000419
3. Source slide page 3: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000420 · EV0000421 · EV0000422

4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 2/22 A sheet of an experimental composite is subjected to a simple tension test to determine its strength along a particular direction. The composite is reinforced by the Kevlar fibers shown, and a close-up showing the direction of the applied tension force F in relation to the fiber directions at point A is shown. If the magnitude of F is 2.5 kN, determine the components Fa,. and Fb of the force F along the oblique axes a and b. Also determine the projections Pa and Pb of F onto the a-b axes. Given F = 2.5 kN To find Fa, Fb, Pa and Pb Solution Angle b/w a and force F = 90 – 50 = 40 S V Venkatesh - EngineeringAngleMechanicsb/wStaticsb- Unitand1 – Chapterforce– 2 – ForceF = 50 - system 40 = 10 EV0000423
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS 2/22 cont. Fb = 2.5 x sin40/sin50 = 2.10 Projection Pa = F x Cos 40 = 2.5 x 0.766 Pa = 1.915 kN Pb = F x Cos 10 = 2.5 x 0.98 Pb = 2.46 kN Components - Sine Rule Fa/sin10 = Fb/sin40 = F/sin50 Fa = 2.5 x sin10/sin50 = 0.567S V Venkatesh kN- Engineering MechanicssystemStatics - Unit 1 – Chapter – 2 – Force ## Page / slide 6 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000424
6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000425

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.516] Dear student, welcome to Engineer Mechanics Statics classes. [00:00:05.220] I am Dr. SV Venkitesh, Civil Engineering, Department of Science and Humanities, PS University. In the last class, that is class number six, we had discussed uh several problems and how do we solve them by both [00:00:22.272] scalar approach as well as vector approach and they were solved problems. Now we will look at some of the unsolved problems from the engineer mechanics textbook eighth edition. [00:00:41.508] The statement of the problem is like this. Two forces are applied to the construction bracket as shown. Determine the angle cheetah, which makes the resultant of the two forces vertical. Determine the magnitude R of the resultant. [00:01:00.228] Now when you read the statement, what is that you need to look at is that they have asked you to determine the angle theta, that is this angle, which the resultant of the two forces uh [00:01:15.716] makes it vertical. Okay, so if you are to find out this angle theta such that the resultant of these two forces would act along the vertical. And they have also asked you to determine the magnitude of the resultant R. Okay. Now what is given? [00:01:34.916] So, one is when you read the statement, we could make out what is that we need to find. Now, read the diagram. From the diagram, what is that we need to look at is they have given you the magnitude of the force F1, which is [00:01:53.028] 800 newtons, and they have also given you what is the direction of this force F1 with respect to the negative x axis, which makes an angle of 70 degrees. No down, this is the angle made by the force F1. [00:02:10.160] With respect to X, okay. The force F1 is in the plane of X and Y only. So how can you make out by looking at this angle that is 70 degrees. [00:02:23.972] Next, they have also given you the magnitude of the force F2, which is equal to 425 newtons, and the angle theta which we have to determine is the angle made by the force F2 with respect to x axis again. [00:02:41.316] Now again this force F2 is also in the plane of XY only. Now the two forces that is given or in the plane of XY itself, so obviously the resultant will also fall. [00:02:57.572] In the XY plane itself. So it's a two-dimensional force system. Even though you see that the diagram that is given, the bracket that is given is three-dimensional,

but the forces acting [00:03:11.936] is in the same plane that is XY plane. So we can treat this as a two-dimensional problem. Now your uh syllabus states that you have engineer mechanic statics and it is only 2D we don't have 3D in that okay now how do we go about solving this [00:03:31.300] Now to get the result of these two forces, first thing that I'm going to do is I'll resolve both the forces F1 as well as F2 along its components, that is along x-axis and y axis. [00:03:45.732] F1x and F1Y. Similarly, I will find out the component of F2, F2X and F2Y. So once I resolve this, what do I get? I get F1x as F1. [00:04:03.268] cos 70 degrees f1 y as f1 sinilarly for this f2 x will be equal to f2 cos theta and f2 y will be f2 sine theta no adding up of [00:04:20.516] The components along x axis, okay, that is the component of f1 along x, that is f1x, and the component of f2 along x, which is f2x, if I add these two, that should be. [00:04:35.556] equal to zero. Why should that be equal to zero? Because what are we supposed to find out is the angle theta which makes the resultant to be vertical. So the vertical component of the resultant of these two forces [00:04:51.840] It should be 0. So that's why we can see here summation of all the forces along X is equal to 0. So we get F1 into cos 70 degrees minus F2 into cos teta should be equal to 0. Why? Because F1 is acting along X, it's positive.

EV0000106 · EV0000107 · EV0000108 · EV0000109

Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.184] The component of F1 acting along X is positive. The component of the force [00:05:15.396] Along X of F2 is negative because the sense of it is opposite to X. So if this component should be equal to 0. By solving it, we are going to get the value of theta which is 49.92. [00:05:31.492] Then, so that is what we have to find out in the first case, theta. Theta is what we have got. The second case, they have also asked you to determine the magnitude of R. So to determine the magnitude of R, take the components of these two forces along acting along y. [00:05:47.940] Find out the summation of that. That will be equal to the resultant force. [00:05:56.804] And the force, the resultant force will be acting along y only, so you get that value. So summation of FY is equal to minus ry it is minus r y. Both F1 Y and F2 Y both of them are acting downwards, so it will be negative. [00:06:13.956] So the component of F1 along y will be minus F1 into sine 70. The component of F2 along y will be minus F2 into sin theta, which is equal to minus r y, which is equal to minus r. That is the resultant. [00:06:30.116] So substituting the values, we will be able to get the magnitude of the resultant R. Obviously, this is acting downwards, only the magnitude is given here, and of course it is acting downwards. I hope all of you have understood this problem. Now we'll move on to the next problem. [00:06:50.084] The next problem that I've chosen is 2 bar 17. The statement of the problem goes like this. [00:06:57.540] The two forces shown act in the XY plane, they have already given that it is acting in the XY plane. So there's no confusion at all. Even the diagram that is given is three-dimensional, but since they are told that the forces acting is in the XY plane, we can treat this as a two-dimensional force system. On the TB. [00:07:18.340] Read out from the beginning. The two forces shown at in the XY plane of the T beam cross section. If it is known that the resultant R of the two forces has a magnitude of 3.5 kilometer, the magnitude of R is given. Okay. And [00:07:36.516] A line of action that lies 15 degrees above the negative x axis. What is this? The resultant magnitude is given as 3.5 kilowatt and it says that the line of action of the resultant is at an angle of 15 degrees above the negative x-axis, that means [00:07:54.308] The resultant will be acting somewhere here. So this is the negative x axis 15 degrees from this would be the resultant line of action. Okay. Now what is asked is determine the magnitude of F1 and the inclination theta of F2. [00:08:12.548] So from the statement what we can make out is we can get

the magnitude of the resultant, we can get the direction of the resultant, and we can also get the sense of the [00:08:22.736] Resultant, of course, the point of application will be the same as the point of application of the two forces. So we are able to get the characteristics of the resultant from the statement. Now looking at the diagram, what is that we have to take from the diagram is what is the magnitude of the force. [00:08:42.116] Or F1, which is which we have to find out, or rather, what is the direction of the force F1, which makes an angle of 30 degrees with the y-axis, then what is the magnitude of F2, which is 2.8 kilonewtons. [00:08:57.056] And the angle made by F2 with respect to x axis is theta. We have to determine the magnitude of F1 and we have to determine the direction of F2. That is with respect to x axis. [00:09:11.780] Okay now drawing the free body diagram for this you can see it here this is a free body diagram and he says that it is in the x y plane so x is this axis y is this axis [00:09:24.932] F2 is acting towards the origin, the magnitude of that is 2.8, and it makes an angle of theta with respect to the positive x-axis. [00:09:37.028] Excuse me, F1 is also acting towards the origin. It makes an angle of 30 degrees with the negative y axis. [00:09:45.860] Now that is also acting towards the origin. So obviously, when we are going to find the magnitude of these two forces, when we merge the two of them, it will act towards the origin, obviously. Okay. So the data that is given us, the resultant.

EV0000110 · EV0000111 · EV0000112 · EV0000113

Transcript block 3 (source interval 000600-000899 seconds): [00:10:03.616] Magnitude is 3.5 kilonton. Alpha is the angle made by the resultant with respect to the negative x-axis of the state is above, that is 15 degrees. What are we supposed to find? Is the magnitude of F1 and the angle theta of the force F2 with respect to x-axis. [00:10:23.684] Let's look at the solution. So here you can see I have represented the force F1 moving away. [00:10:33.060] F2 also moving away and the result end also. So your F1 is like this, F2 is like this. So when I merge these two, I get the result end. Okay. So F1 the angle is 30 degrees and F2 the angle is theta. Now this is 15 degrees that is given. [00:10:51.876] Okay. So obviously the angle made by F1 with respect to R will be 45. How is it 45? So this is 30. So this is y axis, this is x axis. So obviously x and y will be perpendicular to each other, that is 90 degrees. 90 minus 30 minus 15. [00:11:10.148] will be forty five degrees. Okay. Now this I have represented it here so that I can get the triangle [00:11:20.196] Using the law of triangle of forces, so F1, F2 taken in order. So the resultant will be the closing side of the triangle taken in the opposite order. So we have the magnitude of the resultant 3.5. [00:11:33.892] The magnitude of the force F2 as 2.8, and we are supposed to find out the magnitude of F1. We know the angle made by F1 with the resultant R that is 45 degrees. We can also determine. [00:11:50.320] What is the angle made by the force F2 and the resultant which will be equal to theta plus 15? How is it? It's theta plus 15 here. This [00:12:04.324] When I put it here the this would be the alternate angle for this which will also be equal to theta plus fifteen. So whereas this angle will be sum of all the three will be 180. So it will be 180 minus whatever is the angle made by this. [00:12:21.988] Okay. So that would come up to 120 minus Theta. Okay? [00:12:27.236] So using this angles and the values, I can use law of triangle of forces. It states that 3.5 divided by sine of the sine of this angle, that is sine of 120 minus theta, should be equal to [00:12:42.592] Magnitude of this force which is two point eight, opposite angle as forty five, so sine of forty five should also be equal to this force F one. Angle opposite to that is this which is uh [00:12:55.556] Equal to theta plus 45 sine of theta plus 45. So this is using law of triangle of forces. So by this I can get the value of theta from this when I consider 2.8 sin 45. [00:13:14.756] And 3.5 sine 120 minus theta. If I take these two equations, I can rewrite this equation as sin of 120 minus theta

equals 3.5 sine 45 divided by 2.8, from which I can find out the value of theta. [00:13:32.932] By simplifying it. So we get the value of theta as fifty seven point eight eight five. Now [00:13:39.236] Once I get the value of theta, I can take these two equations, okay, substitute the value of theta here, and then find out the value of f1. So we get f1 as 3.784 kilo Newton. Now here in this problem. [00:13:54.240] I have not gone for resolution of forces F1 and F2, because when you go for resolution of forces, the solution will be a lengthier one. So it is easier to solve this problem, especially to our 17 by [00:14:09.344] The law of triangle of forces or using the sine rule. Law of triangle of forces is nothing but which we have got it from sine rule. [00:14:18.372] Hope this is clear to everybody. Okay, let's go on to the next problem. I've taken one more problem, 2 bar 22. The statement of the problem goes like this. A sheet of an experimental composite is subjected to a simple tension test. [00:14:36.036] To determine its strength along a particular direction, the composite is reinforced by the Keplar fibers shore. [00:14:47.492] And a close up showing the direction of the applied tension force F in relation to the fiber directions at point A is shown. If the magnitude of F is 2.5 kilonewt, determine the components FA.

EV0000114 · EV0000115 · EV0000116 · EV0000117

Transcript block 4 (source interval 000900-001199 seconds): [00:15:06.916] And F B of the force F along the oblique axis, A and B also determine the projection P A and P B of the force F onto the A and B axis. [00:15:23.268] Now this problem is the combination of both projection as well as component. What is given is you have a diagram indicating the composite sheet to which a force F is applied, a tensile force F is applied. Now [00:15:42.628] If you take a small enlarged view of the intersection of the fibers, it is got to look like this. So here you have two axes or the fibers, one is along A, the other one is along B. [00:15:57.540] They have also given you what is the angles made by these fibers with respect to the axis. So this is 50 degrees, they have given this as 50 degrees. [00:16:08.964] Now, the angle between the axis A and B is given as 50 degrees. So, since two are vertically opposed to each other, this angle will also be equal to 50 degrees. Okay. So what is given? [00:16:23.908] The tensile force is given as 2.5. What are we supposed to find? The component of the force F along axis A and B, as well as the projection of the force along A and B. So now [00:16:39.396] to solve the problem angle between A and the force F, angle between A and the force F, F is acting along X axis [00:16:49.956] Okay, it will be equal to so this is y axis, this is x axis, they will always be at 90 degrees to each other. Now this angle is given us 50. So this angle made by the axis A and the force would be equal to 90 minus 40, which is 40 degrees. [00:17:08.900] Similarly, angle between B and force F. That is, this is the axis B and the force F is acting along X. So this angle, since they have given this as 50, this will also be 50. So it is going to be 50 minus 40, which is 10 degrees. I hope it is clear. Now. [00:17:30.340] Uh we have got the angles, so now using the sign rule to get the components. Components I have told you will follow law of triangle of forces or law of parallelogram of forces. [00:17:44.740] The law of parallel of forces or the law of triangle of forces is derived from sine rule, so we can use sine rule to get it. So the get the components. Now here the component of the of the force, when you I apply a force here, the component of this along A and along B. [00:18:04.740] I can write it as FA, which is the force along A divided by the sine of the angle between the other two. So if this is the axis, the other two would be this angle, which is the angle made by the force and the axis B, which is sine 10, should be equal to the force along B. [00:18:24.612] Divided by the sine of the angle between the other two, which is x-axis and a that is sine of 40, which should be equal to f, which is the fourth sine of the angle between the other two, which is the total angle between A and B, which is sine 50. So from

this [00:18:40.544] I can get the value of FA and FP. So you get the value of F as 0.567 kilonewtan and F B as 2.1 kilonewtan. The units of the forces are important. What is given? F is given in [00:18:55.088] Kilonewtons, so you will have the components of the forces also in kilonewtons. So you have to mention the units wherever it is required. Okay. Next coming to the projection. [00:19:07.588] Now I have clearly talked about what is the difference between components and projection. Projection is the cause component of the force. Okay? So very clear. So [00:19:21.764] Here you can see the projection, projection of PA, that is projection of the force F. [00:19:29.220] On A, how do I get? If this is A, I should draw a line from the edge of the force to the axis A, such that this line would make an angle of 90 degrees. [00:19:43.460] So this would form a right angle triangle. So this forms a right angle triangle. So this angle we already know it is 45 degrees. So this is the force. F cos 40 would give me the projection of the force on A.

EV0000118 · EV0000119 · EV0000120 · EV0000121

Transcript block 5 (source interval 001200-001499 seconds): [00:20:00.192] That is what is written here. F cost 40. So the value of PA will be 1.915 kilonewt. [00:20:07.908] Now to find out the projection of the force on B, so we take the force and draw a line from the edge of A such that this line would be at 90 degrees. This angle is 90 degrees, so this forms a [00:20:23.808] Right angle triangle. So this angle is given as 10 degrees. So F cos of this angle 10 degrees would be the projection on B. So F cos 10 degrees, we get the value of the projection as 2.46 kilonewton. [00:20:39.716] Okay, so we will stop here because we are running short of time. We'll continue in the next class. Thank you very much. My name is Dr. SV Venpej from Civil Engineering Department of Science and Humanities PS University. Thank you. We'll meet in the next class, that is class number eight.

EV0000122 · EV0000123 · EV0000124 · EV0000125

Full transcript: 3151 normalized words, 83 timestamped segments, 1259.691767 seconds duration. Low-confidence segments flagged: 0.

EV0000106 · EV0000107 · EV0000108 · EV0000109

9 UE26ME142A_Unit 1_Class 5_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 5_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000126 · EV0000127

Graphs and diagrams



Figure G000057. UE26ME142A Unit 1 Class 5 Lecture PPT · source slide page 1.

EV0000427

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ENGINEERING MECHANICS STATICS

Moment

Figure G000058. UE26ME142A Unit 1 Class 5 Lecture PPT · source slide page 2.

EV0000430

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ENGINEERING MECHANICS STATICS

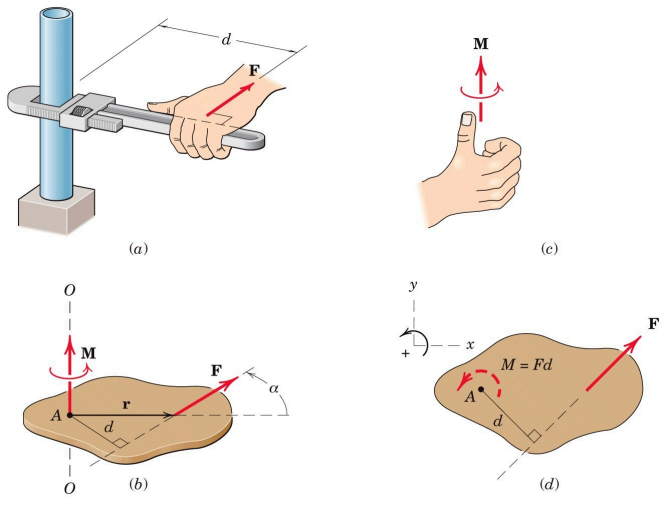


Figure 2-8
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Figure G000059. UE26ME142A Unit 1 Class 5 Lecture PPT · source slide page 3.

EV0000433

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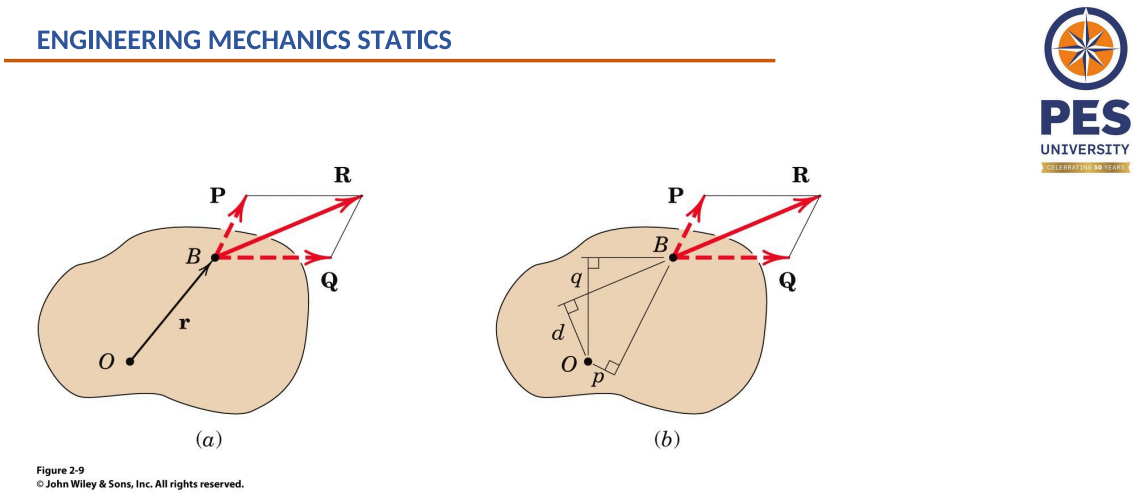


Figure G000060. UE26ME142A Unit 1 Class 5 Lecture PPT · source slide page 4.

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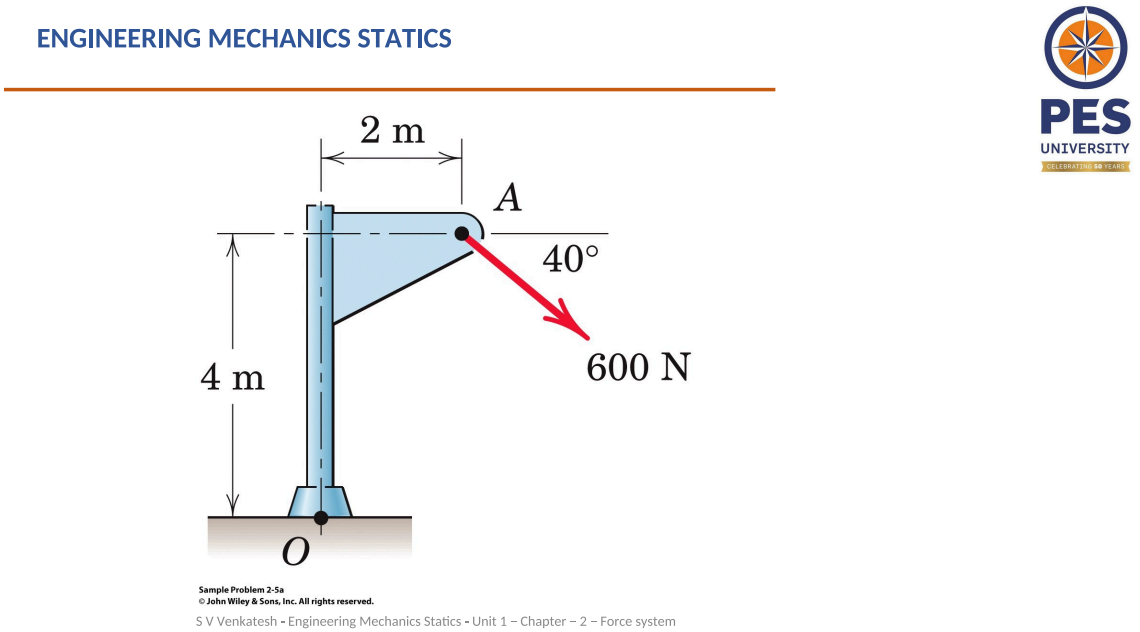


Figure G000061. UE26ME142A Unit 1 Class 5 Lecture PPT · source slide page 5.

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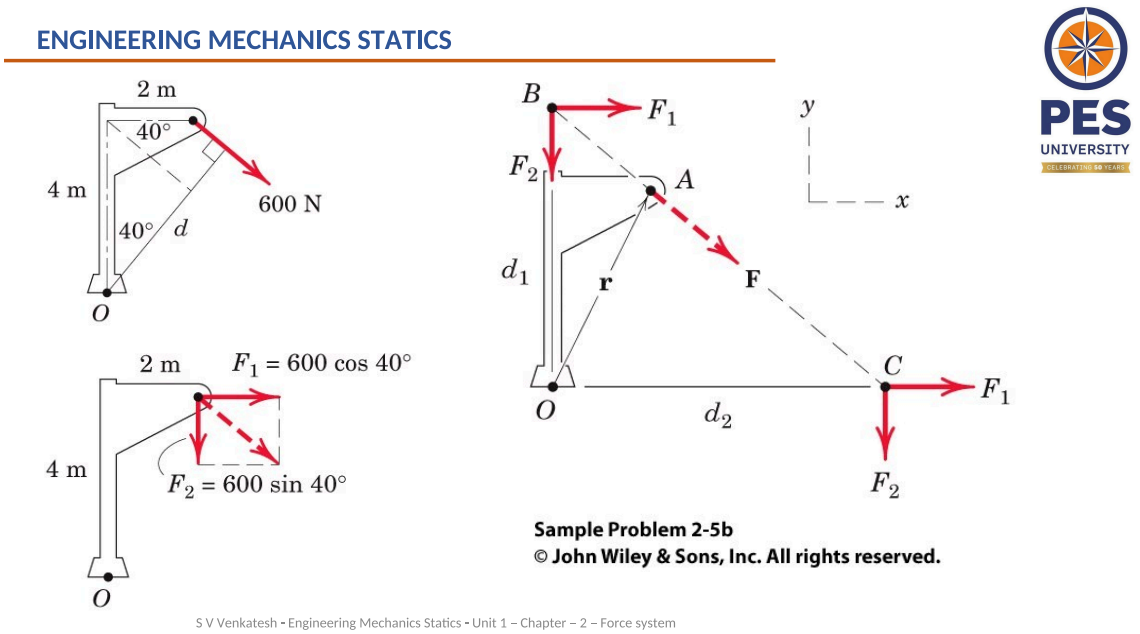


Figure G000062. UE26ME142A Unit 1 Class 5 Lecture PPT · source slide page 6.

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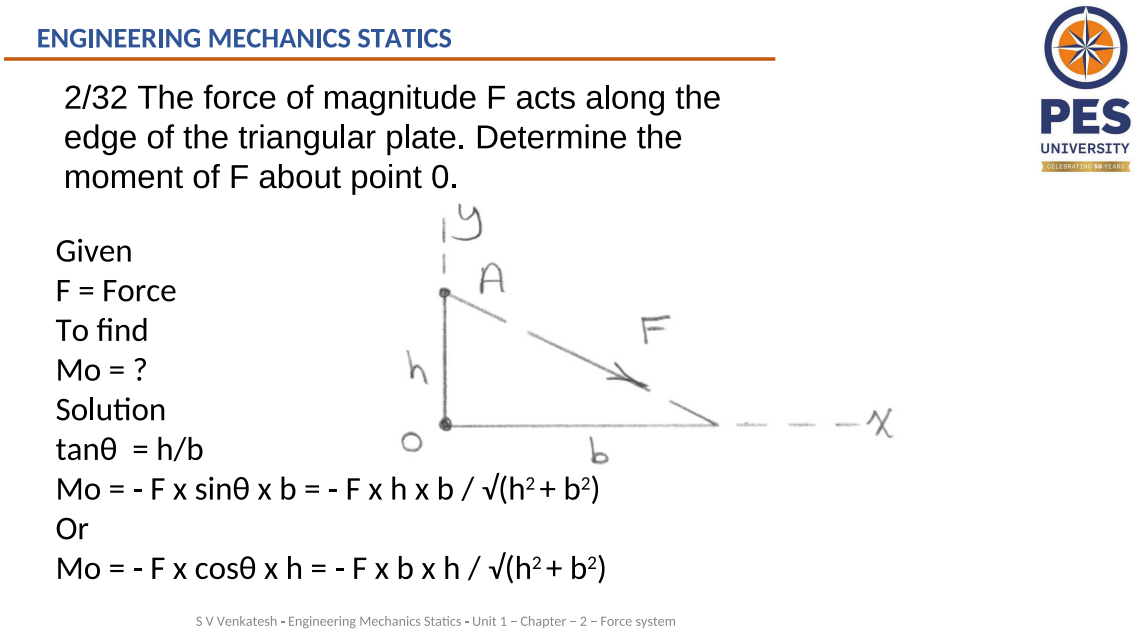


Figure G000063. UE26ME142A Unit 1 Class 5 Lecture PPT · source slide page 7.

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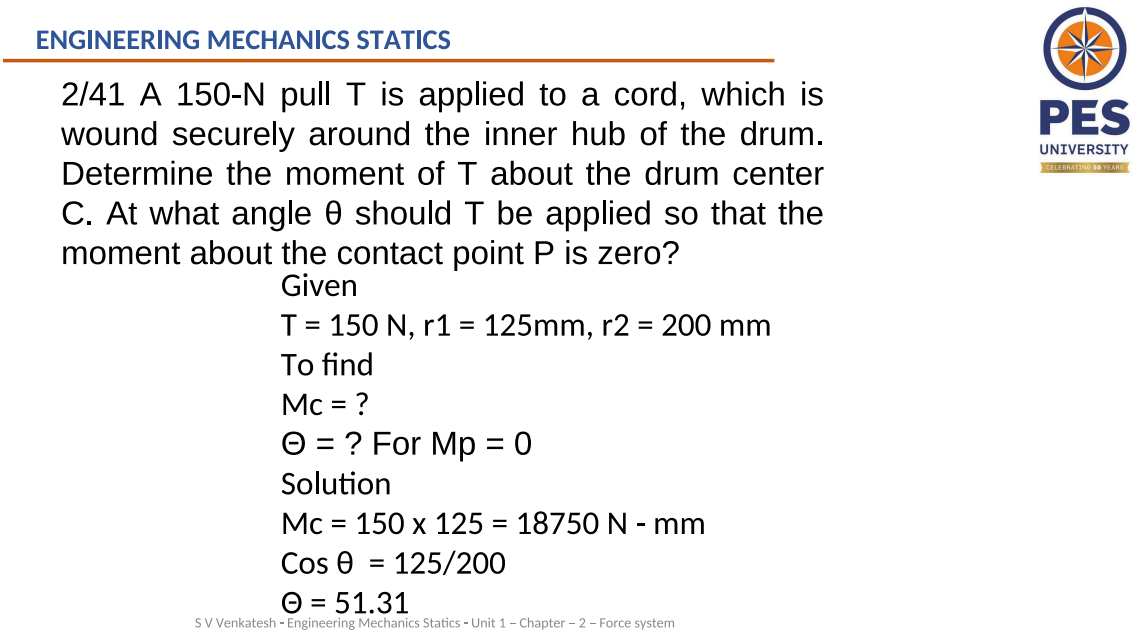


Figure G000064. UE26ME142A Unit 1 Class 5 Lecture PPT · source slide page 8.

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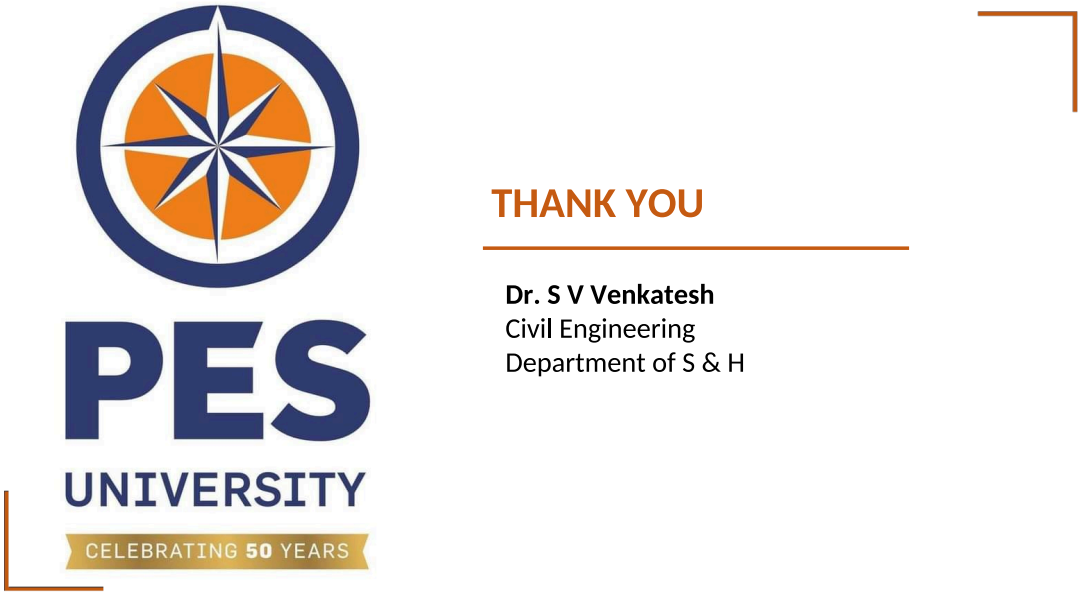


Figure G000065. UE26ME142A Unit 1 Class 5 Lecture PPT · source slide page 9.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000426 · EV0000427 · EV0000428
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Momen t S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000429 · EV0000430 · EV0000431
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000432 · EV0000433 · EV0000434
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000435 · EV0000436 · EV0000437
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ## Page / slide 6 (OCR) ENGINEERINGMECHANICS STATICS 2m B PES 40° UNIVERSITY F CELEBRATING50YEARS 4m 600N x 40° d d 2m F1 = 600 cos 40° H d 4m F2 F2 = 600 sin 40° SampleProblem2-5b JohnWiley&Sons,Inc.Allrightsreserved. Q S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Chapter - 2 - Force system EV0000438 · EV0000439 · EV0000440
6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000441 · EV0000442 · EV0000443
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS 2/32 The force of magnitude F acts along the edge of the triangular plate. Determine the moment of F about point 0. Given F = Force To find Mo = ? Solution tanϑ = h/b Mo = - F x sinϑ x b = - F x h x b / √(h2 + b2) Or S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force Mo = - F x cosϑ x h = - F x b x h / √(h2 + system b2) EV0000444
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS STATICS 2/41 A 150-N pull T is applied to a cord, which is wound securely around the inner hub of the drum. Determine the moment of T about the drum center C. At what angle ϑ should T be applied so that the moment Given about the contact point P is zero? T = 150 N, r1 = 125mm, r2 = 200 mm To find Mc = ? Θ = ? For Mp = 0 Solution Mc = 150 x 125 = 18750 N - S V Venkatesh - Engineeringmm Mechanics Statics - Unit 1 – Chapter – 2 – Force Cos ϑ =system125/200 ## Page / slide 9 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000445
9. Source slide page 9: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000446

Short-note recap

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.164] Dear student, welcome to engineer mechanics classes. I am Dr. Sv. Venkitish from Civil Engineering under Department of Science and Humanities, PS University. [00:00:13.668] This is class number eight. In the last class we had solved a few problems on uh resolution of forces, splitting up of a given force along its ax along its uh x and y axis. [00:00:32.612] And determining the resultant of the given force. Now today we'll continue with that and we will discuss the next topic, which is on moment. What is a moment? The definition of a moment is nothing but [00:00:52.900] The rotating effect produced by

a force on a body. When a force is applied on a body, if the body rotates, that rotation of the body is called as a moment. [00:01:11.620] Now if you remember I have talked about two actions that will happen when a force is apply. One is translatory motion, the other one is the rotatory motion. The the translatory motion is the displacement. [00:01:29.700] When a force is applied onto a body, the body would move from a point A to a point B, or when a force is applied, the body would rotate. [00:01:42.084] Without any displacement, it will be at the same point, but it will rotate. The rotating effect caused by the force is termed as moment. Mathematically, we represent moment as M is equal to R into F. [00:02:00.740] or F into R if we are solving it by scalar. But if we are solving it in vector approach, then it follows what is called as a cross product. [00:02:14.820] In earlier classes I had also discussed about cross product. I have talked about why it should be the position vector cross force vector and not force vector cross position vector. The distributed property is not valid in case of cross product. [00:02:30.980] So if we are trying to find out the moment of a force about any point, we go for what is called as cross product, and cross product is the [00:02:47.876] Position vector cross the force vector. Okay. Now in scalar approach it can be F into R or R into F, it does not matter because there we have to take the rotation into account and then decide whether it is positive or negative. But when we are using cross product [00:03:08.260] By doing or executing the cross product, we would get the sign for the momentum. Now in this diagram, you can see there is a there is a range [00:03:25.252] Ringe is an instrument which is used to rotate a pipe or a bolt or a net. Okay. Now there is a pipe that is here, even, and a range is fixed to it, and a person is applying a force in this direction. [00:03:45.028] Which is perpendicular to this tool, which is called as a range. Okay. So when I apply a force here, what would happen is this pipe would rotate like this. Okay, when I apply this force. [00:04:03.332] So this force is causing a rotation to this point. The magnitude of the rotation is going to be equal to the force that is applied into the perpendicular distance between the point where we want the moment. [00:04:18.564] And the point where the force is applied. That distance in this case is B. Now this angle is 90 degrees. Remember that. This angle [00:04:28.932] Has to be 90 degrees, then only this distance would be the perpendicular distance. Okay. So the movement at this point now or along this axis would be equal to this force F into this perpendicular distance D. [00:04:45.764] Now the sign convention that is going to be used to determine the moment. Okay, the sign convention is followed by using the right hand thumb rule. Now that is what is given here. If you orient your thumb, right hand thumb.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:05.796] Along the positive axis on which or about which the movement is caused, and if the other four fingers, as you can see, is folded like this. If that rotates, if that rotates, then [00:05:23.108] Like this, okay, as indicated in this diagram, then that would be the positive sign for the moment also. So you have to use the right hand only for this, not left hand. So I repeat, how do we decide about the sign convention for moment. [00:05:39.940] Orient the thumb, orient the thumb along the positive direction of the axis about which the moment or the rotation is occurring. If the rotation occurs similar to the other. [00:05:55.876] Fingers of the right hand as indicated here, then we take that as positive. If the rotation occurs opposite to this, that is like this, then we consider it to be negative. Okay. [00:06:12.036] Now, what happens if the force is not acting perpendicular to the axis? That is this axis. Okay? So that is what is given here. [00:06:25.380] Say I will take the same example, an axis is there. I want to find out what is the moment. Now I'm applying a force on this axis to this, but

the force is applied at an angle alpha to this line. [00:06:42.820] Okay, so according to our definition, moment is force into perpendicular distance. So how do I get that? If this distance is R, this angle is alpha, either I can resolve this force. [00:06:58.052] Which is perpendicular to the axis and parallel to the axis, only the component of the force which is perpendicular to this axis would produce a moment. So that would be F sine alpha. F sin alpha into R would be the magnitude of M, that is one way of solving it. [00:07:17.220] The other way of solving it is you extend this line and determine what would be the perpendicular distance of this force. [00:07:28.356] From point A, which is indicated here as D. Now note this, this is the 90 degrees. This is 90 degrees. So I can now write it as moment is equal to this force F into this D. Now how do I get this D? [00:07:43.300] If this is alpha, then this would also be equal to alpha. We know that this is R. Okay, and this forms a right angle triangle. This forms a right angle triangle. So this is R, this is alpha. So I can D I can get D value as R. [00:07:59.600] Sine alpha. So M will be equal to F into D, which is F into R sin alpha. Whether you resolve the force or you determine the perpendicular distance of the force with the point about which it is rotating. [00:08:15.940] Or about which we are finding out the moment, we get the same value. So the same thing is indicated here. Force is F and the perpendicular distance is D. The moment produced is the force into this perpendicular distance D. [00:08:31.844] Now it is again positive. Okay? Now if I take the body, this body to be in XY plane. Okay? So [00:08:45.700] If this is x-axis and this is y-axis, the body is in xy plane, then the movement would be produced in an axis which is perpendicular to xy. So if you take this one x and y, z axis which is coming out of the body, if I orient myself [00:09:02.660] The uh the thumb of my right hand towards me, then you can see that the other four fingers are rotation. That rotation will be the positive direction of the moment, and that is how it is rotating. So this would be positive. I hope this is clear. Right. [00:09:22.020] Next, let's look at what is Warring Nord's theorem. Waring Nord's theorem states that the moment produced by a force is equal to the moment produced by the force. [00:09:42.436] Components about the same point. I'll repeat. Moment produced by a force, in this case, if I take R as the force. [00:09:51.204] The movement produced by this R about any point, now I will take point O. So if I take this force and this as a perpendicular distance F into D, that is the movement produced by the force R with respect to O, then

EV0000130 · EV0000131 · EV0000132 · EV0000133

Transcript block 3 (source interval 000600-000899 seconds): [00:10:07.844] That should be equal to the component of this force R, which in this case is taken as P and Q about some axis. [00:10:18.884] We can resolve a given force along at least along two axis, not lesser than that. We cannot represent a force by about any other axis which will produce the same effect. No, at least there should be two axis about which we can find out the components. So if I take P and Q are the components, then [00:10:35.872] The moment produced by this R about O should be equal to the moment produced by Q about O plus the movement produced by P about O. It is the algebraic sum. Not just the sum, it is algebraic sum. Now in this case, you can refer to this diagram, diagram number two. [00:10:55.748] B you have a movement produced by R above O will be R into B. That would be equal to your now. How what is the sense of this? The R is acting like this. The point O is here, so it makes a rotation. [00:11:15.172] Like this, that is clockwise rotation. Clockwise rotation we take it as negative, and anti-clockwise rotation we take it as positive. [00:11:24.324] Or vice versa, you can take it vice versa also. But usually when we go for vector approach, that is the cross product, we get clockwise as negative and anti-clockwise as positive. To have the consistency, let us follow the same thing, we will have anti-clockwise as positive and

clockwise as negative. [00:11:43.236] So if I take this R into this D, that would be clockwise rotation. So it would be minus. That should be equal to your P into the perpendicular distance of P. Now the P is our Q is extended. [00:12:01.316] This is the perpendicular distance, which is considered as small Q. So this force q into this distance Q would produce again a clockwise movement that would also be negative. Now if I take this force P. [00:12:18.084] This force P and this perpendicular distance are small P would produce an anti-clockwise movement. So that would be positive. So your [00:12:28.900] capital Q into small Q with a negative sign plus capital P into small P with a positive sign should be equal to your capital R into this D. [00:12:42.596] That is what Varingov's theorem states. What is Varinghaus theorem states? Moment produced by a force about any point is equal to the moment produced by the force components about the same point. I hope this is clear. Okay. [00:13:00.388] Now, let's take up a problem. The statement of the problem is calculate the magnitude of the moment about the base O of the 600 newton force in five different ways. [00:13:15.396] How do I get the moment of this force, 600 newtons, about this point O in five different ways. So what is given is the magnitude of the force in the statement, then the point about which we have to find out the moment. Okay. Then if you read the diagram. [00:13:34.692] This diagram gives you the position of A. It is at a distance of 2 meters. If I take this as x-axis. [00:13:43.364] And this has y-axis, okay. 2 meters positive x and uh the origin O is 4 meters from the intersection of these two lines. Okay, that is from here vertical distance is 4 to O. [00:13:59.744] From O to A, the horizontal distance is 2 meters. Then the force is inclined at an angle of 40 degrees and the sense of the force is moving away from A, like this. This is the same line of action of the force. [00:14:15.460] Now how do we go about on solving this? We have to take a moment about this. One way is take this force, determine what would be the perpendicular distance of this force from O. So I if I draw a line here, if this is 90 degrees, then this becomes D. [00:14:34.948] Okay. If I can find that then that will be equal to 600 into D, that is one way of solving it. The next way of solving it is resolve this force along this and along this. Take movement of this about O, take moment of this about O. [00:14:53.476] Next, extend this line and see to it that it cuts this axis at some point. Okay. Resolve this force along horizontal. Determine what is the distance from here to here.

EV0000134 · EV0000135 · EV0000136 · EV0000137

Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.260] Then you can take this for the horizontal component of the force and distance from here to here. Next [00:15:15.140] You can also extend this line downwards till it touches the horizontal line at O. Then resolve this force along vertical. Determine what is the distance from here to here. Likewise, you can solve or find out the moment of this force. [00:15:32.752] About the point O. Now let's look at this. See first method that I had talked about is this. You have 600 newtons, find out this D. Okay. So they have given the angle made, and using that you can find out what is D, then multiply 600 into D, you get the moment about O. [00:15:52.752] That's one way of solving it. The other way of solving it is indicated here. You can resolve the force along X and Y as F1 and F2. You already know the distance 2 meters and 4 meters using this the 600 sine 40 into 2. [00:16:11.204] Would be clockwise negative. This 600 cost 40 will also be clockwise into those four. Combination of these two would give you the moment about O. That's one way another way of solving it. Now next one is you extend this till it touches this axis. [00:16:29.892] Then you resolve the force as we have done it here into F1 and F2. You will get F1 into this distance, if I can find this distance, okay, which is indicated as D1, then you can get the moment which will be clockwise. [00:16:44.996] Next one is

you extend this till it touches this horizontal line. Find out what is D2, then your F2 into D2, you will get the moment about O. So I have explained this is one way of solving it, this is another way of solving it. [00:17:04.804] This is one more way of solving it. This is another way of solving it. Okay? So four different ways I have explained. You can also try and solve it in one more way. [00:17:20.868] Whatever you feel it is correct. I am not going to talk about it right now, but if you have any doubts, you can always meet me and then I will explain about that. Okay? Now let us take another example from the unsolved problem. The problem number is 2 bar 32. [00:17:39.716] The problem statement is this. The force of magnitude F acts along the edge of the triangular plate. Determine the movement of F about O. Okay. So there is a triangulate that is there. [00:17:55.812] And there is a force F that is acting, this is the force that is acting. What is asked is to determine the movement of F about O, that is this point. Now, how do we solve this? We can solve it in different ways. If I consider the point of application of this force as A. [00:18:15.940] Now we know this height and this P. I can determine what is this angle. Okay, tan inverse of H by B. So I'll get that angle. Similar the that angle will be same as this because this is alternate angles. So I can resolve this force along this, which should be F. [00:18:33.060] Cos of this angle theta. So f cos theta into h, I will get the value. Or I can also take this as a point of application. I can resolve this along this axis, which would be a side of this angle theta. Then that would into b will also give you give me the. [00:18:52.068] Moment about O, or I can simply find out what is this perpendicular distance. In whichever way that you want, you can solve it. Okay. The other way of solving it is I can find out what is this. [00:19:05.764] Uh hypotenuse length is in terms of H and B, which is root of H square plus B square, that is what is done here. So instead of F into [00:19:16.740] uh cos teta they have just substituted the value of cos teta as h by or sorry b by root of h square plus b square similarly for this you can solve it in which your way that you want. I hope this is clear. Next we will take up one more problem 2 bar 41 [00:19:36.068] The problem statement is A 150 Newton pull T is applied to a card which is wound securely around the inner half of the drum. Determine the moment of T about the drum center C. So we have to determine the moment about the drum center C. [00:19:57.092] Next is at what angle theta should T be applied so that the moment about the contact point is 0. The moment about this point should be 0. So we have to find out at what angle this should act such that the moment about this point is 0. Now look at the

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:17.060] Diagram. What is that we have to take from this? Okay. First thing is what is that we have to take from this given as the magnitude of the force P is 150 newtons. Then from the diagram I can make out that the inner circle diameter or radius is 125. [00:20:35.364] The outer circle radius is 200 millimeters. What are we supposed to find? First thing is movement about 0.0 because of this. Then second thing is at what angle theta we should apply this for such that moment at P is 0. So let's take first one. What is the moment about [00:20:55.524] MC. So this is the point, this is the center of the circle. If I wound a wire around the circle and pull it, at whatever point that I want to pull, it will always be tangential to the circle. This you have to remember. [00:21:14.980] It will always be tangential to the circle. So the radial line and the tangent to the circle will always be at perpendicular to each other. So moment at C will simply be equal to your. [00:21:29.796] Force that is P into the inner radius R1, and that's what is written here. So you will easily get this. Next, to find out at what angle theta this should act such that movement about this point is 0. If the force simply passes through [00:21:49.956] O P. Or if I if I draw a line such that the line passes through B and also it is

tangent to the inner circle. Okay. [00:22:02.468] Now that should be the angle at which the force should act so that the movement about P is zero. Why we should have that? Because there is no perpendicular distance. If the force passes through any point, the movement produced at that point is zero, that is because there is no perpendicular distance. So if this has to be there, [00:22:22.404] Okay, if this has to be like this, if I take the component of this force along this axis, which will be T cos theta, if this is theta, this will be T cos theta, and this would be the radial line which will be perpendicular to this. [00:22:42.628] The moment produced at C point. Okay, I can equate. And that's what is done here. Moment at C is this, that should be equal to what? Your P cos theta into R. [00:22:58.596] So we know what is uh sorry, t cos teta, we know what is t. If I equate I'll get this. That is one way of solving it. The other way of solving it is if I take join this line, this line, this and the tangent. [00:23:13.668] This, this, and this, this angle will be at night registration. I know this distance is 125, this distance is 200. So cos theta, which is this one, would also be the same as this angle. So cos of theta would be simply equal to [00:23:31.280] 125 divided by 200. So which we will get us 51.31. I hope you have understood. We will stop today at this. We'll continue in our next class. That is in class number 9. Thank you very much. My name is Dr. Sventesh from PS University. Thank you.

EV0000142 · EV0000143 · EV0000144 · EV0000145

Full transcript: 3456 normalized words, 86 timestamped segments, 1430.208 seconds duration. Low-confidence segments flagged: 0.

EV0000126 · EV0000127 · EV0000128 · EV0000129

10 UE26ME142A_Unit 1_Class 6_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 6_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000146 · EV0000147

Graphs and diagrams



Figure G000066. UE26ME142A Unit 1 Class 6 Lecture PPT · source slide page 1.

EV0000448

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

2/41 A 150-N pull T is applied to a cord, which is wound securely around the inner hub of the drum. Determine the moment of T about the drum center C. At what angle θ should T be applied so that the moment about the contact point P is zero?

Given

T = 150 N, r1 = 125mm, r2 = 200 mm

To find

Mc = ?

Θ = ? For Mp = 0

Solution

Mc = 150 x 125 = 18750 N - mm

Cos θ = 125/200

Θ = 51.31

S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

Figure G000067. UE26ME142A Unit 1 Class 6 Lecture PPT · source slide page 2.

EV0000451

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ENGINEERING MECHANICS STATICS

2/49 In order to raise the flag pole OC, a light frame OAB is attached to the pole and a tension of 3.2 kN is developed in the hoisting cable by the power winch D. Calculate the moment M0 of this tension about the hinge point 0.

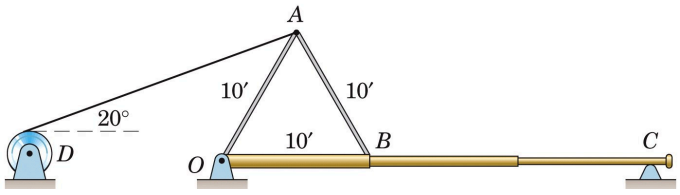
Given

F = Force

To find

Mo = ?

Solution



θ = 60^0

Mo = 3.2 x cos 20 x 3 x sin 60 – 3.2 x sin 20 x 3 x cos 60

Mo = 7.812 – 1.641 = 6.17 kN - m

S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

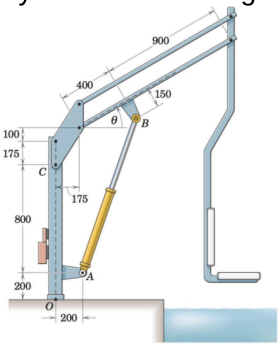
Figure G000068. UE26ME142A Unit 1 Class 6 Lecture PPT · source slide page 3.

EV0000454

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

2/56 The mechanism shown is used to lower disabled persons into a whirlpool tub for therapeutic treatment. In the unloaded configuration, the weight of the boom and hanging chair induces a compressive force of 575 N in hydraulic cylinder AB . Compressive means that the force which cylinder AB exerts on point B is directed from A toward B.) If 8 = 30°, determine the moment of this cylinder force acting on pin B about (a) point 0 and (b) point C.



S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

Figure G000069. UE26ME142A Unit 1 Class 6 Lecture PPT · source slide page 4.

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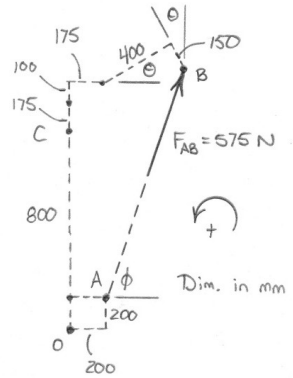
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

2/56

$\phi = \tan^{-1}\left(\frac{800 + 175 + 100 + 400\sin\theta - 150\cos\theta}{175 + 400\cos\theta + 150\sin\theta - 200}\right) = 70.9^\circ$

$\theta = 30^\circ$



S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

Figure G000070. UE26ME142A Unit 1 Class 6 Lecture PPT · source slide page 5.

EV0000460

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

$M_O = 575 \sin\phi\,(0.200) - 575\cos\phi\,(0.200) \longrightarrow M_O = 71.1$ N·m CCW

$M_C = 575 \sin\phi\,(0.200) + 575\cos\phi\,(0.800) \longrightarrow M_C = 259$ N·m CCW

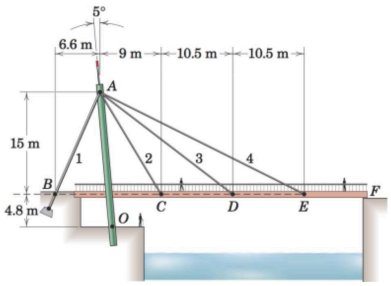
Figure G000071. UE26ME142A Unit 1 Class 6 Lecture PPT · source slide page 6.

EV0000463

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

2/57 The asymmetrical support arrangement is chosen for a pedestrian bridge because conditions at the right end F do not permit a support tower and anchorages. During a test, the tensions in cables 2, 3, and 4 are all adjusted to the same value T. If the combined moment of all four cable tensions about point 0 is to be zero, what should be the value T1 of the tension in cable 1? Determine the corresponding value of the compression force P at 0 resulting from the four tensions applied at A Neglect the weight of the tower.



Given T = T2, T3, T4 ; Mo = 0

To find T1 = ?, P = ?

Solution

Tan θ1 = 15/6.6 , θ1 = 66.25

Tan θ2 = 15/9 , θ2 = 59

Tan θ3 = 15/(9+10.5) , θ3=37.57

Tan θ4 = 15/(9+10.5+10.5) ,

θ4 = 26.57

Figure G000072. UE26ME142A Unit 1 Class 6 Lecture PPT · source slide page 7.

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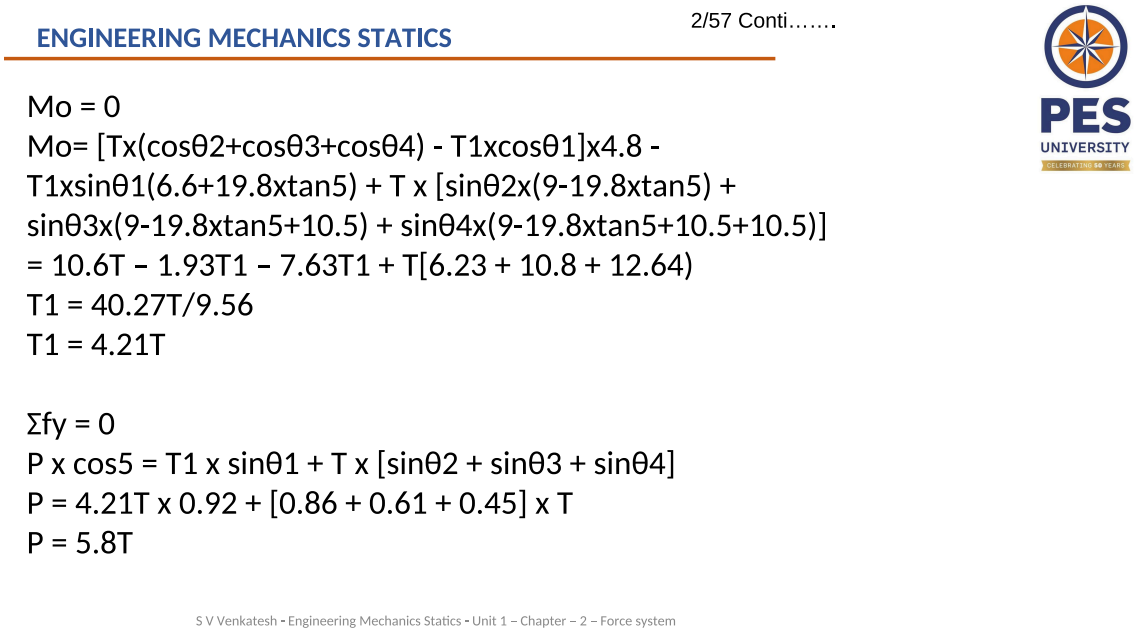


Figure G000073. UE26ME142A Unit 1 Class 6 Lecture PPT · source slide page 8.

EV0000468

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

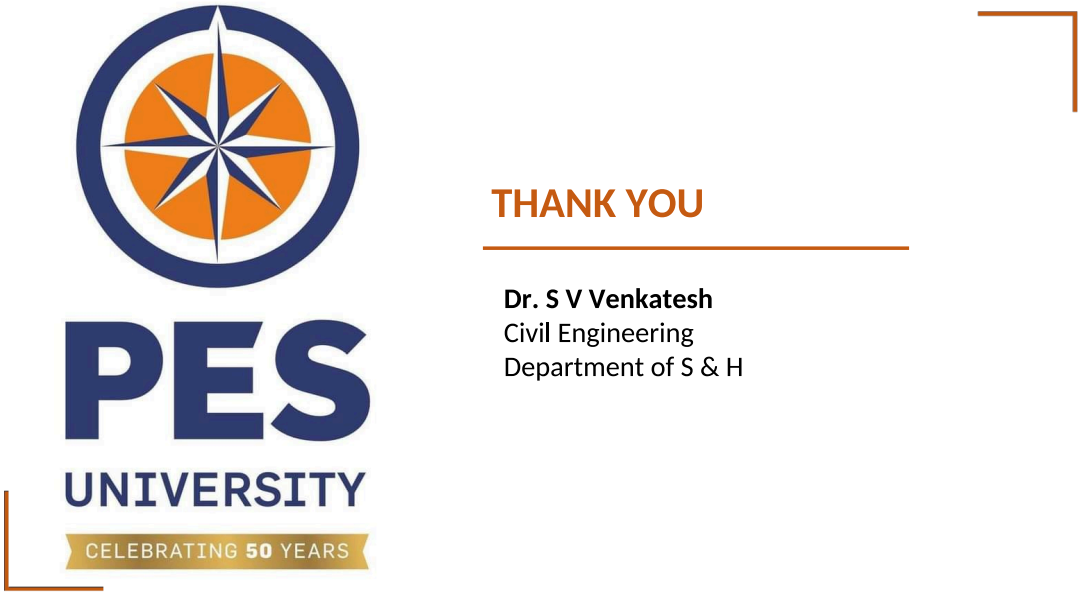


Figure G000074. UE26ME142A Unit 1 Class 6 Lecture PPT · source slide page 9.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000447 · EV0000448 · EV0000449
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 2/41 A 150-N pull T is applied to a cord, which is wound securely around the inner hub of the drum. Determine the moment of T about the drum center C. At what angle ϑ should T be applied so that the moment about the contact Givenpoint P is zero? T = 150 N, r1 = 125mm, r2 = 200 mm To find Mc = ? Θ = ? For Mp = 0 Solution Mc = 150 x 125 = 18750 N - mm S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force Cos ϑ = 125/200 system EV0000450 · EV0000451 · EV0000452
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS 2/49 In order to raise the flag pole OC, a light frame OAB is attached to the pole and a tension of 3.2 kN is developed in the hoisting cable by the power winch D. Calculate the moment M0 of this tension about the hinge point 0.Given F = Force To find Mo = ? Solution ϑ = 600 Mo = 3.2 x cos 20 x 3 x sin 60 – 3.2 x sin 20 x 3 x cos 60 S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – ForceMo = 7.812 – 1.641 = 6.17 kN - m system EV0000453 · EV0000454 · EV0000455
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 2/56 The mechanism shown is used to lower disabled persons into a whirlpool tub for therapeutic treatment. In the unloaded configuration, the weight of the boom and hanging chair induces a compressive force of 575 N in hydraulic cylinder AB . Compressive means that the force which cylinder AB exerts on point B is directed from A toward B.) If 8 = 30°, determine the moment of this cylinder force acting on pin B about (a) point 0 and (b) point C. S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000456 · EV0000457 · EV0000458
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000459 · EV0000460 · EV0000461
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000462 · EV0000463 · EV0000464
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS 2/57 The asymmetrical support arrangement is chosen for a pedestrian bridge because conditions at the right end F do not permit a support tower and anchorages. During a test, the tensions in cables 2, 3, and 4 are all adjusted to the same value T. If the combined moment of all four cable tensions about point 0 is to be zero, what should be the value T1 of the tension in cable 1? Determine the corresponding value of the compression force P at 0 resulting from the four tensions applied at A NeglectGiventheT =weightT2, T3,of theT4 tower.; Mo = 0 To find T1 = ?, P = ? Solution Tan ϑ1 = 15/6.6 , ϑ1 = 66.25 Tan ϑ2 = 15/9 , ϑ2 = 59 Tan ϑ3 = 15/(9+10.5) , S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ϑ3=37.57 EV0000465 · EV0000466 · EV0000467

8. Source slide page 8 (extracted text): 2/57 Conti..... ENGINEERING MECHANICS STATICS Mo = 0 Mo= [Tx(cosϑ2+cosϑ3+cosϑ4) - T1xcosϑ1]x4.8 - T1xsinϑ1(6.6+19.8xtan5) + T x [sinϑ2x(9-19.8xtan5) + sinϑ3x(9-19.8xtan5+10.5) + sinϑ4x(9-19.8xtan5+10.5+10.5)] = 10.6T – 1.93T1 – 7.63T1 + T[6.23 + 10.8 + 12.64) T1 = 40.27T/9.56 T1 = 4.21T Σfy = 0 P x cos5 = T1 x sinϑ1 + T x [sinϑ2 + sinϑ3 + sinϑ4] P = 4.21T x 0.92 + [0.86 + 0.61 + 0.45] x T S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force systemP = 5.8T ## Page / slide 9 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000468
9. Source slide page 9: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000469

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.644] Dear student, welcome to engineer mechanics classes. My name is Dr. Swee Venkadesh from Civil Engineering under Department of Science and Humanities, PS University. In the last class we discussed moment and solved a few problems on movement. [00:00:20.228] That is in class number eight. This is class number nine. We continue to solve a few more problems on moment. So have taken this problem that is 2 bar 49. The statement of the problem is in order to raise the flat pole, OC [00:00:40.516] A light frame OAB is attached to the pole and a tension of 3.2 kN is developed in the hosting cable by the power range D. [00:00:55.012] Calculate the moment M naught this should be M naught suffix not M naught of this tension about the h point O. So if [00:01:07.652] From the statement, what is that we should uh think or we should get is the tension in the cable, that is this cable, uh is 3.2 [00:01:22.660] It is tension, so external forces will be moving away, the internal forces will be moving towards that means the force in this cable internal force will be from A towards D and at D it is being from D towards A. Okay. Now we have to determine [00:01:41.060] What would be the movement caused by this force applied onto this light frame AOB at point O because this flat pole when I apply this tension here. [00:01:57.860] This has to rotate and stand vertical. That's the whole idea of this mechanism. Now, what is given? Force which is 3.2 kilonewton. What are we supposed to find is the moment that oh because of this force that is M naught. [00:02:15.268] Now from the diagram what are we supposed to take? We are supposed to identify that this OA line is inclined at twenty degrees. Similarly, if I draw an horizontal line here passing through A, this angle will also be equal to 20 degrees because these are alternate angles, they will be equal. [00:02:33.156] Then the sides of this triangle OAB is given as 3 meters. That means it forms an isoscelus triangle. [00:02:44.612] So in case of an isosceles triangle, I'm sorry, equilateral triangle, I'm sorry, it's not an isosceles triangle, it's an equilateral triangle. In case of an equilateral triangle, the internal angle of the triangle, all of them will be equal. That means it will be equal to 180 divided by 3. [00:03:02.884] Teta if I take T is the angle made by the internal angle of the triangle that is 180 by 3, which would be 60 degrees. Now to find out the moment about O. [00:03:14.212] For this force, what I need to do is I should take this force, resolve the force along horizontal and along vertical, determine what is the vertical distance from O to A. Similarly, determine what is horizontal distance from O to A. [00:03:32.964] Okay. So the distances that we can find as we have this as three meters if I draw a line here passing through A and perpendicular to O C, then it forms this forms a right angle triangle. So I can determine the vertical distance between OA as three. [00:03:51.268] Sine of this angle which is sixty degrees. Similarly, I can find out the horizontal

distance between OA, that is this distance as three cos of this angle sixty degrees. [00:04:01.924] To find out the component of the force, horizontal component, it is 3.2, this angle will be 20, so cos 20, whereas the vertical component will be 3.2 sin 20. [00:04:15.812] So, using that I can find out M0, which is equal to 3.2 cos 20, which is a horizontal component of the force into the vertical distance that produces an anti-clockwise moment minus the vertical component of the force, which is 3.2 sin 20. [00:04:34.212] into the horizontal distance between OA, which is 3 into cos 20, which will produce a clockwise movement, so that's why it is negative. So you will get the value as 6.14 kilonewton meter. So that would be the moment at point O because of the application of the force [00:04:51.632] equal to 3.2 RTA, which is the tensile form. Right. Next problem is 2 bar 56. The mechanism shown is used to lower disabled persons into a very ball tube or therapeutic

EV0000146 · EV0000147 · EV0000148 · EV0000149

Transcript block 2 (source interval 000300-000599 seconds): [00:05:11.684] Treatment in the uploaded configuration, the weight of the boom and hanging chair induces a compressive force of 5 sub 25 newton in hydraulic cylinder A B. [00:05:28.420] Compressive means that the force which cylinder AB exerts on point B is directed from A towards B. [00:05:40.740] If theta is equal to 30 degrees, determine the movement of this cylinder force acting on pin B about A point O and B point C. Okay. Now what is given here as a mechanism? [00:06:00.100] Which is used for lowering the disabled person into the will pole. Okay. Now the weight of the moment of the person will induce a force in this cylinder. [00:06:13.156] Which is a com which is going to be a compressive force. The magnitude of the force is given as 575 newtons. He says that if it is compressive, means that the force is exerted from [00:06:27.108] If I from A to B at point B that is this is the force that is the internal force. The compressive force means the external forces will be moving towards each other, the internal forces will be moving away. So the force at B will be moving from A to B. [00:06:43.536] That will be the sense of the force, and he also says if the angle theta that is this angle, the boom angle with respect to the horizontal is 30 degrees, what is given here, then determine what is the movement caused by this force about point O and about point A. [00:07:03.908] So what is given as the force, it is compressive force at B, we have to determine it is moving from A to B, then the angle theta is 30 degrees, that's what is given. We have to find out moment about O and moment about C. Now looking at the diagram that is given. [00:07:21.956] We can make out the distances from O to A, the vertical distance from O to A, which is 200, from A to C as 800, from C to this point as 175, from this point to this point as 100. [00:07:36.132] Then the inclined distance along this point to this point is 400, then from this point to this edge is 900, then very important is this distance from B to this. [00:07:51.268] Okay, this is 150. Now we can note that this line is perpendicular to this. Similarly, this line is also perpendicular to this. So they have given you the inclined distances. Okay. So how do we go about solving this? [00:08:09.508] We have a force acting at B. First thing is if I draw a vertical line from B till the horizontal line extended from A, it will cut at some point here. So this forms a right angle triangle. [00:08:27.748] Now I'll take this angle as some angle phi or alpha. Now first thing is to find out this angle. Why should we find out this angle? So that I can resolve this force along vertical and along horizontal. So to do that I should find out this angle. To find this angle, [00:08:44.964] I should get this what horizontal distance, sorry, vertical distance and then horizontal distance. Vertical distance I can find it as 800 plus 175 plus 100 plus this small distance which I can take it as distance from this point to here. [00:09:04.868] Which I'll get it from this 400, from this 400 sine of theta, I'll get this angle, this distance, sorry, this distance. Then I'll subtract from 150 this because this is 90 degrees.

So using this theta, I can find out what is this, so I subtract that. So I get this vertical distance. [00:09:24.900] The horizontal distance between this also I can determine 400 cos theta I'll get this distance from here to here. Similarly 150 and using that angle I'll get this distance, so I'll add that. So I'll get the distance from A to B. So that is what is explained in this diagram. [00:09:45.124] So how do we get the distances? So this angle is 5, okay, which the uh line AB makes with the horizontal extended at A. So this angle is 5. So if you take [00:09:59.684] Or find out 5, if I want to find out 5, which will be equal to 10 inverse of this vertical distance and this horizontal distance. What would be this vertical distance? 800 as I have told you explained, 800 plus 175 plus this is 400.

EV0000150 · EV0000151 · EV0000152 · EV0000153

Transcript block 3 (source interval 000600-000899 seconds): [00:10:18.340] If I take sine theta, I get this vertical distance, but I have to subtract this vertical distance from this to this. So that would be equal to 150. [00:10:29.796] Cause of this angle theta, this would be theta itself because this line and this line are not 90 degrees. If this is theta, if I draw a line here, okay, if this is theta, this will be 90 minus theta and this would be theta. So minus 150 sign theta, I get the vertical distance between A and B. [00:10:49.252] Now to find out the horizontal distance between A and B, I'll take 175. [00:11:00.292] That is this distance, 175 plus 400, which is this, cause of this theta, I get it till this point, so 175 plus this, plus I should take this horizontal distance, which is 150 sin theta, but I will be getting from here to this point. [00:11:19.204] So if I subtract this distance 200 minus 200, I will get the distance, horizontal distance between A and B. So that's what we will get. So you can find out phi from this. Once I get this phi, I can resolve this force as 575 cos of phi, I get the [00:11:39.716] Vertical force 575 sine of 5, I get the horizontal force. Now to find out the moment about C, I'll take this component and this horizontal distance which will be equal to this value. [00:11:56.964] I should not consider this minus 200, only this part, the total. Okay, would be the horizontal distance. The vertical distance would be this, plus I need to add this 200, or in another way, I can take the point of application of this force as A itself. [00:12:15.588] So when I resolve this vertical and horizontal will be acting here. So moment about C will be this vertical component into this distance, which is 200, which will be anti-clockwise. Then this horizontal component is to 200, which will be clockwise. So I will easily get the moment. Similarly, if I want to find out at point C, [00:12:35.812] I'll take this vertical component into this distance 200, this horizontal component into this distance which is 800. So this would produce an anti-clockwise moment, whereas this would also produce an anti-clockwise moment. So I have to sum it up. So that is what is given here. [00:12:54.852] 575 sin 5 into 0.2 minus 575 cos 5 into 0.2 we will get the moment about O M naught. Moment about C will be 575 sin 5 into 0.2 plus 575 cos 5 into 0.8. I get this moment. I hope you have understood this. [00:13:14.692] Next problem is 5 bar or sorry 2 bar 557. So the statement of the problem goes something like this. The asymmetrical support arrangement is chosen for a pedestrian bridge because conditions at the right end F. [00:13:34.436] Do not permit a support tower and anchorage. During a test, the tensions in cable 2, 3, and 4 are all adjusted to the same value T. If the combined movement of all four cable tensions [00:13:53.668] About point O is to be 0. What should be the value T1 of the tension in cable 1? That is what is asked. Then determine the corresponding value of the compressive force P. [00:14:11.844] At O resulting from the four tensions applied at A. Neglect the weight of the tower. So what is given as you have a bridge here, a pedestrian bridge, which is used. Now at point F, we cannot have any. [00:14:32.836] Support, tower, or anchorage. So we cannot have a support reaction at A. Support

reaction at B F is zero. There is no support here. Okay? All the supports has to come through the cables. So he says that at a particular test. [00:14:51.620] The tension in cable two that is this ten, this cable, then three this cable, four that is this cable, all of them have the same tension T, the same tension T in this two, three and four cable.

EV0000154 · EV0000155 · EV0000156 · EV0000157

Transcript block 4 (source interval 000900-001199 seconds): [00:15:07.076] Now for this condition we have to find out what would be the tension in this cable AB, that is cable 1. Then we should also find out what would be the compressive force in this tower AO acting at O. [00:15:25.668] We have to neglect the weight of this tower. Okay? Now what is given, I've already explained. What has to be found also I have explained. [00:15:38.052] Next, we should also find out what is the tension in P1 and 10 and the compressive force that is acting on the tower P. [00:15:47.012] Now all these values that we are going to get will be in terms of T. We should take T as a known quantity that is T. Intention T2, T3, and T4 is equal to T. So we will get the value of T1 in terms of T. Similarly, P also in terms of T. So [00:16:07.204] Now the tension in this cables are given. Now to find out the movement of O, because he says that the combined movement of all the four forces at point O is 0. So this point I have to take movement. So I have to find out what is the inclination of these cables with respect to horizontal. [00:16:27.588] So I have considered this as theta 1, this as theta 2, theta 3 and theta 4, theta 1, theta 2, theta 3, theta 4, or the inclination of the table 1, 2, 3, 4 with respect to the horizontal respectively. So if I take theta 1 tan theta 1, that is this angle theta 1. [00:16:47.780] Now I will draw a vertical line here. Then this is an horizontal line. The horizontal distance is given as 6.6. The vertical distance is given as 15. Note that the uh [00:17:03.812] Tower AO is inclined at an angle of 5 degrees, which is given here. This is not vertical. OA is not vertical, it is inclined to vertical at an angle of 5 degrees. But the distances that is given, these distances are horizontal distances, and this is vertical distance. [00:17:23.236] So to get theta 1, I will take tan theta 1, which will be equal to 15 divided by 6.6, that's what is mentioned here. So I get theta 1. Theta 2 will be this, which will be equal to [00:17:37.092] The vertical distance which is 15 into the horizontal distance 9 or divided by 9. So I get theta 2. Similarly, I will find out theta 3 and theta 4, the explanation is given here. So I will get the values of theta 1, theta 2, theta 3 and theta 4. [00:17:53.508] Corresponding to the inclination of the cable 1, 2, 3, and 4 with the horizontal. Now I will take this tension in this cable, apply it at point C. Similarly, apply the tension P, which is acting at [00:18:12.356] 3 km 3 at D. Similarly tension 4 acting at E which is T again. Then T1 is the tension acting which I will take it at D. I will resolve all these forces along vertical and horizontal. [00:18:27.876] Okay, and then take these values and find out the moment about O. So if I take this T, this would be T sin theta 1, this would be T cos theta 1. Similarly I find out for this and the distances from O to this point. [00:18:46.276] I can find out which will be from this distance and the distance between this. I can find out that. Okay? And that is what is explained here. [00:19:02.020] So, T is the tension in cable 2, 3, and 4 into the cos of the angle made by these cables with the horizontal. So I've taken T out because that is common. Minus T1 cos teta 1. [00:19:16.484] into the vertical distance of all of them are 4.8 itself. This is the vertical distance. So this horizontal component, this horizontal component, this horizontal component, this horizontal component [00:19:32.228] All of them will have the same vertical distance 4.8. So straight away I have taken that. Now coming to the vertical component T1, that is this one. [00:19:43.524] Into I should take this distance from O to B, which will be 6.6.

That is from the O to A to B, the horizontal distance, then I should also take what is the horizontal distance from A to O, which will be equal to 15 plus 4.8, which is 19.8.

EV0000158 · EV0000159 · EV0000160 · EV0000161

Transcript block 5 (source interval 001200-001499 seconds): [00:20:03.460] And this angle is given us 5 degrees. So tan phi I'll get this. That is what is entered here. Similarly, I will find out for all these three vertical forces acting because of these tensions and the distances. [00:20:21.444] But here I have to subtract the horizontal distance from O to A. So to get the distance from O to C, O to D, and O to E, and that's what is mentioned here. So from this I will get the value of T1, which is equal to 4.21 tons, or sorry, tons. Or T [00:20:41.156] Newtons. If T is in Newton or whichever units that T is going to be. T is the tension in the cable, 2, 3, and 4. Next to find out the compressive force in OA, I will take some of all the forces acting along the vertical. Note that the [00:21:00.004] Compressive force that is acting in OA will be acting upwards, but that is at inclined at 5 degrees to the vertical. So I will take this component P into cos of 5 degrees, which will be acting vertically upwards. The rest [00:21:17.636] That should be equal to all these forces which is acting. So that is what I have taken. P1 and this, all this which should be acting downwards. So I from that I will get the value of P. So P will be equal to 5.8 P. [00:21:35.972] Attention in the table two, three, and four. I hope all of you have understood today's class. We will continue in the next class. This is class number nine. And I am Dr. SV Venkatesh from PS University. Thank you very much. We will continue in the next class.

EV0000162 · EV0000163 · EV0000164 · EV0000165

Full transcript: 3177 normalized words, 77 timestamped segments, 1312.448 seconds duration. Low-confidence segments flagged: 0.

EV0000146 · EV0000147 · EV0000148 · EV0000149

11 UE26ME142A_Unit 1_Class 8_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 8_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000166 · EV0000167

Graphs and diagrams



Figure G000075. UE26ME142A Unit 1 Class 8 Lecture PPT · source slide page 1.

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ENGINEERING MECHANICS STATICS

PES UNIVERSITY

Unit - 1
Couple

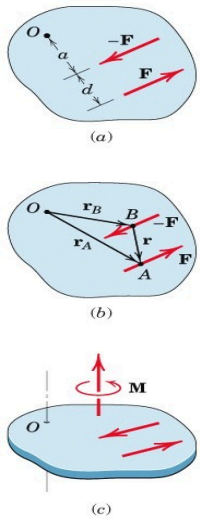
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ENGINEERING MECHANICS STATICS

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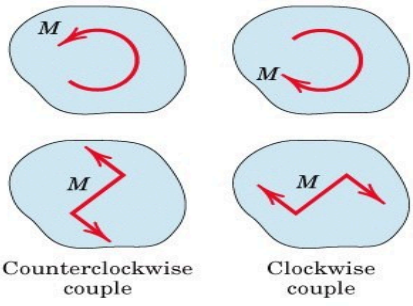


Figure 2-10
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S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

Figure G000077. UE26ME142A Unit 1 Class 8 Lecture PPT · source slide page 3.

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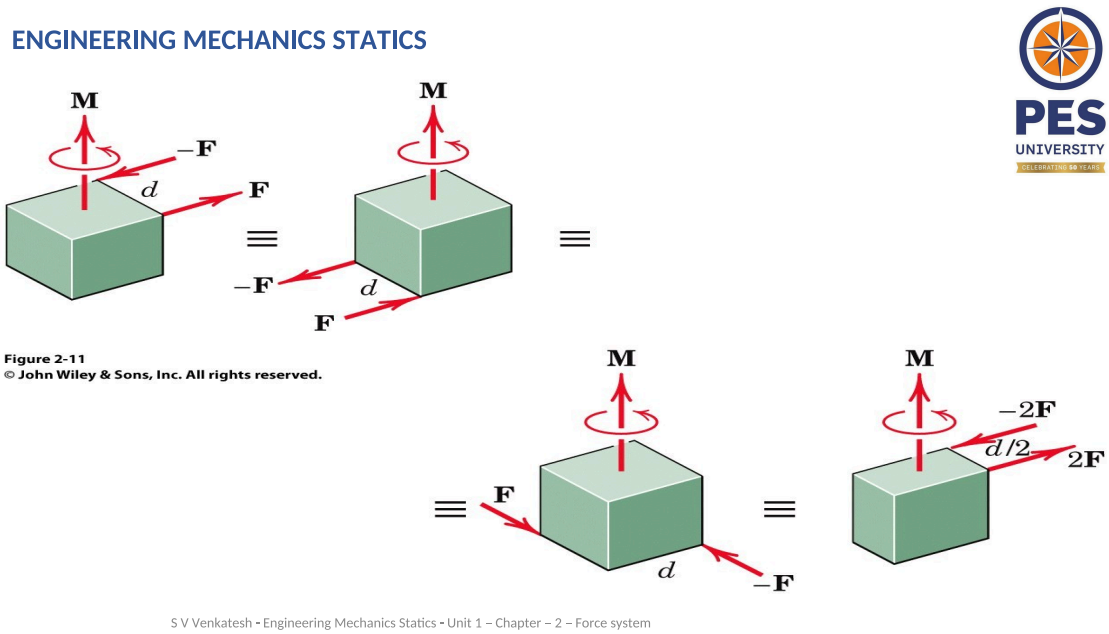


Figure G000078. UE26ME142A Unit 1 Class 8 Lecture PPT · source slide page 4.

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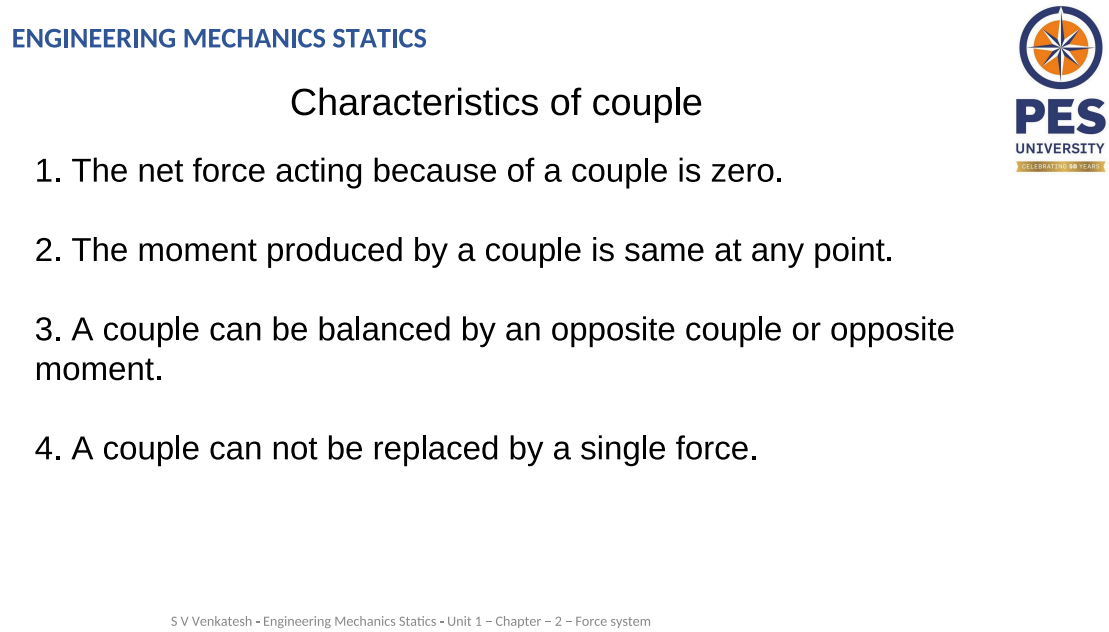


Figure G000079. UE26ME142A Unit 1 Class 8 Lecture PPT · source slide page 5.

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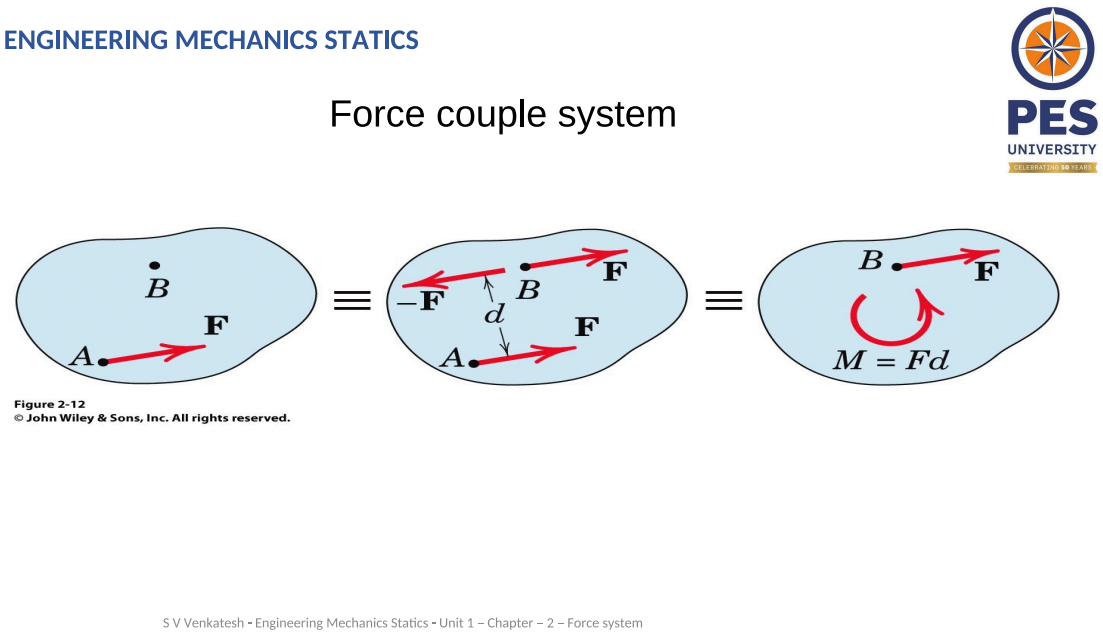


Figure G000080. UE26ME142A Unit 1 Class 8 Lecture PPT · source slide page 6.

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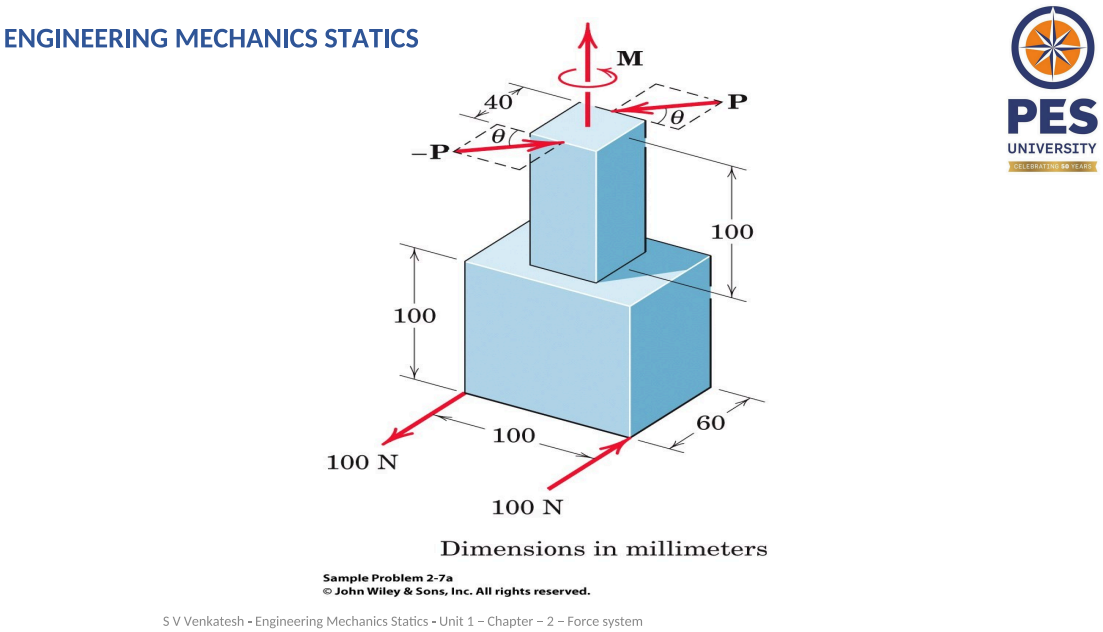


Figure G000081. UE26ME142A Unit 1 Class 8 Lecture PPT · source slide page 7.

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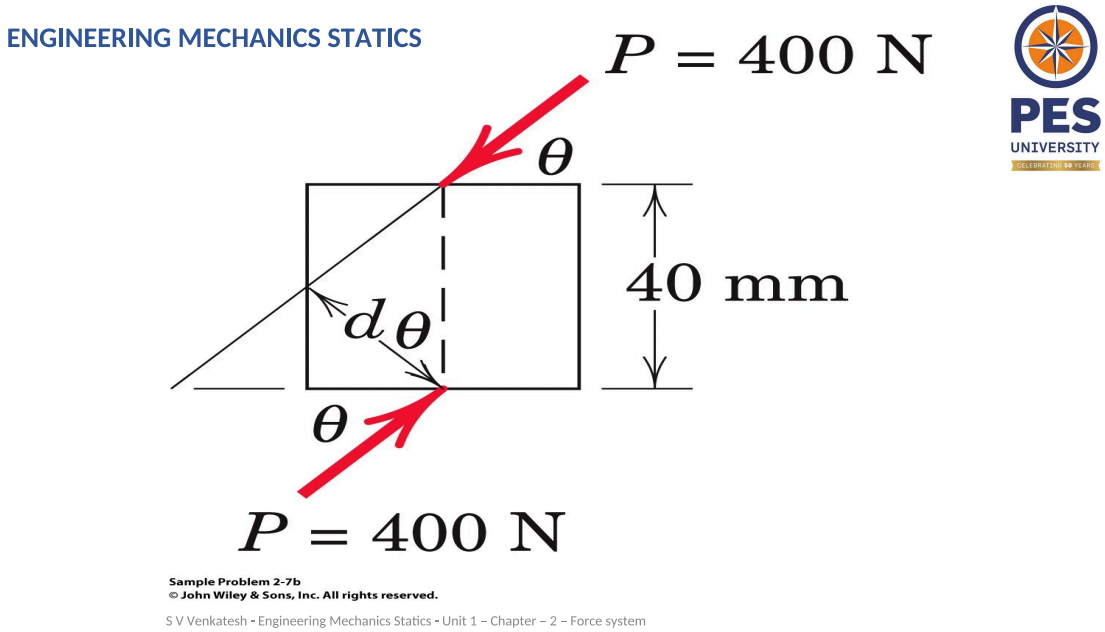


Figure G000082. UE26ME142A Unit 1 Class 8 Lecture PPT · source slide page 8.

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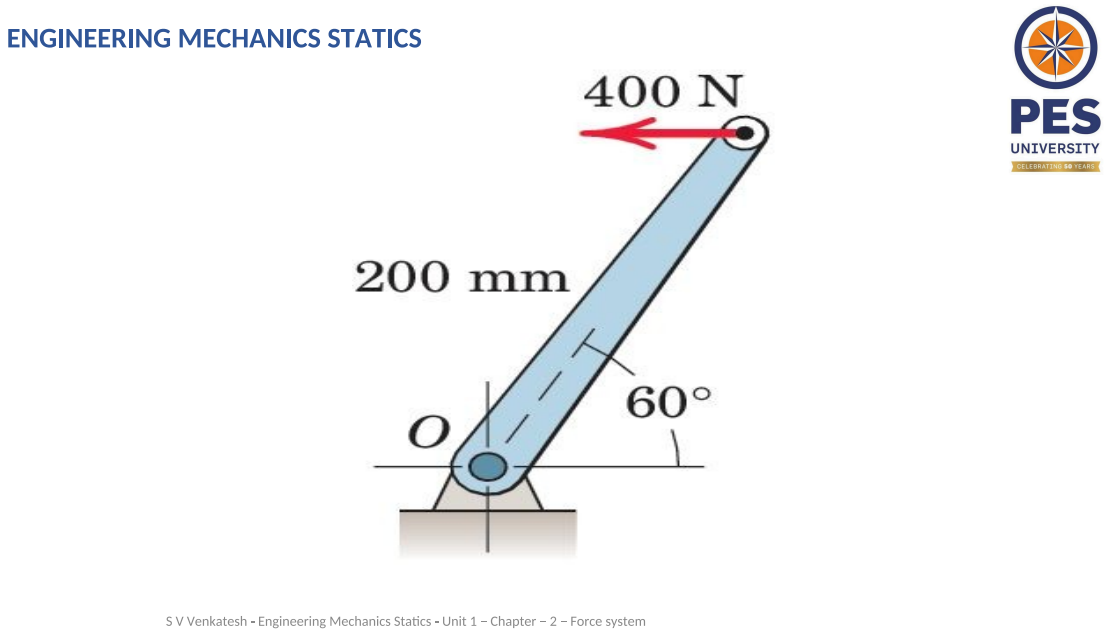


Figure G000083. UE26ME142A Unit 1 Class 8 Lecture PPT · source slide page 9.

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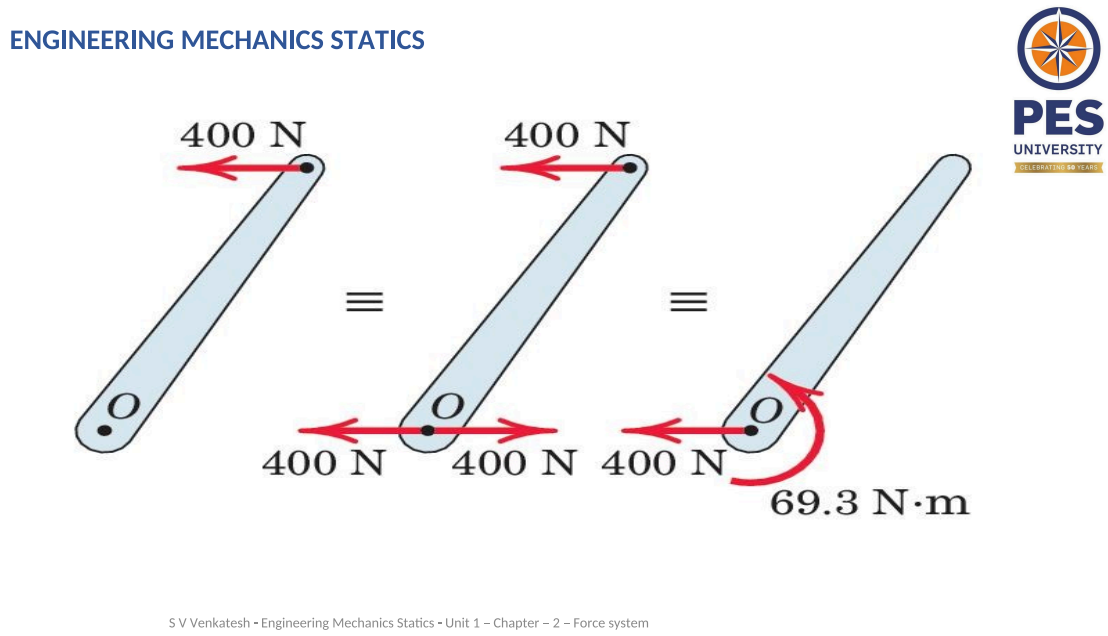


Figure G000084. UE26ME142A Unit 1 Class 8 Lecture PPT · source slide page 10.

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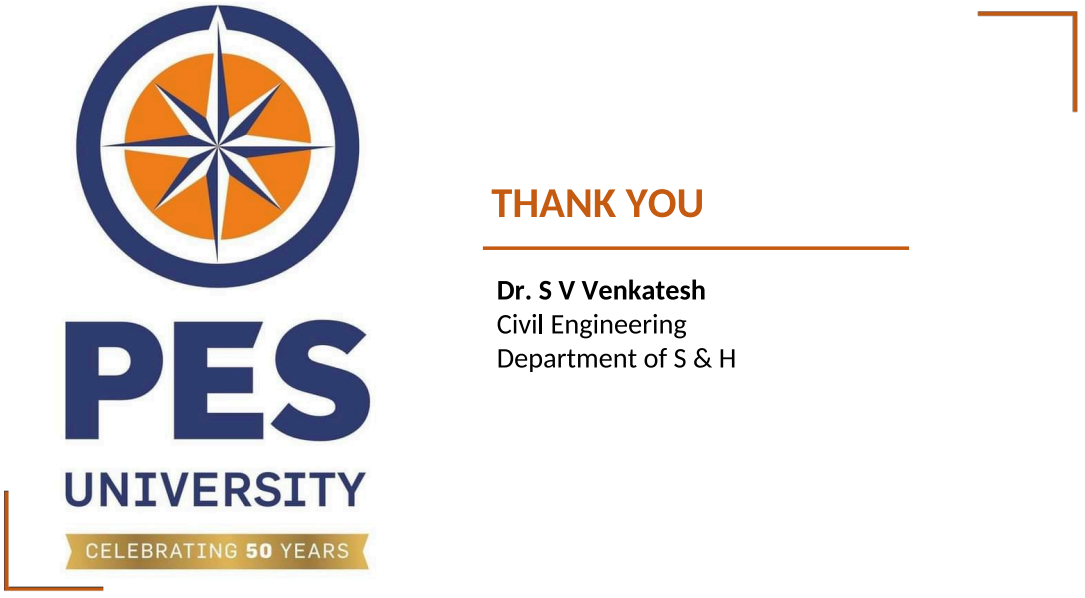


Figure G000085. UE26ME142A Unit 1 Class 8 Lecture PPT · source slide page 11.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000470 · EV0000471 · EV0000472
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Unit - 1 Couple S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ## Page / slide 3 (OCR) ENGINEERINGMECHANICS STATICS PES M O UNIVERSITY CELEBRATING50YEARS M (a) rB B M M Counterclockwise Clockwise (6) couple couple (d) M Figure2-10 JohnWiley&Sons,Inc.Allrightsreserved. S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Chapter - 2 - Force system EV0000473 · EV0000474 · EV0000475
3. Source slide page 3: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000476 · EV0000477 · EV0000478
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000479 · EV0000480 · EV0000481
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS Characteristics of couple 1. The net force acting because of a couple is zero. 2. The moment produced by a couple is same at any point. 3. A couple can be balanced by an opposite couple or opposite moment. 4. A couple can not be replaced by a single force. S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000482 · EV0000483 · EV0000484
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS Force couple system S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000485 · EV0000486 · EV0000487
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000488 · EV0000489 · EV0000490
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000491 · EV0000492 · EV0000493
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000494
10. Source slide page 10 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ## Page / slide 11 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000495
11. Source slide page 11: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000496

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.580] Dear students, welcome to engineering mechanics classes. My name is Dr. SV Vengitesh from Civil Engineering and the Department of Science and Humanities, PS University. This is class number nine. In the last class, we had discussed about [00:00:18.116] The problems how we can solve under the topic moment. Today we will take up the next topic, which is couple. Now what is a couple? Two unlike parallel forces constitute a couple. [00:00:36.516] When I say two, equal and unlike parallel forces

constituble. When I say to equal, what does it mean? Means that the magnitude of the forces are same. That is what it means by equal. When I say unlike parallel, unlike parallel means [00:00:56.484] The sense of the two forces are opposite to each other, that is unlike. Then parallel, they are acting parallel to each other. One is moving into the screen, the other one is coming out of the screen. They are parallel. The magnitude of these two forces are the same. [00:01:16.132] So a couple is defined as two equal, unlike parallel forces constitute a couple. Okay. So you can see here. [00:01:29.796] This is one force which is the center of the forces coming like this. This is another force, the center of the force is going with this. So they are acting in the opposite sense. [00:01:42.948] But the magnitude of the force is F, the magnitude of this force is also F. So this constitutes a couple. Now, what is the effect of a couple? The effect of a couple is to make the body rotate. [00:02:02.020] No, how do you say that? No, since the sense of this force is like this, the sense of this force is like this, the combination of these two, when I sum it up, it will give me zero because the senses are zero, the sense are opposite. [00:02:17.988] The magnitude of the force is the same, so effectively the net force acting is zero. But since they are acting at different [00:02:30.148] Points separated by some distance, it would make the body to rotate. The rotation can happen in the counterclockwise okay, or it can happen in the clockwise direction, depending upon how the sense of the forces are acting. Now, in this case, if you take [00:02:49.828] Uh if this forces to be acting along X and this to be acting along Y, then the Z will be coming out of the plane. So if you orient the right hand indicating the thumb towards U, the other [00:03:05.696] four fingers would be rotating and that is how this force this force would rotate. So it is a positive force. The positive force means it is counterclockwise. It is counterclockwise. [00:03:17.636] And vice versa, if it is going to rotate in the clockwise direction, it is anti-clockwise. So, couple definition I have told, what is the effect of a couple I have told, what are the classifications of a couple, one is counterclockwise, the other one is clockwise. [00:03:33.060] Now, the effect of the couple is to make the body to rotate, which is translative. It is not translatory motion, but it is rotatory motion. Okay. Now, [00:03:47.300] We will take this example and see what would be the movement that is going to be produced at point O. Now this force is acting at a distance of A. [00:04:00.708] Whereas this force is acting at a distance of A plus D. The distance between the two forces is D. So if I take movement about this point, this force into this distance A would produce a clockwise movement. So that is positive, sorry, negative. [00:04:17.764] Whereas this force into this total distance A plus D would produce an anti-clockwise movement. So we get a force or a moment equal to F into D itself. [00:04:29.316] Similarly, I will take a point here. Let's take the perpendicular distance of this force from A to B R A, and perpendicular distance of this force from O to be R B. [00:04:41.700] So the movement produced at O because of this force is this F or it is R B into F in vector approach or R B cross F I should take, then it is plus RA cross.

EV0000166 · EV0000167 · EV0000168 · EV0000169

Transcript block 2 (source interval 000300-000599 seconds): [00:05:01.124] F. Okay, if I sum up this, again, I will get the moment to be acting at this point, which should be equal to F into R. [00:05:09.924] Now again, one more combination is taken here. If I want to find out the movement of these two forces, the sense of this would be acting upwards. If I take the take these two forces to be acting in the horizontal plane, the vertical axis perpendicular to the horizontal plane. [00:05:26.080] Will be the axis about which the movement is going to be produced, and you can see that this would produce a positive movement that is anti-clockwise movement. Okay now, how do we determine the sense of the movement? Also, I have explained. Okay. So what we have defined

till now as what is a couple. [00:05:46.372] What is the effect of a couple? Okay. So a couple is an example for a free vector. That is because the momentum produced by the couple at any point that you take would be the same. So it's a free vector. So we have defined what is a couple. What is the effect of a couple? [00:06:06.404] Then what is the classification of couple? And how do we determine the moment produced by a couple at any point, which should be equal to the [00:06:18.756] Distance between the two forces that is acting or constituting a couple into the magnitude of the force that is acting. Okay. Now coming to what is equivalent couples. Now you can see that here there is a force effecting. [00:06:36.612] There is a force F, negative force F acting at this point, that means the sense are different. They are differentiated or the distance between the two is D. So this produces a moment which is the positive direction. Now instead of applying it here, if I apply it here, [00:06:55.556] The same force. F minus F and F, instead of applying it here, this is the point of application of these two. Whereas if I change it at this point, it would still be equal to M only. Now why only that? If I [00:07:12.772] Change the position to be here and here, provided this distance is still D, again I will get the moment to be equal to M itself, and that is positive. Similarly, in this case also we have the forces acting minus 2F here, and then [00:07:33.220] plus two f here, but the distance between them is D by two. So we will get this as two F into D by two. So we will get still get it as M itself. You can see here this distance is half of this distance. [00:07:51.300] So that is if this is D, this is D by T. Okay. So this is what is equivalent couple means. Now coming to what are the characteristics of a couple. I already explained what is the characteristics, one of the characteristics. [00:08:04.900] The net force acting because of a couple is zero. Because one force is acting in the positive direction, the other force is acting in the negative direction. So combination of these two will give me zero as the magnitude of the couple, or the force acting because of a couple is zero. [00:08:24.772] The movement produced by a couple at any point is same. I have explained this also. The moment produced by a couple is same at any point. Or the movement produced by the couple at any point is same. Okay, that also I have explained it. [00:08:40.260] Next, a couple. How do I balance a couple? I can balance a couple by an equal and opposite couple only. I cannot balance it by a single force because the net force acting is zero. [00:08:54.628] A couple cannot be replaced by a single force. That is also a characteristics. Okay? These are the characteristics of a couple. Now, very important concept is what you mean by a force couple system. What does this mean? [00:09:10.468] For this, I'm taking a simple example of a force acting at a point A. The magnitude of the force is F, it is acting along this line of action, and the sense is moving away from A towards the right side. Now I have to replace this force by some system. [00:09:30.116] At B, what is the condition? The condition is that whatever effect is produced by this force acting at A, the same should be the effect produced by the system of force that's I am going to act at B. [00:09:45.252] Now when I apply a force at A, it can produce a translatory motion, it can produce a rotatory motion, or it can produce the combination of translatory motion as well as rotatory motion. [00:09:56.772] Now if I replace this by a system of force at B, it should also produce the same effect as that of the force you have produced when it is acted up. A that is the condition. That is what is called as a force couple system.

EV0000170 · EV0000171 · EV0000172 · EV0000173

Transcript block 3 (source interval 000600-000899 seconds): [00:10:13.540] So to do this, first thing what has to be done is I have to remove this force. To remove this force, what I'll do is I'll apply an equal and opposite force at B. Okay, so that is this force I'm applying opposite to this. Okay.

[00:10:31.204] So what happens? The net effect of these two would be zero. But I cannot apply simply at P this force because it will change the [00:10:43.652] Effect of a force. So I apply a force F here. So you can see that these two when I apply, they will balance with each other. I get only this again. But now I'll take this force minus F and F acting at A. Those two are now constituting a couple. [00:11:02.340] I'll remove that and replace it by a couple, that is moment M, which is equal to F into D. We know that the moment produced by a couple is. [00:11:11.268] Force into the distance between the two forces constituting a couple. So that would be your so now you can see that this forms a force couple system. The effect of this would be the same as the effect of this. [00:11:25.572] How? Because this will have the same effect as this, because these two will cancel with each other. I will get only this. So that is the same as this. So these two effects are same. Now this and this also are same. How I have replaced these two forces that is minus F and F. [00:11:42.176] acting at A by a moment so this effect and this effect is same so eventually this effect and this effect are also same this is what is called as a force couple system and we will be using exclusively this concept for solving problems. [00:12:01.828] Okay. Next, let's take this simple example. I'll read out the statement of the problem. This is a solid problem in the textbook. The rigid structural member is subjected to a couple. [00:12:17.764] Consisting of the 200 newton forces replace this couple by an equivalent couple constituting of the two forces P and P dash, each of which has a magnitude of 400 newton. [00:12:35.428] Determine the proper angle T tar. So what is given here is there is two force that is acting of magnitude 100 newtons, which is applied at these points. If you refer to the diagram, you will be able to know. [00:12:51.428] The point of application of the two forces of magnitude 100 newtons. One is positive, the other one is negative. They are separated by a distance of 100 millimeters because here it is given dimensions in millimeters. Now we have to replace this by two forces. [00:13:10.660] P and minus P acting at these points at the top of the diagram. Now the distance between the point of application of this force P, P and minus P is given as 40 mm. [00:13:28.228] But the forces P is acting at an angle of theta with respect to this line. Okay? So they would also form a couple only. The effect of the forces 100 newtons, which is separated by [00:13:44.932] 100 millimeter is going to produce a couple positive couple of magnitude 100 into 100. Okay. So 100 into 100. [00:13:58.564] Now that 100 into 100 will be the moment produced, should be the moment produced by this P and minus P also. Because we have to replace that by this. Now how do we do it? So resolve this P. [00:14:13.604] along this line which will be equal to p cos theta. Similarly resolve this p along this line which will be p into cos theta. When I resolve this along the vertical [00:14:25.060] Which will be P sine theta, and this one will be P sin T again. They are acting in the same line and opposite to each other, so they will balance with each other. [00:14:36.740] Whereas this and this, they are not acting in the same line, but they are separated by the distance 40 mm. Okay, so they form a paralyf system. So that would produce a moment which should be equal to 100 into 100. So if I take that. [00:14:56.900] Then get the value of P. Okay. So P cos theta into 40 should be equal to 100 into 100. From that equation I will be able to get the value of P. Okay.

EV0000174 · EV0000175 · EV0000176 · EV0000177

Transcript block 4 (source interval 000900-001199 seconds): [00:15:11.140] Next, let's take this example. So in this he says replace the horizontal 400 newton force acting on the liver by an equivalent system consisting of a force at O and a couple. [00:15:29.348] So what is given in this is we have to replace this force four hundred N which is acting horizontal on this liver. Okay? By a force couple system acting at O. [00:15:44.324] So, magnitude of the force we have to take it from

the statement, but the distance from O to the point, okay, we have to take it from the diagram which is 200 mm, and the inclination of this line with respect to the horizontal is 60 degrees. Okay. Now [00:16:03.140] If I want to replace this force by force couple system at O, what is to be done? Just apply the 400 newton force, whatever is there acting at this point, bring it to this point, which will be 400 newtons acting at O like this, then determine [00:16:22.404] What is the movement produced by this force at O? How do we find that? We take this force, multiply it with the vertical distance between O and this point. [00:16:33.316] Which would be equal to 200 sine of 60 degrees, which would be the vertical distance. So 400 into 200 sine 60 will give me the moment here which will be in the anticlockwise direction. So [00:16:46.564] The force couple system acting at O will be 400 newtons acting at O to the left, then a counterclockwise movement which will be equal to the magnitude of 400 into 200 sin 60. That is what is explained here. [00:17:05.508] For this force, I will apply an equal and opposite force here. Okay, so to balance this, I will apply a force 400 here. So that means simply bring this force here. Now this two would constitute a couple. Replace them by a moment. [00:17:24.132] Which is equal to four hundred, that is this, into two hundred is this distance, sine sixty because this angle is sixty, the vertical distance is sine sixty. So you will get that as sixty-nine point three newton meters. I hope this is clear. So [00:17:42.308] We have explained in this today's class, that is class number nine, what is a couple? What are the characteristics of a couple? [00:17:53.540] What is the effect produced by a couple? Then how do we determine the moment produced by a couple? Then we have looked at what do we mean by force couple system. How do we get it? A force couple system. [00:18:13.700] The force coupled system is nothing but replacing a given force acting at a point A by some system acting at point B by a force couple system such that it produces the same effect, and we also have looked at [00:18:34.020] How do we solve problems using the force couple system? Okay, two problems, example problems I have taken and I have explained these two problems. I hope all of you have understood. This is class number nine. We will continue to solve some more problems from the unsolved. [00:18:54.212] section. Okay, in our next class that is class number 11. Okay, this is class number 10. We will continue in the next class that is class number 11. [00:19:09.124] Thank you very much. My name is Dr. She Vingtej from Civil Engineering under Department of Science and Humanities, PS University. Thank you. See you in the next class.

EV0000178 · EV0000179 · EV0000180 · EV0000181

Full transcript: 2839 normalized words, 72 timestamped segments, 1159.765 seconds duration. Low-confidence segments flagged: 0.

EV0000166 · EV0000167 · EV0000168 · EV0000169

12 UE26ME142A_Unit 1_Class 9_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 9_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000182 · EV0000183

Graphs and diagrams



Figure G000086. UE26ME142A Unit 1 Class 9 Lecture PPT · source slide page 1.

EV0000498

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ENGINEERING MECHANICS STATICS

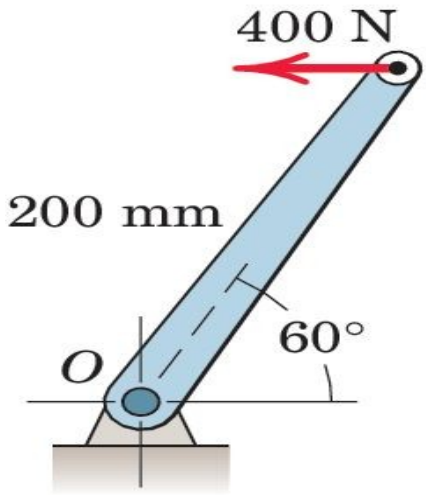
Unit - 1
Couple

Figure G000087. UE26ME142A Unit 1 Class 9 Lecture PPT · source slide page 2.

EV0000501

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ENGINEERING MECHANICS STATICS



S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

Figure G000088. UE26ME142A Unit 1 Class 9 Lecture PPT · source slide page 3.

EV0000504

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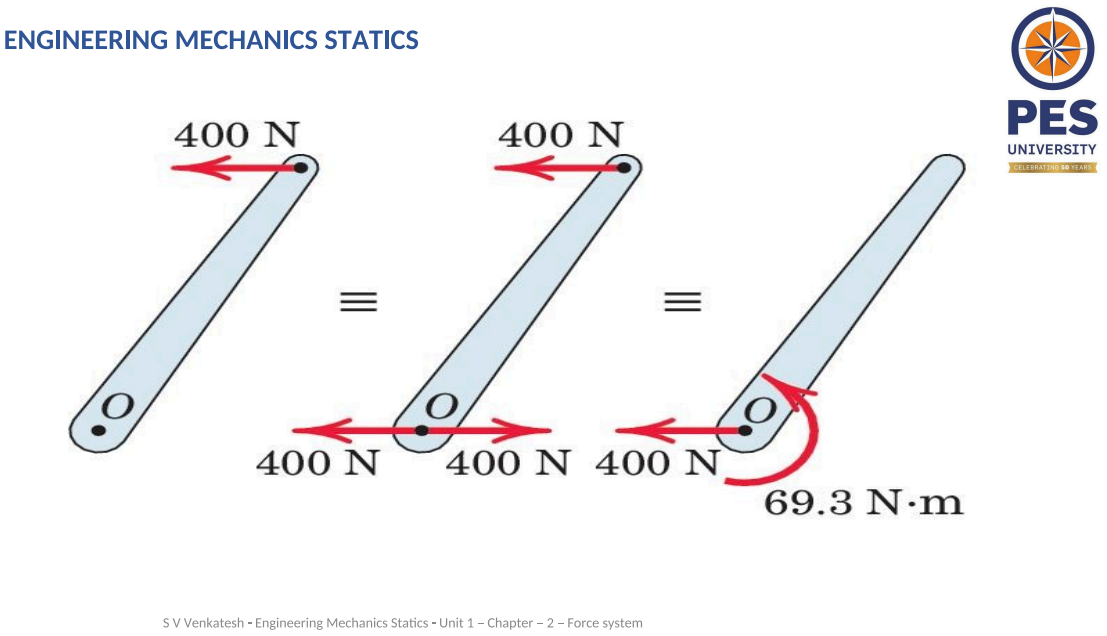


Figure G000089. UE26ME142A Unit 1 Class 9 Lecture PPT · source slide page 4.

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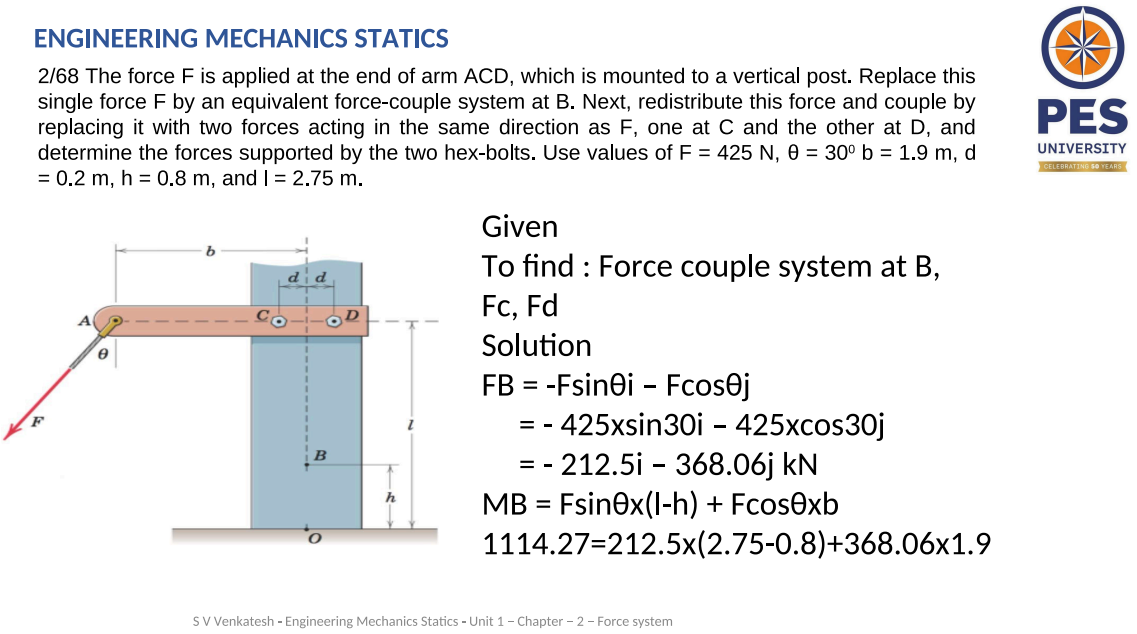


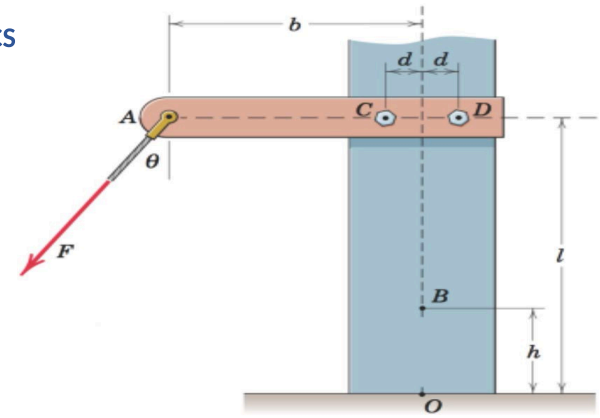
Figure G000090. UE26ME142A Unit 1 Class 9 Lecture PPT · source slide page 5.

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ENGINEERING MECHANICS STATICS

2/68 Conti……



ΣF = 0
Fc - Fd = F = 425
MB = FcxcosθxdFcxsinθx(l-h) + FdxcosθxdFdxSinθx(l-h)
1114.27 = (F + Fd) x cos30 x 0.2 + (F + Fd) x sin30 x (2.75–0.8) + Fd x cos30 x 0.2 – Fd x sin30 x (2.75 – 0.8)
Fd=(1114.27–73.61–414.38)/(0.173+0.97+0.173–0.97)= 1807
Fc = 425 + 1807 = 2232 N

S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

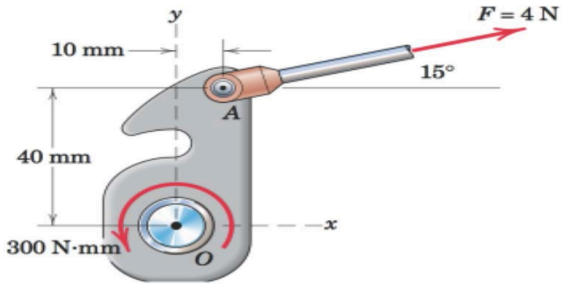
Figure G000091. UE26ME142A Unit 1 Class 9 Lecture PPT · source slide page 6.

EV0000513

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ENGINEERING MECHANICS STATICS

2/76 The device shown is a part of an automobile seatback-release mechanism. The part is subjected to the 4-N force exerted at A and a 300-N · mm restoring moment exerted by a hidden torsional spring. Determine the y-intercept of the line of action of the single equivalent force.



Given F = 4 N,
M = 300 N-mm
Θ = 15⁰
X1 = 10 mm, Y1 = 40 mm
To find : Y-intercept
Solution
Mo = M + FsinθX1 - FcosθY1
- FcosθxY = 300 + 4xsin15x10 – 4xcos15x40
Y = (300 + 10.35 – 154.55) / (- 4xcos15)
Y = 155.8 / - 3.86 = - 40.36 mm

S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

Figure G000092. UE26ME142A Unit 1 Class 9 Lecture PPT · source slide page 7.

EV0000515

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Figure G000093. UE26ME142A Unit 1 Class 9 Lecture PPT · source slide page 8.

EV0000516

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000497 · EV0000498 · EV0000499
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Unit - 1 Couple S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000500 · EV0000501 · EV0000502
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000503 · EV0000504 · EV0000505
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000506 · EV0000507 · EV0000508
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS 2/68 The force F is applied at the end of arm ACD, which is mounted to a vertical post. Replace this single force F by an equivalent force-couple system at B. Next, redistribute this force and couple by replacing it with two forces acting in the same direction as F, one at C and the other at D, and determine the forces supported by the two hex-bolts. Use values of F = 425 N, ϑ = 300 b = 1.9 m, d = 0.2 m, h = 0.8 m, and l = 2.75 m. Given To find : Force couple system at B, Fc, Fd Solution FB = -Fsinϑi – Fcosϑj = - 425xsin30i – 425xcos30j = - 212.5i – 368.06j kN MB = Fsinϑx(l-h) + Fcosϑxb 1114.27=212.5x(2.75- S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system 0.8)+368.06x1.9 EV0000509 · EV0000510 · EV0000511

6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS 2/68 Conti..... ΣF = 0 Fc - Fd = F = 425 MB = Fcxcosϑxd + Fcxsinϑx(l-h) + Fdxcosϑxd - FdxSinϑx(l-h) 1114.27 = (F + Fd) x cos30 x 0.2 + (F + Fd) x sin30 x (2.75–0.8) + Fd x cos30 x 0.2 – Fd x sin30 x (2.75 – 0.8) Fd=(1114.27–73.61–414.38)/(0.173+0.97+0.173–0.97)= 1807 Fc = 425 + 1807 = 2232 N S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000512 · EV0000513 · EV0000514
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS 2/76 The device shown is a part of an automobile seatback-release mechanism. The part is subjected to the 4-N force exerted at A and a 300-N · mm restoring moment exerted by a hidden torsional spring. Determine the y-intercept of the line of action of the single equivalent force. Given F = 4 N, M = 300 N-mm Θ = 150 X1 = 10 mm, Y1 = 40 mm To find : Y-intercept Solution Mo = M + FsinϑX1 - FcosϑY1 - FcosϑxY = 300 + 4xsin15x10 – 4xcos15x40 Y = (300 + 10.35 – 154.55) / (- 4xcos15) Y = 155.8 / - 3.86 = - 40.36 mm S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ## Page / slide 8 (OCR) THANK YOU Dr. S v PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000515
8. Source slide page 8: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000516

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.996] Dear student, welcome back to classes on the subject engineering mechanics statics. My name is Dr. SV Venkitish, Civil Engineering Department of Science and Humanities, PS University. This is class number 11. [00:00:18.564] In the last class we discussed about couple, the effects of couple, the classification of couple, the characteristics of couple, force couple system, then we solved a few problems. [00:00:36.292] Today, that is in class number eleven, we will try to solve few more unsolved problems from the prescribed textbook that is Engineer Mechanics Tactics by Miriam, Craig and Balden. Okay. [00:00:53.988] Use in this problem, which is 2 bar 68, the statement of the problem is the force F is applied at the end of ACD, which is mounted to a vertical post. [00:01:11.524] Replace this single force F by an equivalent force couple system at B. Next redistribute this force and couple by replacing it with two forces. [00:01:29.444] acting in the same direction as F, one at C and the other at D, and determine the forces supported by the two X bolts. [00:01:44.740] Using values of F equals 425 newtons, theta equals 30 degrees, B equals 1.9 meters, D equals 0.2 meters [00:02:01.444] H equals point eight meters and I equals two point seven five meters. Now when you go through the statement of the problem, you should identify that [00:02:16.260] There is a force that is acting at the end of the arm A C D, that is at this point A. We have to replace this force by an equivalent force couple system at B, that is at this point. Then [00:02:35.044] Redistribute this force and couple by replacing it with two forces acting in the same direction as F. Same direction of F means it is at this angle theta. He has not specified the sense of the force. [00:02:52.356] He has specified that the angle of the force that is going to be acting one at C and one at B should have the same direction, which means it is at angle T with respect to the vertical. Okay. [00:03:09.188] Uh and determine the forces supported by the two X bolts. X boles is these are the two bolts that is C and D. What are the forces acting at that? And then he has given you the values of the force F which is 425 newtons. [00:03:26.852] Then the angle theta is 30 degrees. Then the B distance that is from A to the center between the two volts that is C and D is given as

1.9 meters. Then the distance D that is from the center [00:03:43.716] of the bolts C and D till the C, R T is given as D and D. [00:03:50.916] Which is 0.2 meters, then H, which is the distance from O to B the vertical distance as 0.8 meters, and L which is the vertical distance between O to the center of the two bolts, that is given as L. [00:04:08.228] Now, first thing is we have to replace this force by a force couple system at B. So straight away apply this force at B, to straight away apply this force at B. [00:04:20.548] I will find out the component of this force and use that as a force that is going to be acting at B. So that is why indicated as here. P is equal to the component of this force here along X would be to the left. I am taking [00:04:38.692] X axis to the right as positive and upwards y axis as positive. So to the left it becomes negative, so minus F theta is this. So we get F sin theta I which will be this component. Then component of this force along Y will be downwards, so again negative. [00:04:59.396] and it's going to be a cause of this angle theta j. So substituting the value for f1 theta and then finding out we get the force f b represented as a vector as minus 212.5i

EV0000182 · EV0000183 · EV0000184 · EV0000185

Transcript block 2 (source interval 000300-000599 seconds): [00:05:16.580] Minus three sixty eight point zero six J Kilo Newtons because uh sorry, if this is in Newtons, I'm sorry this has to be changed to newtons because what is given here is in newtons. So this also has to be in newtons, can we make that correction? [00:05:34.500] So once we find that, the next thing that we have to do is uh we have to determine the moment about B because of this force that is acting. [00:05:48.964] Okay, so again we will take the component of these two forces. So moment about P because of the horizontal component will be F sine theta into L minus H because the horizontal component vertical distance is L minus H. [00:06:05.668] Then we will take the vertical component which is F cos teta into the horizontal distance is V. Straight away we can put that, then substitute the value for F, L, H and B, we get the moment about B. [00:06:19.908] Okay, that's the first part of the problem to replace the given force which is acting at A point to a force couple system acting at B. So now we have successfully got the force acting at B. [00:06:33.636] which is equal to F sine theta i with a negative sign minus f cos theta j which is acting along the y axis that is the force then movement we have already got because of the two components and multiplying it with the respective distances perpendicular distances. [00:06:52.772] The second part of the problem is to replace this force and the couple by two forces acting at C and D, which will have the same direction, that means this direction. The movement acting at P will be in the anti-clockwise direction. So when I apply a force at C, [00:07:12.804] That should produce a movement at B in the anticlockwise direction. Then when I apply the force at D, that should also produce a movement at B in the anticlockwise direction. [00:07:25.412] So the force acting at C will be at an angle of theta with the vertical, which will be acting downwards from C. So that would that is what would produce a movement at B, which will be anti-clockwise. Similarly, when I apply a force at C, [00:07:44.324] This force when I apply, this force should produce a moment at B which is in the anti-clockwise direction. So if I apply the force upwards at an angle of theta, that would produce an anti-clockwise movement. [00:07:59.984] That would be appropriate as what I feel. So I consider F C to be acting downwards, F D to be acting upwards. Okay? Now those two forces should produce the same force. F [00:08:14.052] Or F B and also the movement at B, whatever is there, should be the movement that is going to be produced by these two forces. Okay. So we will consider sigma F. [00:08:27.268] To be equal to 0. So we will get this force F C which is acting downwards minus F C which is acting upwards should be equal to F because this is the total force

that should be. [00:08:42.692] That we should get when we sum up these two, so that I have equated, and f we know that it is equal to 425 newtons. Now from this equation I can also write F C as F plus F B. Okay. No momentum or B whatever we get because of these two. [00:09:00.772] We are taking so that would be equal to F C, which is this component of this acting along the vertical, which is F C cos theta into D, that is what is given here. [00:09:16.292] That would produce an anti-clockwise movement, so positive. Then the component of F C in the horizontal direction, that is F C sin theta into the vertical distance from B to this point B to C, which is L minus H, and that will also produce an anti-clock movement, so positive. [00:09:32.836] Next, the component of the force along vertical, which is because of Fd, would be Fd. [00:09:40.196] cos teta into D into D which will be positive because that will also produce an anti-clockwise movement. But the component of F T acting in the horizontal direction that is going to be this would be the sense that will present clockwise movement. So it is negative. So Ft [00:09:58.404] Sine theta into the horizontal vertical distance is L minus H, so that is what is indicated here. Now substituting the values for this, we all take determining the value of M B, which is this, then substituting value of F C as F plus M D.

EV0000186 · EV0000187 · EV0000188 · EV0000189

Transcript block 3 (source interval 000600-000899 seconds): [00:10:16.100] Then theta is 30 degrees and D we all know it is 0.2. Similarly for others we are substituting the values. From this we would be able to get the value of F D which is equal eight hundred and seven newtons. [00:10:32.804] Now substitute this here in this equation or this equation and get FC. So F is 425, F D we have got it as 1807, so we will get F C as 2232 Newtons. [00:10:50.276] I hope all of you have understood. Now in this problem, what you need to understand or know clearly is to use the force couple system at B, very clearly, we have a force acting at A. [00:11:04.912] You have to replace that by a force couple system at B. So we have to put the same force acting at A at B, then determine the moment produced by the force acting at A with taking the moment origin as B. So that is what we have done. Then replace this force couple system by uh [00:11:25.124] Forces acting at C and D is what we have done. I hope this is clear. Next, another problem we will discuss, that is problem number 2 bar 76. So the statement of the problem is the device shown is a part of an automobile. [00:11:44.644] Seal seed back release mechanism. The part is subjected to the four newton force exerted at A [00:11:56.164] and a 300 newton millimeter restoring moment exerted by a hidden torsional spring. [00:12:08.484] Determine the y-intercept of the line of action of the single equivalent force. So from the statement, what we can make out is there is a force of four kilon that is acting. [00:12:22.436] And there is a couple or a moment of 300 newton millimeters that is acting. We have to replace this by a single equivalent force. [00:12:37.796] And we should find out where this single equivalent force should act along the y-axis so that that produces the same effect as that of the single force of and the moment 300 newton millimeters. Okay. This one, this problem is. [00:12:57.956] Opposite of that of the force couple system. What we used to do earlier was there was one force given, we used to determine what should be the force couple system, which would produce the same effect when it is applied at some other point, is what we used to determine. [00:13:17.732] So in this problem they have given you the force couple system. So we need to find out the single force. So it is just opposite to what we used to do for force couple system. Okay. Going forward, note down what is given, the force F is given, the moment is given. [00:13:35.204] From the diagram, we can note down that the angle at which the force is acting with respect to the horizontal axis is 15 degrees, that is also given. Then we can also note that the point A is at a

distance of 10 millimeters from O horizontally. [00:13:53.124] and it is at a distance of 40 millimeters from O vertically. Right. So what are we supposed to find? What is that single equivalent force that should be applied on the y axis so that it produces the same effect. [00:14:10.660] Now when we had a force and then we are supposed to find out the force couple system, we used to apply the same force at the point where we have to replace it by a force couple system. Now, since we have the force couple system itself, the force will be the same whatever we have as a force F. [00:14:27.556] 40 newtons, but we have to apply this force on the y-axis such that that force when I apply it on the y-axis would produce a movement which will be the combination of the movement produced by F acting at A and this movement 300 newtons. [00:14:46.948] So, first thing that we need to find is what would be the combined moment by this force F and the moment 300 newtons that would be acting. So first thing is what would be the moment M naught

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:00.720] Combination of this and this is what we have to determine. So that's what is done here. So M naught is equal to M, that is this one, which is acting in the anti-clockwise direction, plus the component of this force along the vertical will produce an anti clockwise movement. So that's why it is plus, it is F. [00:15:19.524] Sine theta by theta is 15 degrees, so this component we will get into this distance 10. Okay. So we will get the moment because of this component along y. So movement because of the component along X would be in the clockwise direction, so it is negative. [00:15:39.364] The component will be of cos theta into the vertical distance which is 40 mm. So substituting the values, we will get the moment M naught. [00:15:52.132] Okay, so once we get the moment M naught, now I will apply this force here on the y-axis, taking the distance from O to the point where we are applying as y. Okay, when I apply that. [00:16:09.860] Okay, the component along the vertical will not produce any moment at all. Only the component at the horizontal component would produce a moment. So it will be yes. [00:16:21.220] Cos 15 into y should be equal to the moment at M naught. So finding out the value of y from that, we get it as negative, which indicates that the force should not be applied above O, it should be applied below O. I hope all of you have understood. [00:16:39.396] The two problems that I have discussed today, which are not solved in the textbook, I have exclusively taken these two problems, which are [00:16:48.836] Tougher and not easy to understand. I hope all of you have got this problems. Okay. So we will stop today's class here. This is class number 11. We will be continuing the class. Next class. [00:17:08.100] So that is class number twelve. Thank you. My name is Dr. Sv Ventej. I am from Civil Engineering under the Department of Science and Humanities, which comes under PS University. Thank you very much. See you in the next class.

EV0000194 · EV0000195 · EV0000196 · EV0000197

Full transcript: 2485 normalized words, 64 timestamped segments, 1045.291 seconds duration. Low-confidence segments flagged: 0.

EV0000182 · EV0000183 · EV0000184 · EV0000185

13 UE26ME142A_Unit 1_Class 10.0_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 10.0_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000198 · EV0000199

Graphs and diagrams



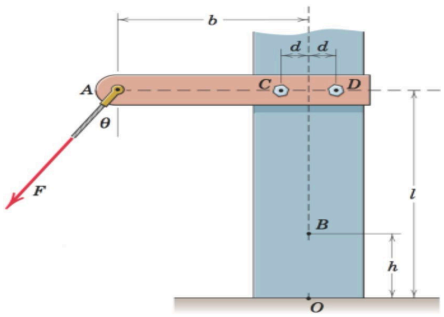
Figure G000094. UE26ME142A Unit 1 Class 10.0 Lecture PPT · source slide page 1.

EV0000518

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

2/68 The force F is applied at the end of arm ACD, which is mounted to a vertical post. Replace this single force F by an equivalent force-couple system at B. Next, redistribute this force and couple by replacing it with two forces acting in the same direction as F, one at C and the other at D, and determine the forces supported by the two hex-bolts. Use values of F = 425 N, θ = 30⁰ b = 1.9 m, d = 0.2 m, h = 0.8 m, and l = 2.75 m.



Given
To find : Force couple system at B,
Fc, Fd
Solution
FB = -Fsinθi – Fcosθj
= - 425xsin30i – 425xcos30j
= - 212.5i – 368.06j kN
MB = Fsinθx(l-h) + FcosθxB
1114.27=212.5x(2.75-0.8)+368.06x1.9

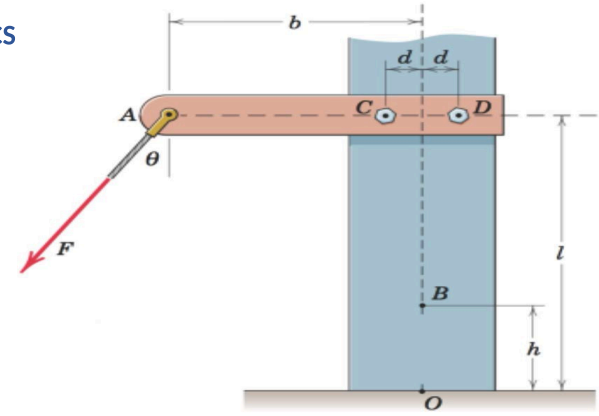
Figure G000095. UE26ME142A Unit 1 Class 10.0 Lecture PPT · source slide page 2.

EV0000521

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ENGINEERING MECHANICS STATICS

2/68 Conti……



ΣF = 0
Fc - Fd = F = 425
MB = FcxcosθxD + FcxsinθxD(l-h) + FdxcosθxD - FdxSinθx(l-h)
1114.27 = (F + Fd) x cos30 x 0.2 + (F + Fd) x sin30 x (2.75–0.8) + Fd x cos30 x 0.2 – Fd x sin30 x (2.75 – 0.8)
Fd=(1114.27–73.61–414.38)/(0.173+0.97+0.173–0.97)= 1807
Fc = 425 + 1807 = 2232 N

Figure G000096. UE26ME142A Unit 1 Class 10.0 Lecture PPT · source slide page 3.

EV0000523

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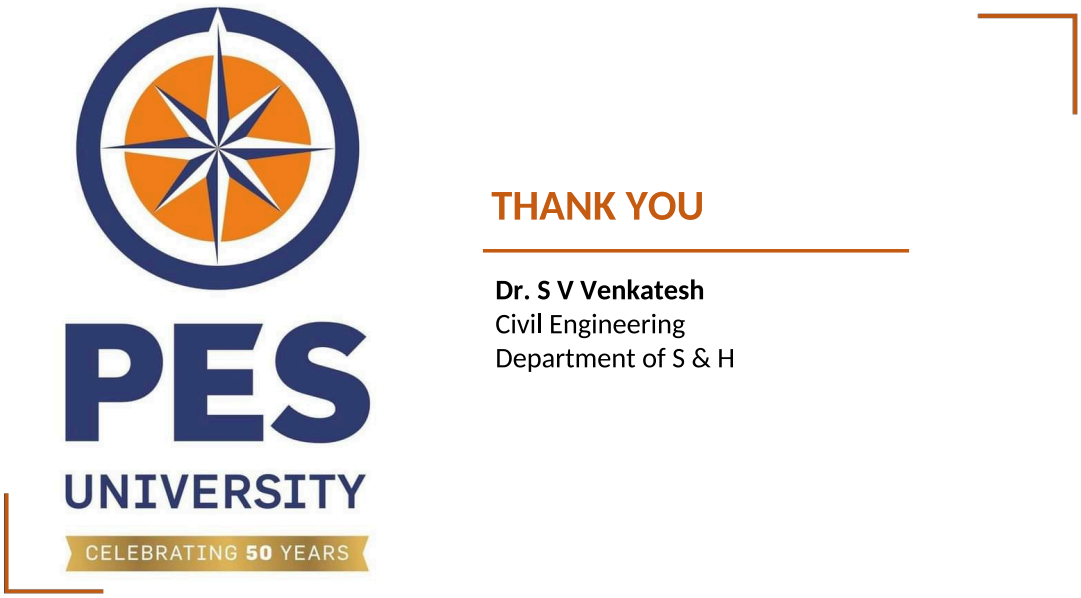


Figure G000097. UE26ME142A Unit 1 Class 10.0 Lecture PPT · source slide page 4.

EV0000524

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000517 · EV0000518 · EV0000519
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 2/68 The force F is applied at the end of arm ACD, which is mounted to a vertical post. Replace this single force F by an equivalent force-couple system at B. Next, redistribute this force and couple by replacing it with two forces acting in the same direction as F, one at C and the other at D, and determine the forces supported by the two hex-bolts. Use values of F = 425 N, ϑ = 300 b = 1.9 m, d = 0.2 m, h = 0.8 m, and l = 2.75 m. Given To find : Force couple system at B, Fc, Fd Solution FB = -Fsinϑi – Fcosϑj = - 425xsin30i – 425xcos30j = - 212.5i – 368.06j kN MB = Fsinϑx(l-h) + Fcosϑxb 1114.27=212.5x(2.75- S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system 0 8)+368 06x1 9 EV0000520 · EV0000521 · EV0000522
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS 2/68 Conti..... ΣF = 0 Fc - Fd = F = 425 MB = Fcxcosϑxd + Fcxsinϑx(l-h) + Fdxcosϑxd - FdxSinϑx(l-h) 1114.27 = (F + Fd) x cos30 x 0.2 + (F + Fd) x sin30 x (2.75–0.8) + Fd x cos30 x 0.2 – Fd x sin30 x (2.75 – 0.8) Fd=(1114.27–73.61–414.38)/(0.173+0.97+0.173–0.97)= 1807 Fc = 425 + 1807 = 2232 N S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ## Page / slide 4 (OCR) THANK YOU Dr. S v PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000523
4. Source slide page 4: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000524

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.612] Dear student, welcome to Engineer Mechanics Statics classes. I am Dr. Svenkitesh from Civil Engineering Department of Science and Humanities, PS University. In the last class, we discussed a few problems on couple. [00:00:18.372] Today we will be discussing more problems. Okay, I have taken this problem 2 bar 73. The statement of the problem is an overhead view of the handlebars on an all-term vehicle is shown. [00:00:37.988] If the indicated forces have a magnitude of F equals to 150 newton, determine the moment created by the two forces about the vertical steering axis through point O. [00:00:55.876] Both n axis are perpendicular to the left handlebar. Treat the problem as two dimensional. Now, from the statement, what you could get is that there is a force of [00:01:14.404] 115 you can acting on the handle of the vehicle at point A and at point B which you can identify from the diagram at point A you have F. [00:01:29.380] The magnitude of which is given in the statement as one fifty newtons. Similarly, there is a force F acting at B, the magnitude of this is also given as one fifty newtons. [00:01:40.900] What is asked is to determine the moment created by these two forces about the vertical steering axis through point O. [00:01:50.660] Point O is here, this is the vertical steering axis. So what is the moment created by these two forces about this axis? The axis is perpendicular to the screen. Remember that. Okay. So if you take the forces to be acting in the plane of the screen. [00:02:08.228] The vertical axis passing through O is perpendicular to the screen. So we have to take or find out what is the moment at this point created by these two forces, and he says that treat the problem as two dimensional. So we can treat this as two dimensional. [00:02:23.780] Now, from the statement we could get the magnitude and we know that we have to find out the moment about point O. Now if you read the diagram, we can look at the force F acting at A. The sense of the force acting at A is towards A, it's going towards A. [00:02:43.812] Whereas the sense of the force F at B is coming out of B. Now if you look at these two, they are parallel to each other. How can I make out that? He says that the N axis, both these axis, M. [00:02:59.748] or perpendicular to this handle axis. He says very clearly both n axis are perpendicular to the left handlebar. So this is the left handlebar. Both n axis are perpendicular to this handle axis. [00:03:18.852] So this n and this n are parallel, and it clearly says that this force F is at an angle of 10 degrees. Here also it is at 10 degrees. So I can easily say that this force and this force are parallel to each other, but the sense of these forces are opposite. Okay. [00:03:36.900] Now when you have a system where you have two forces which has the same magnitude, how can you note down? F and F is the magnitude, and it is 150 newton. The magnitude are same, they are parallel to each other, the sense is opposite. [00:03:56.548] That means what? It is a couple. So what is the definition of a couple? A couple is [00:04:05.412] Defined as when two equal unlike parallel forces are that is going to constitute a couple. So here we have the same thing. We have two equal, unlike parallel forces, so they constitute a couple. [00:04:25.700] So movement produced by a couple about any point is same. That's a property of a couple. So whether you find it about O or you find it about A or you find it about B the movement will not change and it's going to be same. [00:04:40.964] Okay. Next, if you look at this, it says OA is equal to OB. That means the horizontal distance of O to A and O to B are equal and [00:04:56.388] That is why the distance which we need to be, the horizontal distance we need to be, is given as 750 mm. Okay? They were also given this angle as 15 degrees. Well

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.956] We need not have to use this 15 degrees at all. Y means if I could get the component of this force along the vertical, which is equal to how much? Now if you see this is 15 degrees. [00:05:25.604] This axis and this is horizontal. This is a vertical axis. The horizontal and vertical will always be at 90 degrees. So this angle that is the angle made by the handle axis and the vertical. [00:05:40.704] will be ninety minus fifteen, which will be seventy-five degrees. Next we also have y [00:05:50.244] And the axis of the handle, which is at 9 to each other. So this angle, that angle made by the n axis on the vertical will also be at 15 degrees. So the total angle made by the force with the vertical, that is this force. [00:06:07.140] Width of vertical will be equal to this 10 degrees plus this 50 degrees, which will be 25 degrees. So I can easily resolve this force along vertical, which will be F. [00:06:18.116] Cost 25. Similarly, I can resolve this force along this vertical which will be F cos 25. This would be moving upwards, whereas this component will be moving downwards. So they produce a couple. No. [00:06:33.732] What happens to the horizontal component? The horizontal component of F would be F sine 15. Now at A it is moving to the left, whereas at B it is moving to the right, which is also going to be equal to F. [00:06:52.100] Sign 15 degrees. So they will balance with each other. So A and B are at the same horizontal line. So one force is acting to the left, one force is acting to the right. So they will balance with each other. That will not create any movement. [00:07:06.564] Okay, or since they are balancing with each other and the sense are different, one will produce a clockwise moment, other will produce an anti-clockwise moment of the same magnitude, they will balance with each other at any point that you take. So now [00:07:21.732] If I want to find out the moment, I can take only the vertical forces, take moment about A. So we will have it as F cos 25 into 750 mm. We'll give the moment at A. That moment, whatever we are going to get, would be the moment at any point you take it. [00:07:41.572] Okay, so the movement at O will also be equal to the movement that we get because of these two forces about A, or the movement that we get because of these two forces about B. [00:07:52.388] Right? I hope it is clear. So now you can look at this. When I take this force, this is fifteen, this is ten, so this would be twenty-five, so I take it as F. [00:08:03.364] Cost twenty-five, multiply that with this distance that is 0.75. This is in M, so converting it into meters. So I get the moment at O. Whether I take it at A or at B or at O, it is going to be same. So M at O will be equal to 102 point. [00:08:22.628] 2.0 newton meters and it is gonna be clockwise. [00:08:27.844] Okay, so we usually take clockwise as negative. So in this case also you can take the moment to be clockwise. I hope this is clear. No, you can also solve it by considering the point O, resolving this force along vertical and horizontal. [00:08:44.208] Taking moment about all because of it, because of those components, which you will get it as [00:08:50.564] 150 cos 25 into 750 by 2. This is the horizontal distance. Then you can resolve this. That would be 150 sin 25 into the vertical distance would be 750 divided by 2, 10 15 degrees, which you will get here as the [00:09:08.160] Vertical distance. So this would produce a anti uh sorry clockwise moment. This would also produce a clockwise moment. Now if you take this component, this would also produce a clockwise moment. The magnitude of that would be 150 cos 25. [00:09:23.552] into 750 by 2, then we will get this component which will also produce an anticlock as point. So minus it will be minus 150 sine 25 into this vertical distance which will be 750 by 2 times. So you get the same value. So [00:09:41.540] The property of a couple is that moment about any point because of the couple is going to be same. That is what we are going to be using in this equation, and you can get it very simple. [00:09:54.240] one line answer for this

problem even though it looks when you look at the diagram it looks very complicated but if you have understood clearly the properties of a couple then it becomes very simple okay right

EV0000202 · EV0000203 · EV0000204 · EV0000205

Transcript block 3 (source interval 000600-000899 seconds): [00:10:10.080] Let's look at the next problem, which is 2 bar 74. The statement of the problem is the force F is applied to the leg extension exercise machine as shown. [00:10:25.124] Determine the equivalent force couple system at point O. Use values of F is equal to 520 newtons, B equals 450 mm, H equals 2 point. [00:10:41.380] 215 mm, H is equal to 215 mm, R is equal to 325 mm, theta is equal to 15 degrees, and phi is equal to 10 degrees. So whatever system that you is given. [00:10:57.060] So we need not have to worry about the system, whether it is a leg extension exercise machine, or whether it is a dumbbell, or whether it is a pull and push machine, whatever machine that you have, let's not worry about those. [00:11:14.544] Let us worry about only drawing the free body diagram from that. So, to draw the free body diagram for this, we should know where the point of application of the force is there, what is the magnitude of the force, what is the direction of the force, what is the sense of the force. [00:11:31.920] Then what are the distances at which the force is applied with respect to the point where we want to find out the [00:11:39.748] Moment or the couple. So here in this problem, they have given the force F and they have asked us to determine the equivalent force couple system at point O. Now this is the point O, this is the point where the force is acting. So whether we can determine [00:11:57.056] The horizontal and the vertical distance or the distance along this line between point O and A, and the vertical distance perpendicular to this line is what we have to look at. Okay. Now so to look at what is the [00:12:15.780] distance of A to O along this line. Very simple, they have given this as R. So this would also be R. So R plus B would give me the distance between O to A along this line. [00:12:30.180] Similarly, if I want to find out what is the vertical distance between E and O, I can take it as H. [00:12:36.452] Or rather, I can take it as R minus H, I will get this distance that is from A to O. Right? So that's very simple. So to find out what is the force couple system at O, apply the same force at O, that is [00:12:54.724] You take this force, you have the axis, this axis, okay I can draw the same axis here, apply this force at an angle of theta with respect to this, that is this axis, so this is theta, or I can resolve this force. [00:13:12.356] Yes, cos theta would be the component of the force along this axis, so I can apply that. Then I can resolve this force along this, that is the line which is parallel to this. [00:13:26.692] There is this one which will be F sine theta. So that is what I am going to apply here. That will be the four that is acting at O. [00:13:34.756] Now to determine the moment at O, again the component of this force along this would be of cos theta into B plus R would be the clockwise moment. [00:13:47.524] Then the component of this force acting at A, provided R is greater than H would be F sine theta into R minus H. This would be acting in the anticlopped direction, so it is going to be negative. Okay. [00:14:04.544] Now let's look at the free body diagram. This is a free body diagram. We have R, this is angle theta, then this is a point A, this is O R so the [00:14:17.092] Same diagram whatever we have looked at it, the free body diagram is written here. So F is given as 520 newtons acting at an angle of hundred and fifteen degrees counterclockwise above horizontal. So uh [00:14:35.812] 115 is what is given it is given as 15. So this is 15. So he says 115 degrees counterclockwise above horizontal. So that would be the angle is 15. [00:14:56.324] Then this angle phi is given as 10 degrees. So if I draw a horizontal line here, okay, then uh

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:05.860] The angle made by this would be 10 plus 15 is 25 plus 90 is 115 degrees. So that's what we get. However, you can take that into consideration or you can solve it in the way that I have told. So M naught would be equal to F cos theta. [00:15:23.812] which is this force, the component along this into B plus R, as I've told you, B plus R is the perpendicular distance. That would produce a clockwise moment which he has taken as positive. Then we have this force, this component F sin theta, which is this [00:15:41.156] Then into R minus H which will produce an anti-clockwise movement, he has taken it as negative. We can take clockwise as positive, anticlockwise as negative, or clockwise as positive, anti-clockwise as negative, vice versa also is true. [00:15:56.896] Okay. So substituting the value we will get M as this much, which is 374 newton meter clockwise movement. Okay. So [00:16:10.212] In this class, we have solved two problems. That is problem 2 bar 73 and problem 2 bar 74. In the problem 2 bar 73, we have used the properties of the couple, which indicates that. [00:16:28.260] The moment produced by a couple about any point is same. In this case, we have used the concept of force couple system. So that is what we get. Now here this F is acting here, F. [00:16:46.276] Okay, and if you take the horizontal here, this would be 10 degrees, then if I draw a vertical or line here which is perpendicular to this, this would be 90, plus this is 50. So totally this angle would be 115 degrees, that is what is given. [00:17:05.540] Counterclockwise above horizontal [00:17:09.892] Okay, so this is what it is, or if I draw an horizontal line here, so I can get that value of 115 degrees. So if I draw the force F here, so this angle can also be 115 degrees. Where F is moving from O to [00:17:25.860] Upwards. Okay. So if I apply F here and an horizontal line here. [00:17:32.036] This line if I extend this angle will be 10 plus this angle perpendicular to this line will be 90. So this 10 plus 90 and if I draw this line this angle will be 15. So totally I will get it as 115 degrees. [00:17:46.480] In the counterclockwise direction from the positive x axis okay I hope everybody has understood these two problems. Okay. So this we are going to end this class here. This is class number [00:18:02.148] uh 12 we will continue in the next class and we will look at solving more problems on couple okay so till now we have taken chapter one where we have discussed about a few basic uh definition and concepts [00:18:21.252] In the second chapter, we have discussed so far three topics. First one was the resolution of forces. [00:18:29.988] Where we have looked at the rectangular components, and what is the difference between projection and component, and how to get the component of the force if the two axis are perpendicular to each other, and where [00:18:45.312] The two axis are not perpendicular to each other. In the second topic, we have discussed about couple. [00:18:51.172] Okay, sorry, we have discussed about movement. In the third topic, we have discussed about couple. Okay. So with this, I will end this class. We'll see you in the next class. Thank you very much. My name is Dr. SV Vengtesh. [00:19:07.568] From Civil Engineering in the Department of Science and Humanities, PS University. Thank you. Let's meet in the next class.

EV0000210 · EV0000211 · EV0000212 · EV0000213

Full transcript: 2818 normalized words, 78 timestamped segments, 1156.331 seconds duration. Low-confidence segments flagged: 0.

EV0000198 · EV0000199 · EV0000200 · EV0000201

14 UE26ME142A_Unit 1_Class 10.1_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 10.1_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000214 · EV0000215

Graphs and diagrams

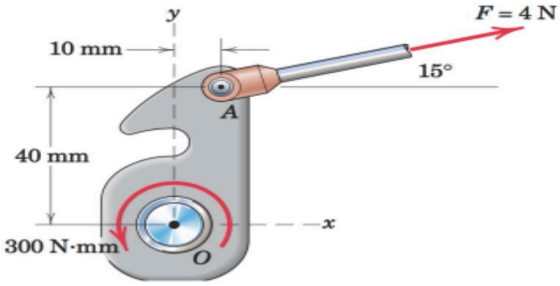


Figure G000098. UE26ME142A Unit 1 Class 10.1 Lecture PPT · source slide page 1.

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2/76 The device shown is a part of an automobile seatback-release mechanism. The part is subjected to the 4-N force exerted at A and a 300-N · mm restoring moment exerted by a hidden torsional spring. Determine the y-intercept of the line of action of the single equivalent force.



Given F = 4 N,
M = 300 N-mm
Θ = 15⁰
X1 = 10 mm, Y1 = 40 mm
To find : Y-intercept
Solution
Mo = M + FsinθX1 - FcosθY1
- FcosθxY = 300 + 4xsin15x10 – 4xcos15x40
Y = (300 + 10.35 – 154.55) / (- 4xcos15)
Y = 155.8 / - 3.86 = - 40.36 mm

S V Venkatesh - Engineering Mechanics – Unit 2 – Chapter – 2 – Force system

Figure G000099. UE26ME142A Unit 1 Class 10.1 Lecture PPT · source slide page 2.

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PES UNIVERSITY
CELEBRATING 50 YEARS

THANK YOU

Dr. S V Venkatesh
Department of Civil Engineering

Figure G000100. UE26ME142A Unit 1 Class 10.1 Lecture PPT · source slide page 3.

EV0000529

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000525 · EV0000526 · EV0000527
2. Source slide page 2 (extracted text): 2/76 The device shown is a part of an automobile seatback-release mechanism. The part is subjected to the 4-N force exerted at A and a 300-N · mm restoring moment exerted by a hidden torsional spring. Determine the y-intercept of the line of action of the single equivalent force. Given F = 4 N, M = 300 N-mm Θ = 150 X1 = 10 mm, Y1 = 40 mm To find : Y-intercept Solution Mo = M + FsinϑX1 - FcosϑY1 - FcosϑxY = 300 + 4xsin15x10 – 4xcos15x40 Y = (300 + 10.35 – 154.55) / (- 4xcos15) Y = 155.8 / - 3.86 = - 40.36 mm S V Venkatesh - Engineering Mechanics – Unit 2 – Chapter – 2 – Force system ## Page / slide 3 (OCR) THANKYOU Dr. S V Venkatesh PES Department of Civil Engineering UNIVERSITY CELEBRATING 5O YEARS EV0000528
3. Source slide page 3: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000529

short-note recap

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.420] Dear student, welcome to Engineer Mechanics Statics classes. My name is Dr. Sviven Kitish, Civil Engineering, Department of Science and Romantics, PS University. In the last class, we had discussed [00:00:15.204] Uh few problems the unsolved problems from uh the prescribed textbook, Engineer Mechanics Statics by Miriam, Craig and Holton. Today is the thirteenth class, we'll be continuing to solve few more problems on the topic couple. [00:00:33.540] Okay. So we'll start here while considering this problem number two bar seventy five. The statement of the problem goes something like this. The system consisting of the bar OA [00:00:48.132] Two identical pulleys and a section of thin tape is subjected to the two 180 newton tensile forces shown in the figure, determine the equivalent force couple system at point O. So from the statement, what we can identify is [00:01:09.028] There is a force of hundred and eighty newton force which is tensile force applied on a bar OA [00:01:21.380] Which has two identical pulleys, and if you observe the diagram that is given from this diagram, you can note on that. This is bar OA. There are two pulleys, one is here, the other one is here. The force of 180 newton tensile is applied on these two pulleys. [00:01:40.612] Which are both of them are horizontal, then he has asked us to determine the equivalent force couple system at point O. So equivalent force couple system at point O would be comprising of the net force because of these two forces plus the movement produced by these two forces at this point. [00:02:00.612] So that's what we need to find. Now if you observe the diagram, in the diagram it is also given the inclination of the line OA with the horizontal, which is 45 degrees, that is this one. [00:02:16.164] Then they have also given you what is the distance from O to the center of the lower pulley along the line OA, which is 50 mm, then the distance between the centers of the two pulleys along OA, which is 100 mm. [00:02:34.916] He is also given the radius of the pulley as R and the value of R is also given as 25. Now this is what we can make out from the statement and from the diagram that is given. Now to solve this problem. [00:02:52.740] If you observe the given two forces, both of them are acting horizontal, and the two forces are acting opposite to each other. Both of them are horizontal. The upper force is acting to the left, the lower force is acting to the right. [00:03:12.580] So from this, what can you conclude? You can look at that they form two equal unlike parallel

force system, which is nothing but a couple. Okay. Now we know that. [00:03:29.284] From the characteristics of a couple or properties of a couple that the net force acting because of a couple is zero. And that's what we will get here also. 180 newton force is acting to the left. The upper one, 180 newton force, the lower one is acting to the right. [00:03:49.028] So since the magnitudes are same and the sense is opposite, they will balance with each other, the net force that is acting would be zero. Now we also know from the property of couple that the movement produced by a couple at any point is zero. [00:04:06.340] So to find out the force couple system at O, because of these two forces, there won't be any force. So force acting at O will be zero. Now to find out the moment produced by these two forces at O, simply I will take the movement produced by any of these force. [00:04:25.860] About if I take this force, I'll take the moment about this point. If I take this force, I'll take the moment about this point. That would be the moment at or at the center or at a point or anywhere you take, that would be the moment that is produced. Because it's a couple. [00:04:43.620] So to to find out the movement of this force about this point, what is that I want? This is horizontal, so I would like to know what is the vertical distance between the two forces, which is 180 and 180, these are the two forces. So [00:04:59.780] Excuse me. How do I get the vertical distance between these two forces? Now we know that from O to this point, that is the center of the lower pulley is 100. And we also know that OA is inclined at 45 degrees. So if I take 100,

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:17.540] Sine 45 would get the vertical distance between the two centers of this pulley. We also know that this is radius, which is 25. So this 25 plus this 100 sign 45 T here. [00:05:32.708] Okay, plus this would also be the radius R. So again 25 I have. So it becomes 25 plus 100 sine 45. [00:05:44.708] Plus 25 will be the vertical distance. So I'll set if I multiply 180 into this vertical distance, I get the moment about O. So the force couple system acting at O will be comprising of a force which is of magnitude 0 and a couple [00:06:00.228] Which is acting in the anti-clockwise direction, which should be equal to the magnitude 180 newtons, cross or into the vertical distance between the two forces. [00:06:12.452] And you can see the same thing here. Now you can see here this is 180, 180, this is 45. This is the radius. This is also the radius. So obviously this would be the radius because this is the tangent, this would be acting tangent to this circle. [00:06:27.840] So radial distance would always be at perpendicular to the tangent, so this would also be R. Similarly, here this force is tangent, so this would also be R. So this R plus the vertical distance between A to this point, which will be 100 sin 45, plus this R will be the vertical distance between these two. [00:06:46.852] Forces. So the system at O is a couple. So moment is equal to force into the perpendicular distance, which is equal to 180 into as I have told you, I have explained, this hundred sign 45 plus 25 plus 25. You will get this as [00:07:02.564] 21700 newton millimeters or 21.7 newton meters, which will be acting in the anti-clockwise direction, and we take anti-clockwise direction as positive. I hope all of you have understood this, right? [00:07:19.396] Now, we had solved a pro similar kind of problems earlier also, where I had taken the problem comprising of a scooter on which [00:07:29.764] Who forces are acting parallel to each other on the sides or the edges of the handle of the two forces, there were acting parallel and horizontal distance was given. We resort the force along the horizontal and multiply with the distance to get the movement at the pivotal point of the handle that is an O. [00:07:47.280] Okay, because they are also a couple. So movement produced by a couple up at any point would be same. So I have multiplied the horizontal distance with the component of the force

along vertical. Similarly, here we have the horizontal force, I multiplied with the [00:08:03.908] Vertical distance, which is the perpendicular distance to these forces. So I get the moment about any point, and that would be the moment that's going to be acting at point O. I hope all of you have understood this, right? Okay, next. I think of this problem. This is a very important problem, and this problem. [00:08:22.500] Which is 2 bar 78, the statement goes something like this. The force F acts along line MA, where M is the midpoint of the radius along the x-axis. [00:08:36.676] Determine the equivalent force couple system at O if theta is equal to 40 degrees. From the statement, what we can know is there is a force that is acting. The line of action of the force is MA. [00:08:55.716] The point M is the midpoint of the radius along the x axis, that is this is the x axis. [00:09:03.556] This is the radius and this M is the midpoint of this radius, and they have also given that the radius is R or they have given here this distance is R by 2, this is also R by 2, so the radius would be capital R [00:09:18.820] Then they have also says that the angle theta is what it is. What is this angle theta? This is angle theta, which the radius OA makes with the x-axis. Now we have to determine. [00:09:36.100] The force couple system of this force which is acting at this point, A, along M, a force couple system acting at so how do we solve it? Now I have to resolve this force first along horizontal and vertical. [00:09:51.428] To resolve this force along horizontal and vertical, I should get this angle, say alpha. Now how do I get this angle alpha? To get this angle alpha, this problem is majority of geometry and very less mechanics part is there. So listen carefully the geometry.

EV0000218 · EV0000219 · EV0000220 · EV0000221

Transcript block 3 (source interval 000600-000899 seconds): [00:10:11.716] So to find out this angle alpha, what I am going to do is I will drop a vertical line here which makes an angle of 9 degrees to this axis x or I will draw a line parallel to y axis passing through A, which will cut the x axis at some point. [00:10:30.404] Okay, say I'll take that point as B. I can find out what is the horizontal distance OB and the vertical distance AB. How? Because I know that this is R and this angle is theta. So R cos theta I get OB and R sin theta I will get [00:10:50.980] AB. Now I can also determine M B, which will be equal to OB, which we have just now found, which is equal to R cos theta minus R by 2. So I get M B, I also know AB. So I can find out alpha as tan inverse of [00:11:10.724] AB by M B. So once I get this angle alpha, I can resolve this force as F, cos alpha, and this force as F sine alpha. Now I can apply this F cos alpha along X, F sine alpha along Y would be the force. [00:11:30.692] Or I can apply this force F here, which makes an angle of alpha with the x-axis which we have found. Next to determine the moment about O, I can take this force which is a component of F along x axis, which is F. [00:11:48.676] Cos alpha multiplied with AB, so which will produce a clockwise moment which will be negative. The component of this force along y will be F sin alpha into this distance which is OB, which we have already found, which will produce an anti-clockwise moment which will be positive. [00:12:07.268] So that would be the movement that's going to be acting at O. Right? So that's the solution now. So the free body diagram. This is just the arc. Then we have OA. Then we have [00:12:21.860] Uh M A. Now as I told you this angle is considered as alpha, which will be the same as this, because they are corresponding angles. Okay. Now they have also given you this is R by 2, this is R by 2. So AM, which is A M. [00:12:40.356] I can get this to be equal to. Now in this case, what is being done here is to get this distance, they have got they have tried to get this distance by using this triangle. I you can solve it by the approach that I have explained earlier, or by this approach, you take this triangle and use. [00:12:59.268] The cosine rule. What does the cosine rule state? The side, any

side, if I want to find out, a m that is this distance, a m square is equal to [00:13:13.348] This the other side R square and one more side plus R by 2 square minus 2 times R into R by 2 the other two sides. [00:13:25.796] Cause of the angle opposite to the side on which we are we are supposed to find out that is AM. Opposite to AM is theta, that was that is what is mentioned here. So using this we can get the distance AM okay in terms of R, which we will get as 0.696 R. [00:13:46.468] So using the sine rule we can find out what is beta. So sin beta pi sine beta by the side opposite to it is R is equal to sine 40, which is this theta sine 40 divided by the side opposite to this is AM. [00:14:06.660] Okay, which is point six nine six R, which we have found. So from this I can find out what is beta. So once I get beta. [00:14:15.396] I can find out alpha as 180 minus beta. This is another way of solving or determining the angle alpha. I have explained two different ways of solving alpha. You can use n number of ways to solve alpha. [00:14:31.268] Determining of alpha is left to your knowledge that you have gained till now. You can solve it in n number of ways to determine alpha. Now once I get alpha by whatever means and method that I get, but [00:14:46.948] Make sure that the angle that you get is correct one, 67.5. Use correct logic to determine the angle alpha. Once you get angle alpha, now I will find out what is the moment about O by resolving this force along horizontal and vertical.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:05.700] So the what vertical component of this force F is F sine alpha, this is F sine alpha, into the distance. If I take this force to be acting at this point M itself, I can take this as F. [00:15:23.780] Sin alpha into r by 2, that's what is taken here, which produces an anti-clockwise moment. Then I'm taking this as a point of application, so this component will be F cos alpha, F cos alpha, or I can also take the component of this by taking [00:15:43.620] Instead of alpha I'll take uh uh [00:15:50.660] Alpha is 65. Okay. M naught is equal to F sin alpha. Okay. This component, whatever you get cos alpha into there is no perpendicular distance because this four this component will pass through O itself. So he has not considered that. [00:16:09.828] Mind it, what he has taken is F sin alpha is sine 67.5. He has not considered this component because it passes through O itself. Okay. So you will straight away get the movement produced by this force about this point, if I take this as the point of application with only one component and multiplying that with the [00:16:29.028] Perpendicular distance. So the force of the system that's going to be acting at O will be F. Magnitude of F is same as F, the sense of the force will be the same as this, which is moving away from O or moving upwards. [00:16:48.676] Then the direction of this force with respect to the x axis measured in the anti-clomage action positive x axis would be 67.5 plus the movement that's going to be acting at O will be equal to the value whatever we get from here which is 0.462 F into R [00:17:07.108] Now if I could get the value of the force, the radius R, then I can substitute here and get the value of moment. Not exactly. No. [00:17:24.228] Earlier, what I explained was considering this as a point of application, I had explained how to get the moment about O. I had also explained how to get this angle alpha by taking or constructing a triangle. [00:17:38.884] uh a b m and determining the distance mb and ab and then finding out alpha once you get alpha you take the point of application of this force as a itself determine the movement about O by resolving the forces of F along horizontal and vertical [00:17:56.388] Now you can use that approach to solve this problem, or whatever is explained here. Both the ways you can solve it. Of course, you can use n number of ways to solve this problem. That is, you can also resolve this force along this line. [00:18:16.228] OA provided you can find out what is the angle made by the force of with respect to OA, and then

resolving this along O and perpendicular to O A, then you will get F. [00:18:30.052] component along this axis which is perpendicular to OA into this distance R, you will still get this answer provided you find out that angle whatever F makes with the line OA. Right? So you can solve in that approach also. [00:18:48.004] So there are n number of ways to solve this problem, but finally, what are we supposed to do is we are supposed to determine the force couple system at point O. [00:19:00.004] And that is what we have done. So you can use n number of ways to solve this. Okay. So with this, we have come to the end of. [00:19:09.316] The topic couple. This is the third topic under unit one, chapter two, we have discussed first topic was on resolution, rectangular, then projection, then N and T components. [00:19:25.744] Then what is the difference between projection and component? Okay. Then the second topic was on moment. We have discussed how to determine moment, what is the positive and negative sign of moment. The third topic was couple. We have discussed couple characteristics of couple. [00:19:41.444] The sense and classification of couples. Then we have discussed what is force couple system, and we have solved few problems. [00:19:50.532] uh from the solid examples and few problems on from unsolved examples. Okay. So this is class number thirteen. In the next class that is class number fourteen, I'll start with the last topic of unit one that is resulted. Mind

EV0000226 · EV0000227 · EV0000228 · EV0000229

Transcript block 5 (source interval 001200-001499 seconds): [00:20:07.376] We have already discussed few things about resultant. Okay. I will discuss more about resultant in my next class. That is class number 14. Okay. My name is Dr. SV Venkage from the Department of Civil Engineering under Science and Humanities Department, PS University. Thank you. See you in the next class.

EV0000230 · EV0000231 · EV0000232 · EV0000233

Full transcript: 3065 normalized words, 74 timestamped segments, 1226.496 seconds duration. Low-confidence segments flagged: 0.

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15 UE26ME142A_Unit 1_Class 11_AVSummary

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Graphs and diagrams



Figure G000101. UE26ME142A Unit 1 Class 11 Lecture PPT · source slide page 1.

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ENGINEERING MECHANICS STATICS

Unit - 1
Resultant

Figure G000102. UE26ME142A Unit 1 Class 11 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS STATICS

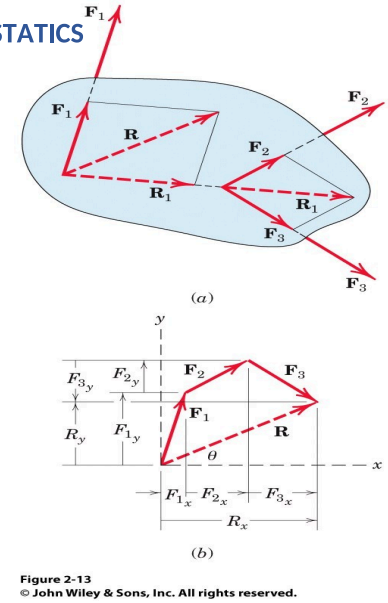


Figure 2-13
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Figure G000103. UE26ME142A Unit 1 Class 11 Lecture PPT · source slide page 3.

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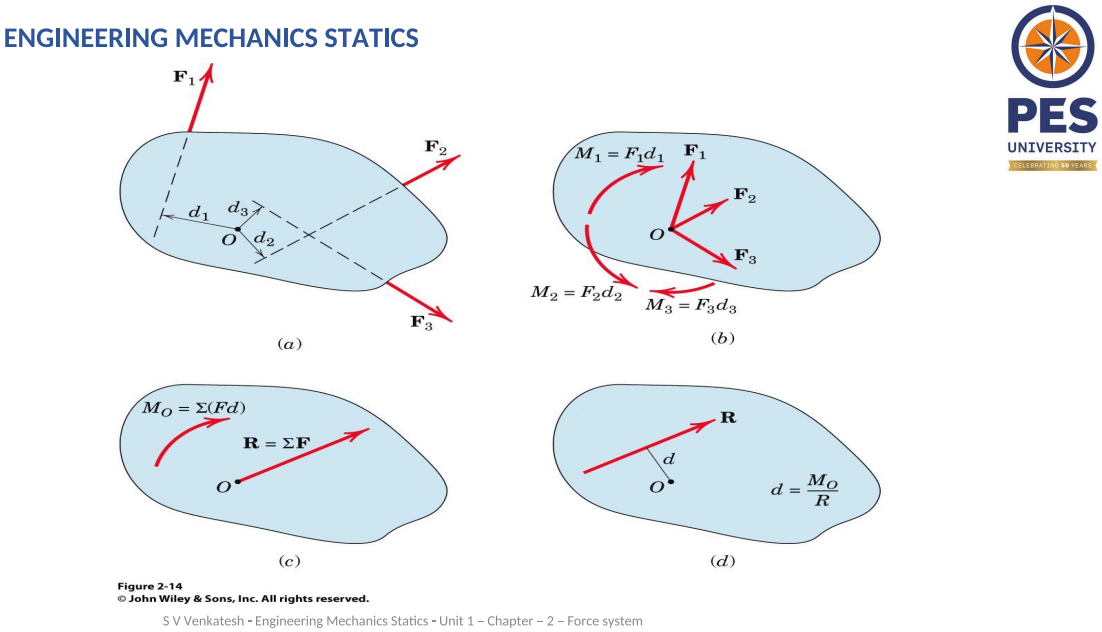


Figure G000104. UE26ME142A Unit 1 Class 11 Lecture PPT · source slide page 4.

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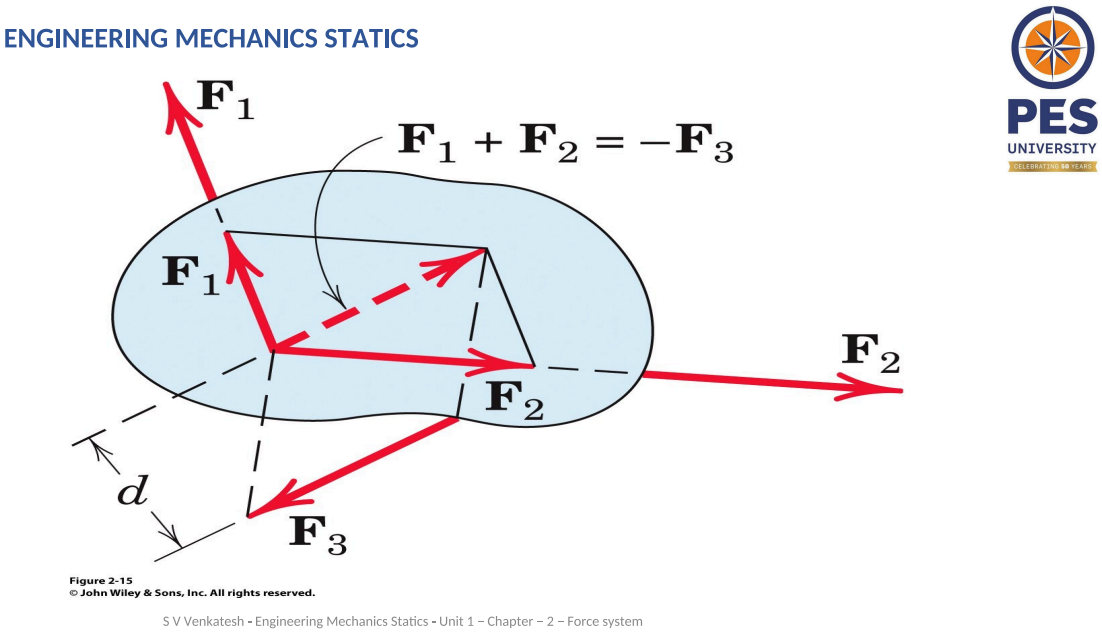


Figure G000105. UE26ME142A Unit 1 Class 11 Lecture PPT · source slide page 5.

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Figure G000106. UE26ME142A Unit 1 Class 11 Lecture PPT · source slide page 6.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000530 · EV0000531 · EV0000532
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Unit - 1 Resultan t S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000533 · EV0000534 · EV0000535
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000536 · EV0000537 · EV0000538
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000539 · EV0000540 · EV0000541
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ## Page / slide 6 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000542
6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000543

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.484] Dear student, welcome to Engineering Mechanics Statics. I am Dr. Svinkitesh from Civil Engineering under Department of

Science and Humanities BS University. [00:00:12.068] In the last class, we had discussed about the solving of problems on couples, and I had also informed you that in the next class, that is class number fourteen, I would be starting the last topic of unit one, that is resultant. Let us look at this. What is resultant? [00:00:29.764] Resultant, I had also uh mentioned you regarding resultant in the very few first classes. Resultant is that single force which would produce the same effect as that of us system of forces. For example, you have a system of forces. [00:00:47.780] If you apply the system of forces on a physique body, there could be some effect. The effect could be a translatory motion or a rotatory motion or the combination of these two. It can be translatory as well as rotatory. [00:01:03.012] Now, instead of applying this entire system of forces on the body, if I could replace this entire system by one single force, now when I apply this single force on the body, [00:01:18.308] This single force would also produce the same effect as that of that system of forces, then this single force would be called as the resultant. [00:01:33.092] Okay now I had also explained how to get the resultant also earlier. So we will go for this resultant and see. [00:01:43.204] I talked about dot product and when I talked about dot product I told you if you have vectors or if you represent the forces as vectors and get the dot product of this vectors, that would give you the resultant. [00:02:03.140] That means it would give you the vector form of the resultant. That means it will give you the magnitude of the component of the resultant along x axis, y axis. It would also give you the sense of the components of the resultant along X along y. [00:02:23.396] Then it would also tell you what is the direction of these components, that is, it is along X and along Y. Now that is what is mentioned here. So resultant R is equal to the dot product of several forces. [00:02:39.556] Okay, and that is nothing but summation of all the forces, the algebraic sum. You should also take the sign of the forces or the vector into consideration. Now [00:02:52.964] This f sigma f x will be in the form of Fx i or RX i plus R y j or Rx I minus R y j. What about the sign convention according to the forces that is acting. Now we can also get the resultant in another way that is you find out. [00:03:13.028] The sum of all the forces or the component of all the forces acting along X, algebraic sum of all the components of the forces acting along X, which should be nothing but your Rx. Similarly, find out the algebraic sum of all the [00:03:30.308] Components of the forces acting along y, which will be r y, and then I can find out the magnitude of the resultant R is equal to root of summation of R square plus summation of R y square. Okay. [00:03:46.148] Now the angle at which the resultant is acting we can find out theta which will be equal to tan inverse of r y by rx. Now this theta is with respect to the x axis, mind that theta is with respect to x axis. [00:04:01.828] So teta will be equal to tan and also teta will be equal to 10 inverse of Ry is nothing but summation of R F and Rx is nothing but summation of Fx. Okay, so that is what how we find out the resultant. Now this [00:04:19.396] Would give us the magnitude of the resultant. The theta will give us the direction of the resultant with respect to the x-axis. [00:04:31.076] Based on whether this is positive or negative, I would get the sense of the component of the resultant along x axis positive or negative, or y axis positive or negative. When we club these two, I get the sense of the resultant. The same thing is explained here. Say if you have a rigid body. [00:04:49.828] And there are four three forces acting F1, F2, and F3. Now, because of these two forces F2 and F3, I can find out the resultant as R1 using the law of parallelogram of forces. Now I'll apply this R1 along with this F1 and take these two as the two.

EV0000234 · EV0000235 · EV0000236 · EV0000237

Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.604] Adjacent sides of the parallel and then find out the resultant R. This is one way of finding out the resultant. The other

way of finding out the resultant is explained here. Each force you find out the component of the forces. [00:05:25.316] X and Y similarly for F2 again F2x and F2Y for F3 also again you find out F3x and F3Y. Now we also have to take the sign of this component F1x is positive, F2x is positive. [00:05:44.388] F3x is also positive, whereas F1 y is positive. F2y is also positive, whereas F3Y is negative. So adding all this, I get Rx, adding all the components along X, I get Rx. Adding all the components. [00:06:04.260] Algebraically I get R y. So using this I can find out the resultant R by using this formula, and then this theta would be equal to this formula, where this would be equal to R y and this would be Rx. So tan inverse of R y by Rx I get this theta. [00:06:23.716] Okay now very important is the fourth characteristics of the resultant. We know that for a force there will be four characteristics. What are these four characteristics? I had explained this earlier also, I will repeat it. First one is the magnitude of the force. [00:06:42.404] Or the quantity of the force is the magnitude. The second one is the direction, which is the angle made by the force with respect to a reference axis. The sense or line of action of the force, whether the force is moving away from the point of application or moving towards the point of application, and the fourth one is the point of application. [00:07:02.308] Now using this, whatever I have explained here, I can find out the three characteristics of the four of the resultant, which is force itself, or the vector, which are that the magnitude of the force, that is this one for using this formula, the direction of the resultant. [00:07:20.964] Okay, and then the sense of the resultant I can find out by merging Rx and R by together, whatever is going to be the sense. Now to find out the fourth characteristics that is the point of application of the resultant. If all the three forces have the same point of application, or if it is a [00:07:41.764] Concurrent force system, the resultant will also have the same point of application. But if it is not a concurrent force system, as you can see here, this is one force, this is another force, this is another force. [00:07:56.068] When we produce these three forces, all the three of them will not meet at the same point. So it forms a non-concurrent force system. In such case, where am I supposed to apply the resultant force is the question. [00:08:12.708] So, where should I apply? To find out the point of application of the resultant, we go for what is called as principle of movement, or it is also called as extension of varignos theorem. [00:08:28.036] Or it is also called as law of movements. I would define all this. Okay? Now what is Varingos theorem? If you remember I defined this when I was talking about the second topic in unit one, chapter two, that is movement. [00:08:43.268] What does Varing North theorem state? Varing North theorem states that if you have a force, the moment produced by the force would be equal to the moment produced by its force components about the same point. [00:09:02.436] Very importantly, about the same point is very important. Okay. The moment produced by a force. [00:09:09.380] About a point is equal to the moment produced by the component of the force about the same point. Okay. Now coming to principle of moments or law of moments or extension of Varingon's theorem means. [00:09:29.540] The moment produced by a system of forces about any point will be the same as the moment produced by their resultant about the same point. I'll repeat. The moment produced by a system of forces about any point. [00:09:47.428] Is equal to the moment produced by their resultant about the same point. So again, here the same three forces are taken: F1, F2, and F3. Now there is a point O that is mentioned here.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.036] And they have mentioned the distances D1, D2, and D3, which are the perpendicular distance of this force F1 and F2 and F3 from point O. That means this angle is 90 degrees, this angle is also 90 degrees, this

angle is also 90 degrees. [00:10:23.908] Okay? So now why we have taken this point? To find out what is the movement of all these three forces about this point O, which is given here. So I will take these three forces at at this point. [00:10:36.708] Along with this, I will also apply this moment M1, M2, and M3, which is nothing but the force couple system. If you remember, so instead of applying this force at this point A, F1, F2, and F3, I can apply these forces at O by a force couple system, and that's what is indicated here. [00:10:56.676] So we get M1 is equal to F one D1, M2 is equal to F2 D2, M3 is equal to F3 D3, D and [00:11:04.068] The sense of that. So here if I take the moment for this force it will be clockwise, that is what is indicated here. If I take this about this point the moment will be anti-clockwise and that is what is indicated here. [00:11:18.212] Then if I take this force, the moment about this point will be this force into this distance, which will again be clockwise, and that is what is indicated here as M3. Now you add all these three forces, you get the moment. Okay, M naught. That is equal to summation F into D. [00:11:36.516] Then you add all these three, that is dot product, you get R, which is sigma F. Okay. So now this is forming a force couple system. [00:11:46.308] Now I have to find out what would be the single force which represent this. So that we can get us R at some distance D from O. So this is producing a clockwise moment, so I'll apply this force. [00:12:02.084] Somewhere away from O at a distance perpendicular to this R, that is D. So but this should produce the same movement as this. So if I do that, I would be able to get the distance D. [00:12:18.916] Okay? So that would define the point where the resultant needs to be applied. Okay? So what we have looked at now till now is how do we get this resultant? Okay. [00:12:36.516] The four characteristics. One is the magnitude, which is the dot product of all the forces acting on the body. Then how do we get the direction of the resultant? We can either find out Rx and R y. [00:12:52.452] And then find out theta which is tan inverse of R y by R x where theta is the angle made by the resultant with the x axis. Then [00:13:04.772] To find out the sense of the resultant, you have Rx and R Y, merge these two, whatever you are going to get will be the sense of the resultant. Then to determine the [00:13:20.452] Point of application of the resultant we should go for principle of movement or what we call it as the extension of variaus or law of movements. Okay, I hope this is clear. Now whenever we have to find out the resultant [00:13:39.780] There could be a situation where we get only a moment as a resultant. We may not get the force as a resultant. Okay now for that I have taken this example. Say for example we have F1, F2, and F3, three forces acting on a body. [00:13:57.988] I would take F1 and F2 and find out the resultant of these two, which would be this. Which would be this. That is, I just have completed the parallelogram. The diagon passing through the intersection of the two forces will give the resultant. So we have got this resultant, which is nothing but F1 plus F2. This is the resultant R. [00:14:18.020] But if this resultant that we get, okay, if that resultant of F1 and F2 that we are going to get as is will be equal to the magnitude of F3, which is acting as an external force on the rigid body. [00:14:37.252] But in the opposite sense, as indicated here as minus F3, this would be the resultant of F1 and F2, which would be the same as the magnitude of F3, but the sense would be opposite. Then now you can see that these two. [00:14:55.172] forms a couple and we know that the characteristics of a couple is the characteristics of a couple is to make the body rotate only and we also know that one of the

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:10.916] Uh characteristics of a couple S the net force acting because of a couple S zero. So when I take the net force of these two, one is acting [00:15:21.444] Like this, the other one is acting like this. So they are opposed to each

other, they have the same magnitude, so net force would be zero, but it is going to produce a moment, and that moment would be in the clockwise direction. Okay. The magnitude of that would be equal to F3 into D. F3 into D. [00:15:41.508] Now where am I supposed to apply this? So we we don't get a force, we get it as a moment, moment is also a vector. So because now these two when I take it the moment produced will be a clockwise moment of magnitude which will be equal to F3 into D. [00:16:00.484] So we know what is the magnitude of the moment, we know the direction of the moment, which is clockwise, okay? And we also know about which axis it is acting, it is acting about the z axis which is coming out of the plane. Okay [00:16:19.940] No, where should we apply? We have already discussed that moment is a [00:16:25.732] Free vector. Mom is a free vector. You can apply the moment anywhere you want. Okay. So you can apply this moment anywhere you want. That would be the resultant of these three forces. [00:16:40.740] Which is going to be acting. The resultant of these three forces now is just a moment of magnitude F3 into D, which would be acting in the clockwise direction along the z axis. You can apply it anywhere you want. Now when I say anywhere you want, [00:16:59.460] The law of transmissibility of forces will apply, or the principle of transmissibility of forces will apply. [00:17:06.788] The point of application of the vector can be anywhere, provided it is in contact with the body. Then only the body will experience that. Right? So it should be within or it should be somehow connected to this body, then we can apply that. [00:17:24.932] I hope it is clear to everybody. In today's class, we have discussed the last topic of unit one, which is under chapter two. That is resulant. Resultant is that single force which produces the same effect as that of a system of forces. [00:17:45.156] Okay. How do we find the resultant? The magnitude of the resultant is nothing but the dot product of all the forces. The direction is nothing but uh tan inverse of R y by Rx, provided the direction is with respect to the x axis. [00:18:03.236] Where R is the component of all the forces along Y, and Rx which is the component of all the forces along X, R it is the component of the resultant about X and Y. Then we also have seen how to get the sense, you merge the Rx and R y together, you get the sense of the resultant R. [00:18:20.804] Then to determine the point of application of the single force, you have to go for extension of Arianov's principle or principle of moments or law of moments, which states that the moment produced by a system of forces is equal to the moment produced by the [00:18:39.332] Resultant about the same point. And we have also looked at what could be the result. Resultant could be that only one single force or it could be a movement also. That what how do we get only one single movement is what I have explained in this last slide in this class. [00:18:59.044] Okay this is class number fourteen. [00:19:02.724] We will be continuing our class in the next class, that is class number 15, where I will take up and solve few problems related to resultant. Okay? So with this, we have come to the end of class number 14. Thank you very much. We will meet in the next class.

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Full transcript: 2895 normalized words, 71 timestamped segments, 1167.744 seconds duration. Low-confidence segments flagged: 0.

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16 UE26ME142A_Unit 1_Class 12.0_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 12.0_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000250 · EV0000251

Graphs and diagrams



Figure G000107. UE26ME142A Unit 1 Class 12.0 Lecture PPT · source slide page 1. EV0000545

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ENGINEERING MECHANICS STATICS

Unit - 1
Resultant

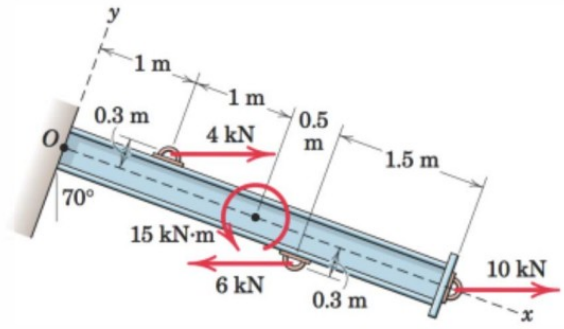
Figure G000108. UE26ME142A Unit 1 Class 12.0 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS STATICS

2/81 Replace the three horizontal forces and applied couple with an equivalent force-couple system at 0 by specifying the resultant R and couple M_0. Next, determine the equation for the line of action of the stand-alone resultant force R.



Problem 2/81

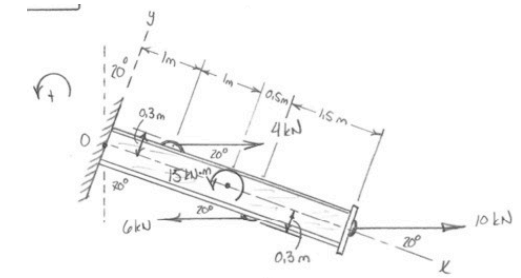
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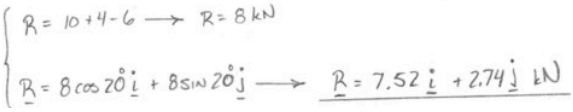
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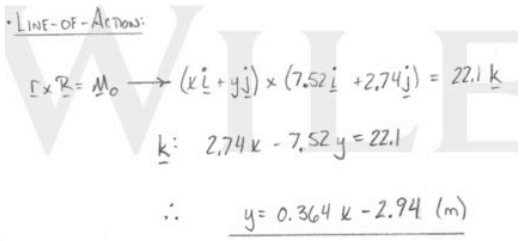
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ENGINEERING MECHANICS STATICS

2/81 cont..







Mo = 15 + 4sin20(1) - 6sin20(2.5) + 10sin20(4) – 4cos20(0.3) – 6cos20(0.3)
Mo = 22.1 kN-m CCW

S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

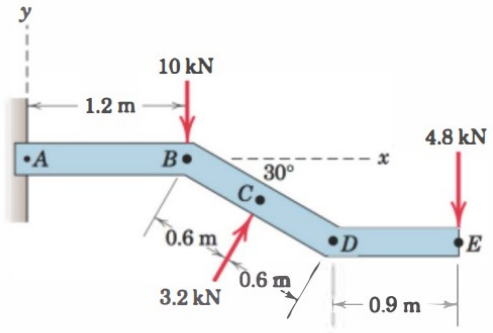
Figure G000110. UE26ME142A Unit 1 Class 12.0 Lecture PPT · source slide page 4.

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ENGINEERING MECHANICS STATICS

2/91 Replace the three forces which act on the bent bar by a force-couple system at the support point A. Then determine the x-intercept of the line of action of the stand-alone resultant force R.



Problem 2/91

S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system

Figure G000111. UE26ME142A Unit 1 Class 12.0 Lecture PPT · source slide page 5.

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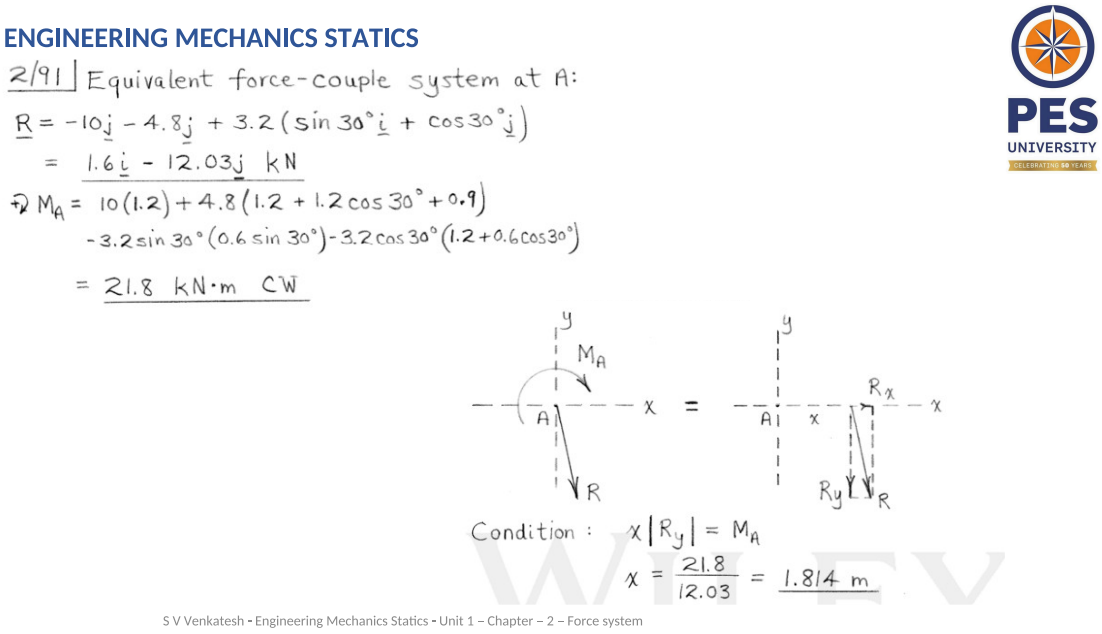


Figure G000112. UE26ME142A Unit 1 Class 12.0 Lecture PPT · source slide page 6.

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Figure G000113. UE26ME142A Unit 1 Class 12.0 Lecture PPT · source slide page 7.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000544 · EV0000545 · EV0000546
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Unit - 1 Resultan t S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000547 · EV0000548 · EV0000549
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS 2/81 Replace the three horizontal forces and applied couple with an equivalent force-couple system at 0 by specifying the resultant R and couple M0. Next, determine the equation for the line of action of the stand-alone resultant force R. S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000550 · EV0000551 · EV0000552
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 2/81 cont.. Mo = 15 + 4sin20(1) - 6sin20(2.5) + 10sin20(4) – 4cos20(0.3) – 6cos20(0.3) Mo = 22.1 kN-m CCW S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000553 · EV0000554 · EV0000555
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS 2/91 Replace the three forces which act on the bent bar by a force-couple system at the support point A. Then determine the x-intercept of the line of action of the stand-alone resultant force R. S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000556 · EV0000557 · EV0000558
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ## Page / slide 7 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000559
7. Source slide page 7: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000560

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.036] Dear student, welcome to Engineering Mechanics Statics classes. I am Dr. SV Venkatesh, Civil Engineering, Department of Science and Humanities, PS University. [00:00:12.260] This is class number 15. In the last class, I had introduced you to the last topic of unit 1, chapter 2, that is resultant. We had discussed about what is resultant, how do we find the characteristics of resultant, that is the magnitude. [00:00:32.164] The direction, the sense, and the point of application. [00:00:38.244] The point of application can be determined by using the principle of movements. I had defined what is principle of moments and explained how do we find out the point of application of the resultant force for our system of forces. Now today I would like to [00:00:58.180] Solve a few problems taken from the unsolved problem category from the prescribed textbook that is Engineering Mechanics Statics by Miriam Craig and Bolton published by John Wiley Publications. Okay so [00:01:18.500] The problem that I have chosen is 2 bar 81. The statement of the problem goes something like this. Replace the three horizontal forces and apply the couple with an equivalent force couple system at O. [00:01:36.548] By specifying the resultant R and the couple M naught, next determine the equation for the line of action of the standalone resultant force R. Now, according to the statement.

[00:01:54.948] There are three horizontal forces acting and there is a couple that is acting. We have to replace this system, that is this three [00:02:04.804] Forces plus one couple by an equivalent force couple system. What we had earlier discussed under the topic couple, force couple system at point O. So there are three forces, one is this, this is another one, this is one more. Then we have a couple here. [00:02:25.124] So these four vectors need to be replaced by a single vector acting at O and a couple. Okay, or a moment we can say. Okay? Then we have to determine [00:02:43.684] The line of action of the standalone resultant force. Line of action of the standalone resultant force is what is the line of action of the single force where we need to apply that. Okay, we have to find out the equation. For any value of X and Y. [00:03:02.404] Where this single resultant force needs to be acting, so that we can get the same effect as that of these three forces and the couple. Okay. Now if you read the diagram that is given, the first force. [00:03:20.004] Which is acting at this point is of magnitude 4 newton. The sense of this is towards the right. It is acting at a distance of 1 meter along x axis. Look at the axis. X axis is not the horizontal axis here, but this axis, x axis [00:03:39.492] is inclined to the vertical at an angle of 70 degrees. So if I draw an horizontal line here and if I draw a vertical line here this would be 70 degrees [00:03:52.644] So horizontal vertical will always be at 90 degrees. So this angle will be the remaining angle that is 90 minus 70, which will be 20 degrees. So this is 4 kilonet, which is acting horizontally from left to right, whereas this one, the point of application is this. This is acting from right to left. [00:04:12.644] The magnitude of that is 6 kilonewtons. Then we have the another force of magnitude 10 kilonewtons acting on the x-axis at this point. Okay, that is at a distance of 1 plus 1 plus 0.5 plus 1.5. [00:04:32.356] which is equal to four meters still here. Whereas this six kilometers it act is acting along the x axis at a distance of two point five meters. This four kilometers is acting at a distance of one meter along the x axis. [00:04:49.572] Now this force ten kilometers is acting on the x axis, whereas this four kilometer is acting at a distance of 0.3 meters along y axis. That is above x axis. Whereas this six kilometer is acting below x axis by a distance of 0.3 meters.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:08.004] Be careful. This small distances also would play a very important role. Okay? So for this we have to find out a force of system or two. First thing that we need to do is find out the dot product of all the forces. [00:05:25.380] Now, since here all the forces are acting along the horizontal line itself, so I can simply take the algebraic sum of all these three forces along horizontally. So we get it as 4 plus 10, which is 14 minus 6. [00:05:41.988] Okay, so that would be 8 kilonottons acting from left to right. That would be the single force which is acting. I can straight away apply that at home. So this single standalone resultant force. [00:05:58.692] Or the force couple system when we are finding out at O would be the force would be equal to 8 kilonewtons acting from left to right along the horizontal. The next thing is determine the movement of all these four vectors at O. So for that [00:06:17.444] We have to resolve this four kilometers. [00:06:20.708] Along x axis and y axis, similarly 6 kilonoton also along x axis and y axis, along this axis and along this axis, and this 10 kilonoton also along x axis of and y axis, of course the x-axis component of this 10 kiloton will not produce any movement. [00:06:39.396] But these two both X both the component along X axis and Y axis would produce a moment at W. So that gives us the force of our system at war. Then having this [00:06:53.924] Force couple system at what I will now apply that single resultant force which is 8 kilonoton at some point, okay, some point here, such that this force should produce the same effect or same moment, whatever we would have got because of this four. [00:07:12.676] When

does not go? So that would give us the line of action of the single resultant force. Okay. So let's look at the solution now. So the free body diagram is this, as I told you, this is 20 degrees, this is also 20 degrees, because they have given this as 70 degrees. [00:07:32.612] Now R the single resultant force as I have told you 10 plus 4 minus 6 which is 8 kilonewtons and it is positive because it is active to the to from left to right along the horizontal. [00:07:46.820] But if I resolve that along x axis and y axis, so I'll get this as 8 cos 20i which will be along x, then plus eight sin twenty j which will be along y axis. So this is that single force that should be our applied at O. [00:08:06.308] Now to find out moment about O, first we will take this moment which is acting in the anti-clockwise direction. So it will be plus this is kilonewton meters, so 15 kilonewton meters. Then we will take the component of this force along vertical. [00:08:24.900] Which will be four cause of this angle twenty degrees. Sorry, sine of this angle which is twenty degrees. So four sine twenty degrees would be this force into the perpendicular distance would be one. [00:08:36.964] The advantage of resolving this force along X and Y would be simply that we can determine the perpendicular distance of those components and then we can multiply. Okay. So that would produce an anti-clockwise movement again at O. So it would be plus. [00:08:52.100] Then we will take this component along y, which will be 6 sin 20, which will produce an clockwise movement, so it is negative. Into the distance at which this is acting is 2.5, then plus the component of this force. [00:09:08.740] Which will be 10 sin 20, which will be acting along vertical, which will produce an anti-clockwise movement. So plus into the distance is a 4. Okay. Next the component of the 4 kilometer acting along x axis. [00:09:26.404] Will be 4 cos 20 into this perpendicular distance will be 0.3. That would produce a clockwise moment, so it is negative. Similarly, this 6 cos 20, which will be acting along x-axis into this distance is 0. That would also produce a clockwise moment, so that is also minus 6 cos. [00:09:45.572] 20 into 0.3. Of course point three is unable to be seen here because of my photo. So that is 0.3. So if you take out the total of this, you will get it as 22.1 kilonewton meters positive, which means that it is in the counterclockwise direction.

EV0000254 · EV0000255 · EV0000256 · EV0000257

Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.516] So, what we have now got is the single force and the moment about O that forms a force couple system. Now we have to replace this force couple system acting at O by the single resultant force R acting, and we have to find out what is the line of action of that. So [00:10:25.956] If you take the resultant force, if I find out the cross product of the position vector of R [00:10:38.500] Cross the R vector that should be equal to the moment produced at O. So that is what is indicated here. So R is the position vector of the small R is the position vector of the resultant force and capital R is the [00:10:57.636] vector mutation of the resultant magnitude. Okay. So that should be equal to M. So we will write the position vector as Ri cross yj cross or Rij cross the [00:11:16.260] Value of this that is R, whatever we have got 7.52i plus 2.74 J. So that should be equal to the momentum that we have got here, which is 21, 22.1 kilonoms. So from this [00:11:33.188] I can get when I cross multiply this, I'll get this as 2.74 minus 7.52 y, which is equal to 22.1. This is about the z axis. Okay. [00:11:48.900] So now if I want to find out what would be the position of the single resultant force acting along the x axis, that is what would be this distance when I if I apply it if I apply the force on the x axis itself, what would be this distance. [00:12:04.676] So I will simply take y component y to be equal to 0, put this here, then I will get x to be equal to 22.1 divided by 2.74, which I will get it as 0.364k. So that means I have to apply [00:12:21.572] Sorry, x to be equal to minus 2.64. That means at apply the single resultant force to the left of O at this point. So when I apply it, that resultant uh I will

get the moment. Okay. [00:12:41.636] Then to find out what would be the y component, if I apply that single resultant force along y, what would be that? So I will substitute x as 0, so I get y to be equal to 22.1 divided by 7.52. [00:12:58.852] So I get that as 0.364. Okay. I hope it is clear for all of you. Right? Okay. So we will move on to the next one. Or [00:13:14.148] If you look at this, I will get an equation as y equals 0.364x minus 2.94. Okay, simply taking y and substitute simply cross-multiplying this, I get this value. This is the equation of the line of action of the resultant force. [00:13:34.052] So that is very clear, I hope. Okay. So to find out the x distance, I will substitute y equals 0, I get the x value. If I want the y distance, I will substitute x equals 0, I get the y component. Or if I just re-assemble this equation. [00:13:53.764] In as y equals in terms of x, I get this as equation. This is the line of action of the single resultant force that I need to apply. I hope it is clear. Okay. Coming to the next problem, I have considered 2 bar 91. [00:14:10.756] So the problem statement is replace the three forces which act on the bent bar by a force couple system at the support point A, then determine the X-intercept of the line of action of the standalone resultant force R. [00:14:28.388] Again, in this, we have three forces, and we are first to find out what is the force couple system at A, that is this point, and then we have to determine what is the X-intercept, that means what should be the distance along the x-axis from A. [00:14:47.364] At which I can apply the single resultant force such that it produces the same effect. So again in this problem, first we need to find out the force couple system at A for that resolve this force along horizontal and vertical.

EV0000258 · EV0000259 · EV0000260 · EV0000261

Transcript block 4 (source interval 000900-001199 seconds): [00:15:03.236] Take all the component along X, so we get only one component that is 3.2. So this is 30 degrees. So if I draw an horizontal line here and a vertical line here. [00:15:20.004] This would be 30 degrees. So the angle made by the vertical with C D would be 60. So now this force, the angle made by this force with the vertical drawn at C, this would be 30. So the horizontal component acting at C will be 3.2 [00:15:40.100] Sine 30, okay that would be the Rx value. Then 3.2 cos 30 would be the vertical component of 3.2 acting at C. So R y will be minus 10. [00:15:57.188] Plus the component of this 4, which is 3.2 cos 30 minus 4.8. So we get R x value and R y value. So we can find out R as well as the angle theta. Then to find out the moment at this point. [00:16:14.532] Take this force, multiply with this distance, so 10 into 1.2, which will produce a clockwise moment. Then you take this force, which would produce an anti-clockwise moment. Into we have to take this distance which would be equal to 0.6. Okay, if I complete this triangle. [00:16:34.564] This is 0.6, this is 30 degrees, so sin 30 would be the vertical distance, so I multiply that. Then 3.2 cos 30 into 1.8 plus this distance would be 0.6 cos 30 degrees plus 1.2 would be the distance. So I get the [00:16:54.340] Counterclockwise movement, so I add that. Then I will take this force which will be clockwise, so minus 4.8 into this total distance would be 1.2 plus 1.2 cross 30 degrees plus 0.9 will be the movement acting at E. [00:17:10.852] Okay. So once I get the force couple system acting at A, now I will take R cross R equal to the moment acting at A, and then find out what is the x-intercept. Right. So [00:17:27.620] So equivalent force couple system acting at A. So R as I told you 10 kilon 10 kilon is acting downwards. So 10 J and it is minus minus 4.8 J again the one which is acting at the free end of the V, that is acting downwards. [00:17:44.900] Plus 3.2 sin 30i plus cos 30 j. So we get this as the vector notation or vector representation of the resultant force R. That is 1.6i minus 12.03 J. [00:18:03.876] So x component is positive acting from left to right, y component is acting downwards, so it is negative. So to find out the moment at here, 10 which is acting downwards into

1.2. So that is negative minus or we can take it as positive. So clockwise is taken as positive. [00:18:23.556] Then 4.8 into 1.2 plus 1.2 cost 30 plus 0.9 as I have explained. That would also be clockwise, so if we have taken it as positive. Then minus 3.2 cost 30 into 0.6 sine 30, that is the [00:18:42.820] This is the horizontal component, and this is the vertical distance. So that would produce an anti-clockwise movement. So minus. Now this is 3.2 cross 30 will be the [00:18:52.548] horizontal component into the sorry this is the vertical component into this is the horizontal distance 1.2 plus 0.6 cros thirty. So if I take the entire thing then I will get it as 21.8 kilonewton meters as the movement acting at a which is in the clockwise direction. [00:19:12.036] So you can see here the notation is given that clockwise direction is taken as positive. That's why this is positive, this is positive. Counterclockwise is taken as negative. I hope it is clear. So we caught the force couple system acting at point A. This is R. [00:19:29.508] X component is positive, y component is negative, so we get this as R. And the moment MA which is clockwise, we have got this. Now to find out the x-intercept, I will apply this force along x axis at some distance, that is what is indicated here. [00:19:48.516] Now this force should produce the same movement at A as what we have got here. [00:19:55.492] So equating that you have the components Rx and Ry, Ry is the only one which produces the moment at A. So Ry into this x distance should be equal to MA. So I get x to be equal to 21.8 divided by 12.03, which I get as 1.814 meters.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:14.148] Now we have got this as positive, that indicates that whatever that we have taken that x to be acting to the right of A is correct. If you would have got a negative sign here, then it would have indicated that instead of taking it to the right, we should have taken it to the left. Okay. I hope this is clear. [00:20:34.116] So we will stop today's class here and continue in the next class. I'll be solving some more problems. Thank you very much. My name is Dr. Sv Vengtesh from Civil Engineering, Department of Science and Humanities, PS University. We'll meet in the next class.

EV0000266 · EV0000267 · EV0000268 · EV0000269

Full transcript: 3018 normalized words, 73 timestamped segments, 1250.182267 seconds duration. Low-confidence segments flagged: 0.

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17 UE26ME142A_Unit 1_Class 12.1_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 1_Class 12.1_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000270 · EV0000271

Graphs and diagrams



Figure G000114. UE26ME142A Unit 1 Class 12.1 Lecture PPT · source slide page 1.

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ENGINEERING MECHANICS STATICS

Unit - 1
Numerical on Resultant

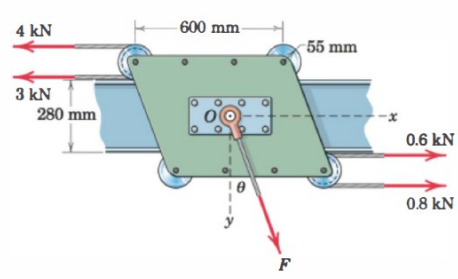
Figure G000115. UE26ME142A Unit 1 Class 12.1 Lecture PPT · source slide page 2.

EV0000565

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ENGINEERING MECHANICS STATICS

2/98 Five forces are applied to the beam trolley as shown. Determine the coordinates of the point on the y-axis through which the stand-alone resultant R must pass if F = 5 kN and θ = 30°.



Problem 2/98

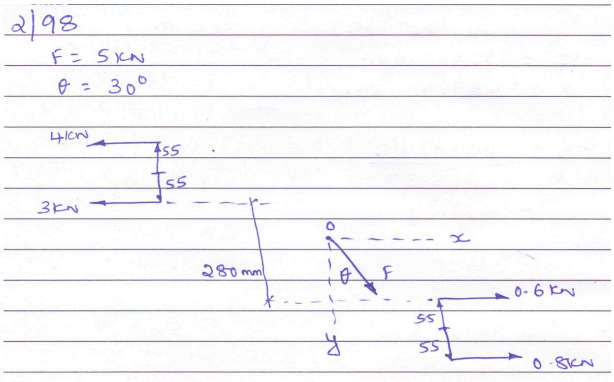


Figure G000116. UE26ME142A Unit 1 Class 12.1 Lecture PPT · source slide page 3.

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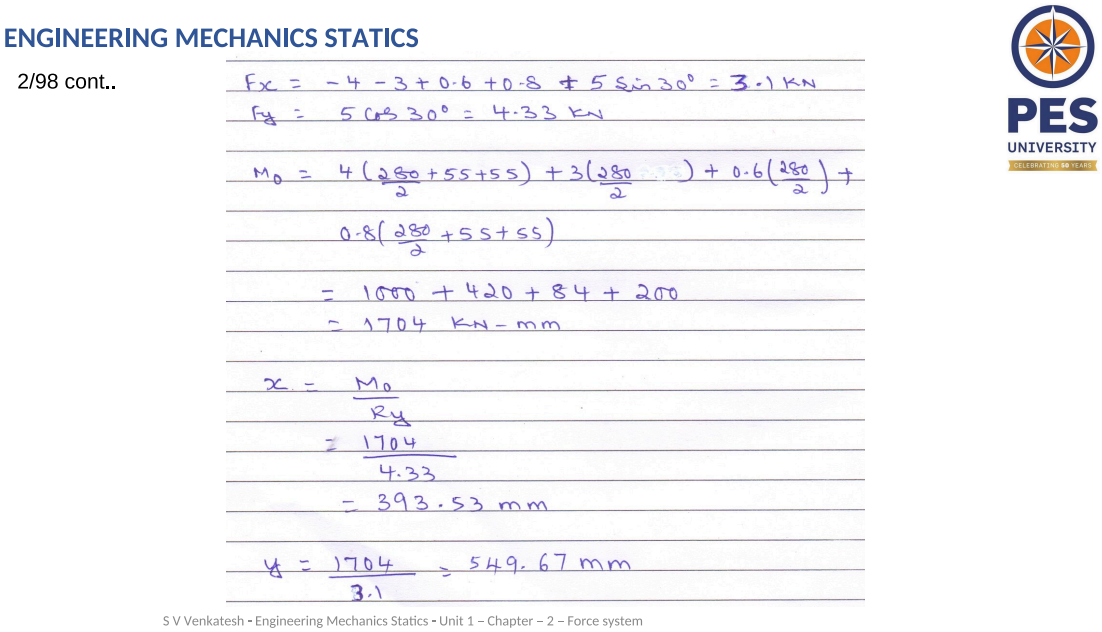


Figure G000117. UE26ME142A Unit 1 Class 12.1 Lecture PPT · source slide page 4.

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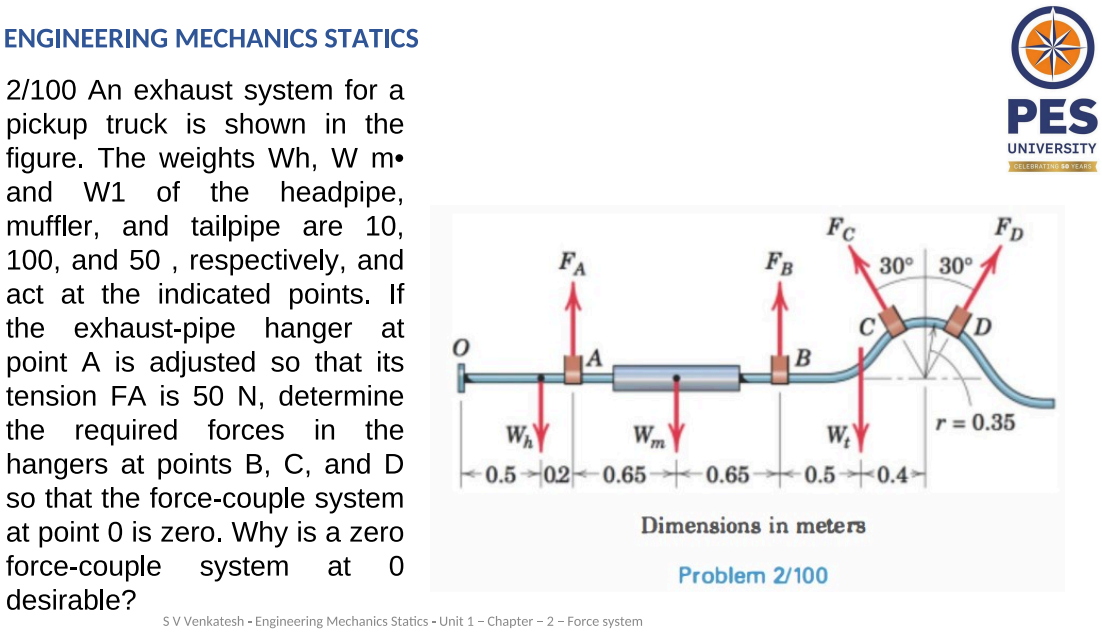


Figure G000118. UE26ME142A Unit 1 Class 12.1 Lecture PPT · source slide page 5.

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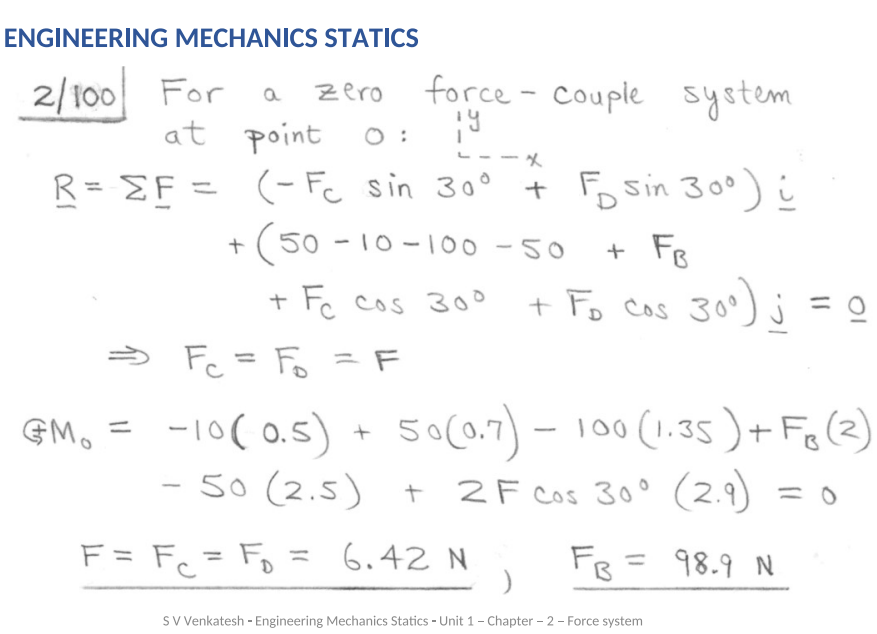


Figure G000119. UE26ME142A Unit 1 Class 12.1 Lecture PPT · source slide page 6.

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Figure G000120. UE26ME142A Unit 1 Class 12.1 Lecture PPT · source slide page 7.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000561 · EV0000562 · EV0000563
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Unit - 1 Numerical on Resultant S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system EV0000564 · EV0000565 · EV0000566
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS 2/98 Five forces are applied to the beam trolley as shown. Determine the coordinates of the point on the y-axis through which the stand-alone resultant R must pass if F = 5 kN and ϑ = 30°. S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force system ## Page / slide 4 (OCR) ENGINEERINGMECHANICS STATICS 2/98 cont.. =-4-3+06+0845S30°=3.KN PES 5Cb3o°= 4.33Z UNIVERSITY 4 (280+ 55+55)+3 (80 +0.6(280 CELEBRATING50YEARS Q 2 0-8C +ss+ss 100+420+84+200 704 Kz-33 1704 4.33 393.53mm 549.67mm =) 704 31 S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Chapter - 2 - Force system EV0000567 · EV0000568 · EV0000569
4. Source slide page 4: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000570
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS 2/100 An exhaust system for a pickup truck is shown in the figure. The weights Wh, W m• and W1 of the headpipe, muffler, and tailpipe are 10, 100, and 50 , respectively, and act at the indicated points. If the exhaust-pipe hanger at point A is adjusted so that its tension FA is 50 N, determine the required forces in the hangers at points B, C, and D so that the force-couple system at point 0 is zero. Why is a S V Venkatesh - Engineering Mechanics Statics - Unit 1 – Chapter – 2 – Force zero force-couple system atsystem ## Page / slide 6 (OCR) ENGINEERINGMECHANICSSTATICS 2/100 For force-Couple zero a system PES at point O UNIVERSITY F.sin3o CELEBRATING50YEARS R=-MF= (-Fcsin 30 ┼ 50-10-100-S0 ┼ + F +FCos30° Coswao Fc=F 二F GM ◦= -10(o.S) + 5oCo.7 -100(1.3S)+ F(2) -S0 (2.s) (2.9) +2FCos3° F=F 6.42 N F 98.9N B 川 S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Chapter - 2 - Force system ## Page / slide 7 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000571
6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000572
7. Source slide page 7: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000573

Short-note recap

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.772] Dear student, welcome to the classes on engineering mechanics statics. My name is Dr. SV Venkitish from Civil Engineering under Department of Science and Humanities, PES University. This is class number sixteen. In the last class, that is class number fifteen, we had discussed [00:00:20.964] Two problems taken from the unsolved problems from the prescribed textbook that is Engineering Mechanics Statics by Miriam, Craig and Bolton. And it is the eighth edition. Today we will be continuing with [00:00:40.484] Some more numericals. So on the resultant that is from unit one. And this is the last

topic from unit one. Okay. For today's discussion, I have considered a problem on 2 bar 98. The statement of the problem goes something like this. [00:01:00.068] Five forces are applied to the beam, trolley as shown, determine the coordinates of the point on the y-axis through which the standalone resultant R must pass. [00:01:14.948] If F is equal to 5 kilonewton and theta is equal to 30 degrees. So from the statement we can make out that there are five forces acting, and we are supposed to determine [00:01:29.348] The Y intercept of the standalone resultant [00:01:39.556] The force F magnitude is given as 5 kilo Newton and the angle T at which this force is acting is given as 30 degrees. Now if you look at the given data from the statement F is equal to 5 kilonewton and T is equal to 30 degrees. [00:01:56.196] What are we supposed to find is the y-intercept or the coordinate of the point on the y-axis at which the standalone resultant needs to be applied. Now, if I look at or read the diagram that is given here. [00:02:14.820] So I can find out the magnitude of the other four forces. So this force the magnitude is four kilonewton acting to the left along the x axis. This force the magnitude is three kilonewton again, it is acting along the x axis to the left. [00:02:33.892] Then coming to this force, the magnitude of the force is 0.6, again it is acting along y, but it is in towards the right, so positive x axis, then this force is 0.8 kilometers, again it is acting along the x axis, again it is positive because it is acting to the right. [00:02:52.708] Now the force F is acting at the center of the trolley. F force, the magnitude of F is given as 5 kilometer. The angle at which the F force is acting with respect to the y-axis. [00:03:08.036] Is given as 30 degrees. Now look at the sign convention of the x axis and y axis. They have given x to be positive along this, that is to the right, whereas y is positive downwards. So keep this in mind. Okay then. [00:03:26.884] If they have not given the access or not mentioned the access, then you are at liberty to take whichever access that you want. But in case if they have given you the [00:03:39.332] Axis, you are supposed to take that as the positive axis. I hope this is clear to everybody. Right? Okay. X axis positive is to the right and downwards is the positive y axis. Okay. [00:03:55.076] Now you can also look at what is the distances of these forces. Okay. With respect to the center of O, they have not given you the distances, but they have given you the distance. [00:04:08.260] Between this pulley and this pulley, which is 600 mm, and they have also given you what is the distance between these two uh forces, which is 280. Of course, your x-axis is the center of this, which is at 140 mm. Okay. Now [00:04:27.204] Uh how do we go about this? Now we can consider the four Kronutan force and three kilonot and force. This is a free body diagram. [00:04:36.388] Then the F force is acting here. Then of course this also is considered, which you are unable to see here. This is the free body diagram here. You can look at the PPT and you can make out that diagram that is given. Okay. So this is what is given. [00:04:53.664] This distance is 280 mm. Okay. Now how do we go about this? First take all the forces along acting along x axis, algebraic sum of all the forces. So you get the component of the resultant along x axis.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.540] Then take all the forces acting along y, algebraic sum of that, then you get the component of the resultant about the y-axis. So once you get that, probably you get uh [00:05:22.852] With the values that we have here, you you get the component of the resultant force acting towards the left and along the x-axis and component of the force [00:05:33.764] Along y-axis will be acting downwards. So this would be in the third quadrant. So you can make out that the resultant would be acting in the third quadrant. Okay. So you can find out the magnitude of the resultant. You can also find out the [00:05:48.464] angle at which the resultant is acting with respect to the negative x axis or the positive y axis.

Okay. So turning works off either R y by Rx or Rx by R y as the case may be with respect to whichever axis that you are going to take. I hope this is clear. Once you get that [00:06:06.788] The next thing is we have to find out where are we supposed to apply this force along the y-axis for which we will go for usage of principle of movements or extension of Baring North principle, that is [00:06:20.624] Moment produced by all the forces about one point, where in this problem we will take it as a point O should be equal to the moment produced by the resultant force. If I take the resultant to be acting here, I will get the X component of the resultant into the distance at which it is acting. If I take it as y, then I can find out y. [00:06:39.716] So let's look at the solution here. So here you can see sum of all the forces acting along the x-axis is what is taken here. So we have one force acting to the left, four kilometers, so minus four, another force acting to the left, so that is minus three. [00:06:56.068] Then we have one force acting towards the right, that is point six, another force acting towards the right again, that is point eight. Then we also have the component of the [00:07:07.620] F force that is acting along X axis, which is B 5, sine 30 degrees, that will be acting to the right, so that is positive. So sum of this you will get as 3.1 kilometer. Next, to take all the forces acting along y, f p will be [00:07:23.012] The component of the F4 which is acting along the y axis which will be downward that is positive, which will be phi cos 30 degrees, you will get that as 4.33 kilonutons. Now take the moment about the point O because of all the forces. So [00:07:42.116] You take the force 4 kilometer that is acting to the left side along the x-axis, the distance would be 280 divided by 2, which was the distance which I had talked about, plus we should take the radius of the [00:08:00.996] uh pulley that we have or the diameter of the pulley at the top as well as at the bottom. So that is why it is 555 plus 55. Okay if you look at the [00:08:14.308] Uh figure that is given you can make out what is the radius of the pulley. So radius on the top from the center and the bottom work together plus 280 divided by 2 till the center, you will get that value. Then plus [00:08:29.520] 3 kilonutan is acting at the bottom of the pulley, top pulley. So you will have only it is 280 by 2 only, right? [00:08:38.084] Plus we have 0.6, which is also going to be acting at the downward, which is at a distance of 280 by 2 itself. Now if you look at this would produce a anticlockwise movement, this would also produce an anti-clockwise movement. This would also produce an anti-clockwise movement. [00:08:54.244] Plus 0.8 would also produce an anti-clock point moment, but the distance would be 280 by 2 plus 55 plus 55. So sum of all this you get the moment about O which will be 1704 kN millimeters. [00:09:10.724] Because I have taken this in the millimeters on this force magnitude in kilone, so it is kilonewton millimeters. Now to find out what is the x-intercept, of course, you can find out x-intercept, which is not asked in the problem, but I have formed the x-intercept also. Applied this resultant force. [00:09:30.020] On the x-axis, okay. So you get the component acting along the y, which would be downwards. So if I apply this. [00:09:38.916] To the left of O, then you will have a negative movement, so you get that to be equal to 393.53 mm. So if you take it to the left, you get it to be positive, so that would be correct. Now to find out the y intercept, I will take the moment. [00:09:57.508] Or I'll apply this resultant force, this resultant force on the y-axis, which is below O. So you have the uh X component, this is X component, which we will get it as

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:16.740] Minus this would be minus 3.1, which will be acting to the left. [00:10:24.836] Okay. So if I take that and divide it or take or equate the moment, y will be equal to m naught by your component. So we will get that as 549.64 millimeters. That would be the [00:10:40.452] Y intercept of the resultant from the point O.

I hope this is clear. Okay. Next, I have taken the last problem in this unit, unit 1, chapter 2. That is problem 2 bar 100. [00:10:56.324] The problem statement goes something like this. An exhaust system for a pickup truck is shown in the figure. The weights WH, WM under W1 of the head pipe. [00:11:11.524] Moffl and tailpipe are 10, 100, comma 50 respectively. And at the indicated points, if the exhaust pipe hanger [00:11:28.228] At point A is adjusted so that its tension FA is 50 newtons. Determine the required forces in the hangar at point B, C and D. [00:11:45.028] So that the force couple system at point O is zero. Why is a zero force couple system at O is desirable. Now if you identify from the statement what they have given as [00:12:03.204] The weight of WH, which is this weight, the weight of W M, which is this one, the weight of W1, which is this one. [00:12:16.548] Or it is W T it should be, I think I'm sorry, this is W T, so this should be W T are given respectively as 10, 100 and 50. They have also given the force acting at A, that is this force F A to be equal to 50 newton. [00:12:36.068] And they have asked you to determine the force at B, that is this force F B, F C, and F such that the force couple system at this point O is equal to 0. Now it also says that. [00:12:54.820] If the exhaust pipe hanger at point A is adjusted so that its tension FA is 50 newton, why is the zero force couple system what most desirable? That is what is asked. No. [00:13:11.620] This would be a candile. If the forces are balanced with each other, there won't be any forces that is acting at war. So the chances of this uh this system coming out or falling would be reduced. [00:13:31.812] That is the first thing why a force couple system, zero force couple is desirable. So we should look at that this system does not come out of this. [00:13:43.876] And there is no force acting at this point. So if it is a zero system, if this can balance itself, that would be very nice. And it should also not rotate. That is why your couple at this point should also be zero. That is the explanation for why we should not. [00:14:01.296] Have or have or why we should have zero force couple system. I hope this is clear. Now looking at the diagram, we can make out what is the distances between these forces from O and between them. [00:14:17.636] And also we can note down that the force acting at AB, then your WH, WM, and WT, all of them are acting vertically. Except that F C and F D are acting. [00:14:31.908] At an angle to the vertical, which is 30 degrees for both of them. So they are symmetrically placed with respect to the y-axis. Now, if a zero force couples system is desirable here, the resultant force acting at O should be zero. [00:14:51.940] That means the component of the resultant force along X and along Y should be zero. So if I take these two forces, only these two forces would produce a force along the x-axis, the F force will produce a negative force

EV0000278 · EV0000279 · EV0000280 · EV0000281

Transcript block 4 (source interval 000900-001199 seconds): [00:15:10.820] X axis force, whereas Fd will produce a positive x axis force. So if I take some of all the forces acting along the x axis, I can easily say that your force F C should be equal to F D. Very simple. Next, if I take all the forces acting along the vertical [00:15:30.820] I can say that this component F C and F D plus this component F C minus these three forces would give me the force in F B. Okay. So that we can determine easily. [00:15:47.940] Then to determine the value of F C and F D, I will take the moment about O, which we will get it in terms of F P, F C and F D. So taking these three equations, I can solve the problem. Now the other way of looking at the problem is [00:16:04.996] As per the statement given, we are supposed to find out the force at B, force at C, and force at D. So there are three unknowns. One is force at B, which we have to find, force C we have to find, force at B we have to find. [00:16:21.188] There are three unknowns, so at least we require three equations. Which are these three equations? Since the resultant at O is 0, the component of the

resultant along X should be equal to 0. So with that, we get the [00:16:37.156] relationship between between these two which is which we can say that F C is equal to F D. So that would reduce now to two unknowns. Now if I take sum of all the forces acting along the vertical to be equal to zero I get the F B in terms of F C or F D. [00:16:56.036] Okay, then if I take the moment about O, I can easily find out either FP or F C and then substituting the values of F C here I get F and here I get F. I hope this is clear. Now let's look at the solution. [00:17:14.212] So for a zero force couple system at point O, X and Y axis is considered to be this, the positive X and Y axis. So the resultant force, sum of all the forces, if you represent it as a vector. [00:17:28.164] So your F C will have a negative sign, sine 30 is F C sin 30, that is acting along the x axis, negative x axis, F T sine 30 which is acting along the positive x axis, so those two is acting along the x, so it is i. [00:17:45.060] The rest is acting along the vertical. So 50 is acting at A, that is positive. Minus 10 is your one force that is W acting. Then one more force. [00:18:00.452] W this is WH this is W M and this is W T all the three of them are acting downwards so it is negative Your F B is acting upwards [00:18:11.236] The component of the force F C along the vertical will be acting upwards, so positive, that is F C cos 30 degrees. The component of the force F T will also be acting upwards, that is F T cos 30 degrees acting upwards. [00:18:25.188] So this forms the component of all the forces acting along the vertical. So it is J. So the total has to be equal to 0. From this, if I take all the components acting along the y, I can easily make out that F C is equal to F D, which is equal to F. [00:18:44.964] If I take all the component acting along the y direction, substituting this here for F C and F D as F, I can get the value of F P in terms of F. So this equation would now reduce. [00:19:02.660] To all the unknowns to be acting or in terms of F, I can make out. Next you take movement about O of all the forces. So we get this equation. Okay. So this is your WB, which will produce a clockwise movement into the distance is 0.5. [00:19:23.300] 50 is the force acting at A, which is acting upwards. So that will produce an anti-clockwise movement. So the distance is 0.5 plus 0.2, which is 0.7. 100 is M B which is acting downward, which will produce a clockwise movement, the distance from O to B or where the force is acting F. [00:19:43.364] uh f um you will have the distance as 1.35 then your f p force is acting upward which will produce an anti clockwise movement the distance from o to that force is two kilon two meters so that is what is taken here then your f

EV0000282 · EV0000283 · EV0000284 · EV0000285

Transcript block 5 (source interval 001200-001499 seconds): [00:20:01.892] T is acting downward, which will produce a clockwise movement. So that is 15 into the distance from O is 2.5. Then we have the two forces F C and F D, the component of those two, you can take it as 2F, cos 30 degrees acting vertically, the distance of that would be 2.9. [00:20:21.348] So from this we have only one unknown, or we will have an unknown in terms of F. Okay, that is equal to 0. So from this I will get the relationship between F B in terms of F. [00:20:38.212] Here I can substitute this in this equation, and then I can get the value of F B. Okay. Then substitute that whatever relationship that we have got, F B in terms of F, then substitute. Here we get the value of F C and F D. [00:20:58.404] Okay, so your FC and F P value is equal to 6.42 Newton, and your F value is equal to 98.9 Newton. I hope this is also clear. Okay. So with this we have come to the end of unit one. [00:21:15.012] In the next class, we will have an experiential component class, that is class number 17. Next to that, we will start with the next unit, that is unit 2. That unit is equilibrium chapter. [00:21:31.428] Number three. Okay. So we will stop today's class here. Thank you very much. I am

Dr. SV Ventesh from Civil Engineering under Department of Science and Humanities, PS University. Thank you.

EV0000286 · EV0000287 · EV0000288 · EV0000289

Full transcript: 3262 normalized words, 80 timestamped segments, 1305.365 seconds duration. Low-confidence segments flagged: 0.

EV0000270 · EV0000271 · EV0000272 · EV0000273

18 Source slide deck: UE26ME142A Unit 1 Class 1.0 Lecture PPT

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Graphs and diagrams



Figure G000001. UE26ME142A Unit 1 Class 1.0 Lecture PPT · source slide page 1.

EV0000291

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Engineering Mechanics

Statics

Eighth edition
SI Version

J. L. Meriam and L. G. Kraige

Chapter 1
Introduction to Statics

S V Venkatesh
Civil Engineering

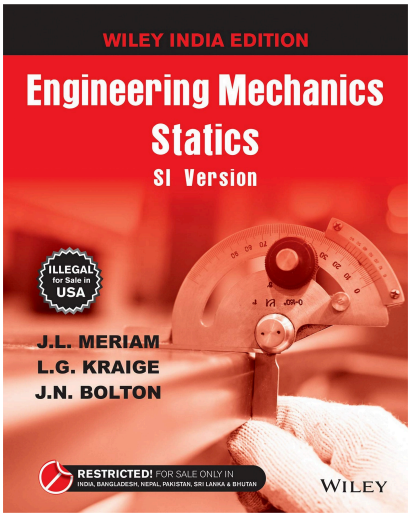
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Figure G000002. UE26ME142A Unit 1 Class 1.0 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS STATICS



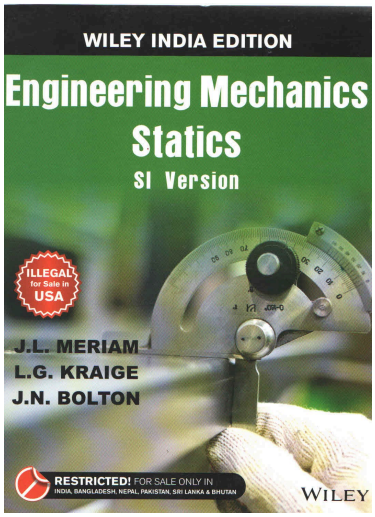


Figure G000003. UE26ME142A Unit 1 Class 1.0 Lecture PPT · source slide page 3.

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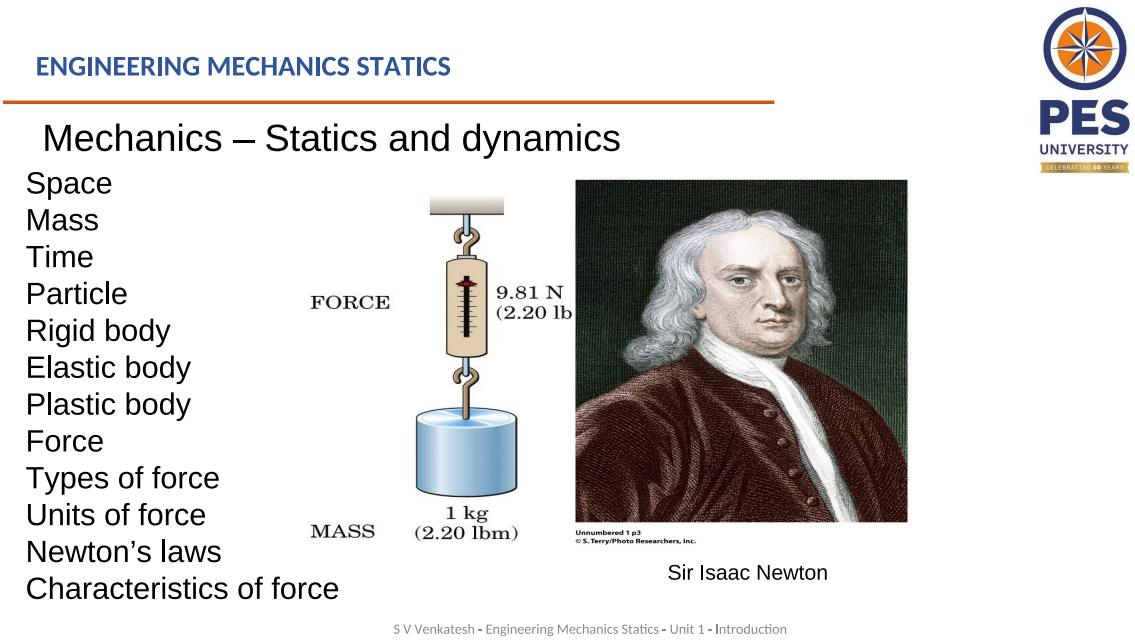


Figure G000004. UE26ME142A Unit 1 Class 1.0 Lecture PPT · source slide page 4.

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Figure G000005. UE26ME142A Unit 1 Class 1.0 Lecture PPT · source slide page 5.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000290 · EV0000291 · EV0000292
2. Source slide page 2 (extracted text): Engineering Mechanics Statics Eighth edition SI Version J. L. Meriam and L. G. Kraige Chapter 1 Introduction to Statics S V Venkatesh Civil Engineering Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved. S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction EV0000293 · EV0000294 · EV0000295
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction EV0000296 · EV0000297 · EV0000298
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS Mechanics – Statics and dynamics Space Mass Time Particle Rigid body Elastic body Plastic body Force Types of force Units of force Newton's laws Sir Isaac Newton Characteristics of force S V Venkatesh - Engineering Mechanics Statics - Unit 1 - Introduction ## Page / slide 5 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000299
5. Source slide page 5: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000300

SHORT-NOTE RECAP

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EV0000290

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.

Source	Path	Use
UE26ME142A_Unit 1_Class 1.1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1.1_AVSummary__bea4975b08967ab4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1.1_AVSummary__bea4975b08967ab4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1.1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1.1_AVSummary__bea4975b08967ab4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1.1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1.1_AVSummary__bea4975b08967ab4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1_AVSummary__1ad410489823544f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
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Source	Path	Use
UE26ME142A_Unit 1_Class 1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1_AVSummary__1ad410489823544f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/001_UE26ME142A_Unit_1_Class_1_AVSummary__1ad410489823544f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.0_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.0_AVSummary__254d366ff7b73c45.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 2.0_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.0_AVSummary__254d366ff7b73c45.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
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Source	Path	Use
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UE26ME142A_Unit 1_Class 2.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.1_AVSummary__3c66109b736f8c89.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
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UE26ME142A_Unit 1_Class 2.2_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/002_UE26ME142A_Unit_1_Class_2.2_AVSummary__80ec6ef9f2cc872e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

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UE26ME142A_Unit 1_Class 3_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/003_UE26ME142A_Unit_1_Class_3_AVSummary__eee232e254abeb39.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
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UE26ME142A_Unit 1_Class 5_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/005_UE26ME142A_Unit_1_Class_5_AVSummary__86fffcdbe06dbb6b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 5_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/005_UE26ME142A_Unit_1_Class_5_AVSummary__86fffcdbe06dbb6b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
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UE26ME142A_Unit 1_Class 5_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/005_UE26ME142A_Unit_1_Class_5_AVSummary__86fffcdbe06dbb6b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 6_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 6_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 6_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 6_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/006_UE26ME142A_Unit_1_Class_6_AVSummary__f35d743d3b761113.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 8_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 8_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 8_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 8_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/008_UE26ME142A_Unit_1_Class_8_AVSummary__f49bcc7a576db09d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 9_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 9_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 9_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 9_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/009_UE26ME142A_Unit_1_Class_9_AVSummary__866a19feaac2a3bf.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.0_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.0_AVSummary__4d79e3bc52cb3c4f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.0_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.0_AVSummary__4d79e3bc52cb3c4f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.0_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.0_AVSummary__4d79e3bc52cb3c4f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.0_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.0_AVSummary__4d79e3bc52cb3c4f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 10.1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/010_UE26ME142A_Unit_1_Class_10.1_AVSummary__c39186c34db5b2f5.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 11_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 11_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 11_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 11_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/011_UE26ME142A_Unit_1_Class_11_AVSummary__3d8f98864b87c9b2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.0_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.0_AVSummary__ff48ce3f9a237d73.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.0_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.0_AVSummary__ff48ce3f9a237d73.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.0_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.0_AVSummary__ff48ce3f9a237d73.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.0_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.0_AVSummary__ff48ce3f9a237d73.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 12.1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/012_UE26ME142A_Unit_1_Class_12.1_AVSummary__4e48a92aba0ceae0.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: UE26ME142A Unit 1 Class 1.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 1.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 1.0 Lecture PPT (converted PDF; original hash 1adeec92bd8b55d28a301ab05f3ac6211855294e88abc2dc8c745e99c5a081a1)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 1.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 1.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 1.1 Lecture PPT (converted PDF; original hash 1653a8ffbc63c6f1cebca6688bc6f0c7cee6632d7af40bab1d99597d8dbeaedb)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 2.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 2.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 2.0 Lecture PPT (converted PDF; original hash 3661bb44d6ba64ef223560e46dbb47020bbbbd8adba40bdebe6abc98c692f8ea)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 2.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 2.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 2.1 Lecture PPT (converted PDF; original hash 36e0e092882d543954d5d699ebdfc177ff83bb355782a7a41956ee4b6874289d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 2.2 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 2.2 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 2.2 Lecture PPT (converted PDF; original hash a6f22c4d1cac09e09f87cfafbf1e99a0ddc24850719cb5b5ce4ed918d7d5ab9e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 3 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 3 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 3 Lecture PPT (converted PDF; original hash b0f98082451c388673f5a78dde1aea22a173f2b5d16752ebf90f8207e81c713d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 4 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 4 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 4 Lecture PPT (converted PDF; original hash e14ad2ad44deec3d7e8473a5dcb210d1d4719a1ab45ba54c227fc13ef666ddb7)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 5 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 5 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 5 Lecture PPT (converted PDF; original hash 3b5cb4980cdc4b41b4f2e156137ff96a23026d13ab15abde89149d8d720351d9)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 6 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 6 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 6 Lecture PPT (converted PDF; original hash 22dd11615d21af38626b1a33b7b1d8446e3a14b0733b41776fbeab98198f7285)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 8 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 8 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 8 Lecture PPT (converted PDF; original hash 1c1d191c0ed477478825d748b36f9d8c40417a73e583b877f3773c82d759991c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 9 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 9 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 9 Lecture PPT (converted PDF; original hash 568f6e5391b7defd2b7d1afe3863f7da14ce323d0de3008413f995d3c130cdb4)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 10.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 10.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 10.0 Lecture PPT (converted PDF; original hash f9dc99f62b1b462ef1ce3968961567 17fad583b69f6eefddb607e7fcf04a1c15)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 10.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 10.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 10.1 Lecture PPT (converted PDF; original hash 3dd0700564b31ad9ccb79167f4be5e9b2d3f520ab58af273220b2f5400baf9cb)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 11 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 11 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 11 Lecture PPT (converted PDF; original hash 8de1359871fb43079aea2ca316a794b5f1f0ff2e8b79bf892009259a7d41b25d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 12.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 12.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 1 Class 12.0 Lecture PPT (converted PDF; original hash f358c92b19e16b9e23427a6f5aa33c6a2d5811e70282189b6a221316ec336963)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 1 Class 12.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 1 Class 12.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 1 Class 12.1 Lecture PPT (converted PDF; original hash ec9d7e8c8c05cfbfda48f6154f40ac51ad134a3db1ca4c0c31537c2fe1f2670a)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.

4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.
9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.

18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV00000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.
EV00000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0040. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000014	SRC00002/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000015	SRC00003/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000016	SRC00004/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00005/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000019	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000020	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000021	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000022	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000023	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000024	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000025	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000026	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000027	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000028	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000029	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000030	SRC00006/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0053 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000031	SRC00007/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0053 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000032	SRC00008/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0053 to s0069. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000033	SRC00009/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0053 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000034	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000035	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000036	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000037	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000038	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000039	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000040	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000041	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000042	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000043	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000044	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000045	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000046	SRC00010/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000047	SRC00011/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000048	SRC00012/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0081. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000049	SRC00013/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000050	SRC00010/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000051	SRC00011/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000052	SRC00012/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0087. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000053	SRC00013/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000054	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000055	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV000056	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000057	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000058	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000059	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000060	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000061	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000062	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000063	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000064	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000065	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000066	SRC00014/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000067	SRC00015/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000068	SRC00016/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0077. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000069	SRC00017/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000070	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000071	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000072	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000073	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000074	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000075	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000076	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000077	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000078	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000079	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000080	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000081	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000082	SRC00018/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000083	SRC00019/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000084	SRC00020/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0077. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000085	SRC00021/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000086	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000087	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000088	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000089	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000090	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000091	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000092	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000093	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00022/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00023/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00024/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00025/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000098	SRC00022/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000099	SRC00023/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001000	SRC00024/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001001	SRC00025/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001002	SRC00022/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001003	SRC00023/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001004	SRC00024/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0087. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000105	SRC00025/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000106	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000107	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000108	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000109	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000110	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000111	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000112	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000113	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000114	SRC00026/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000115	SRC00027/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000116	SRC00028/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000117	SRC00029/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000118	SRC00026/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000119	SRC00027/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000120	SRC00028/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000121	SRC00029/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000122	SRC00026/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000123	SRC00027/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000124	SRC00028/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0083. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000125	SRC00029/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 26	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 27	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 28	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 29	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 30	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 31	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 32	SRC00032/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000133	SRC00033/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000134	SRC00030/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000135	SRC00031/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000136	SRC00032/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000137	SRC00033/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000138	SRC00030/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000139	SRC00031/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000140	SRC00032/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0074. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000141	SRC00033/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000142	SRC00030/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0075 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000143	SRC00031/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0075 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000144	SRC00032/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0075 to s0086. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000145	SRC00033/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0075 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000146	SRC00034/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000147	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000148	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000149	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000150	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000151	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000152	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000153	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000154	SRC00034/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000155	SRC00035/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000156	SRC00036/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000157	SRC00037/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000158	SRC00034/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000159	SRC00035/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000160	SRC00036/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0071. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000161	SRC00037/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000162	SRC00034/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0072 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000163	SRC00035/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0072 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000164	SRC00036/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0072 to s0077. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000165	SRC00037/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0072 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000166	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000167	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000168	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000169	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000170	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000171	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000172	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000173	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000174	SRC00038/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000175	SRC00039/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000176	SRC00040/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000177	SRC00041/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000178	SRC00038/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000179	SRC00039/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000180	SRC00040/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0072. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000181	SRC00041/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000182	SRC00042/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000183	SRC00043/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000184	SRC00044/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000185	SRC00045/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000186	SRC00042/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000187	SRC00043/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000188	SRC00044/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001189	SRC00045/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001190	SRC00042/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001191	SRC00043/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001192	SRC00044/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001193	SRC00045/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001194	SRC00042/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001195	SRC00043/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 96	SRC00044/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0064. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 97	SRC00045/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 98	SRC00046/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 99	SRC00047/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 00	SRC00048/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 01	SRC00049/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 02	SRC00046/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000203	SRC00047/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000204	SRC00048/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000205	SRC00049/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000206	SRC00046/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000207	SRC00047/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000208	SRC00048/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000209	SRC00049/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000210	SRC00046/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000211	SRC00047/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000212	SRC00048/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0078. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000213	SRC00049/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000214	SRC00050/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000215	SRC00051/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000216	SRC00052/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000217	SRC00053/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000218	SRC00050/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000219	SRC00051/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000220	SRC00052/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000221	SRC00053/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000222	SRC00050/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000223	SRC00051/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000224	SRC00052/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000225	SRC00053/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000226	SRC00050/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000227	SRC00051/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000228	SRC00052/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0073. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000229	SRC00053/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000230	SRC00050/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0074 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000231	SRC00051/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0074 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000232	SRC00052/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0074 to s0074. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000233	SRC00053/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0074 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000234	SRC00054/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000235	SRC00055/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000236	SRC00056/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000237	SRC00057/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000238	SRC00054/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000239	SRC00055/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000240	SRC00056/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000241	SRC00057/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000242	SRC00054/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000243	SRC00055/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000244	SRC00056/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000245	SRC00057/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000246	SRC00054/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000247	SRC00055/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000248	SRC00056/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0071. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000249	SRC00057/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0056 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000250	SRC00058/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000251	SRC00059/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000252	SRC00060/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000253	SRC00061/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000254	SRC00058/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000255	SRC00059/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000256	SRC00060/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000257	SRC00061/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000258	SRC00058/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002259	SRC00059/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002260	SRC00060/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002261	SRC00061/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002262	SRC00058/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002263	SRC00059/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002264	SRC00060/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0071. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002265	SRC00061/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002 66	SRC00058/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0072 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 67	SRC00059/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0072 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 68	SRC00060/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0072 to s0073. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 69	SRC00061/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0072 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 70	SRC00062/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 71	SRC00063/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 72	SRC00064/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000273	SRC00065/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000274	SRC00062/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000275	SRC00063/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000276	SRC00064/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000277	SRC00065/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000278	SRC00062/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000279	SRC00063/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000280	SRC00064/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000281	SRC00065/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000282	SRC00062/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000283	SRC00063/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000284	SRC00064/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0074. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000285	SRC00065/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000286	SRC00062/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0075 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000287	SRC00063/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0075 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000288	SRC00064/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0075 to s0080. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000289	SRC00065/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0075 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000290	SRC00066/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000291	SRC00068/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000292	SRC00067/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000293	SRC00066/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000294	SRC00068/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000295	SRC00067/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000296	SRC00066/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000297	SRC00068/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000298	SRC00067/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000299	SRC00068/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000300	SRC00068/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000301	SRC00069/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000302	SRC00071/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000303	SRC00070/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000304	SRC00069/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000305	SRC00071/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00003 06	SRC00070/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 07	SRC00071/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 08	SRC00071/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 09	SRC00072/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 10	SRC00074/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 11	SRC00073/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000312	SRC00072/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000313	SRC00074/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000314	SRC00073/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000315	SRC00072/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000316	SRC00074/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000317	SRC00073/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000318	SRC00074/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000319	SRC00074/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000320	SRC00075/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000321	SRC00077/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000322	SRC00076/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000323	SRC00075/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003324	SRC00077/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003325	SRC00076/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003326	SRC00075/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003327	SRC00077/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003328	SRC00076/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003329	SRC00075/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000330	SRC00077/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000331	SRC00076/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000332	SRC00075/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000333	SRC00077/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000334	SRC00076/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000335	SRC00075/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000336	SRC00077/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000337	SRC00076/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000338	SRC00075/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000339	SRC00077/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000340	SRC00076/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000341	SRC00075/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000342	SRC00077/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000343	SRC00076/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000344	SRC00075/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000345	SRC00077/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000346	SRC00076/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000347	SRC00077/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000348	SRC00077/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000349	SRC00078/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000350	SRC00080/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000351	SRC00079/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000352	SRC00078/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000353	SRC00080/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000354	SRC00079/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000355	SRC00078/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000356	SRC00080/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000357	SRC00079/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000358	SRC00078/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000359	SRC00080/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000360	SRC00079/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000361	SRC00078/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000362	SRC00080/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000363	SRC00079/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000364	SRC00078/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000365	SRC00080/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000366	SRC00079/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 67	SRC00078/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 68	SRC00080/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 69	SRC00079/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 70	SRC00078/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 71	SRC00080/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 72	SRC00079/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000373	SRC00078/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000374	SRC00080/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000375	SRC00079/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000376	SRC00078/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000377	SRC00080/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000378	SRC00079/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003379	SRC00078/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003380	SRC00080/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003381	SRC00079/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00003382	SRC00080/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003383	SRC00080/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003384	SRC00081/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 85	SRC00083/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 86	SRC00082/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00003 87	SRC00081/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 88	SRC00083/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 89	SRC00082/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 90	SRC00081/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000391	SRC00083/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000392	SRC00082/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000393	SRC00081/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000394	SRC00083/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000395	SRC00082/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000396	SRC00081/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 97	SRC00083/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 98	SRC00082/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 99	SRC00081/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 00	SRC00083/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 01	SRC00082/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 02	SRC00081/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004 03	SRC00083/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 04	SRC00082/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 05	SRC00081/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 06	SRC00083/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 07	SRC00082/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 08	SRC00081/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000409	SRC00083/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000410	SRC00082/p9	covered	Alternate extracted slide text page 9; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000411	SRC00083/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000412	SRC00083/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000413	SRC00083/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000414	SRC00084/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000415	SRC00086/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000416	SRC00085/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000417	SRC00084/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000418	SRC00086/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000419	SRC00085/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000420	SRC00084/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004021	SRC00086/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004022	SRC00085/p3	covered	Alternate extracted slide text page 3; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00004023	SRC00086/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004024	SRC00086/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004025	SRC00086/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004026	SRC00087/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000427	SRC00089/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000428	SRC00088/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000429	SRC00087/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000430	SRC00089/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000431	SRC00088/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000432	SRC00087/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000433	SRC00089/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000434	SRC00088/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000435	SRC00087/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000436	SRC00089/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000437	SRC00088/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000438	SRC00087/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000439	SRC00089/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000440	SRC00088/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000441	SRC00087/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000442	SRC00089/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000443	SRC00088/p6	covered	Alternate extracted slide text page 6; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000444	SRC00089/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000445	SRC00089/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000446	SRC00089/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000447	SRC00090/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000448	SRC00092/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000449	SRC00091/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000450	SRC00090/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000451	SRC00092/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000452	SRC00091/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000453	SRC00090/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000454	SRC00092/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000455	SRC00091/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000456	SRC00090/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004 57	SRC00092/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 58	SRC00091/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 59	SRC00090/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 60	SRC00092/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 61	SRC00091/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 62	SRC00090/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004063	SRC00092/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004064	SRC00091/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00004065	SRC00090/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004066	SRC00092/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004067	SRC00091/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00004068	SRC00092/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000469	SRC00092/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000470	SRC00093/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000471	SRC00095/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000472	SRC00094/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000473	SRC00093/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000474	SRC00095/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000475	SRC00094/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000476	SRC00093/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000477	SRC00095/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000478	SRC00094/p3	covered	Alternate extracted slide text page 3; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000479	SRC00093/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000480	SRC00095/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000481	SRC00094/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000482	SRC00093/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000483	SRC00095/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000484	SRC00094/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000485	SRC00093/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000486	SRC00095/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000487	SRC00094/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 88	SRC00093/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 89	SRC00095/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 90	SRC00094/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 91	SRC00093/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 92	SRC00095/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 93	SRC00094/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000494	SRC00095/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000495	SRC00095/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000496	SRC00095/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000497	SRC00096/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000498	SRC00098/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000499	SRC00097/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000500	SRC00096/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000501	SRC00098/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000502	SRC00097/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000503	SRC00096/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000504	SRC00098/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000505	SRC00097/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000506	SRC00096/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000507	SRC00098/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000508	SRC00097/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000509	SRC00096/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000510	SRC00098/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000511	SRC00097/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000512	SRC00096/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000513	SRC00098/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000514	SRC00097/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000515	SRC00098/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000516	SRC00098/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000517	SRC00099/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000518	SRC00101/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000519	SRC00100/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000520	SRC00099/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000521	SRC00101/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000522	SRC00100/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000523	SRC00101/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000524	SRC00101/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000525	SRC00102/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000526	SRC00104/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000527	SRC00103/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000528	SRC00104/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000529	SRC00104/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000530	SRC00105/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000531	SRC00107/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000532	SRC00106/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000533	SRC00105/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000534	SRC00107/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000535	SRC00106/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000536	SRC00105/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000537	SRC00107/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000538	SRC00106/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000539	SRC00105/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000540	SRC00107/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000541	SRC00106/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000542	SRC00107/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000543	SRC00107/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000544	SRC00108/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000545	SRC00110/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000546	SRC00109/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000547	SRC00108/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000548	SRC00110/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000549	SRC00109/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000550	SRC00108/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000551	SRC00110/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000552	SRC00109/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000553	SRC00108/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000554	SRC00110/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000555	SRC00109/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000556	SRC00108/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000557	SRC00110/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000558	SRC00109/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000559	SRC00110/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000560	SRC00110/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000561	SRC00111/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000562	SRC00113/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000563	SRC00112/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000564	SRC00111/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000565	SRC00113/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000566	SRC00112/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000567	SRC00111/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000568	SRC00113/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000569	SRC00112/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000570	SRC00113/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000571	SRC00113/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00005 72	SRC00113/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 73	SRC00113/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/Slides/012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.ppt; visual page included as vector figure. Human layout/content review is pending.

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